

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

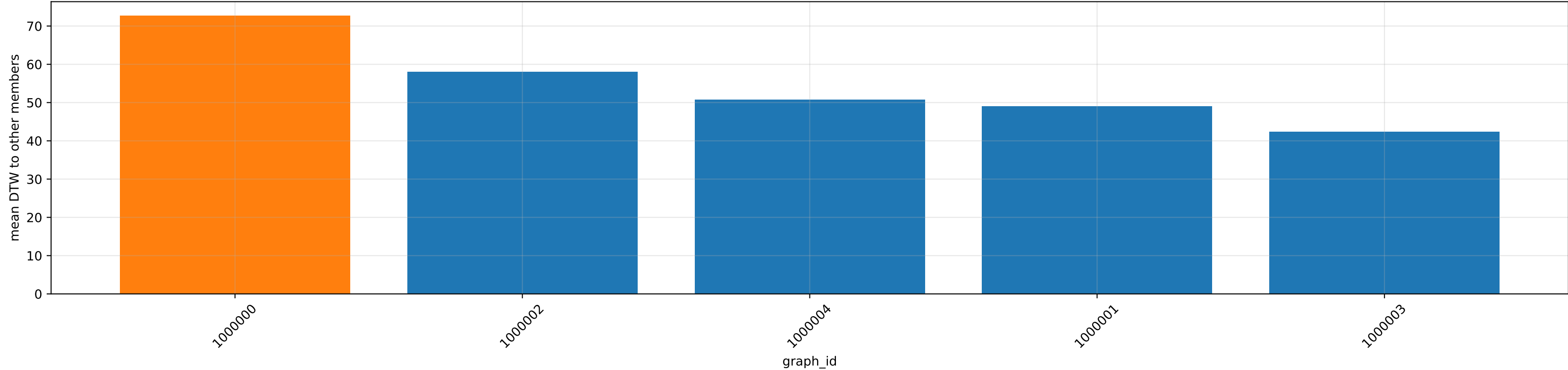
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	26	0	4 vs 1	1000000	1000000	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
72.732	42.463	nan

Per-member mean DTW



Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

First applied split pairwise DTW metadata | seed=0

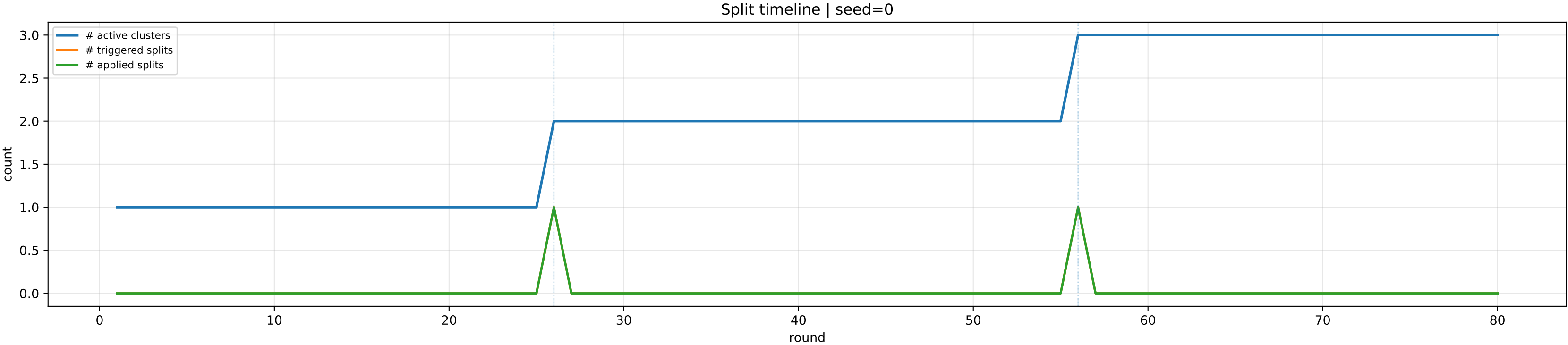
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	26	0	5	1000000.0	0.055	72.732	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 85.7 at (1000000, 1000001)

graph_id	1000000	1000001	1000002	1000003	1000004
1000000	0.0	85.7	75.9	70.2	59.2
1000001	85.7	0.0	45.0	22.9	42.5
1000002	75.9	45.0	0.0	43.2	68.0
1000003	70.2	22.9	43.2	0.0	33.1
1000004	59.2	42.5	68.0	33.1	0.0

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003



Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	26, 56	26, 56	2, 3, 4	2:1, 3:1, 4:3

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Split lineage story | seed=0

Cluster 0
5 clients

round 26

Cluster 1
4 clients

round 56

Cluster 4
3 clients

split off -> Cluster 2
1 client
specialized_cycle2:1

split off -> Cluster 3
1 client
specialized_cycle6:1

Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
26	0	5	166.876 < 169.079 , 242.614 > 241.960	C2 (1 client)	specialized_cycle2:1	C1 (4 clients)	specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1
56	1	4	170.319 < 171.521 , 251.966 > 227.443	C3 (1 client)	specialized_cycle6:1	C4 (3 clients)	specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

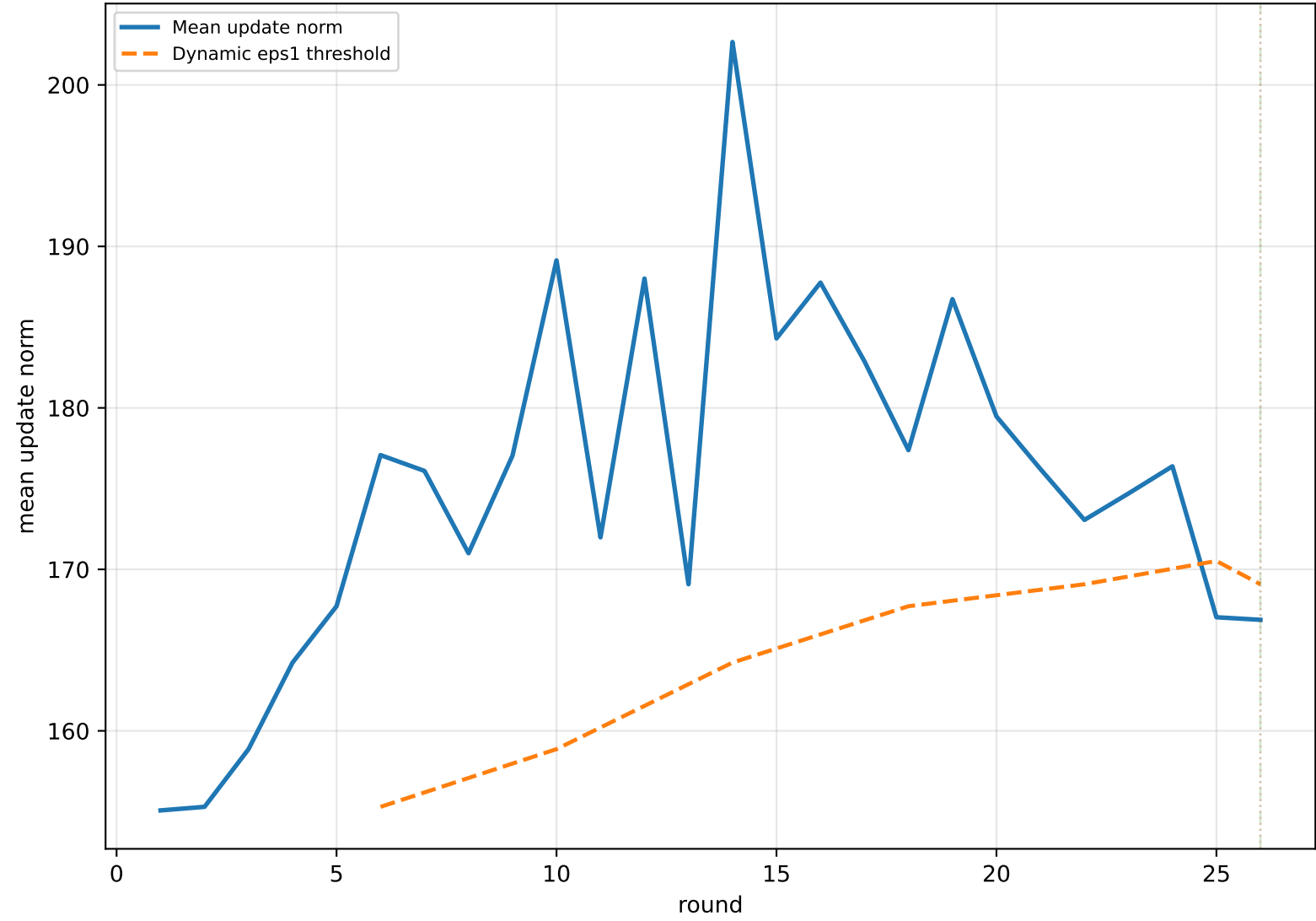
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
2	1	1000000	0	specialized_cycle2:1
3	1	1000004	4	specialized_cycle6:1
4	3	1000001, 1000002, 1000003	1, 2, 3	specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

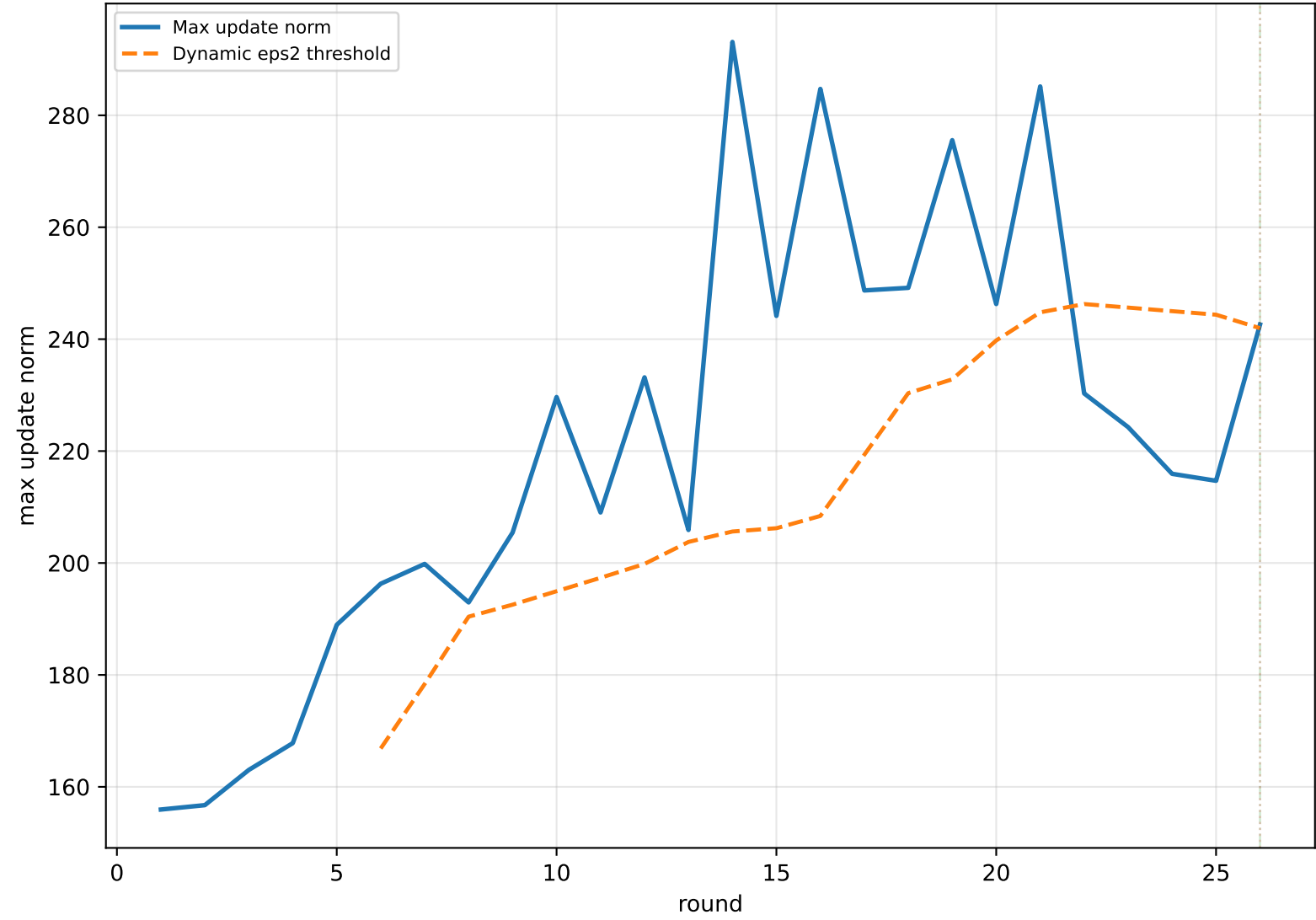
Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (0-1)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

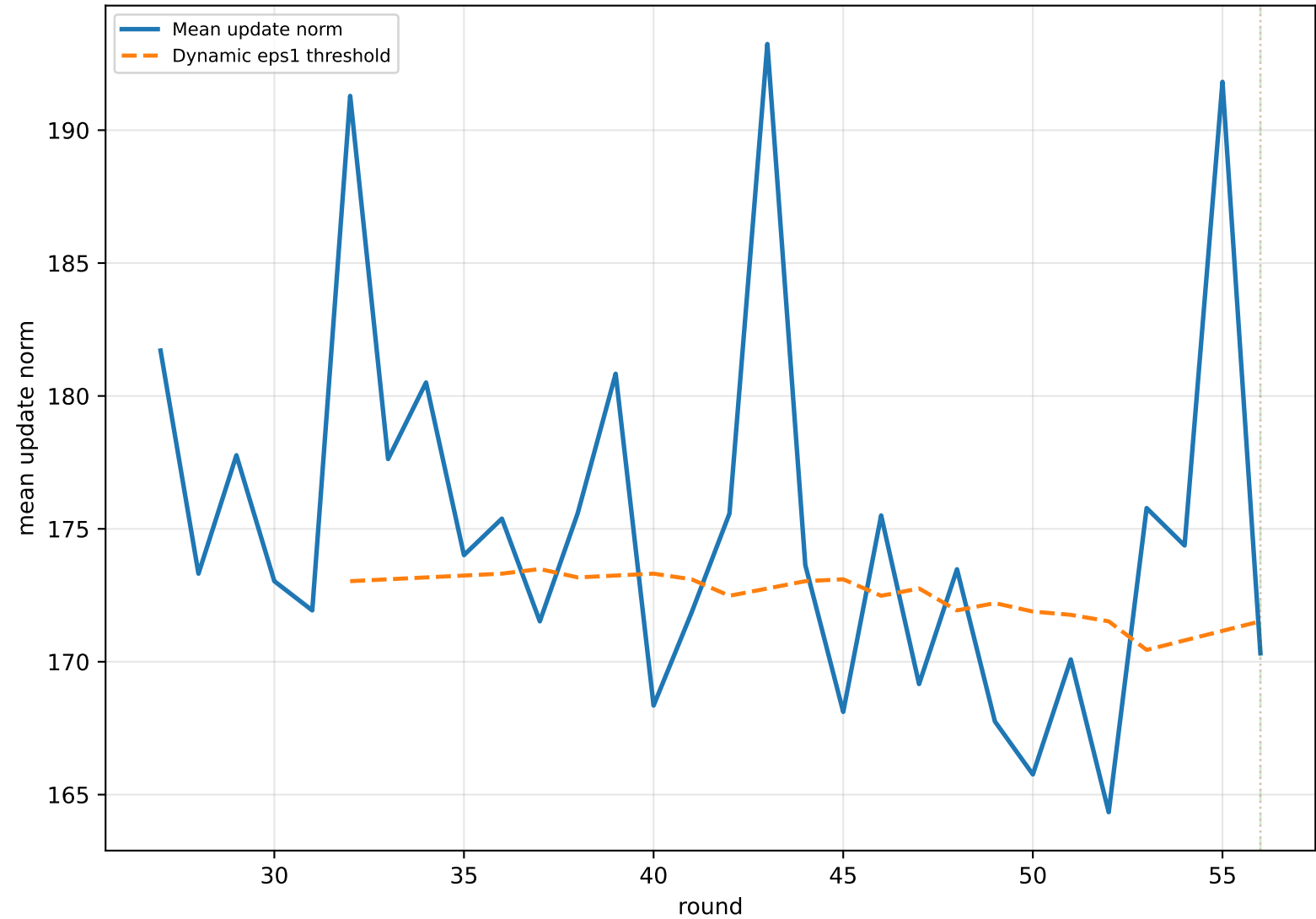
Cluster 0 | Mean update norm



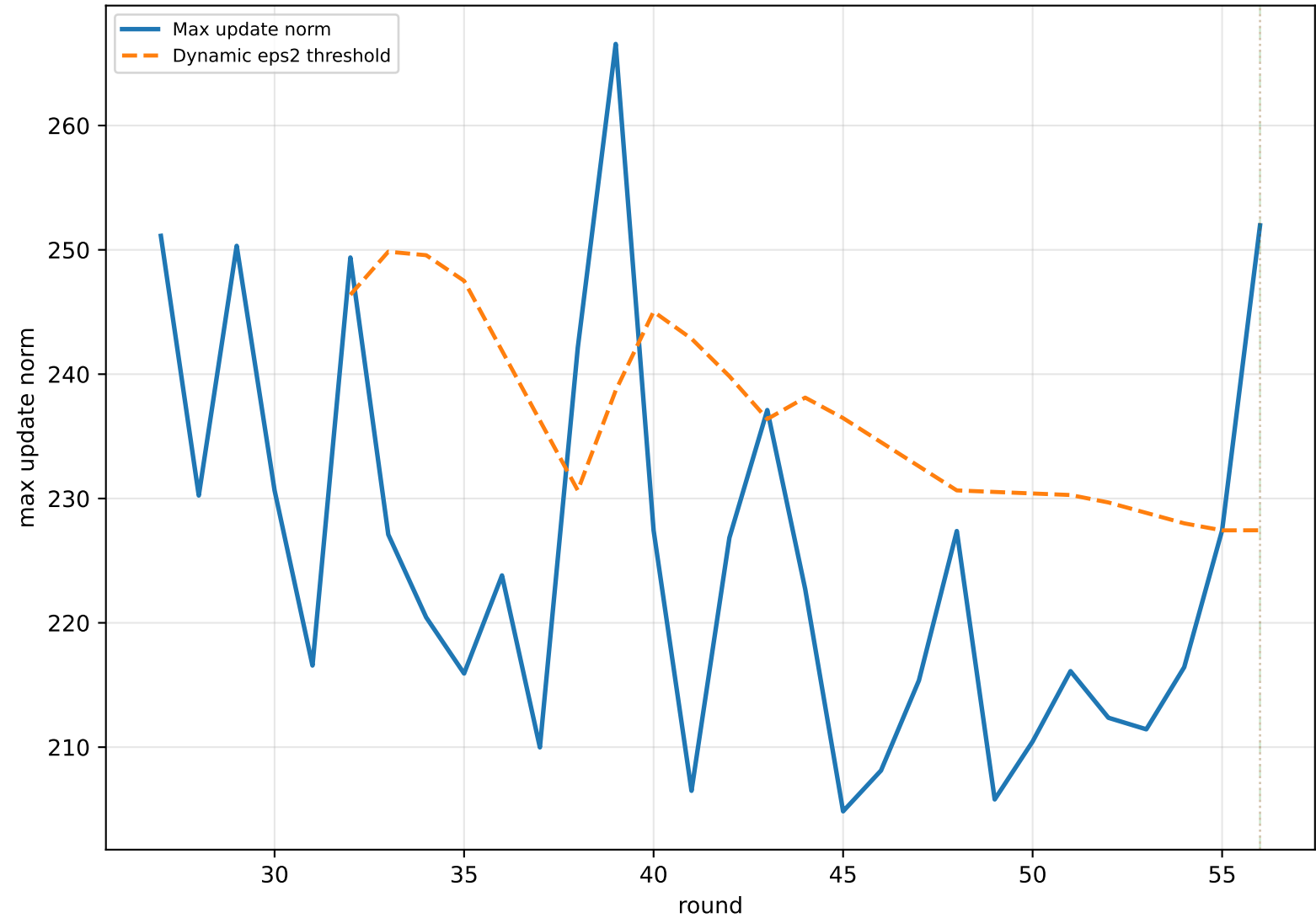
Cluster 0 | Max update norm



Cluster 1 | Mean update norm



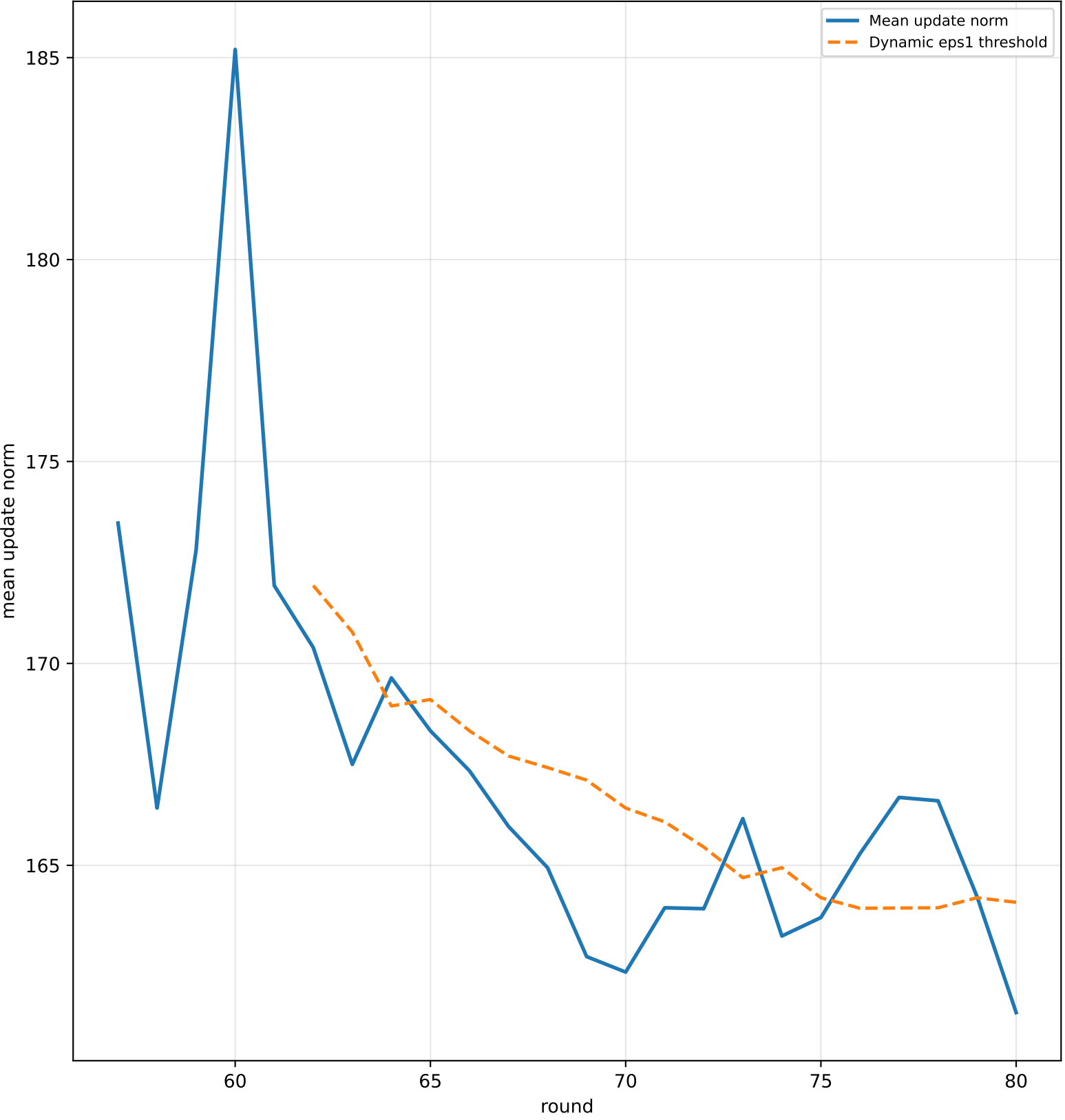
Cluster 1 | Max update norm



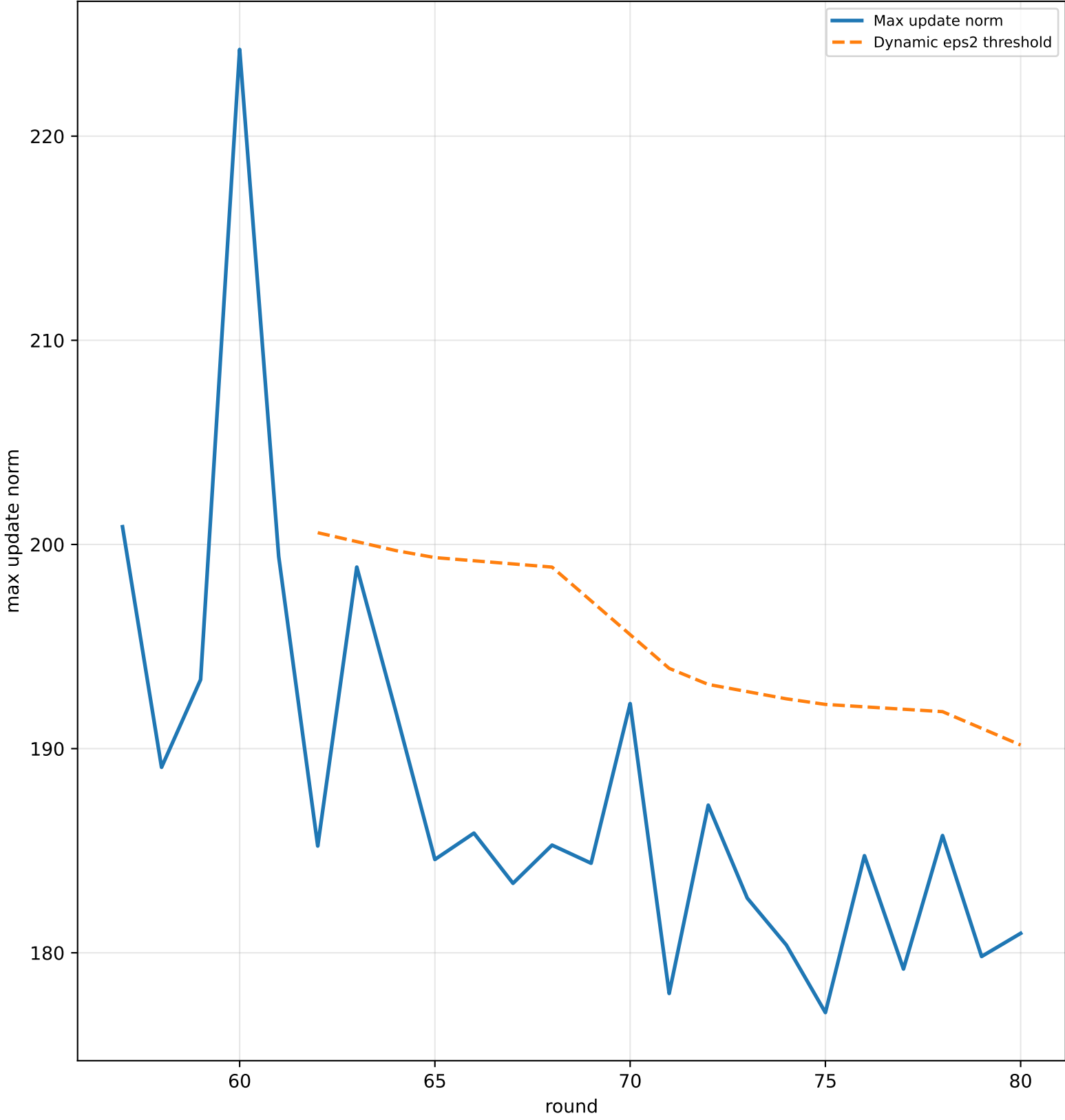
Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (4-4)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

Cluster 4 | Mean update norm



Cluster 4 | Max update norm



Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

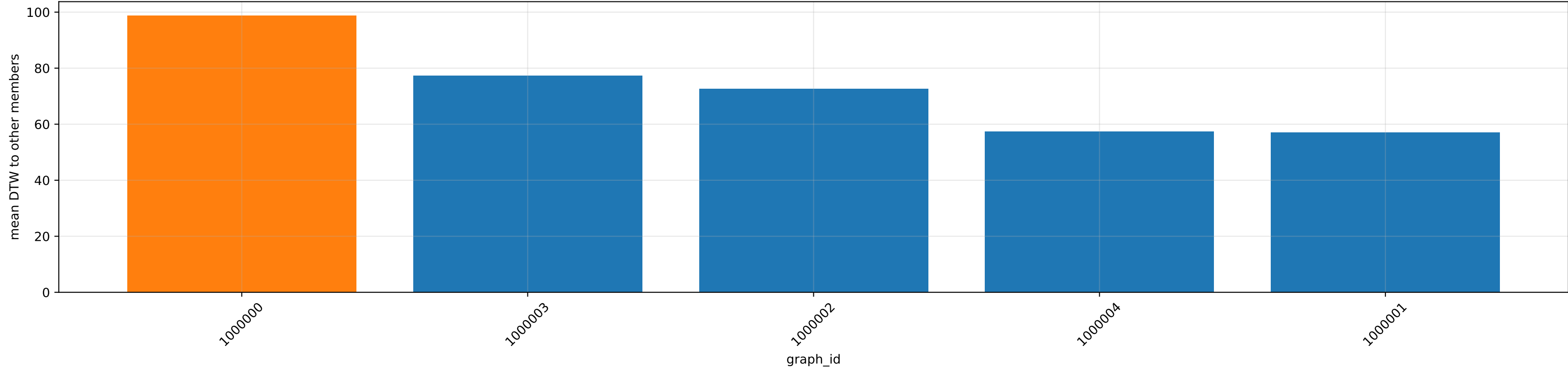
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	22	0	4 vs 1	1000000	1000000	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
98.797	55.170	nan

Per-member mean DTW



Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

First applied split pairwise DTW metadata | seed=1

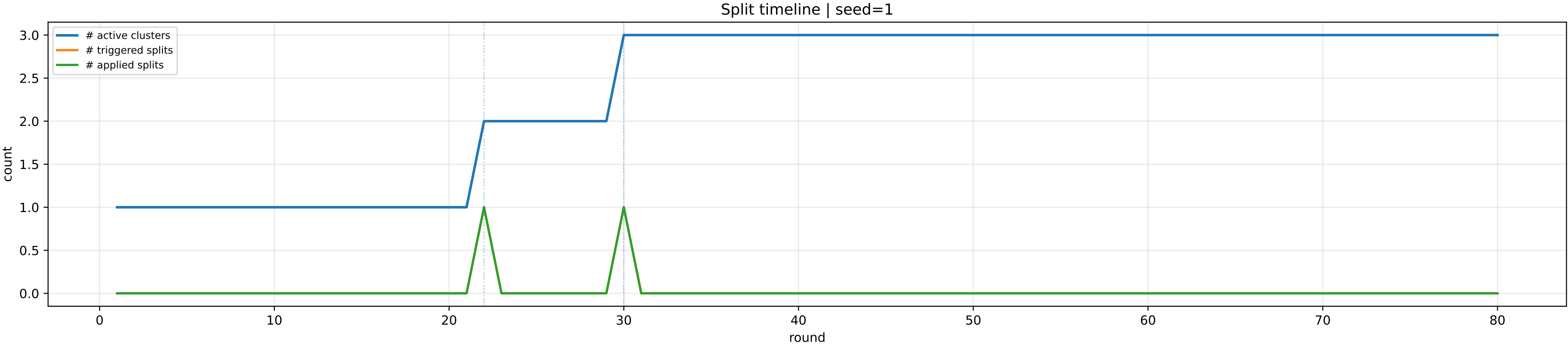
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	22	0	5	1000000.0	0.041	98.797	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 116.3 at (1000000, 1000002)

graph_id	1000000	1000001	1000002	1000003	1000004
1000000	0.0	79.2	116.3	116.2	83.5
1000001	79.2	0.0	59.0	63.5	26.3
1000002	116.3	59.0	0.0	62.3	52.9
1000003	116.2	63.5	62.3	0.0	67.1
1000004	83.5	26.3	52.9	67.1	0.0

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003



Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	22, 30	22, 30	2, 3, 4	2:1, 3:3, 4:1

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Split lineage story | seed=1

Cluster 0
5 clients

round 22

Cluster 1
4 clients

round 30

Cluster 3
3 clients

split off -> Cluster 2
1 client
specialized_cycle2:1

split off -> Cluster 4
1 client
specialized_cycle3:1

Applied split story | seed=1

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
22	0	5	172.731 < 174.492 , 256.134 > 243.426	C2 (1 client)	specialized_cycle2:1	C1 (4 clients)	specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1
30	1	4	179.015 < 181.208 , 280.352 > 266.822	C4 (1 client)	specialized_cycle3:1	C3 (3 clients)	specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | final active clusters

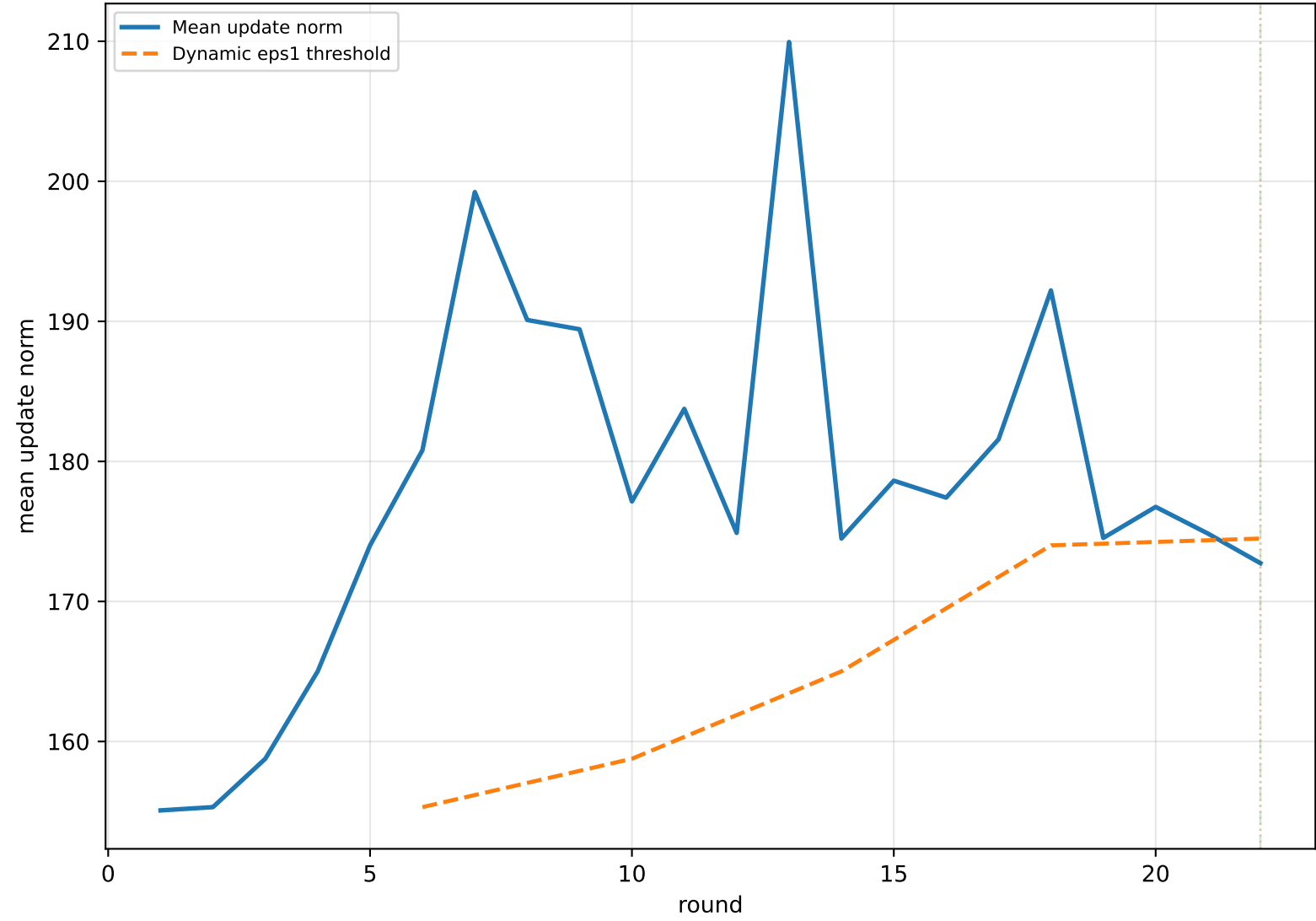
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Final active clusters | seed=1

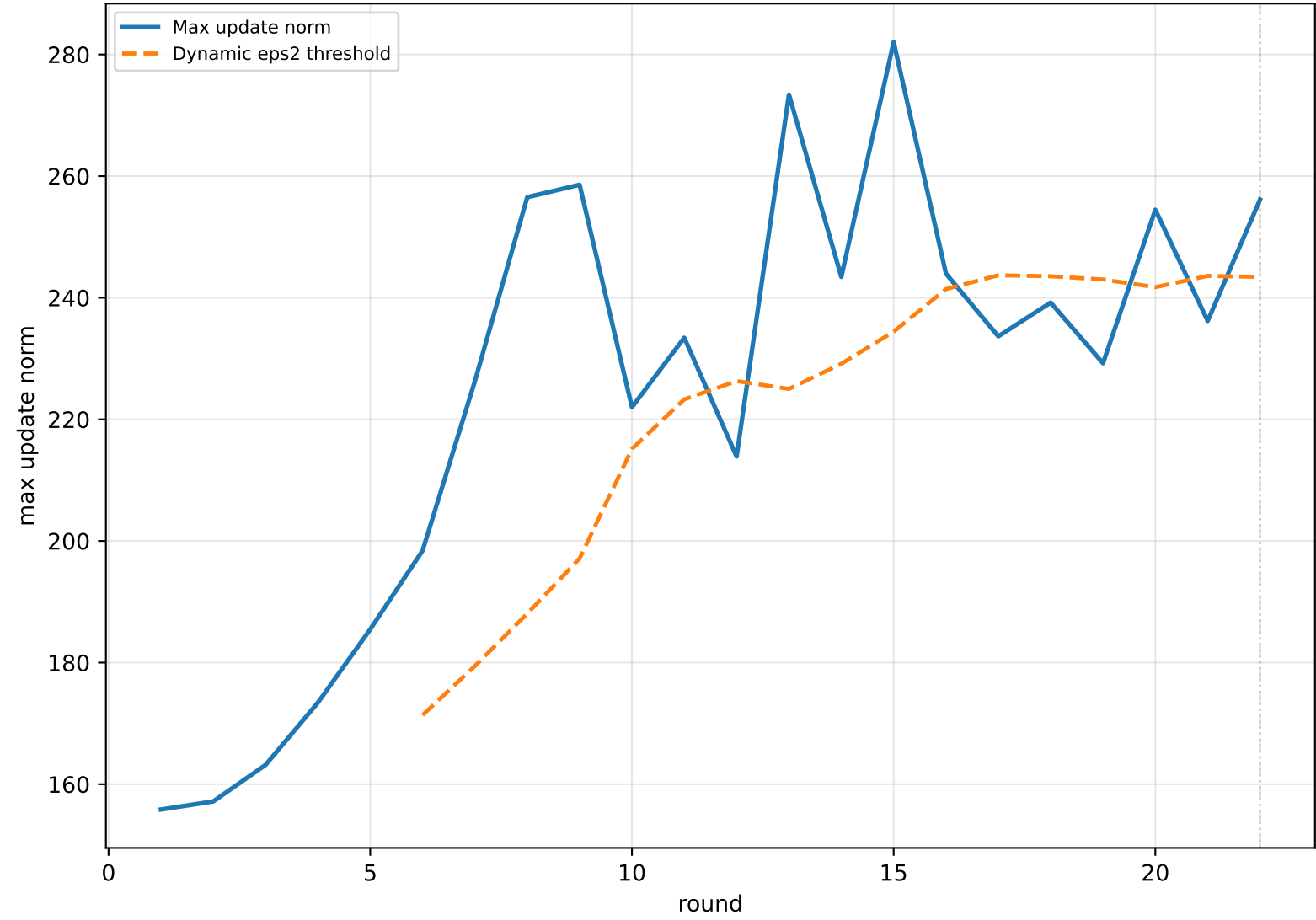
cluster_id	cluster_size	members	member_indices	family_mix
2	1	1000000	0	specialized_cycle2:1
3	3	1000002, 1000003, 1000004	2, 3, 4	specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1
4	1	1000001	1	specialized_cycle3:1

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

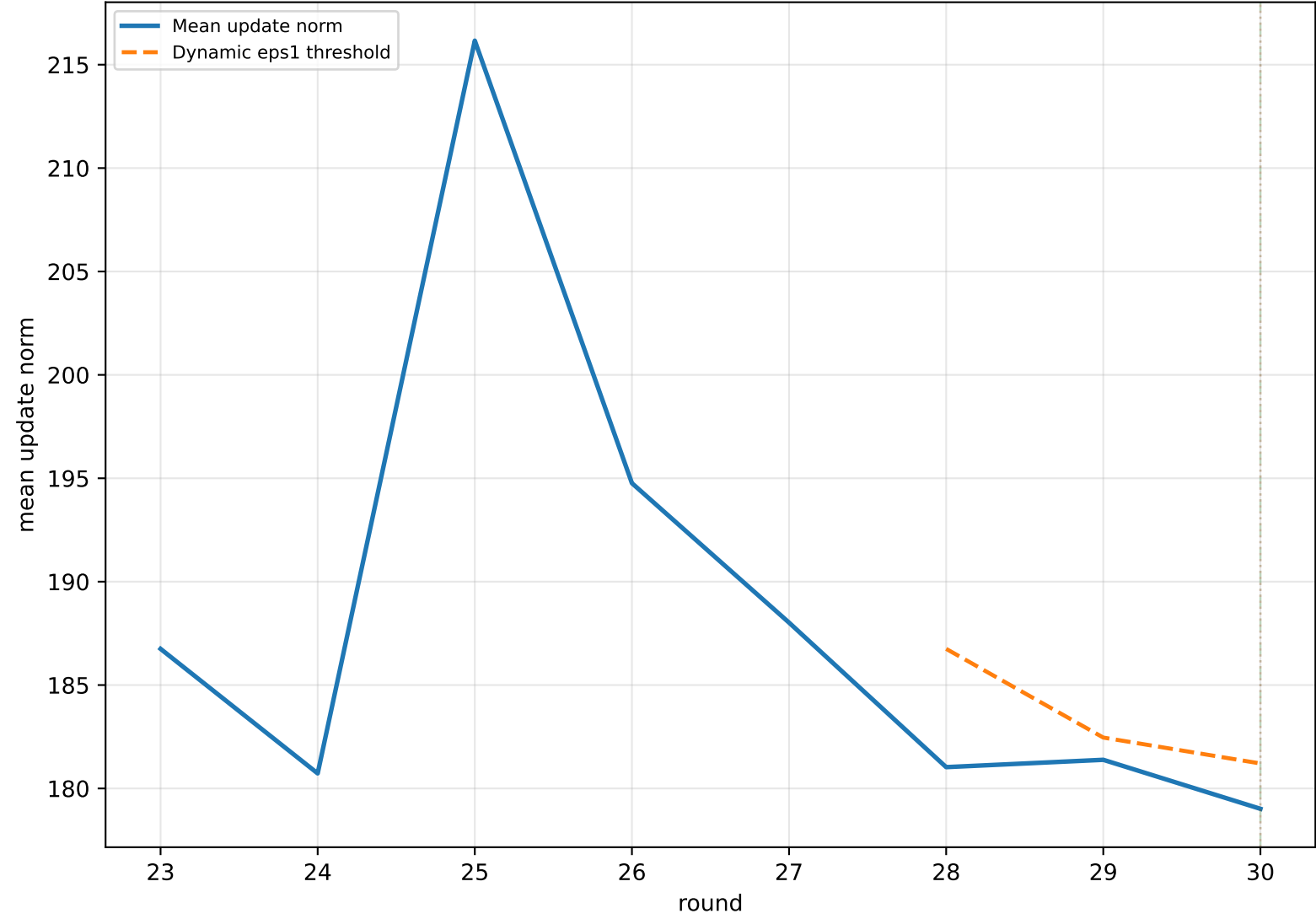
Cluster 0 | Mean update norm



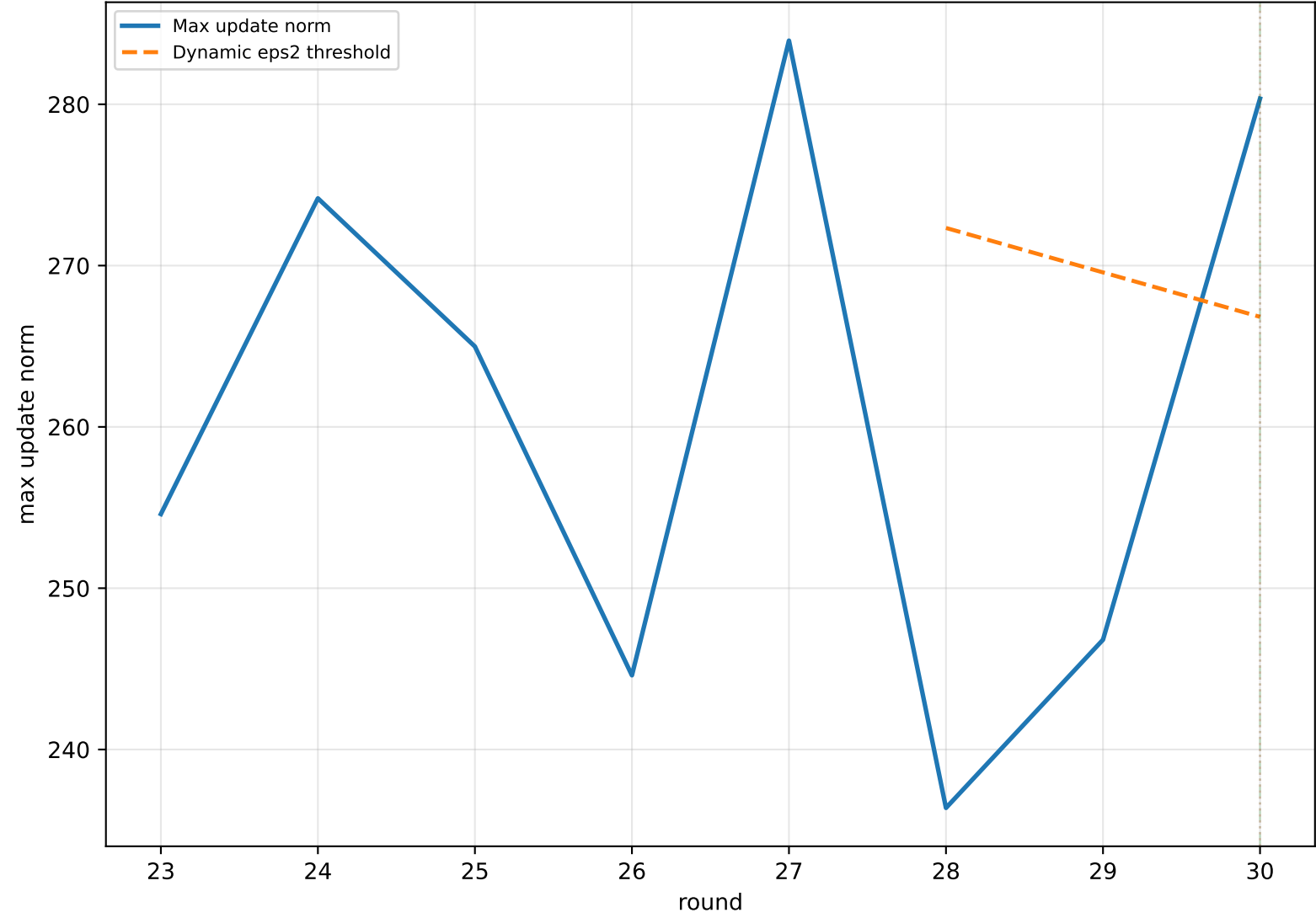
Cluster 0 | Max update norm



Cluster 1 | Mean update norm



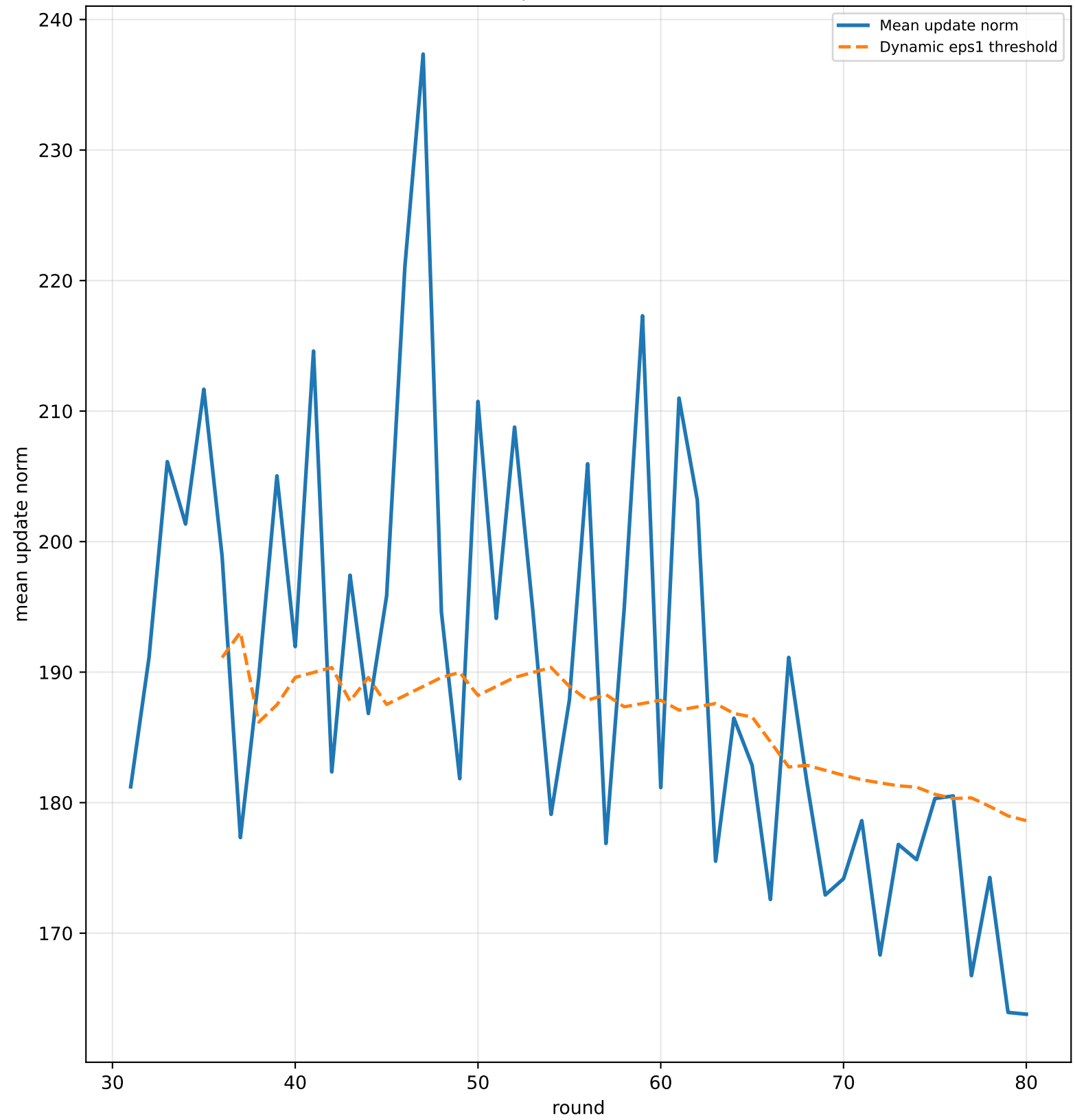
Cluster 1 | Max update norm



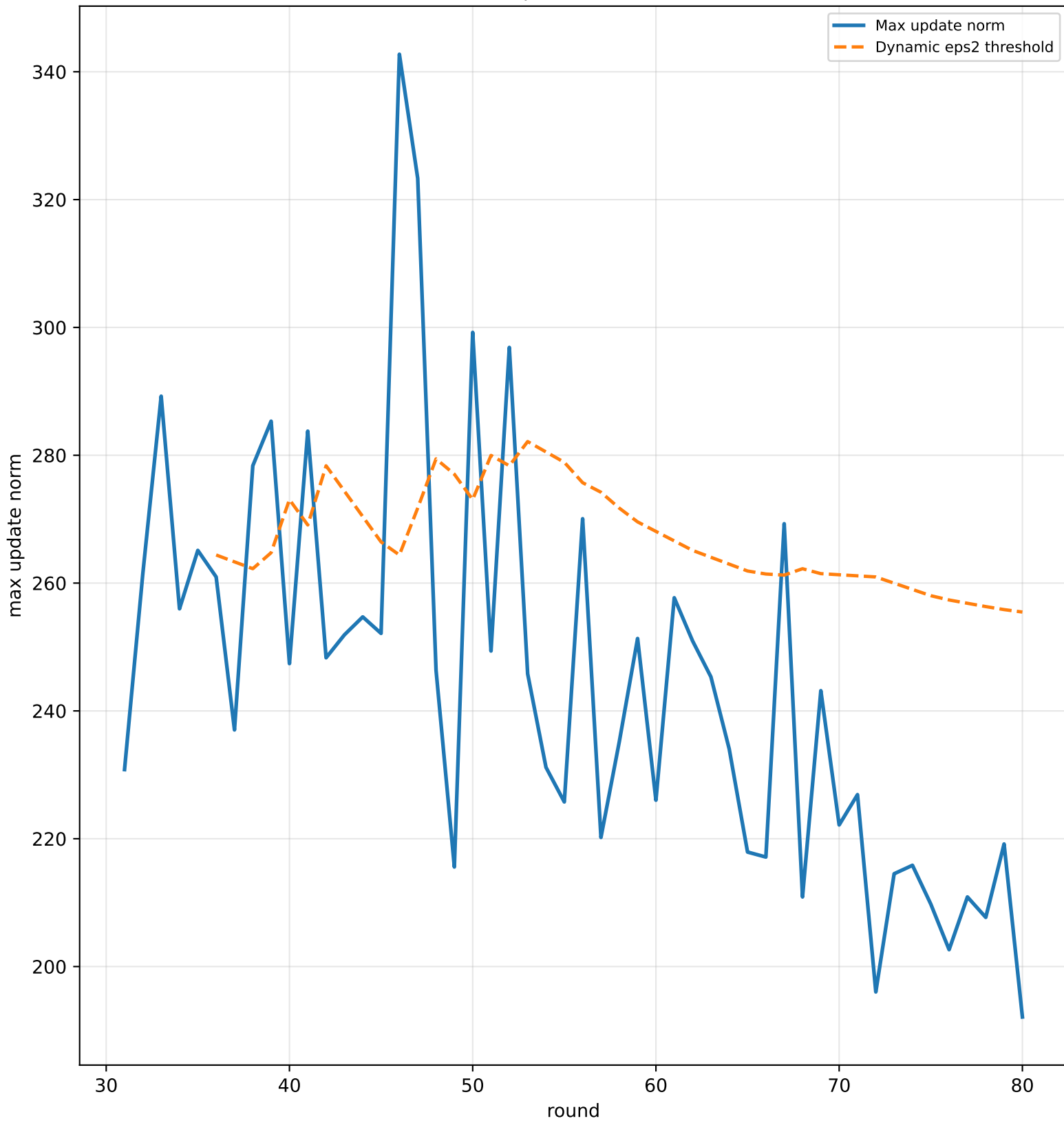
Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | norm / threshold panels (3-3)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

Cluster 3 | Mean update norm



Cluster 3 | Max update norm



Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

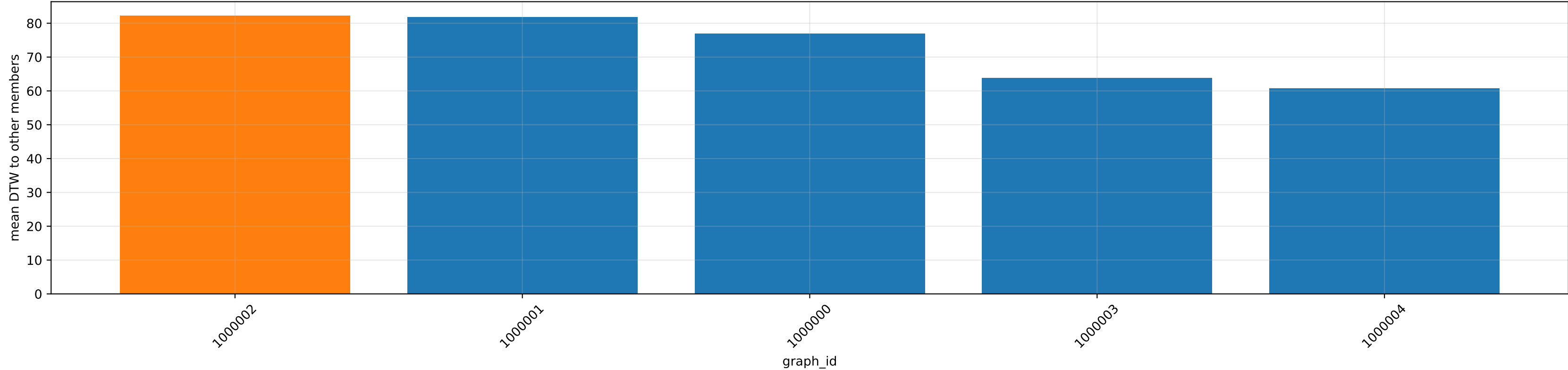
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	24	0	1 vs 4	1000002	1000002	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
82.260	nan	67.004

Per-member mean DTW



Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

First applied split pairwise DTW metadata | seed=2

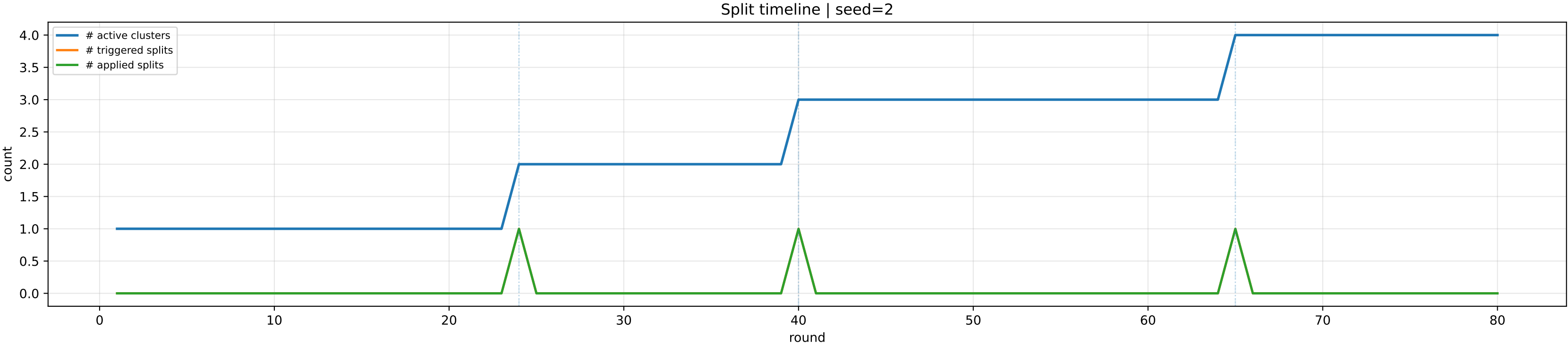
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	24	0	5	1000002.0	0.054	82.260	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 125.4 at (1000001, 1000002)

graph_id	1000000	1000001	1000002	1000003	1000004
1000000	0.0	97.8	51.9	75.9	82.0
1000001	97.8	0.0	125.4	76.9	27.1
1000002	51.9	125.4	0.0	60.4	91.3
1000003	75.9	76.9	60.4	0.0	42.3
1000004	82.0	27.1	91.3	42.3	0.0

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

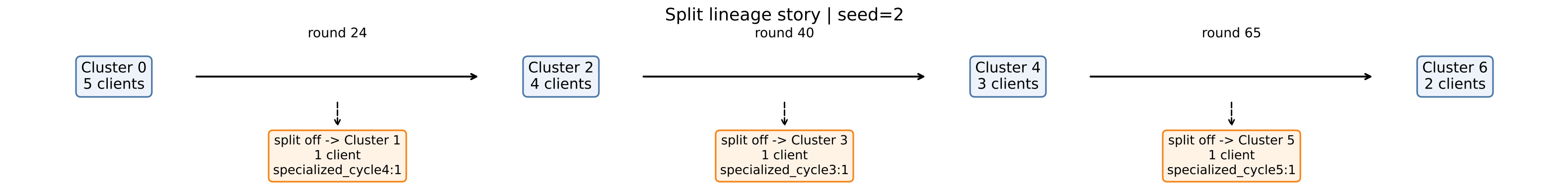


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	24, 40, 65	24, 40, 65	1, 3, 5, 6	1:1, 3:1, 5:1, 6:2

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003



Applied split story | seed=2

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
24	0	5	162.934 < 164.998 , 217.927 > 209.279	C1 (1 client)	specialized_cycle4:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle5:1 , specialized_cycle6:1
40	2	4	164.937 < 166.444 , 224.939 > 212.910	C3 (1 client)	specialized_cycle3:1	C4 (3 clients)	specialized_cycle2:1 , specialized_cycle5:1 , specialized_cycle6:1
65	4	3	164.293 < 164.479 , 200.386 > 197.982	C5 (1 client)	specialized_cycle5:1	C6 (2 clients)	specialized_cycle2:1 , specialized_cycle6:1

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | final active clusters

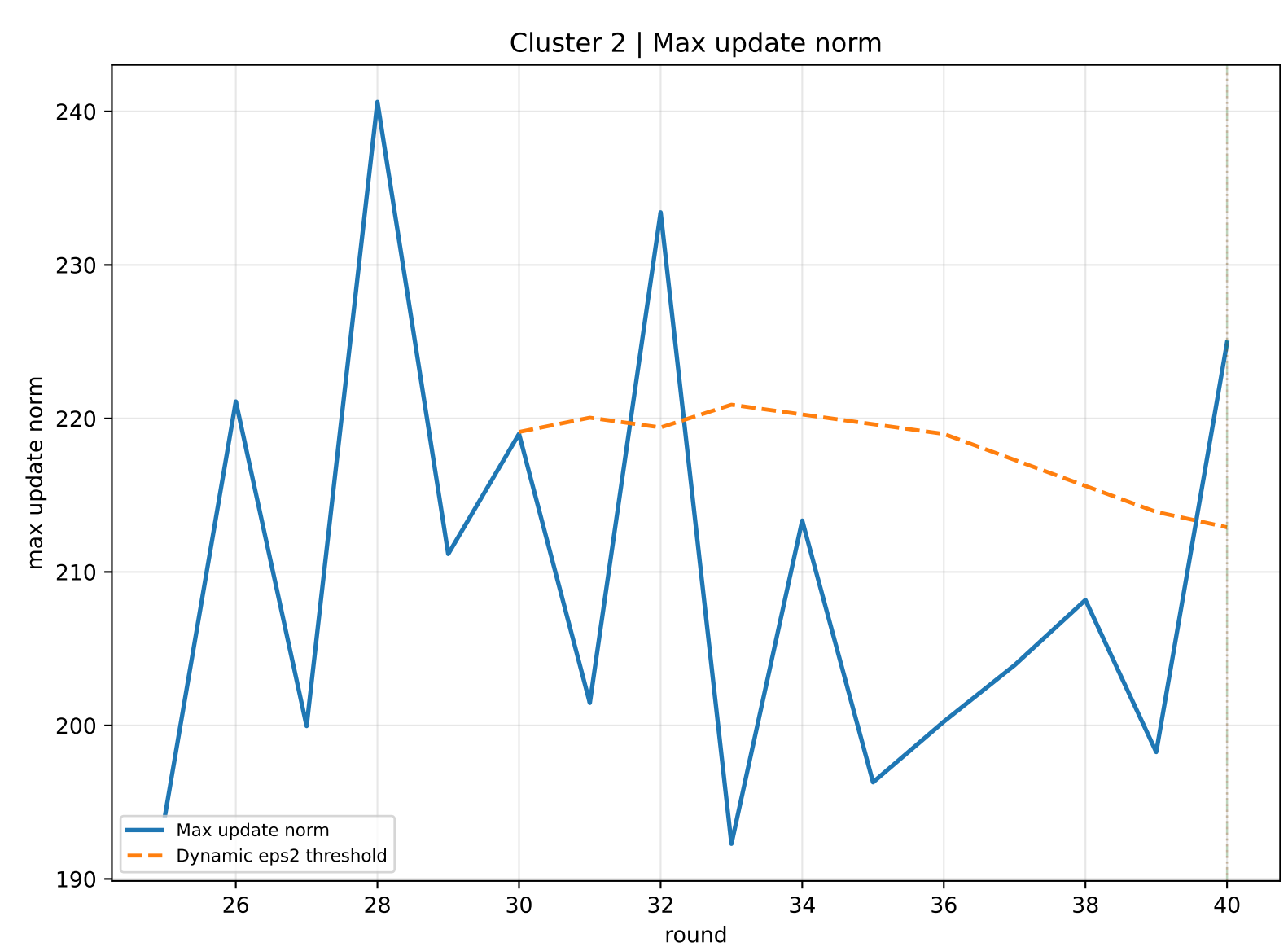
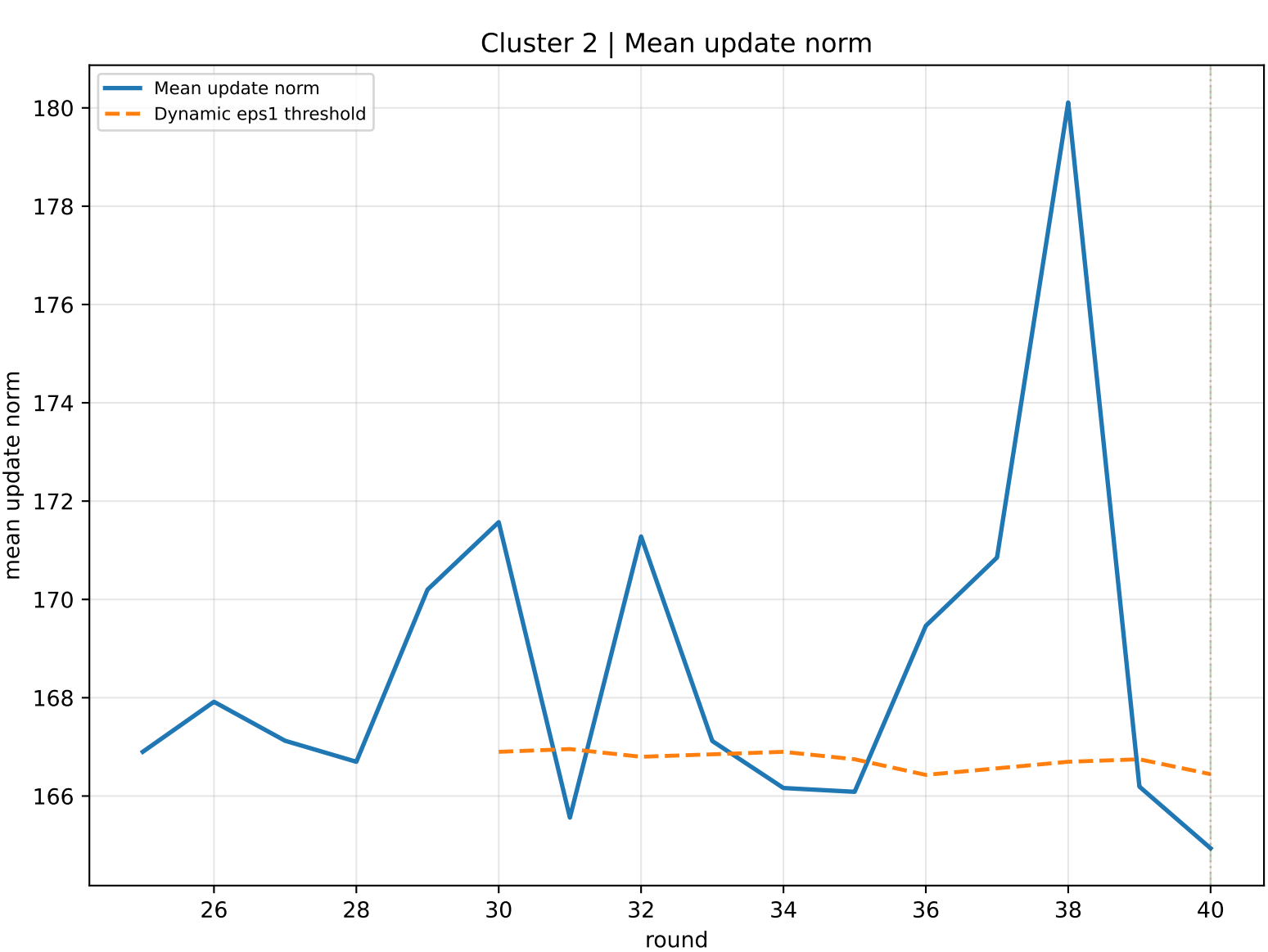
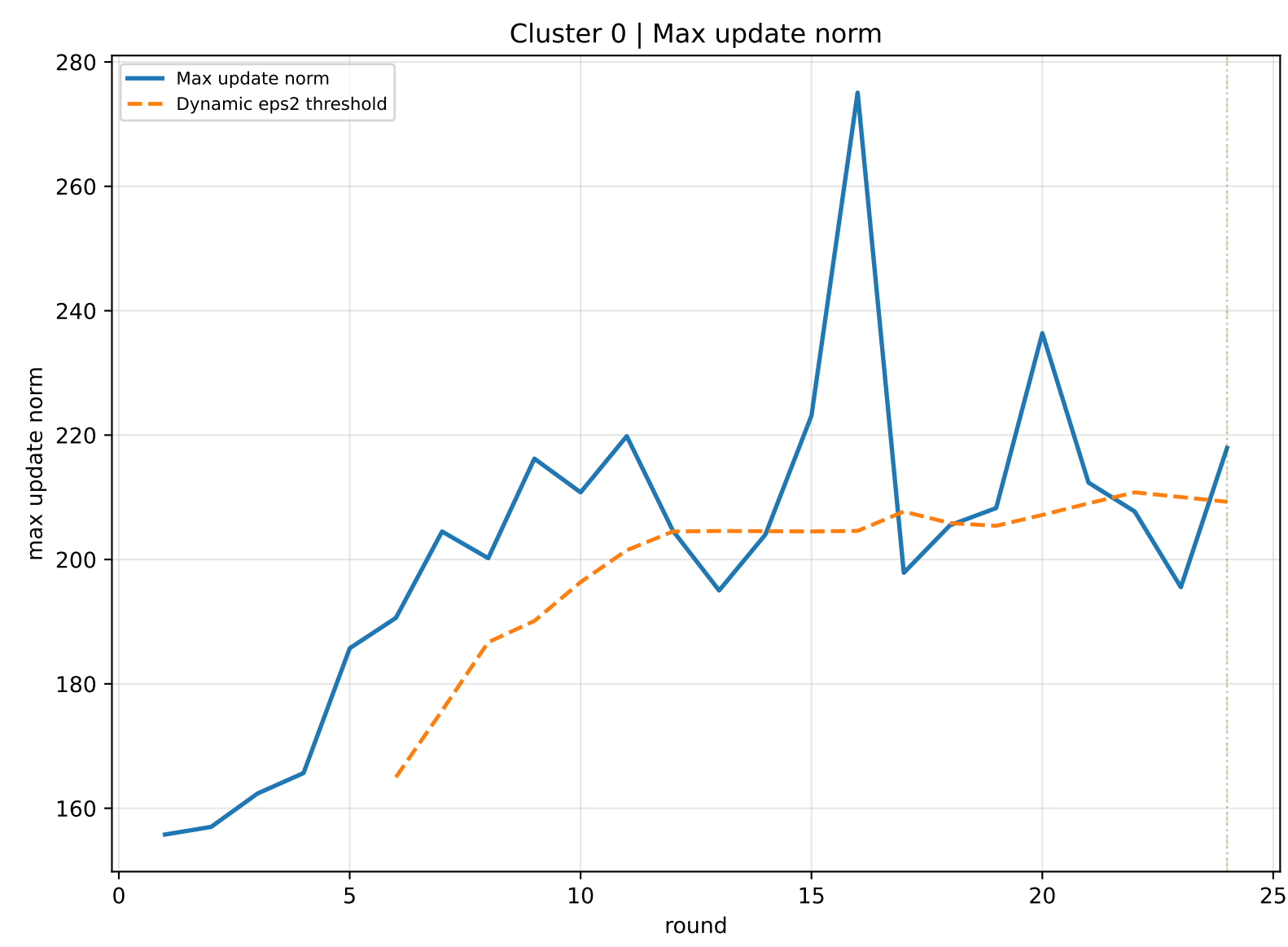
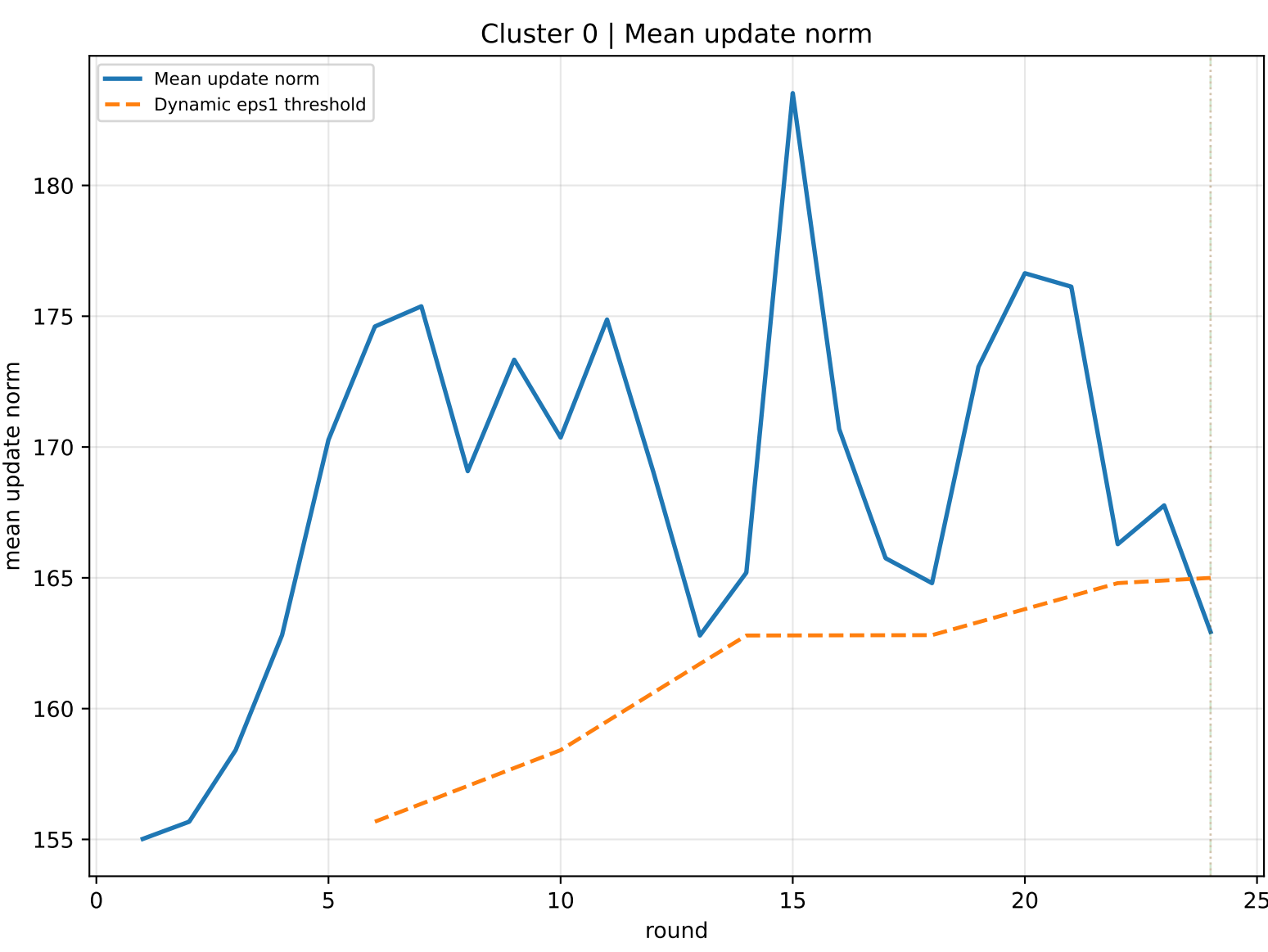
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000002	2	specialized_cycle4:1
3	1	1000001	1	specialized_cycle3:1
5	1	1000003	3	specialized_cycle5:1
6	2	1000000, 1000004	0, 4	specialized_cycle2:1 , specialized_cycle6:1

Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (0-2)

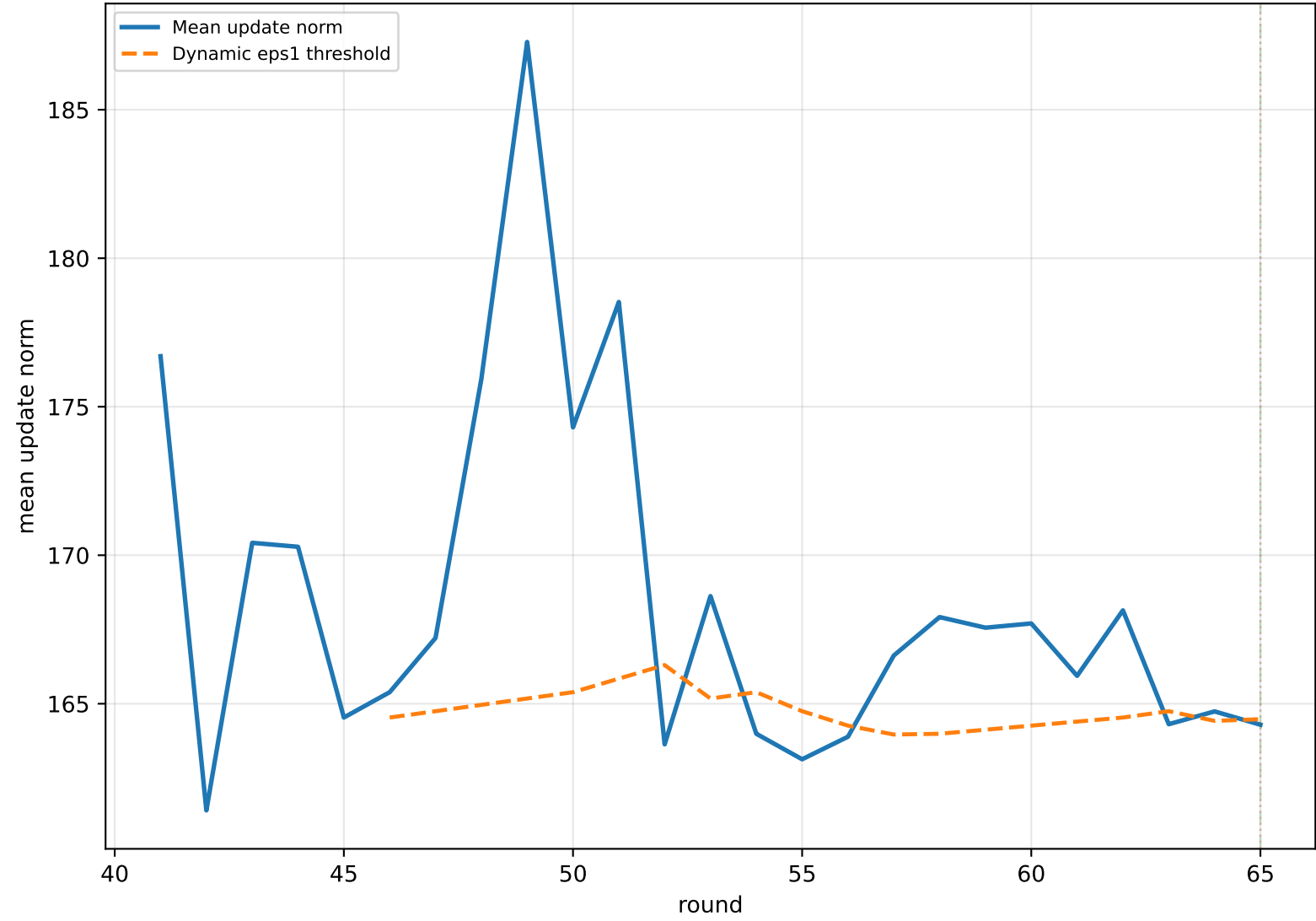
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003



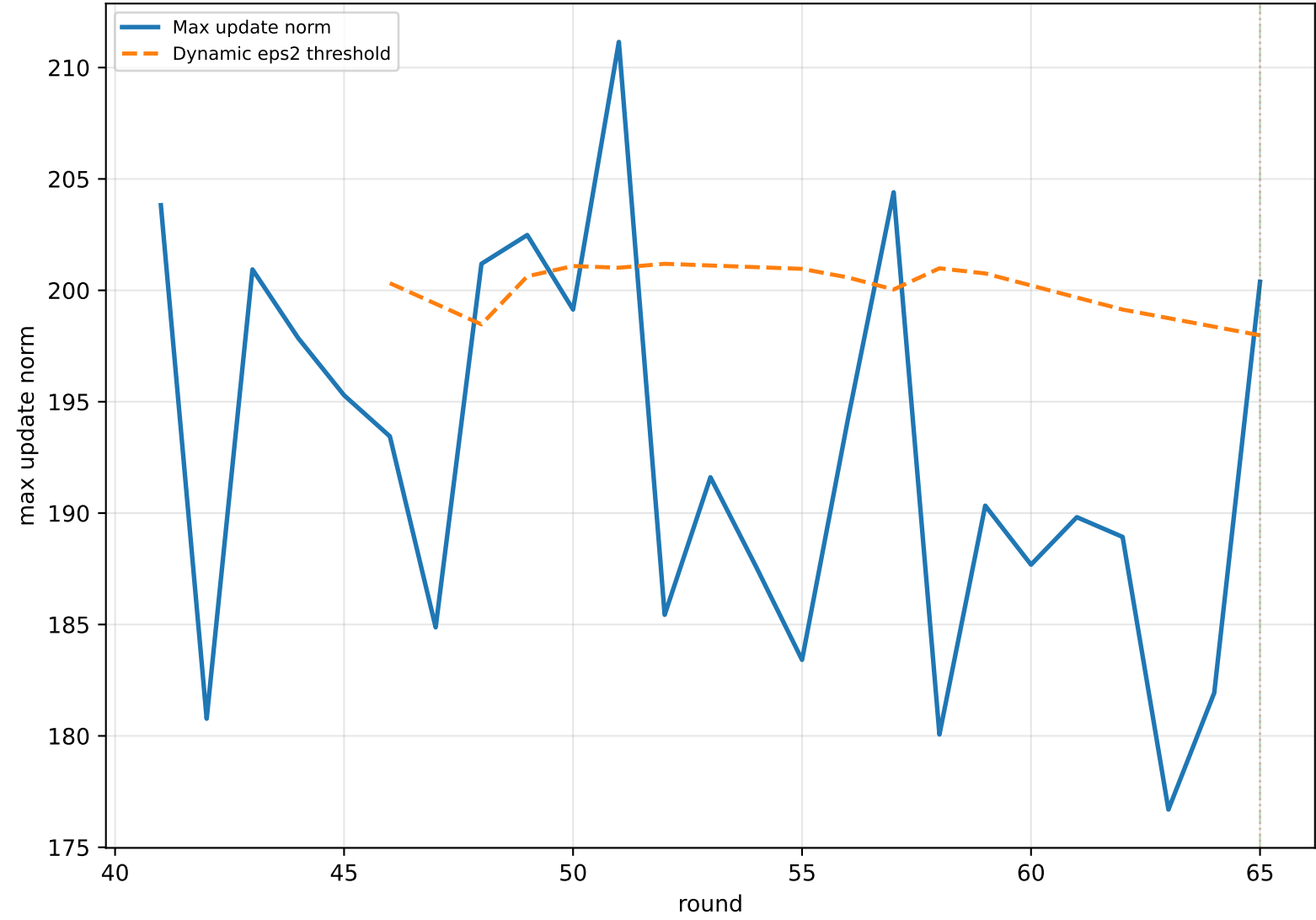
Controlled task-specialization level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

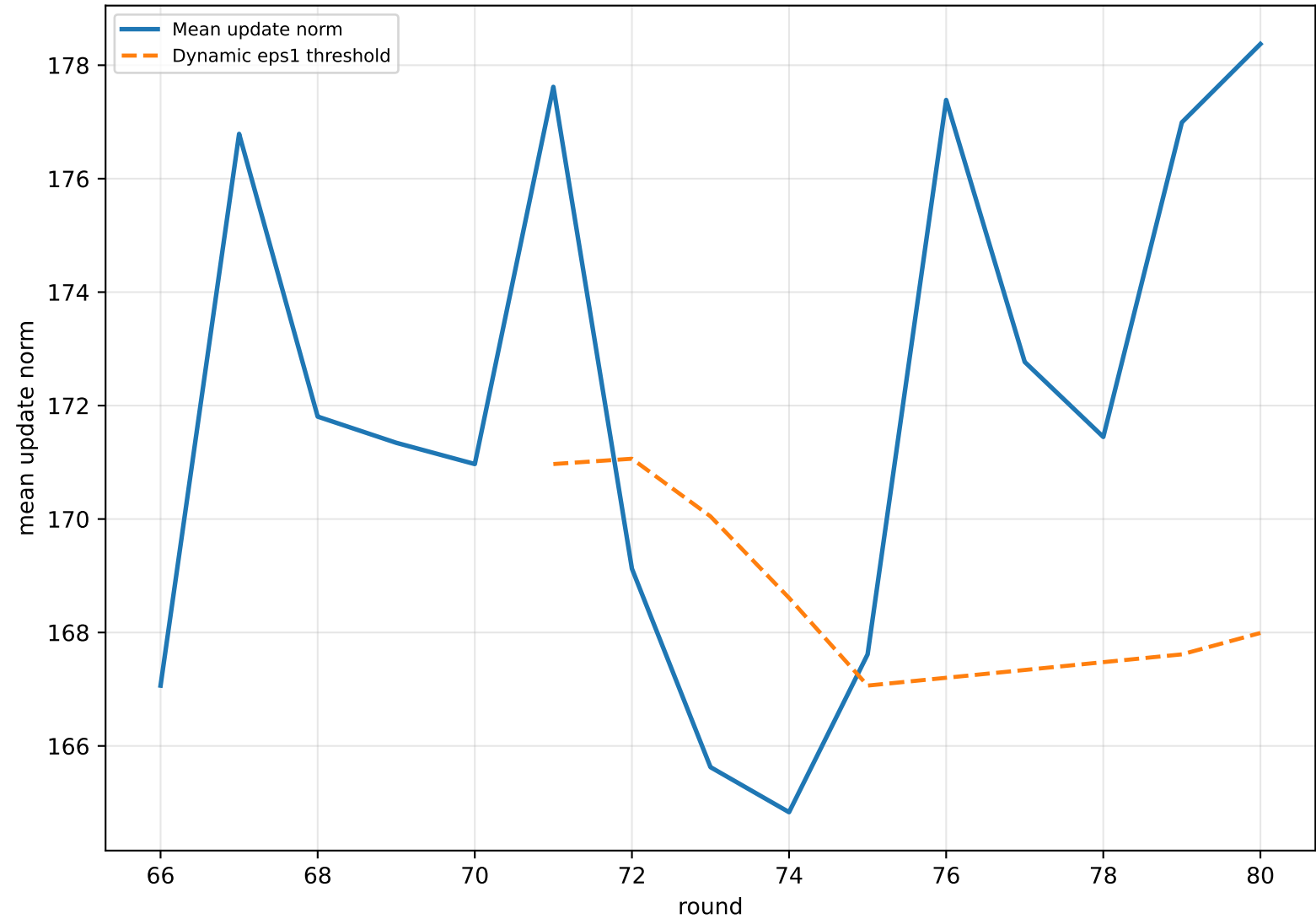
Cluster 4 | Mean update norm



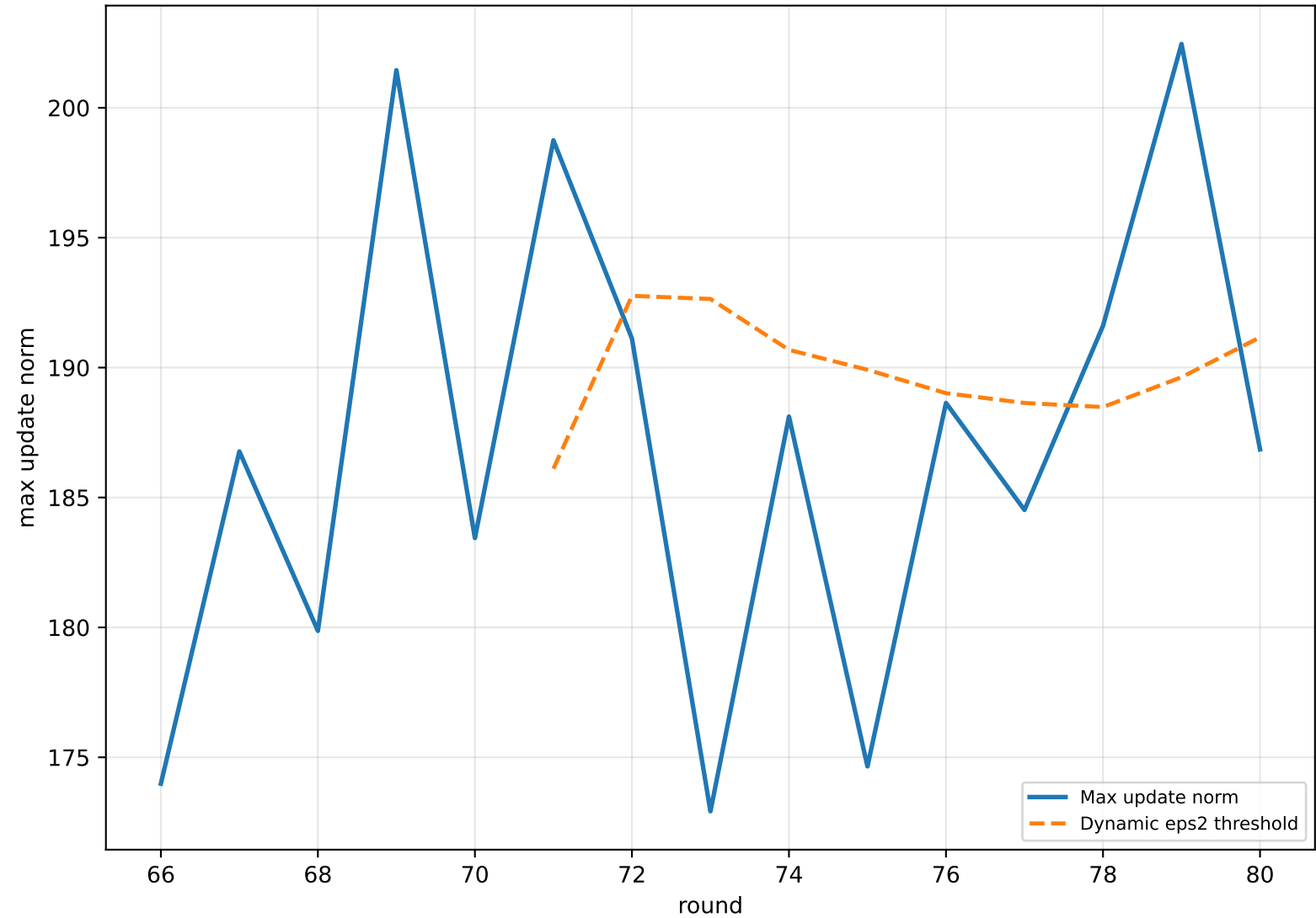
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



Cluster 6 | Max update norm



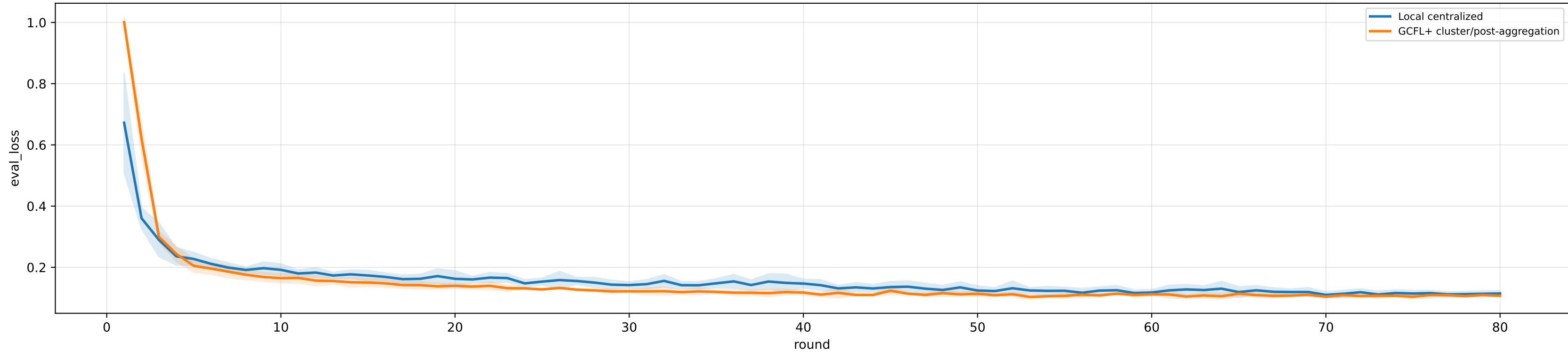
Controlled task-specialization level: Low | GCFL+ diagnostics / aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

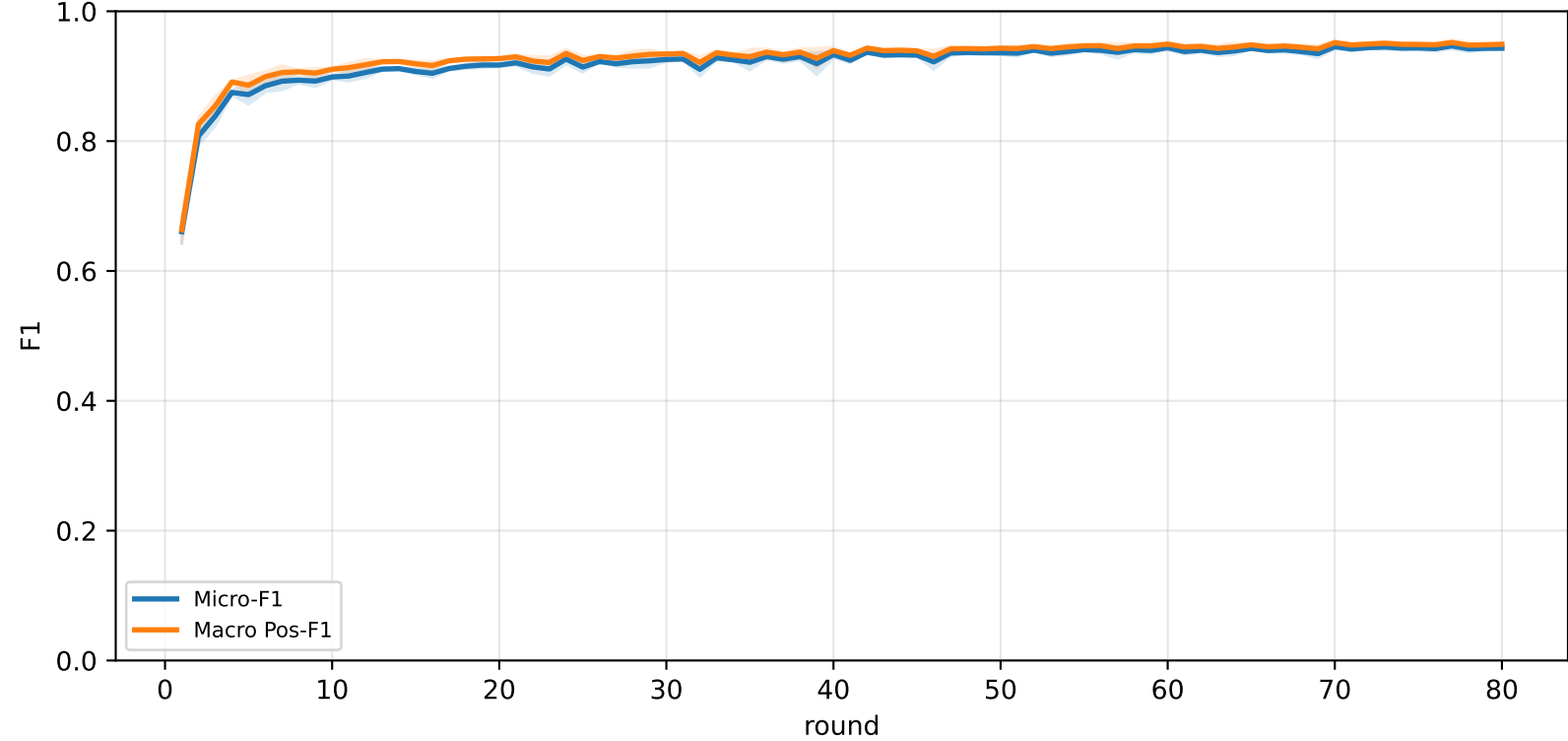
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.1%	94.7% ± 0.1%	99.0% ± 0.2%	98.3% ± 0.3%	96.2% ± 0.3%	93.3% ± 0.4%	86.9% ± 0.2%
GCFL+	94.3% ± 0.1%	94.9% ± 0.1%	98.8% ± 0.1%	98.5% ± 0.2%	95.9% ± 0.3%	93.0% ± 0.5%	88.2% ± 0.6%

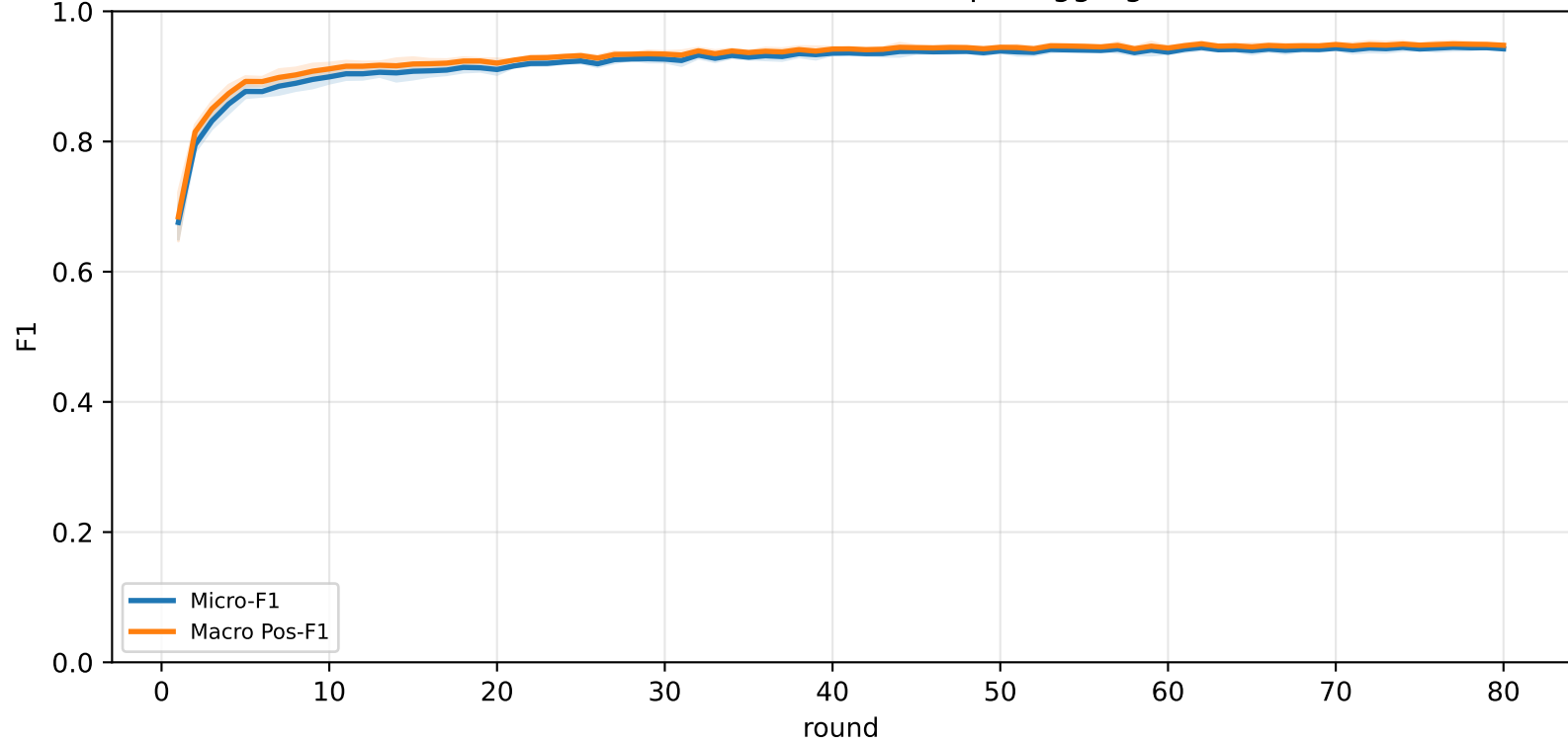
Validation loss comparison



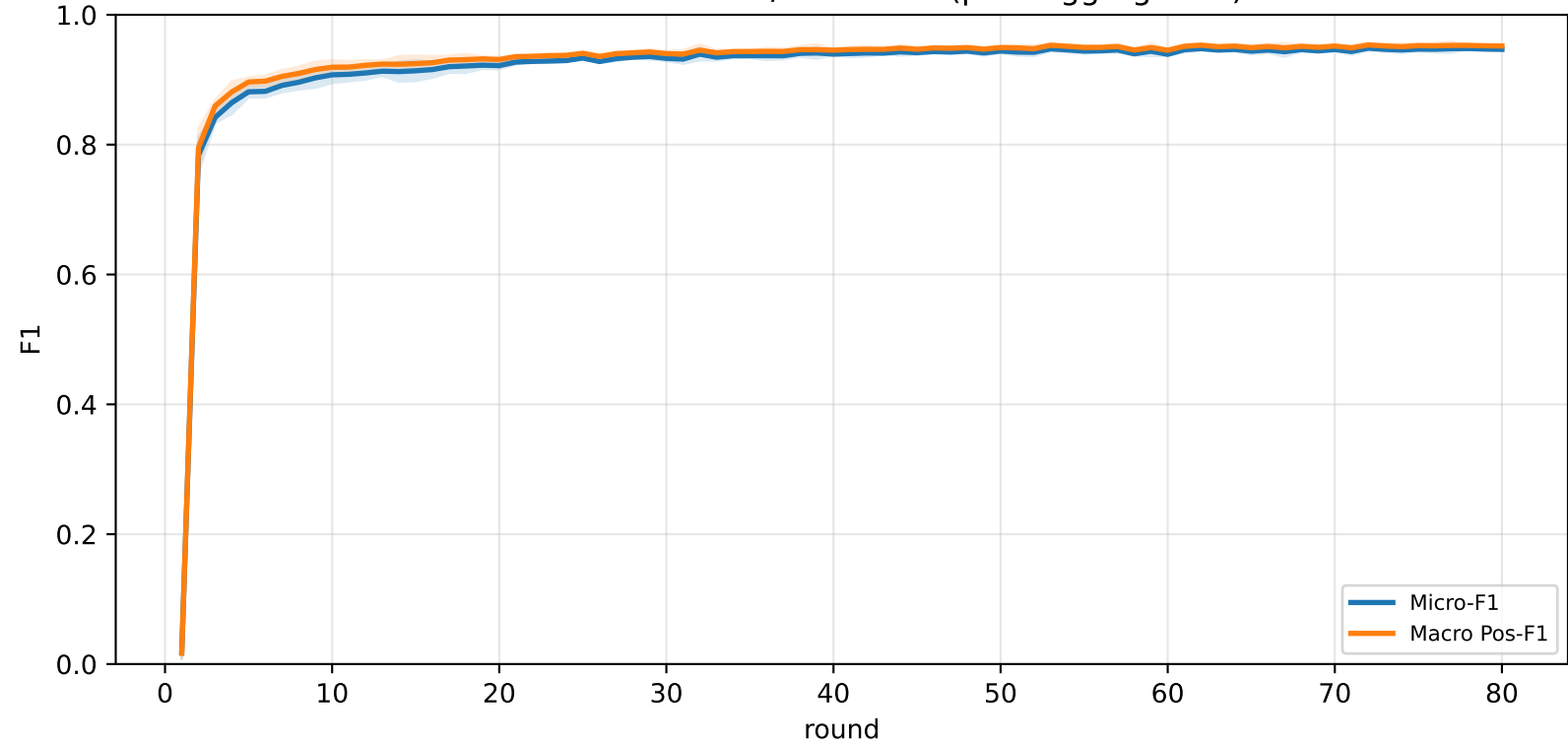
Pooled local centralized micro / macro F1



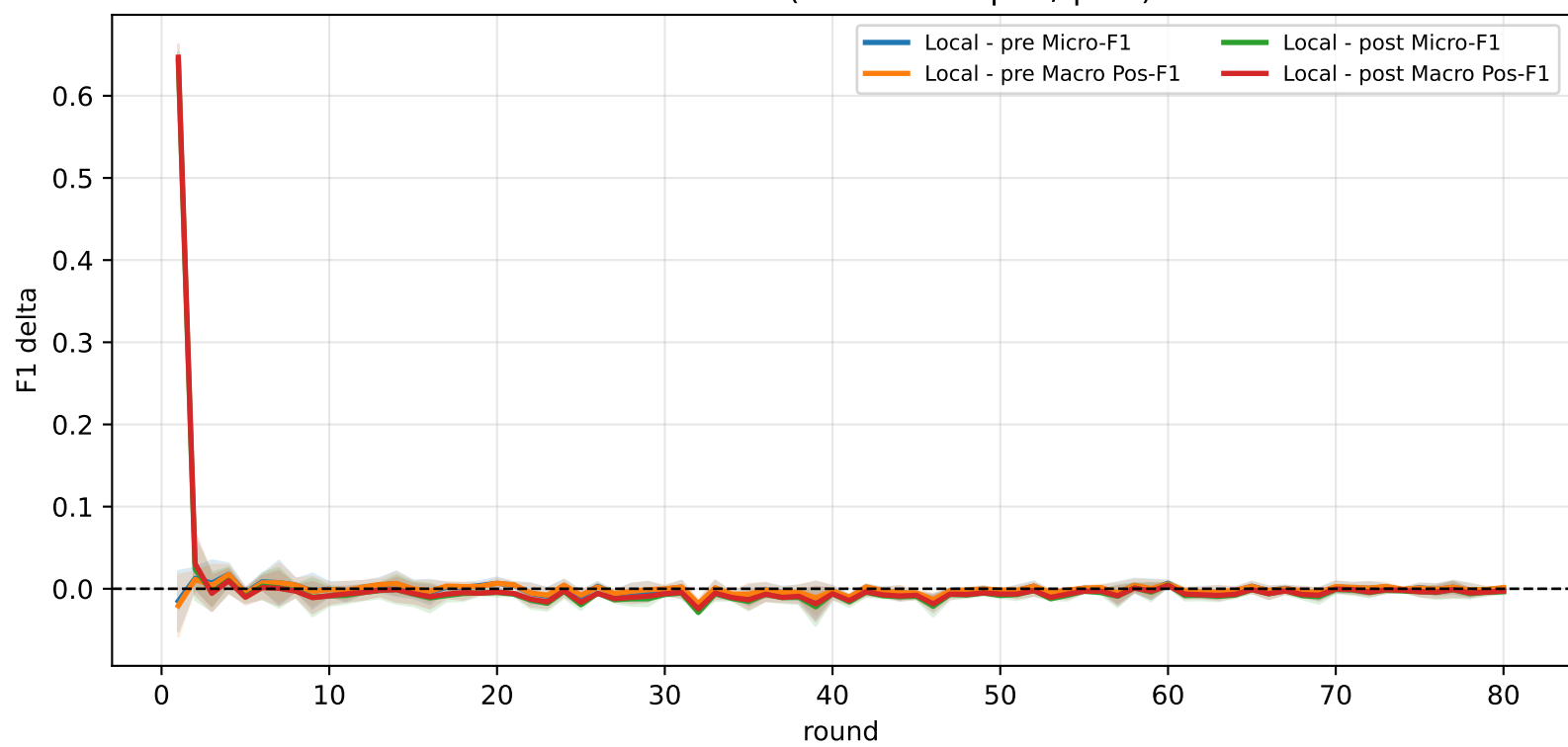
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



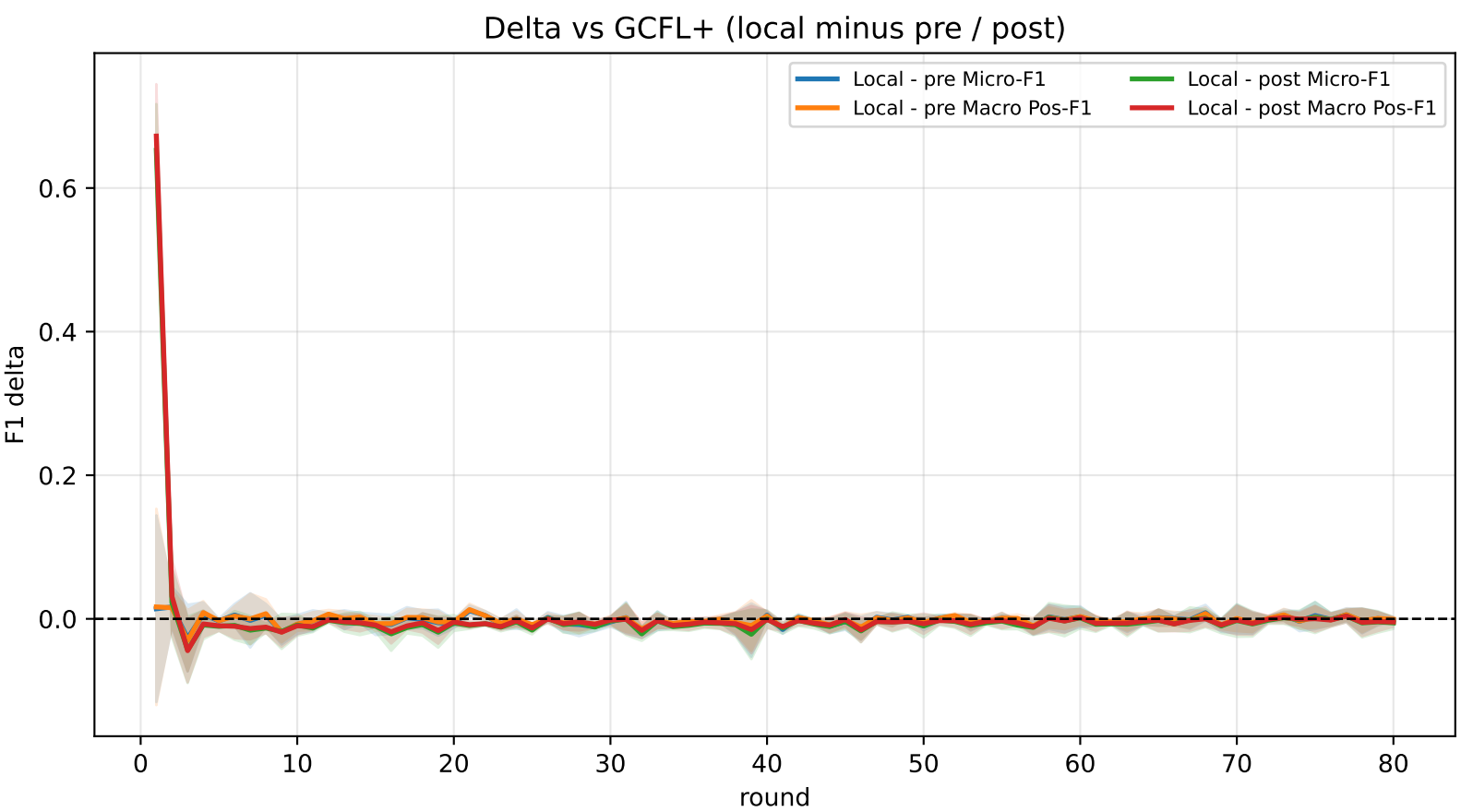
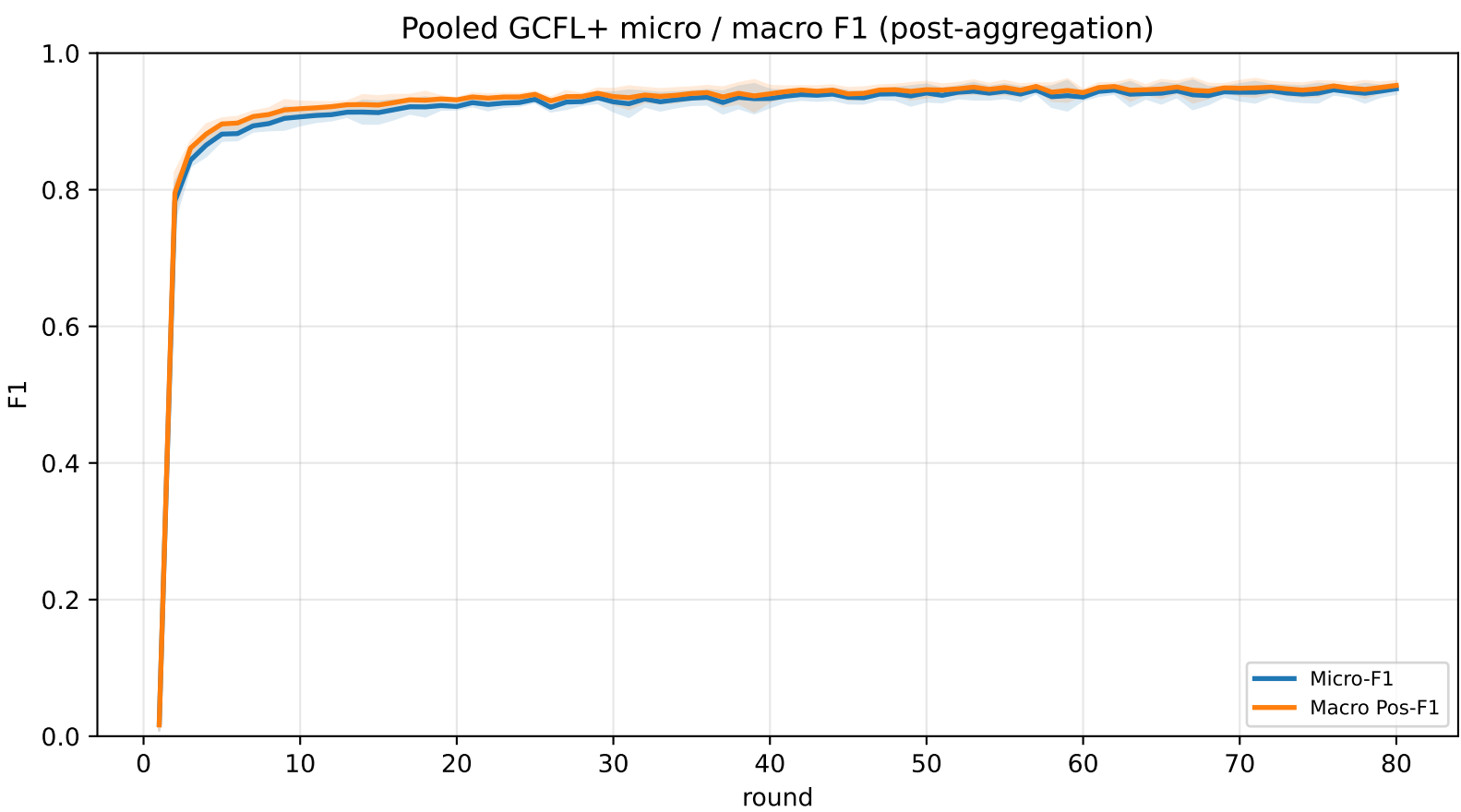
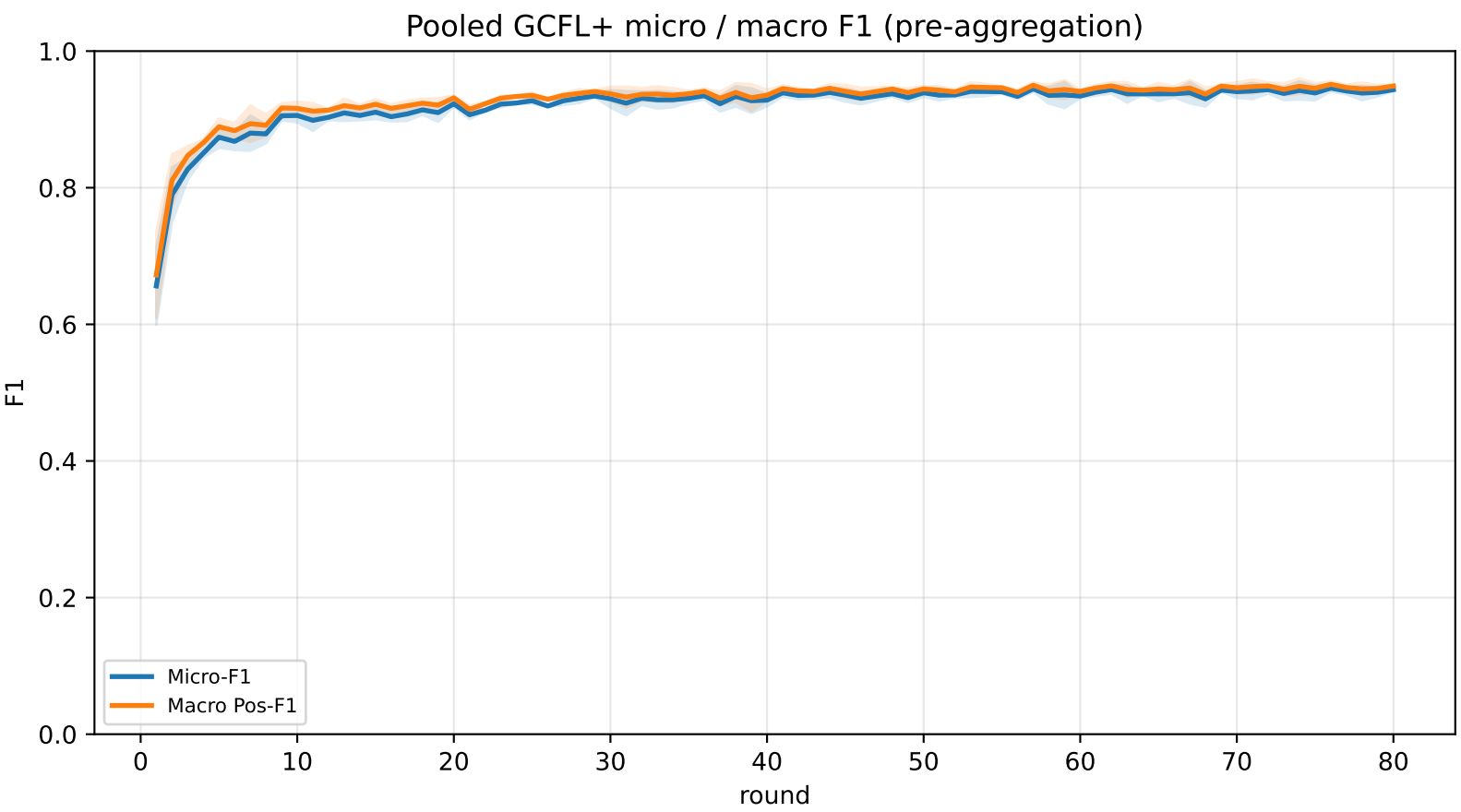
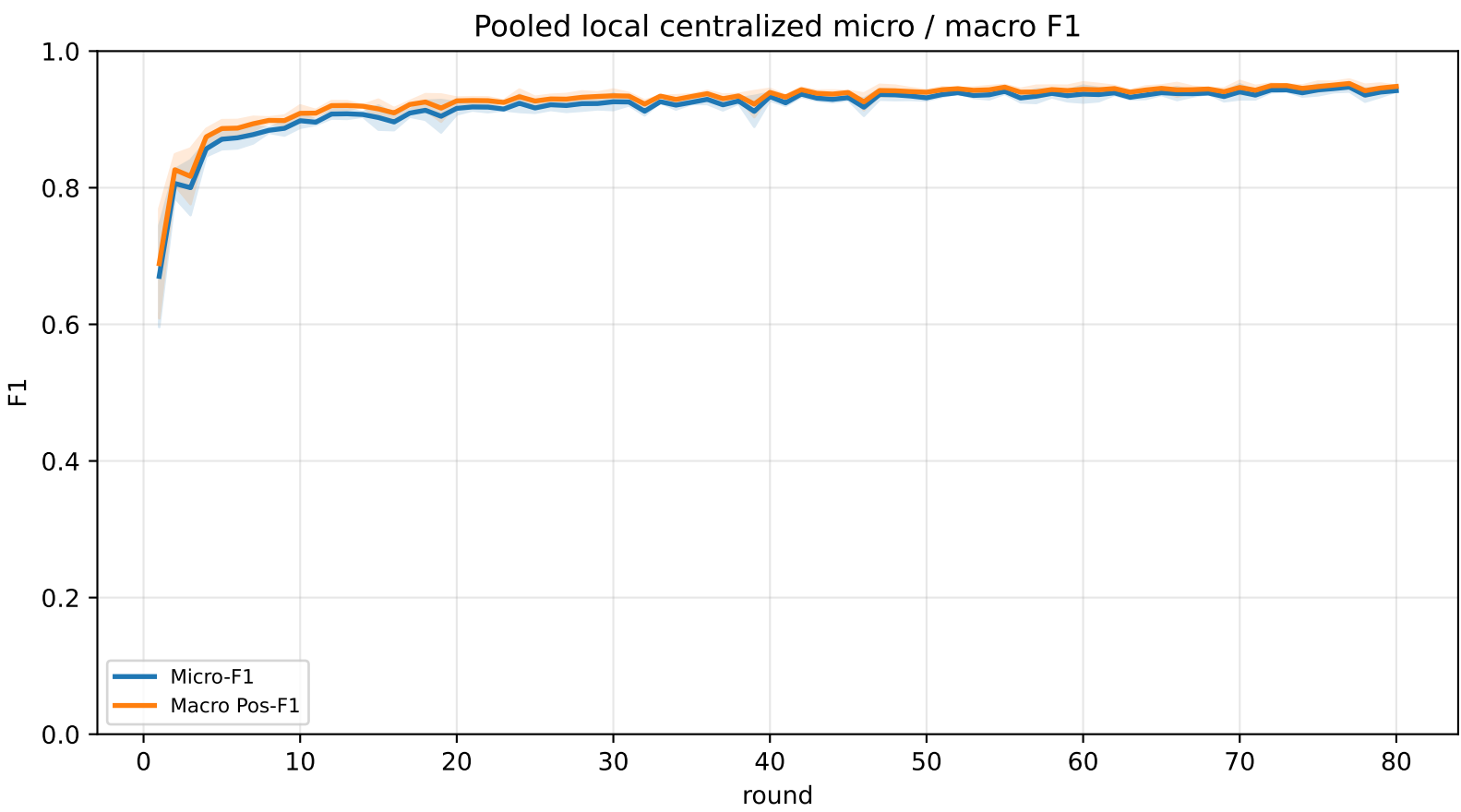
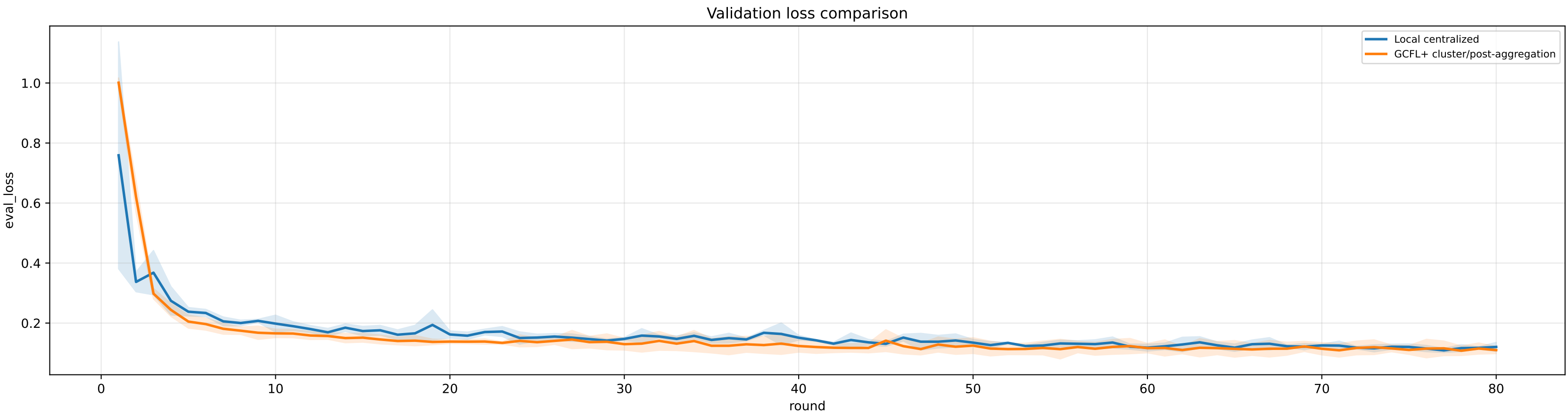
Delta vs GCFL+ (local minus pre / post)



family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (GCFL+ vs Local centralized)

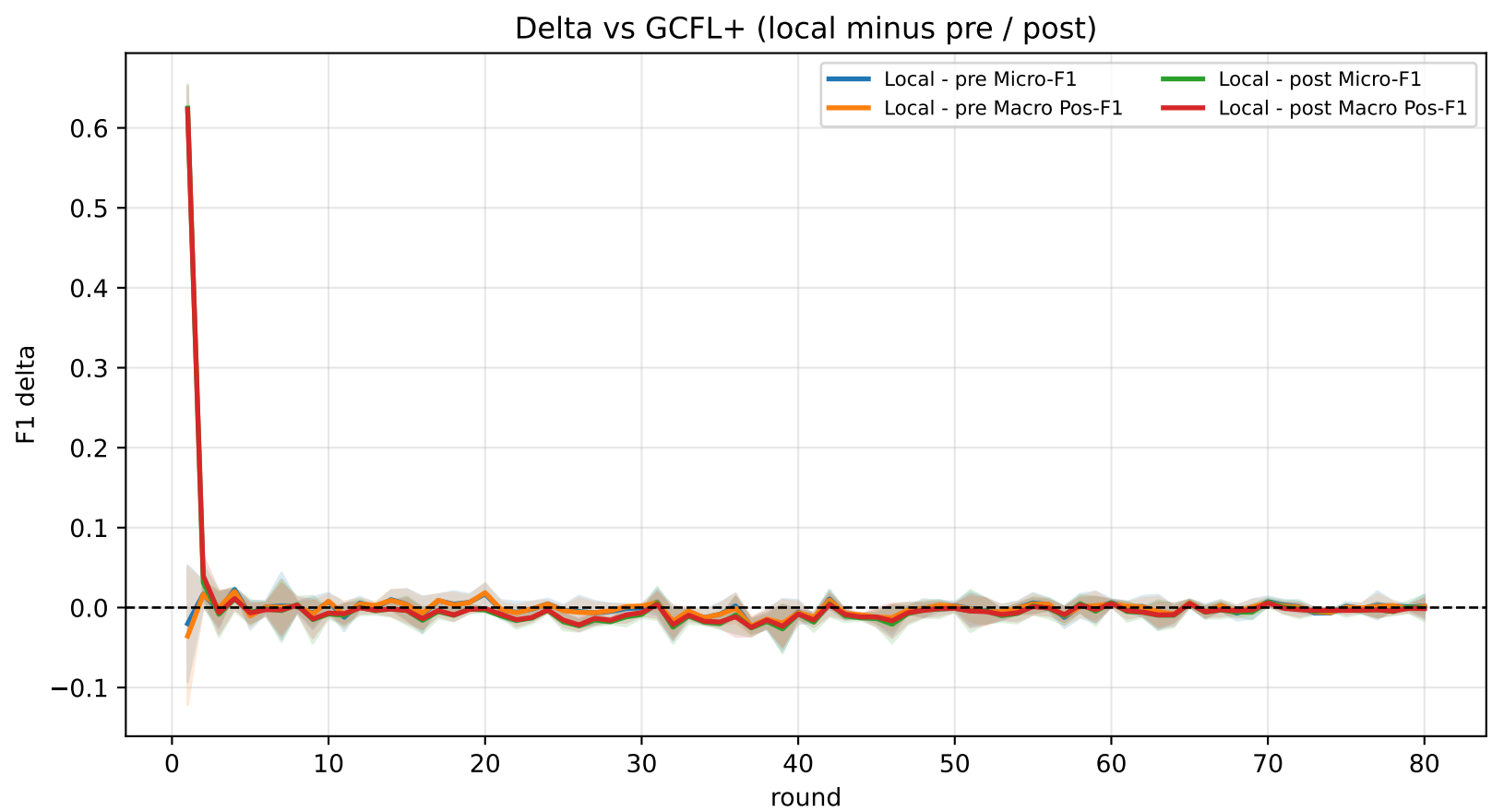
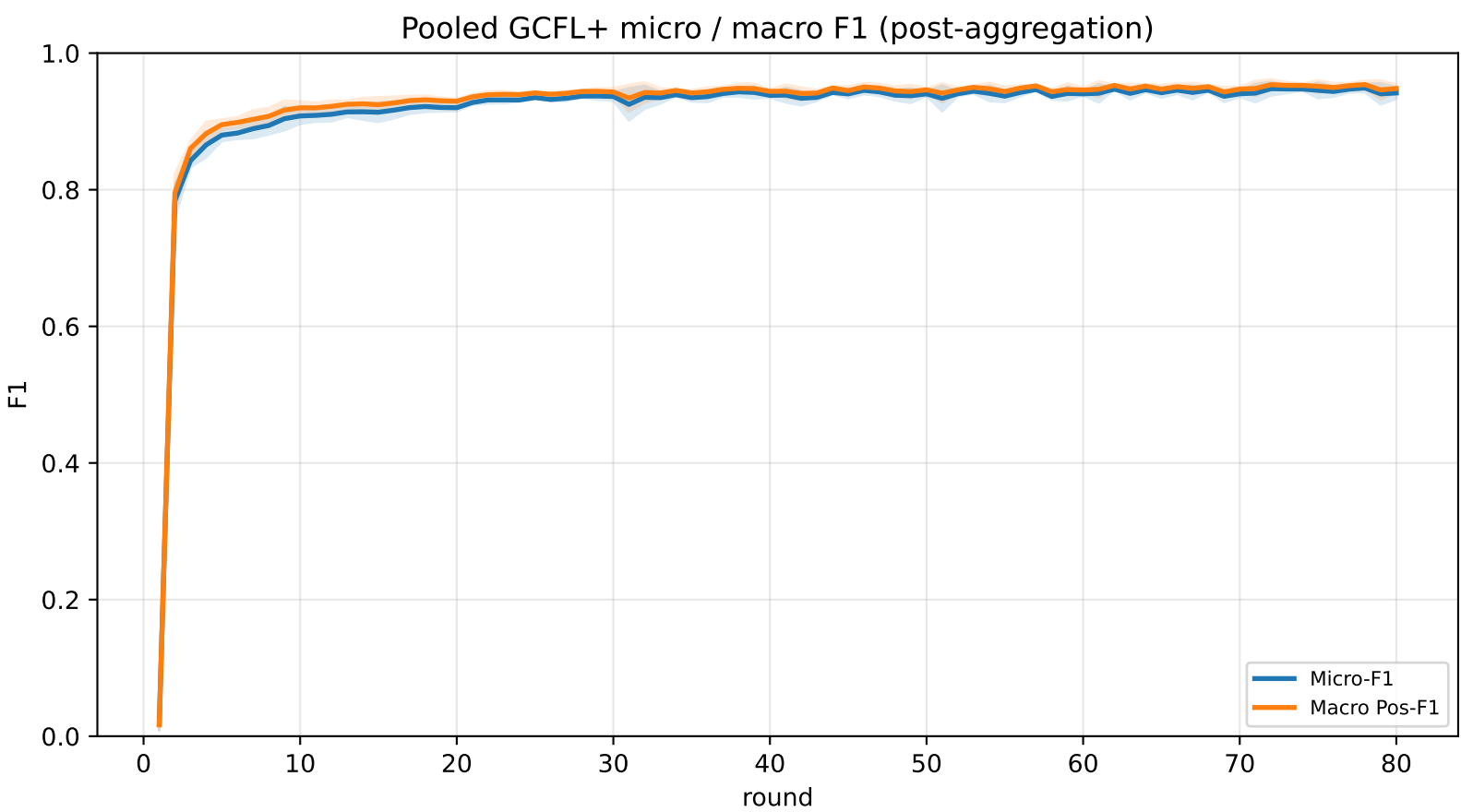
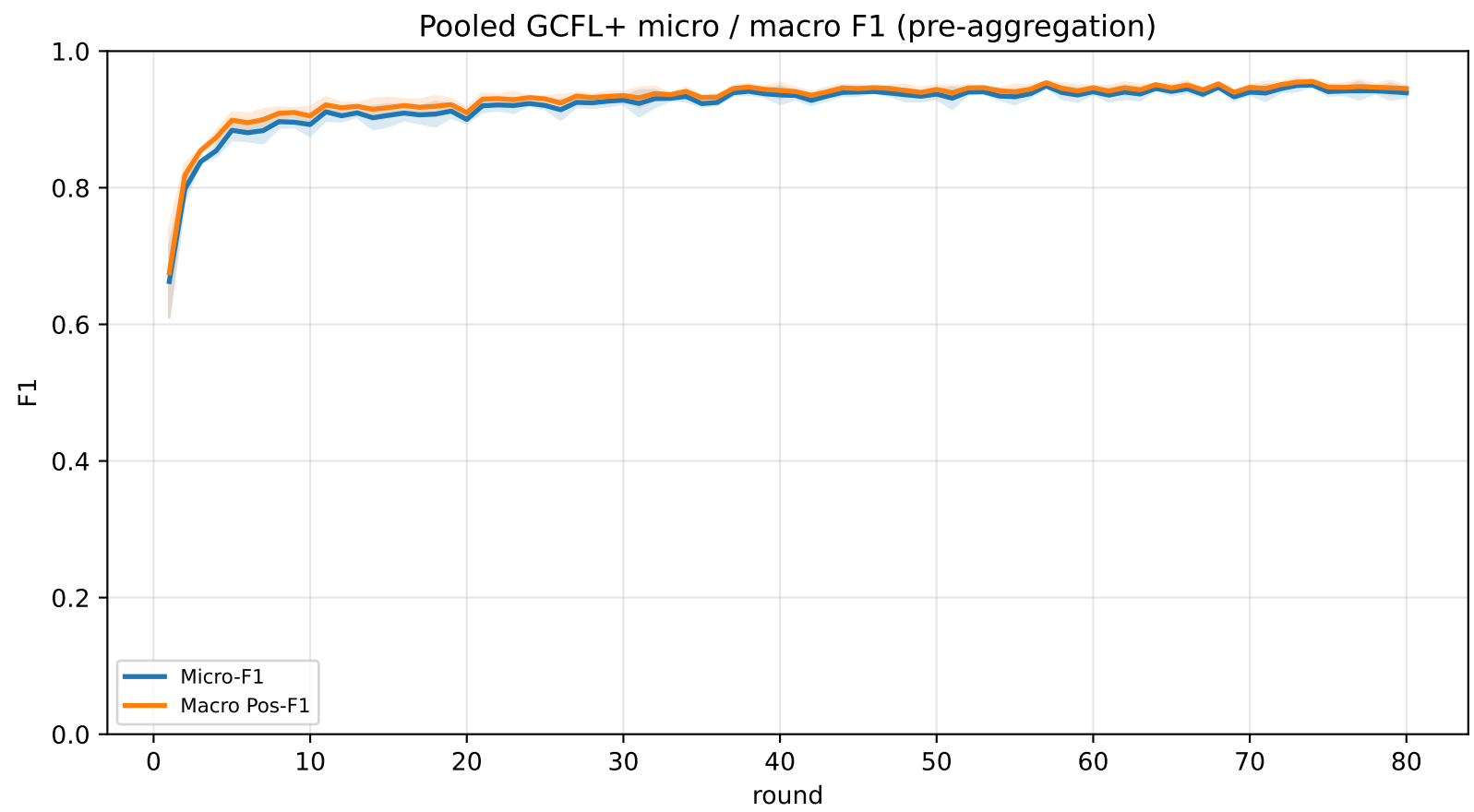
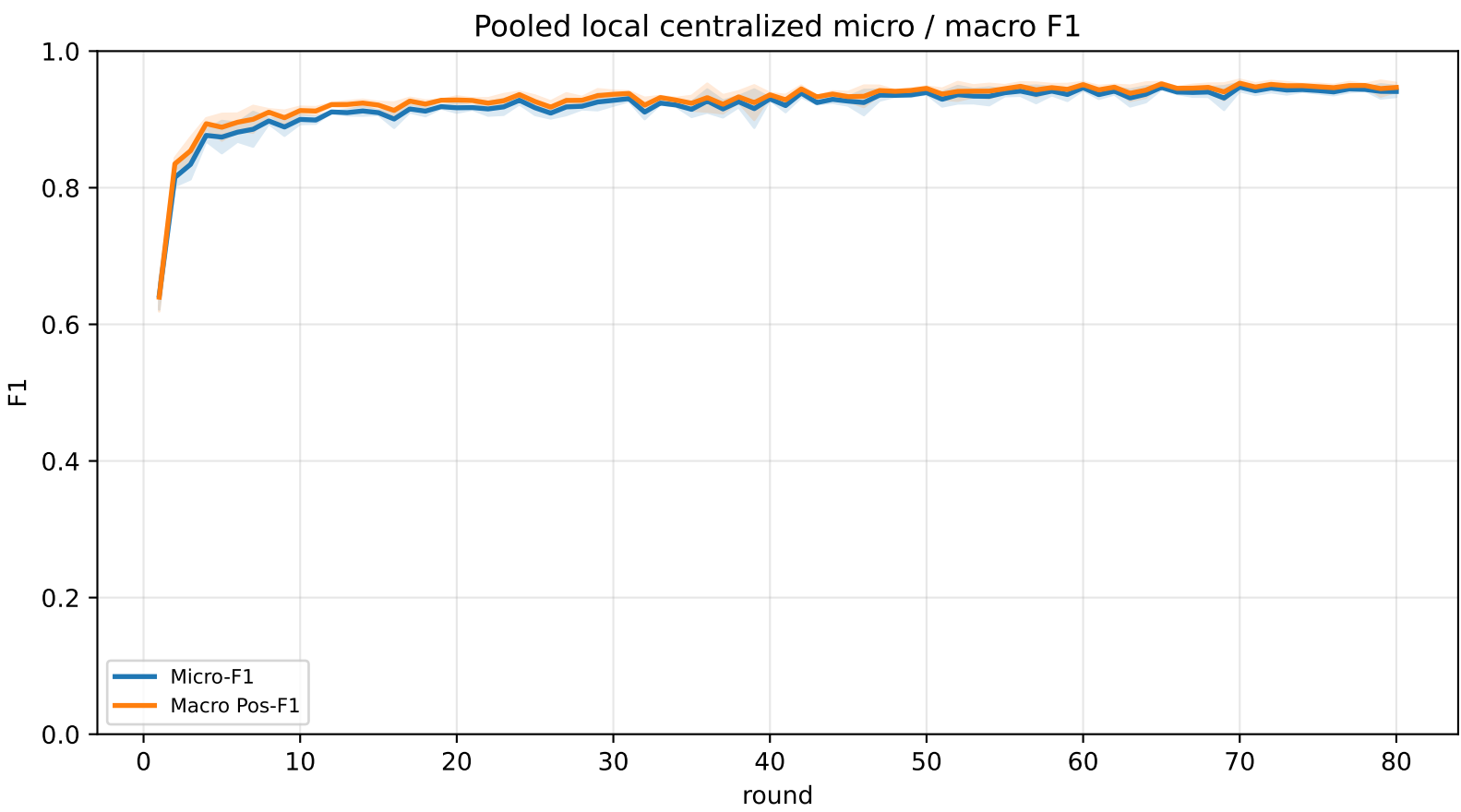
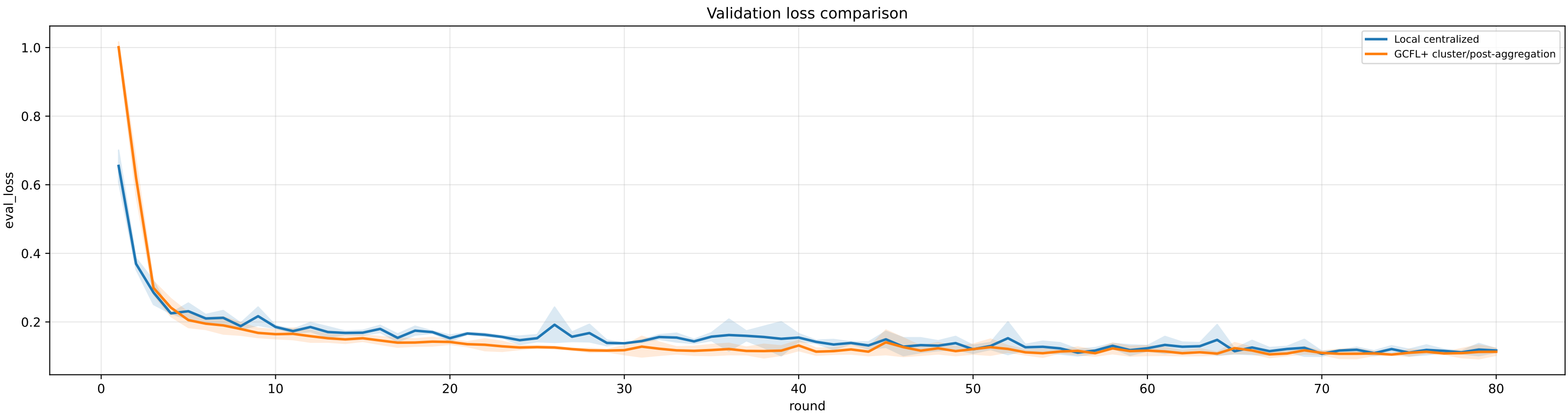
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.4%	94.7% ± 0.3%	99.0% ± 0.3%	98.4% ± 0.1%	96.1% ± 0.9%	92.7% ± 0.7%	87.4% ± 1.3%
GCFL+	94.1% ± 0.5%	94.7% ± 0.4%	98.7% ± 0.4%	98.8% ± 0.1%	95.6% ± 0.3%	92.7% ± 0.6%	87.7% ± 1.4%



family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.1%	94.7% ± 0.1%	99.0% ± 0.2%	98.2% ± 0.3%	95.9% ± 0.1%	93.2% ± 0.3%	86.9% ± 0.5%
GCFL+	94.3% ± 0.4%	94.8% ± 0.3%	98.4% ± 0.2%	98.1% ± 0.4%	96.4% ± 0.5%	93.0% ± 1.1%	88.2% ± 0.4%

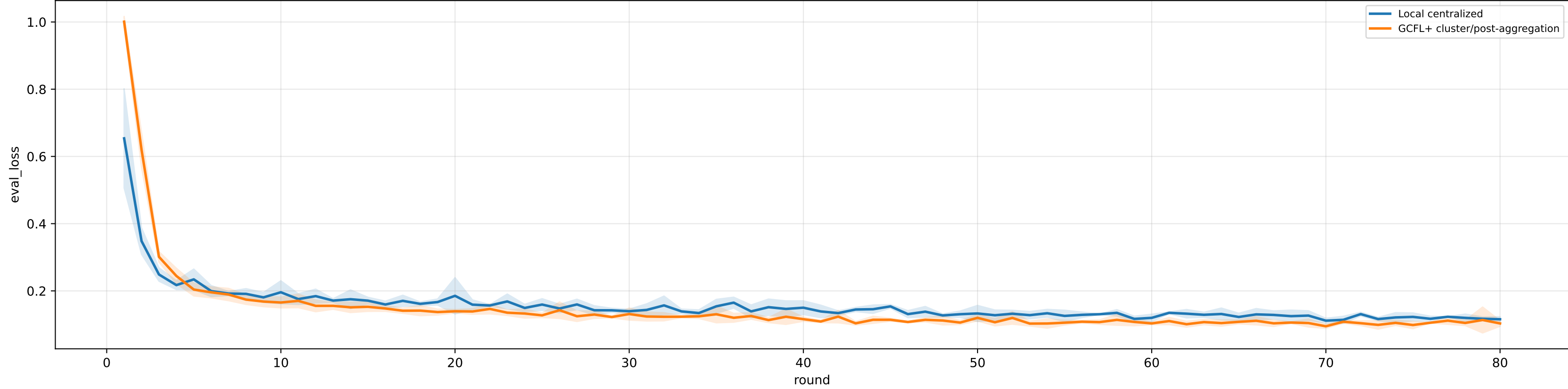


family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

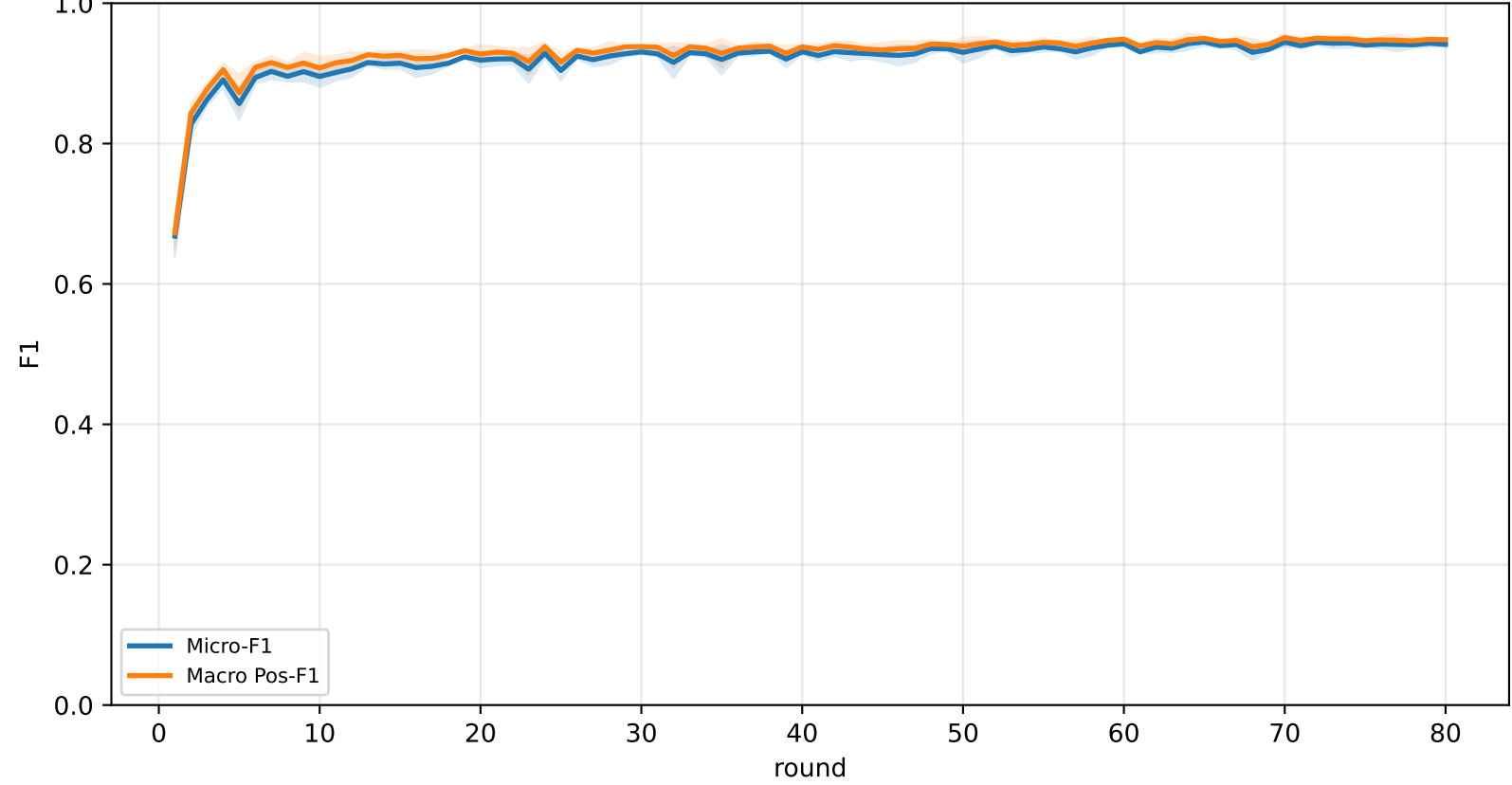
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% ± 0.7%	94.3% ± 0.6%	98.9% ± 0.2%	98.1% ± 0.6%	96.4% ± 0.4%	92.7% ± 0.6%	85.4% ± 2.5%
GCFL+	94.2% ± 0.4%	94.8% ± 0.4%	98.7% ± 0.6%	98.6% ± 0.5%	96.1% ± 0.3%	92.8% ± 0.2%	87.6% ± 1.0%

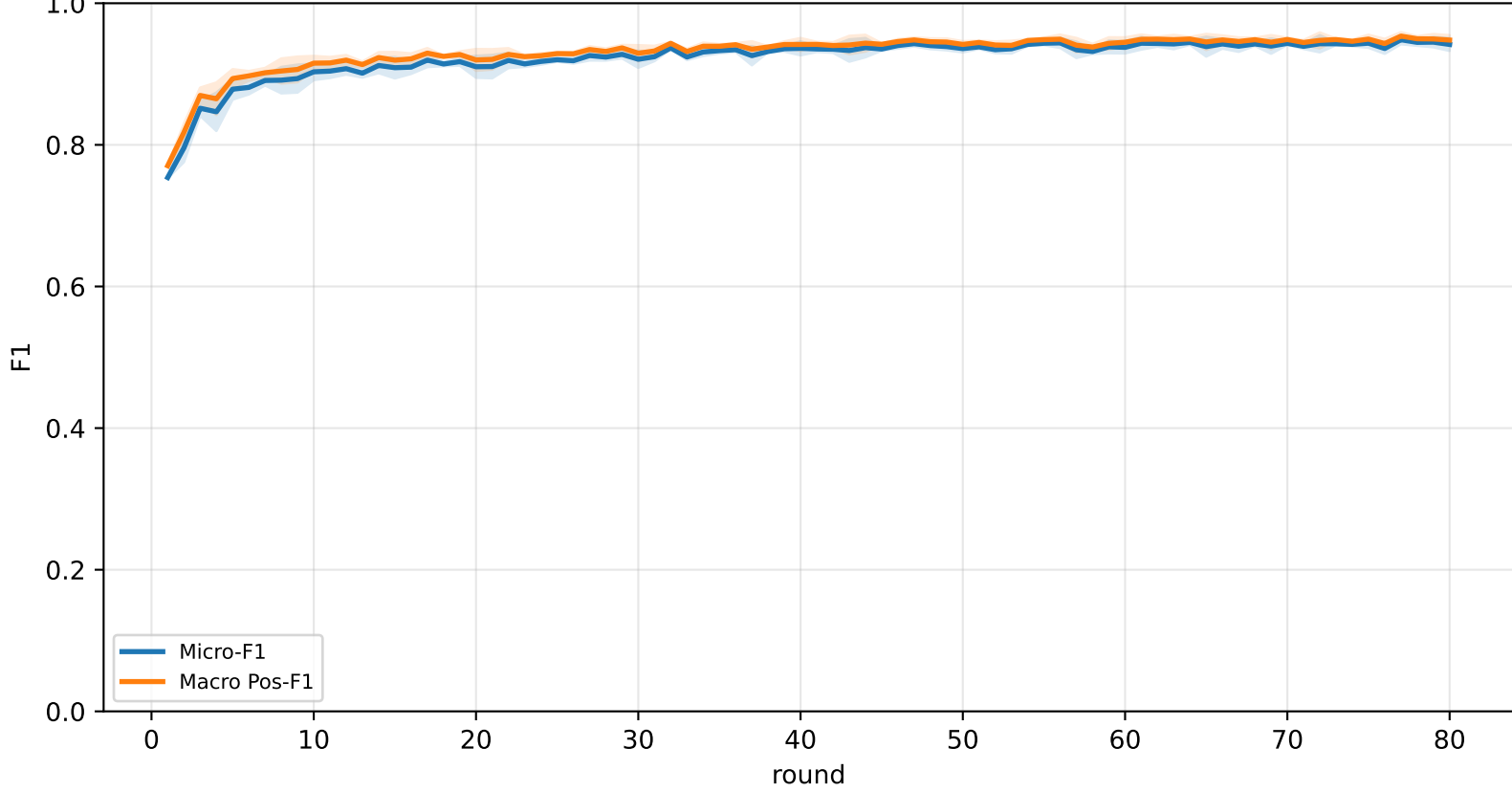
Validation loss comparison



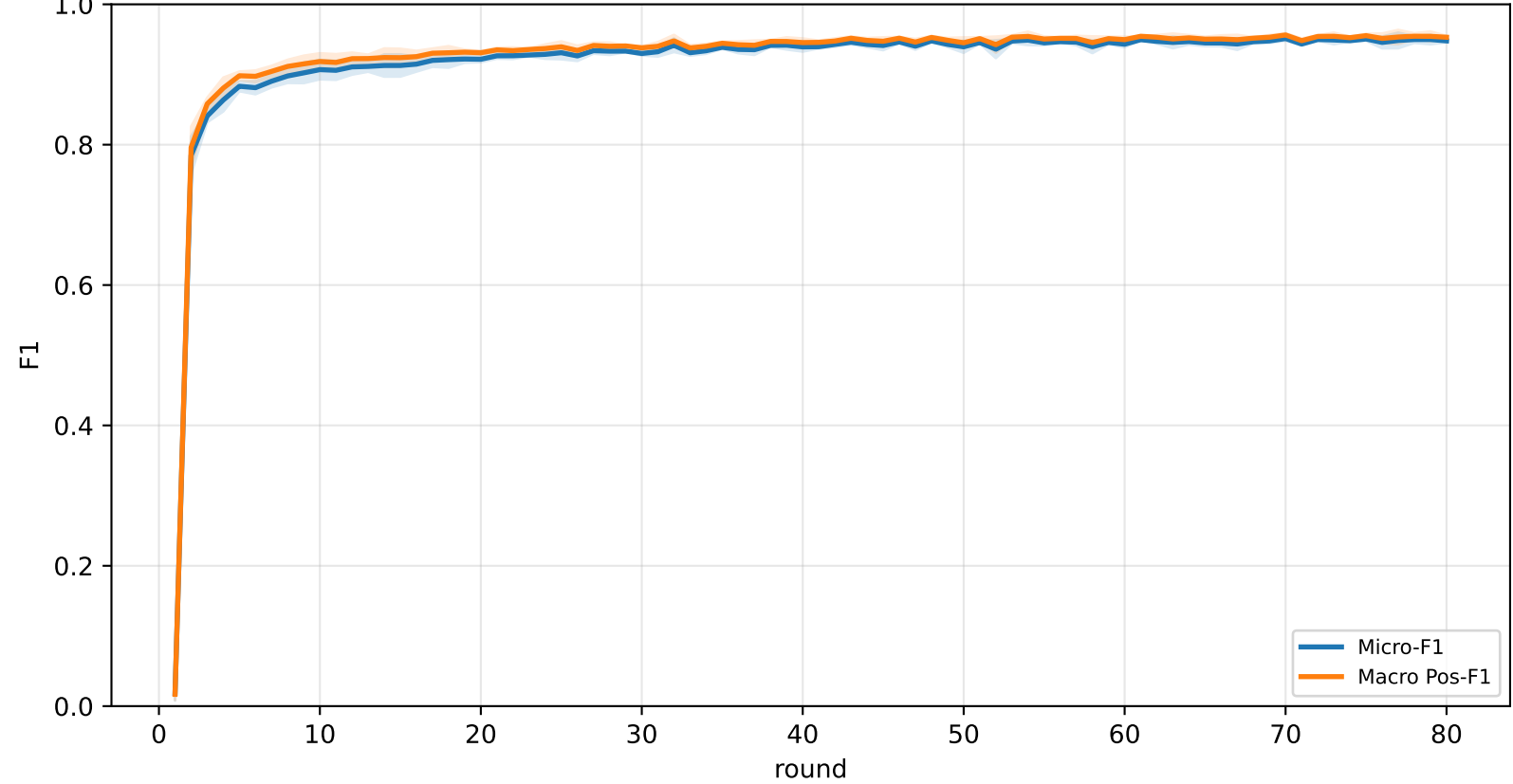
Pooled local centralized micro / macro F1



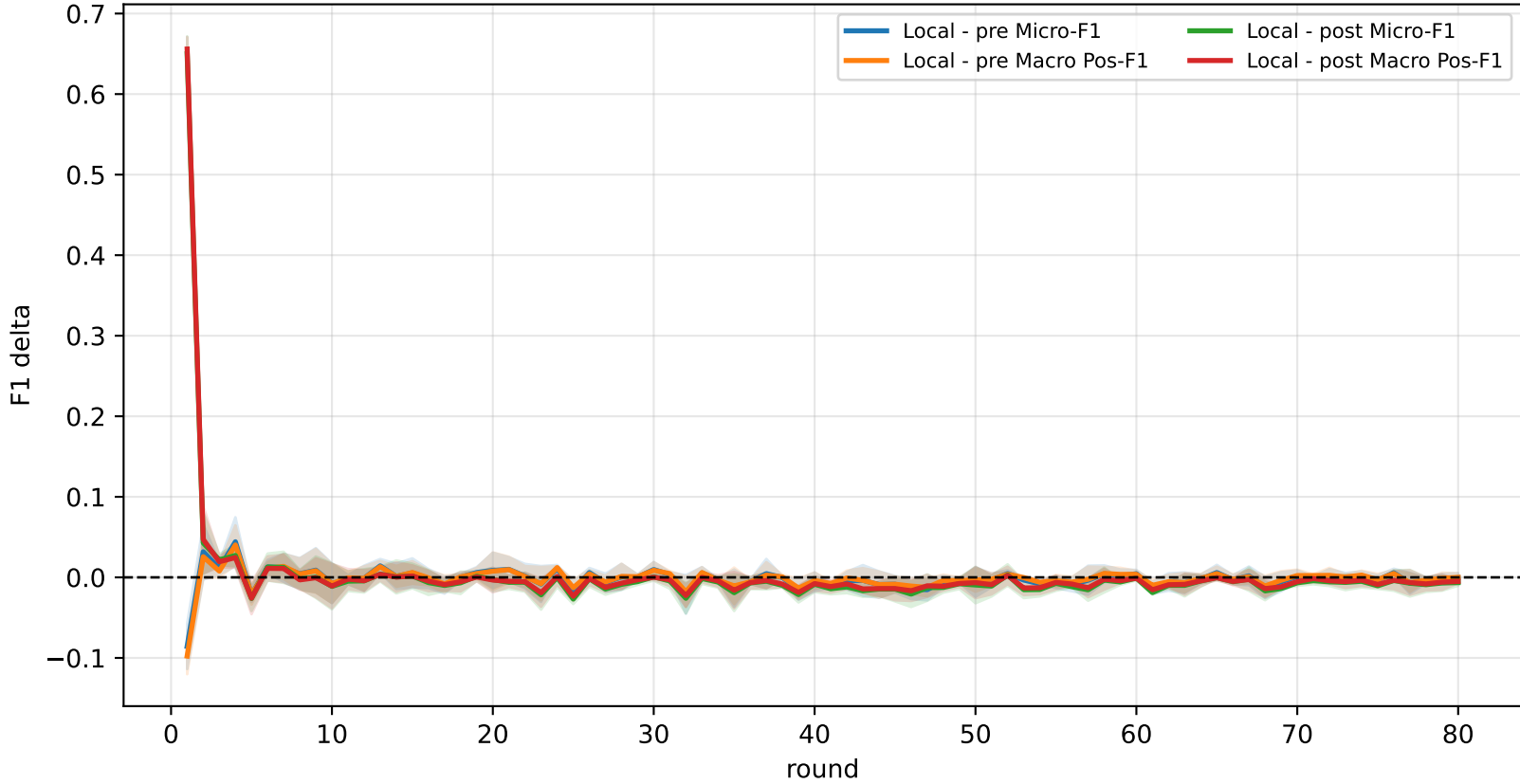
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

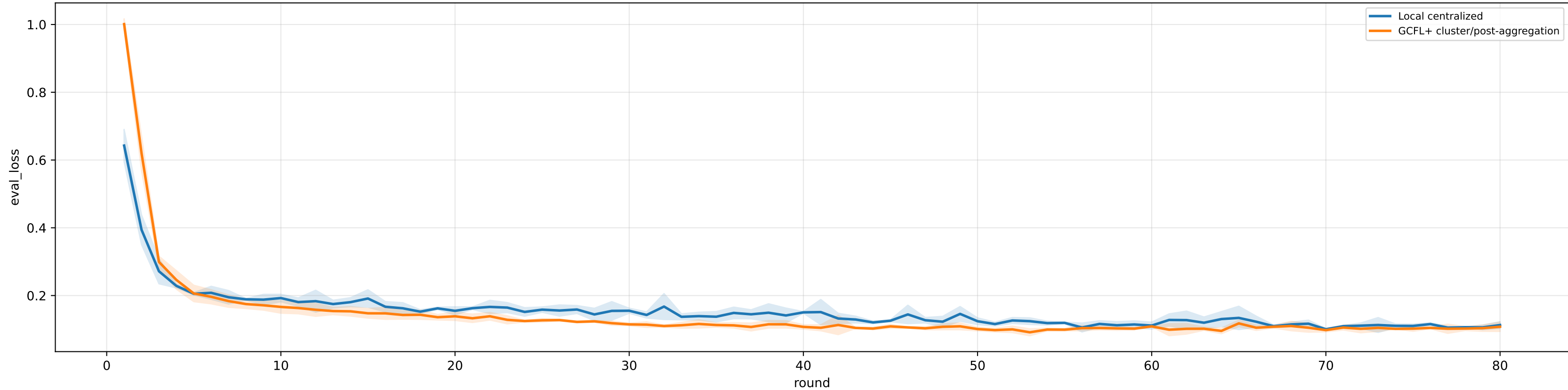


family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

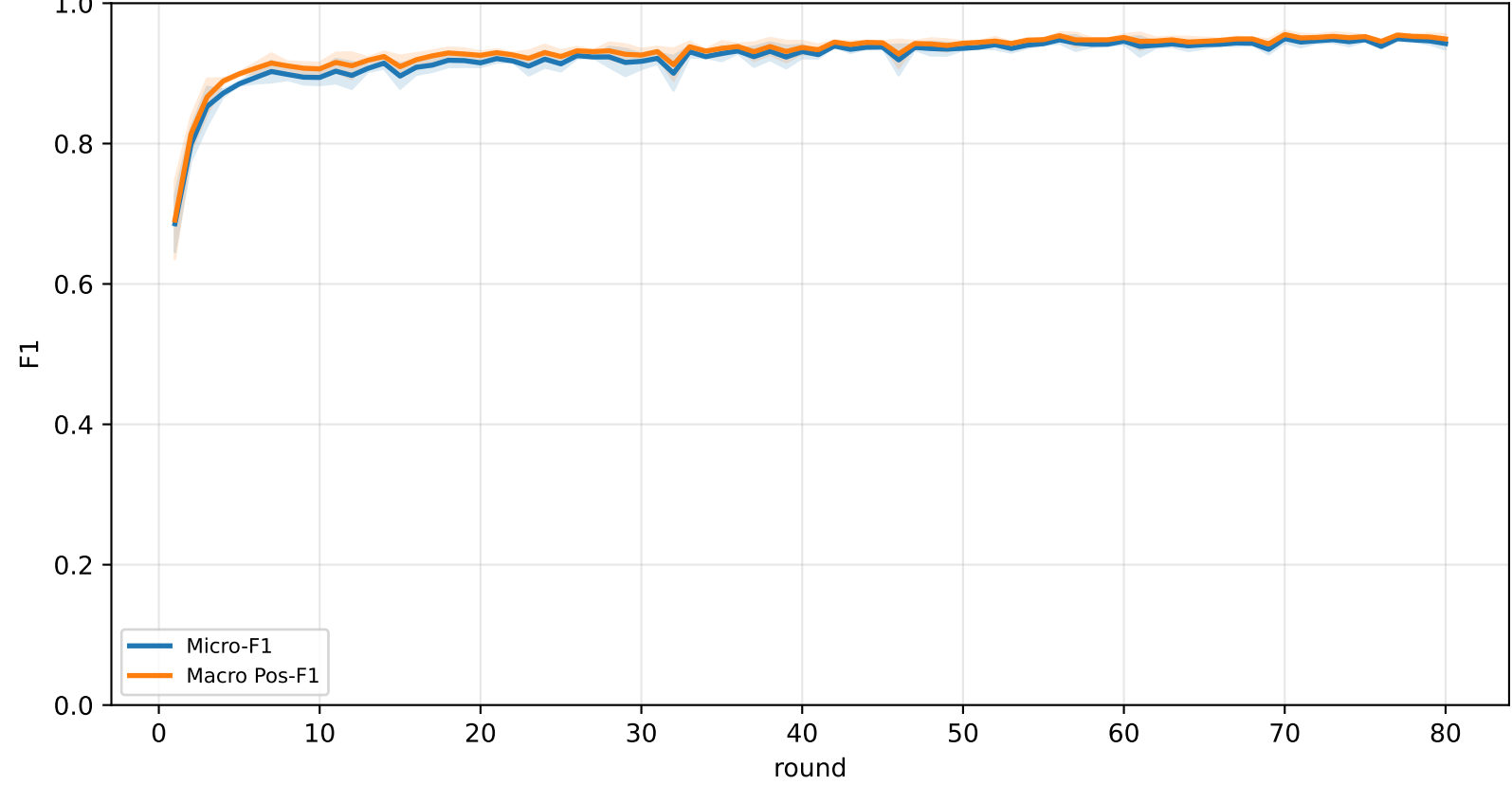
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.4% ± 0.4%	95.0% ± 0.3%	99.0% ± 0.2%	98.5% ± 0.3%	96.6% ± 0.2%	93.9% ± 1.0%	87.1% ± 0.8%
GCFL+	94.4% ± 0.1%	94.9% ± 0.1%	98.7% ± 0.1%	98.3% ± 0.1%	95.4% ± 1.0%	93.2% ± 0.4%	88.7% ± 0.7%

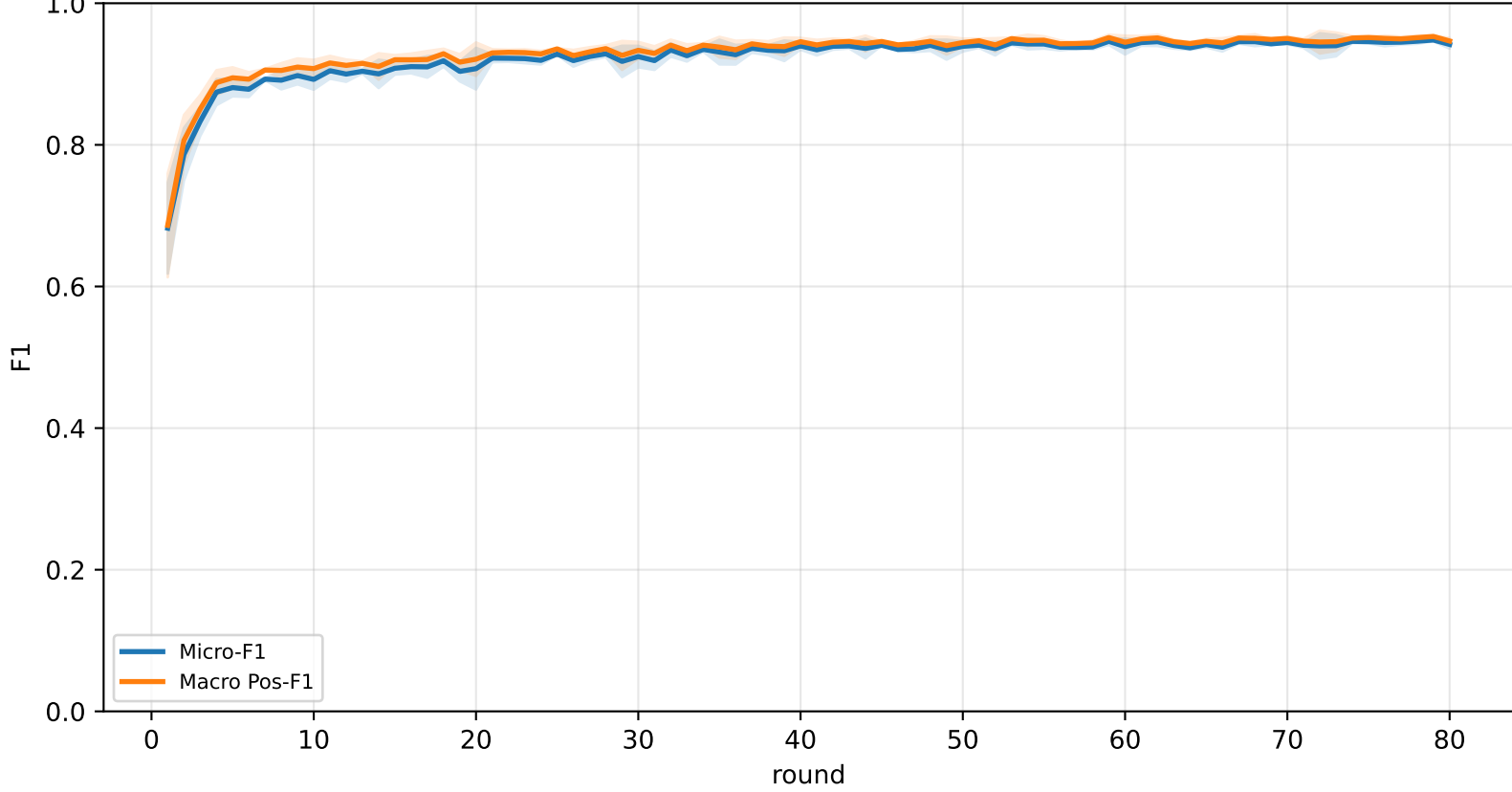
Validation loss comparison



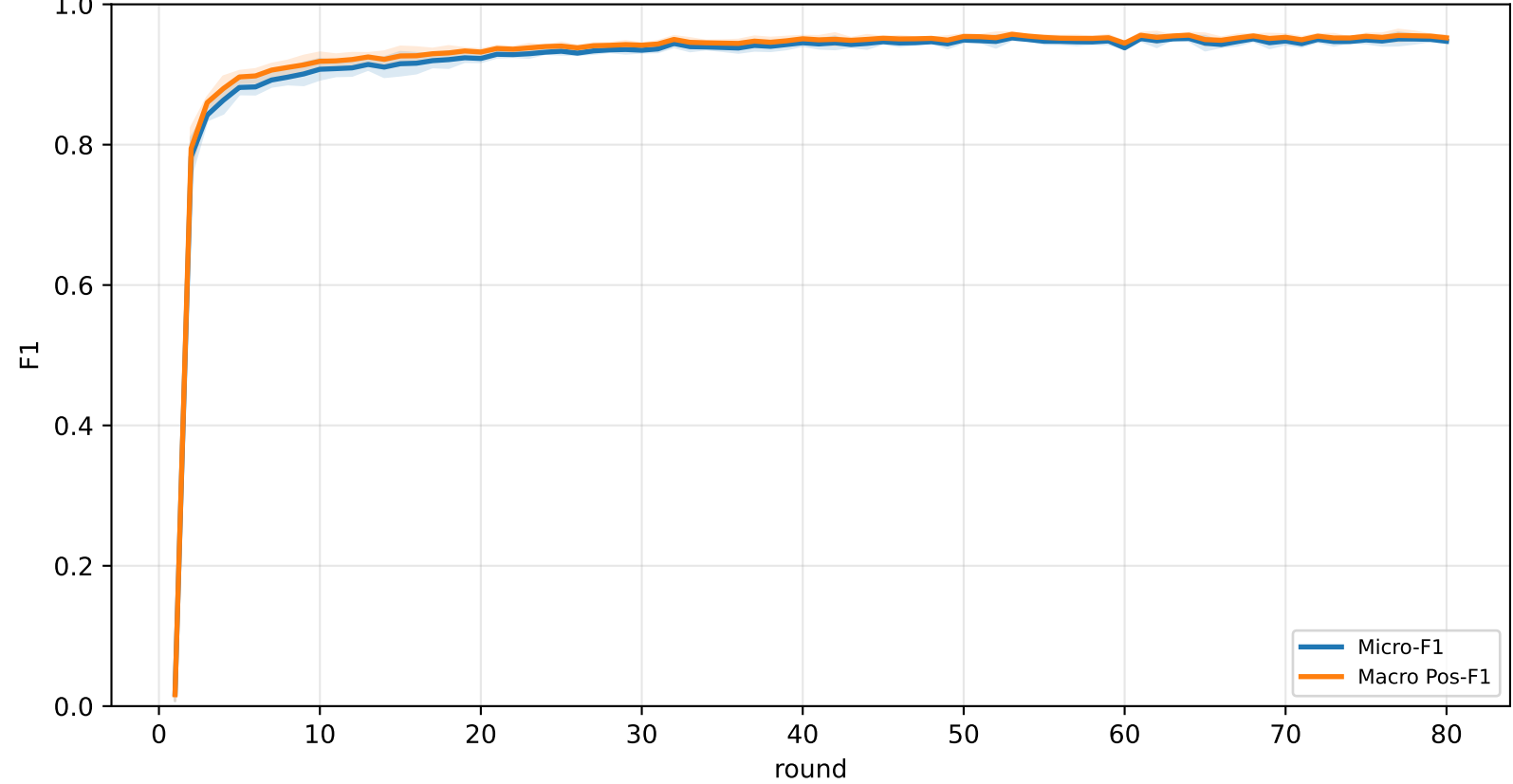
Pooled local centralized micro / macro F1



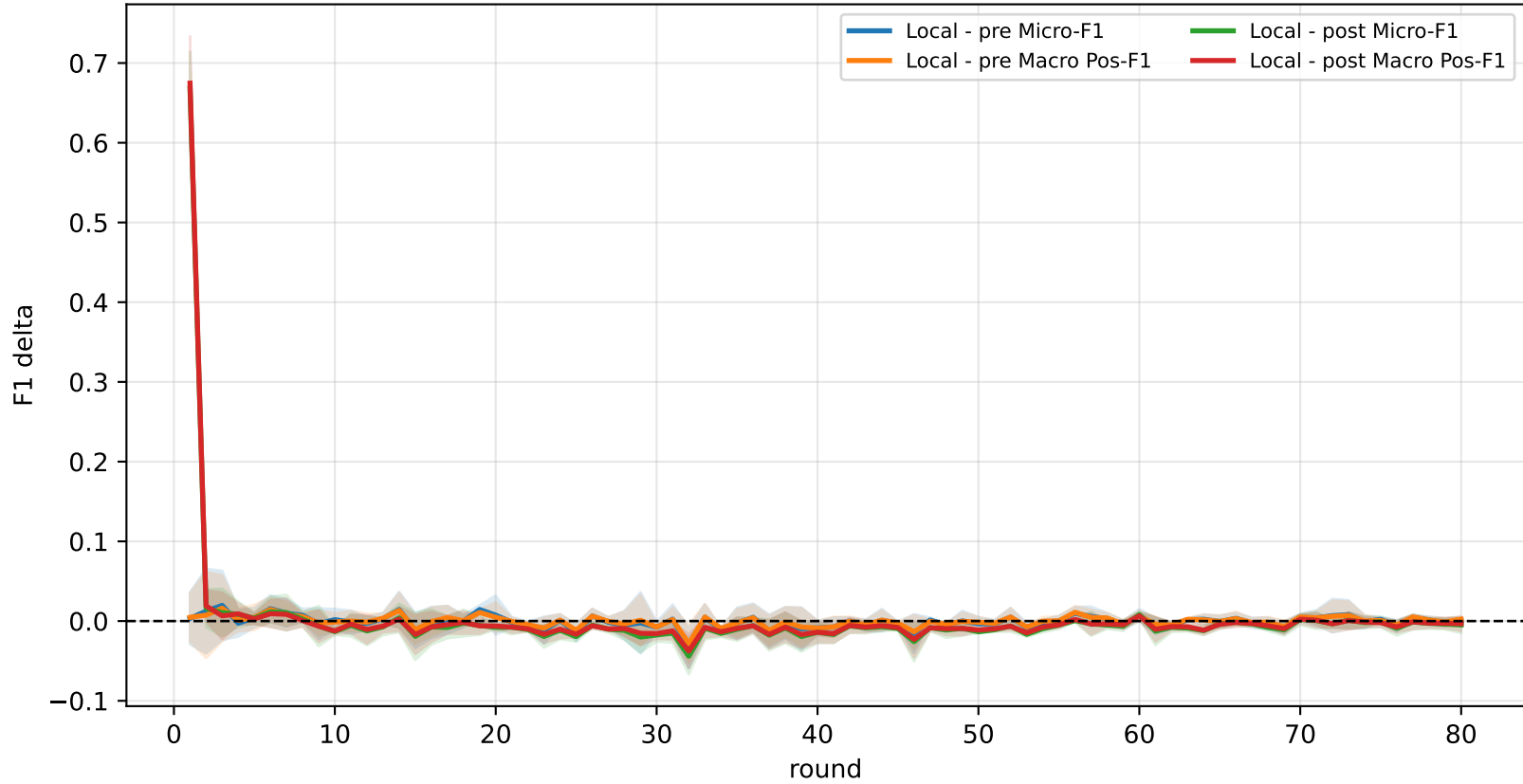
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

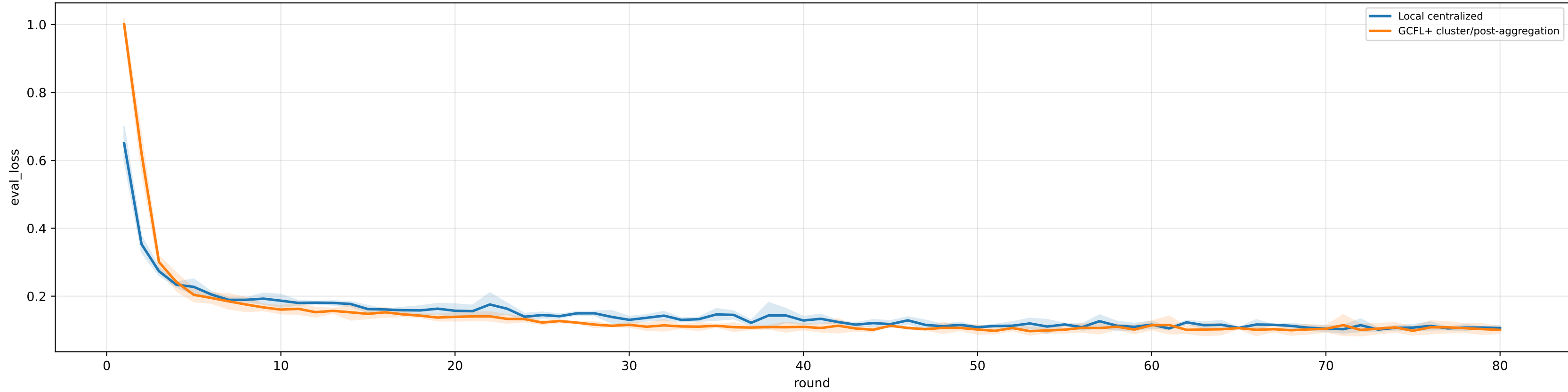


family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

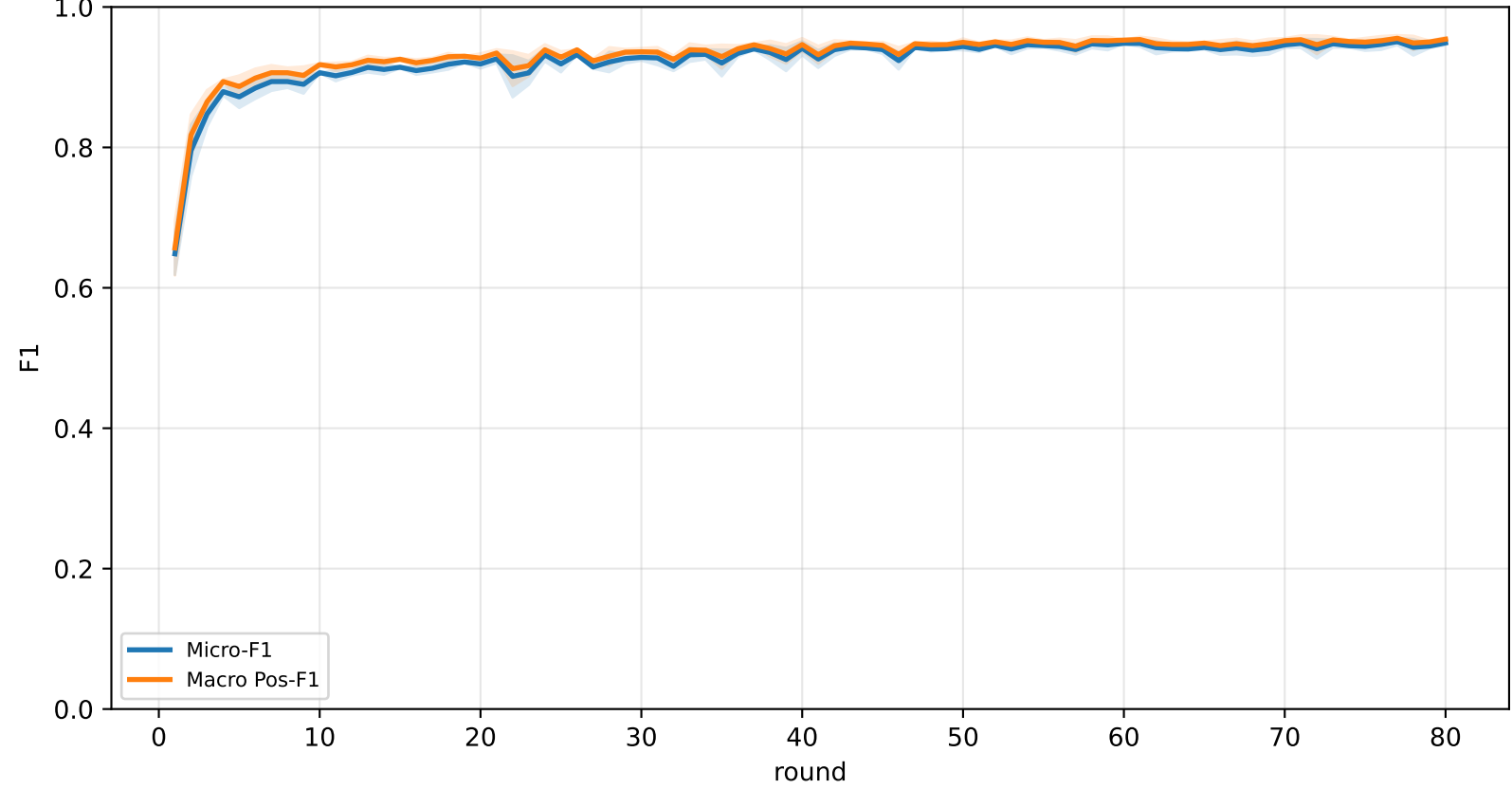
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.3% ± 0.5%	94.9% ± 0.4%	98.9% ± 0.1%	98.2% ± 0.5%	96.0% ± 0.4%	93.8% ± 1.0%	87.6% ± 0.5%
GCFL+	94.7% ± 0.1%	95.2% ± 0.1%	99.3% ± 0.1%	98.7% ± 0.0%	96.1% ± 0.9%	93.2% ± 0.3%	88.8% ± 1.0%

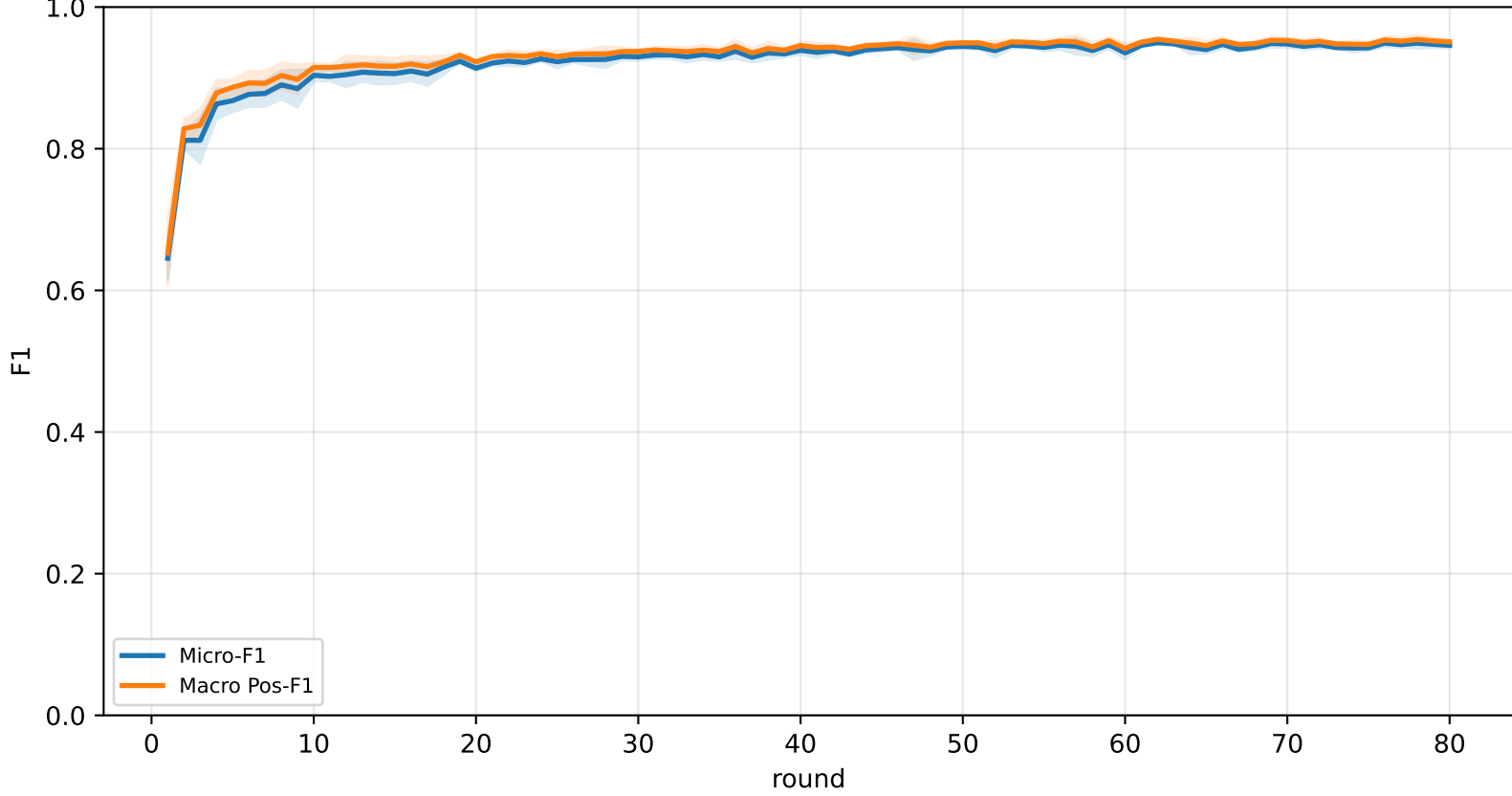
Validation loss comparison



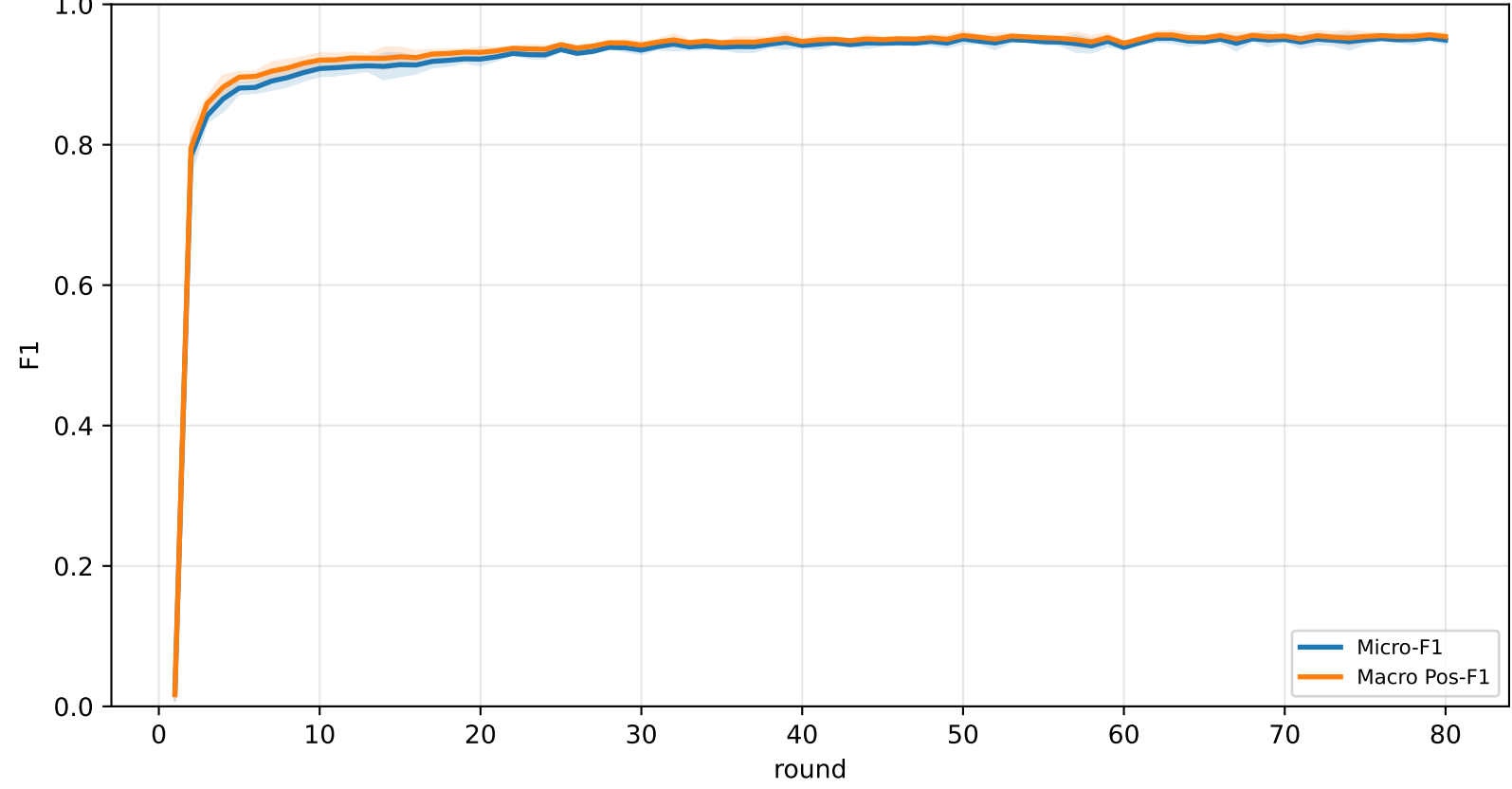
Pooled local centralized micro / macro F1



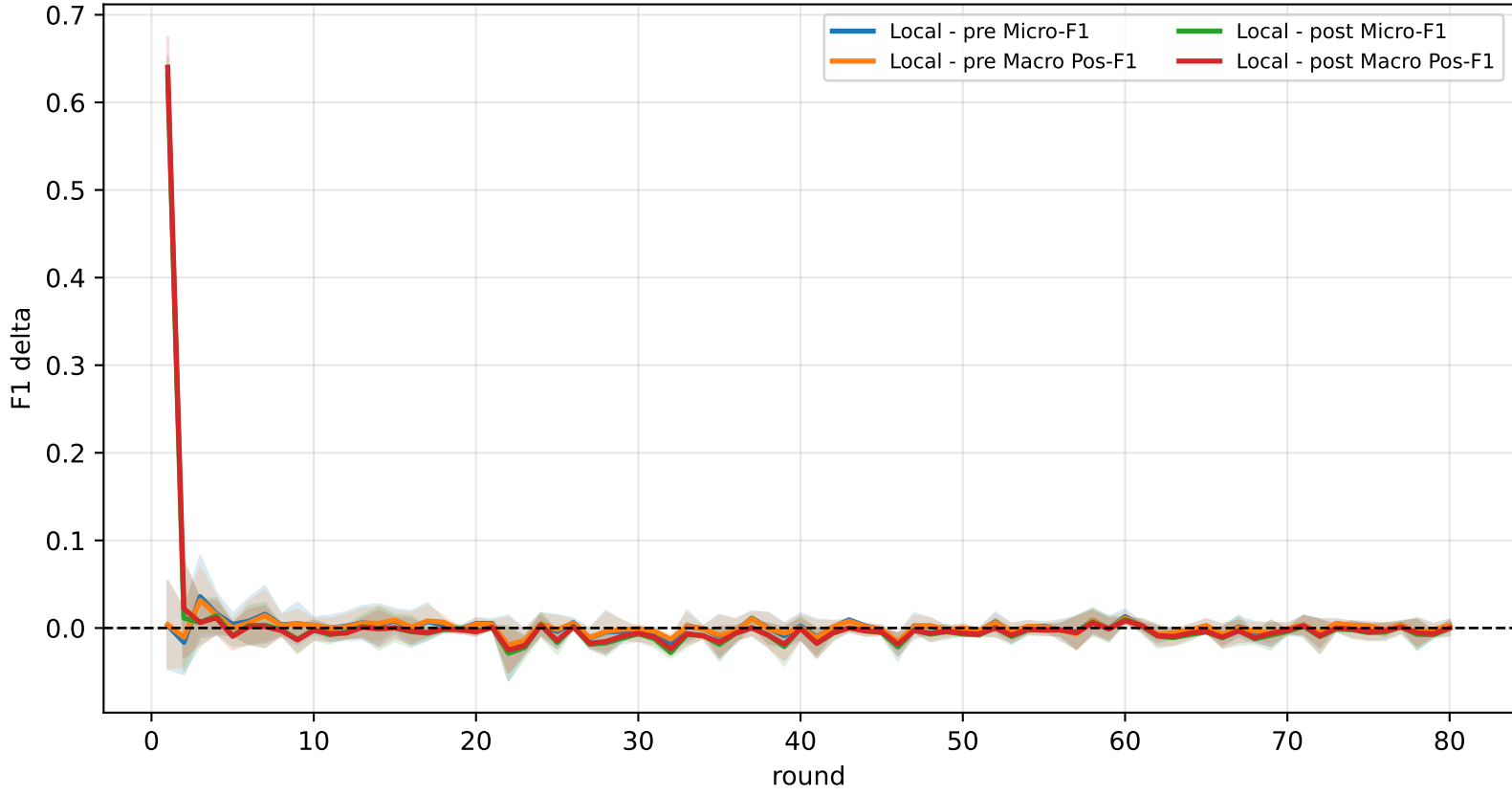
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

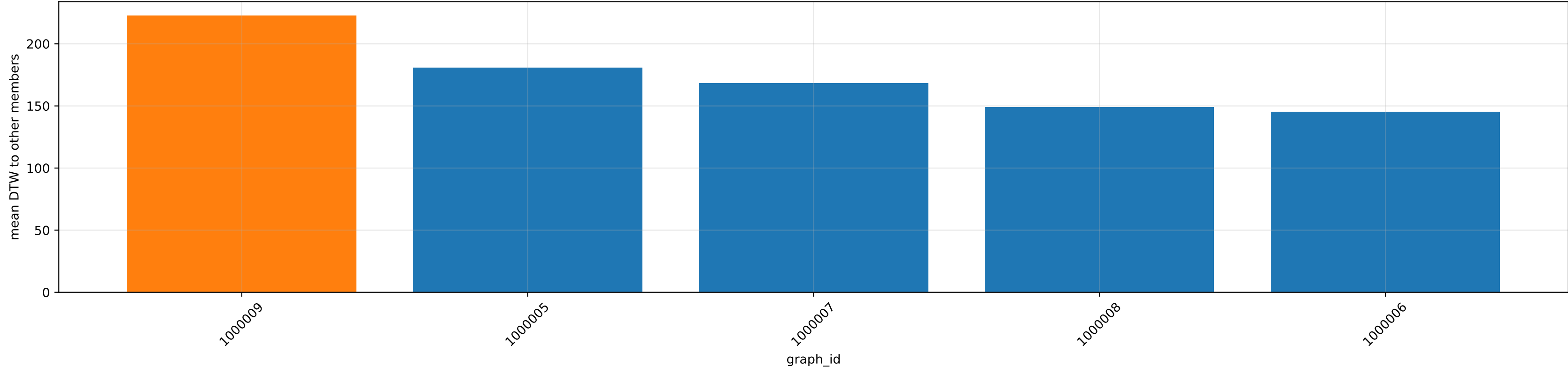
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	51	0	1 vs 4	1000009	1000009	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
222.771	nan	140.122

Per-member mean DTW



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

First applied split pairwise DTW metadata | seed=0

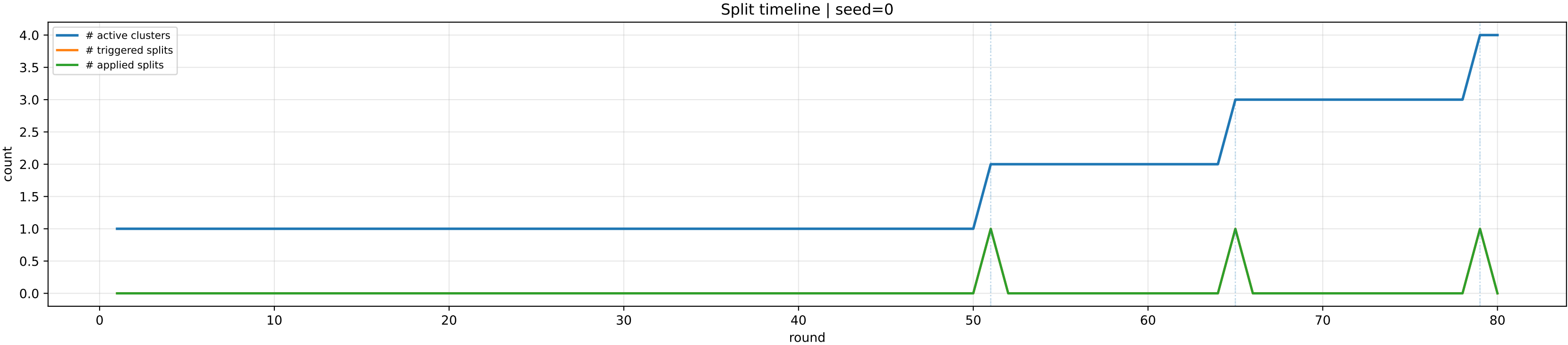
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	51	0	5	1000009.0	0.018	222.771	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 248.9 at (1000005, 1000009)

graph_id	1000005	1000006	1000007	1000008	1000009
1000005	0.0	177.6	191.1	105.0	248.9
1000006	177.6	0.0	103.2	85.3	214.3
1000007	191.1	103.2	0.0	178.6	200.0
1000008	105.0	85.3	178.6	0.0	227.8
1000009	248.9	214.3	200.0	227.8	0.0

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

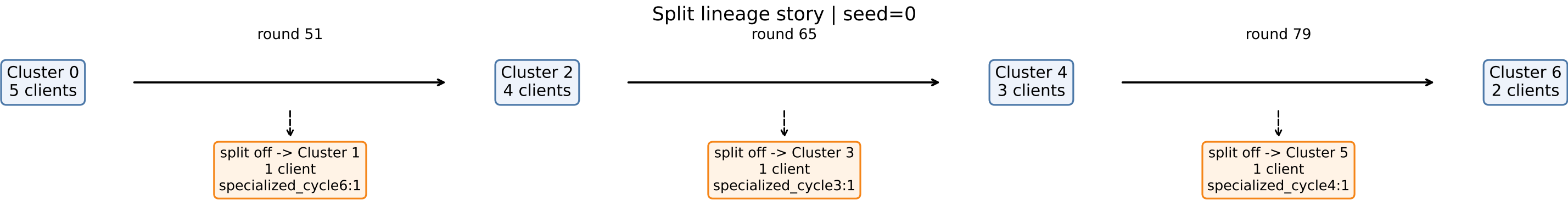


Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	51, 65, 79	51, 65, 79	1, 3, 5, 6	1:1, 3:1, 5:1, 6:2

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028



Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
51	0	5	176.044 < 186.033 , 355.774 > 345.711	C1 (1 client)	specialized_cycle6:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1
65	2	4	177.892 < 191.196 , 336.698 > 326.036	C3 (1 client)	specialized_cycle3:1	C4 (3 clients)	specialized_cycle2:1 , specialized_cycle4:1 , specialized_cycle5:1
79	4	3	188.078 < 197.007 , 373.586 > 299.772	C5 (1 client)	specialized_cycle4:1	C6 (2 clients)	specialized_cycle2:1 , specialized_cycle5:1

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

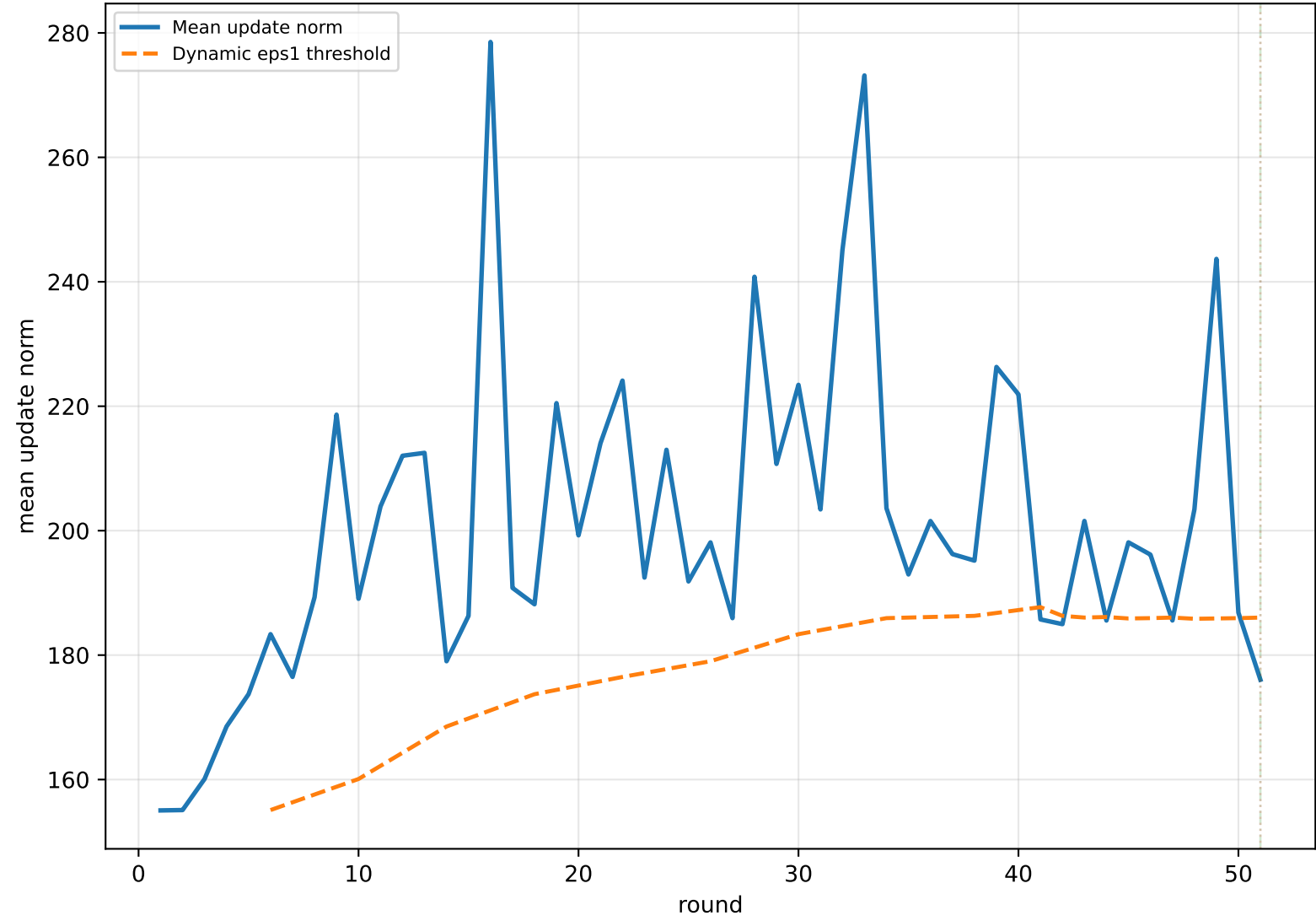
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000009	4	specialized_cycle6:1
3	1	1000006	1	specialized_cycle3:1
5	1	1000007	2	specialized_cycle4:1
6	2	1000005, 1000008	0, 3	specialized_cycle2:1 , specialized_cycle5:1

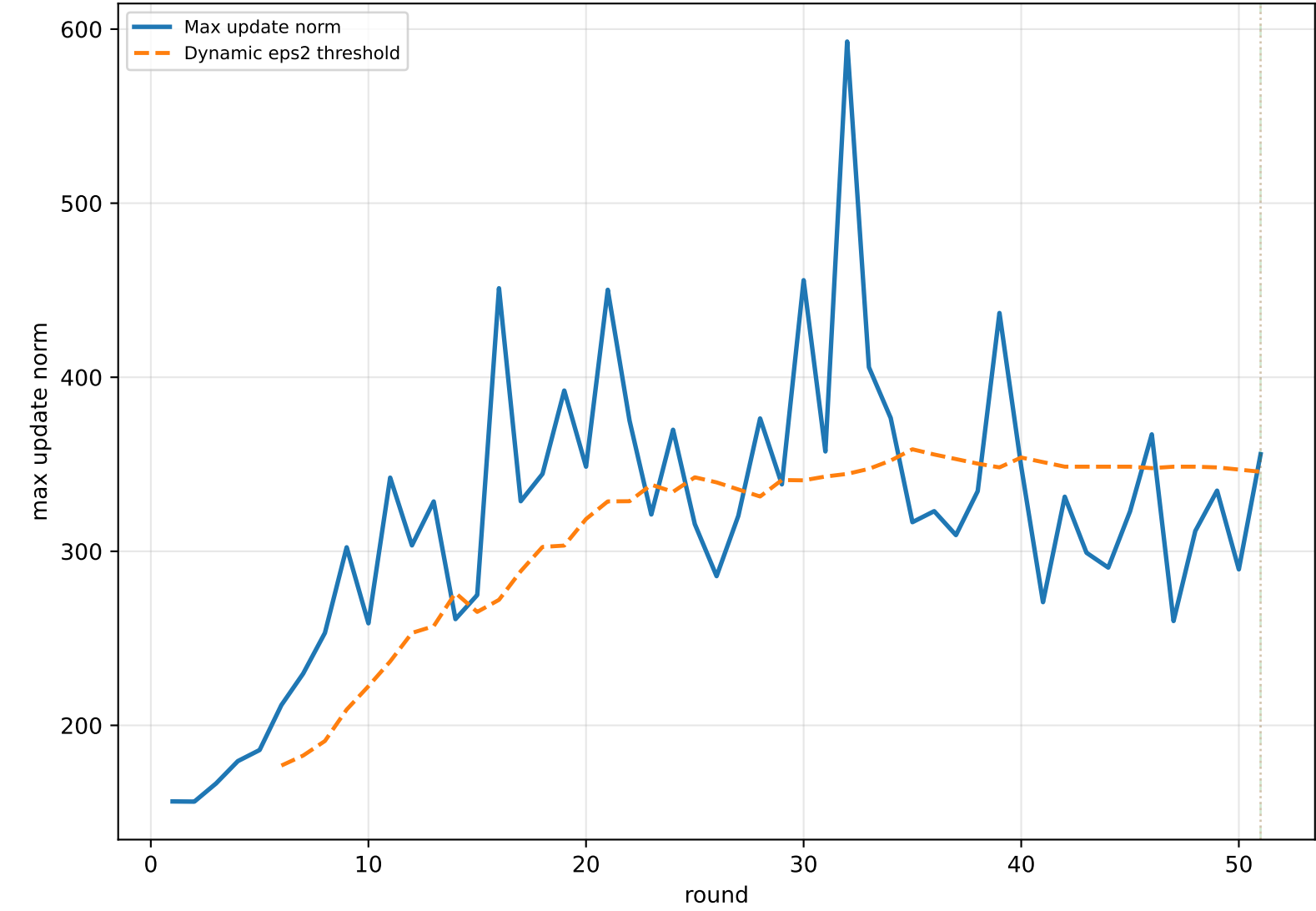
Controlled task-specialization level: Mid | GCFL+ specialized | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

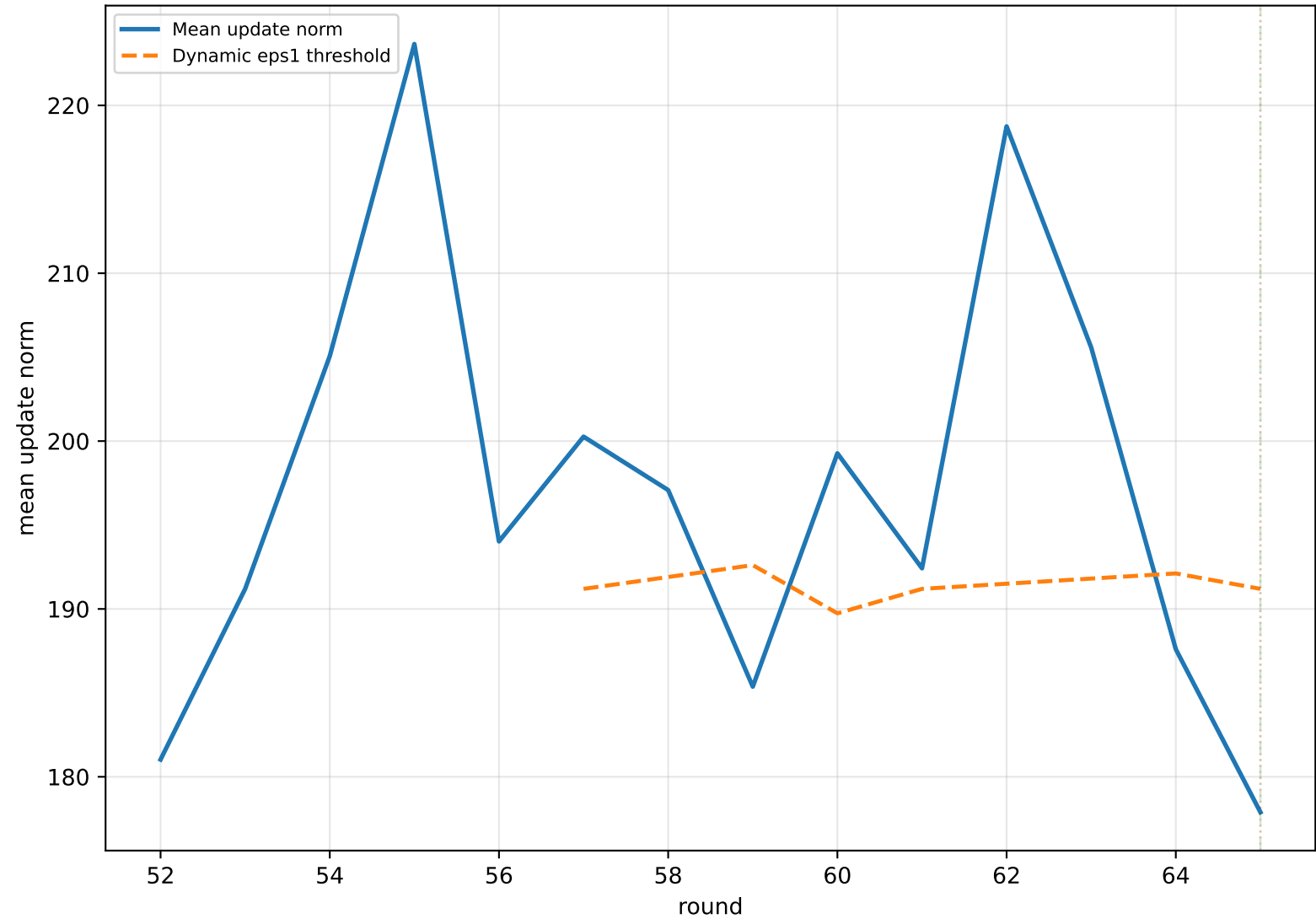
Cluster 0 | Mean update norm



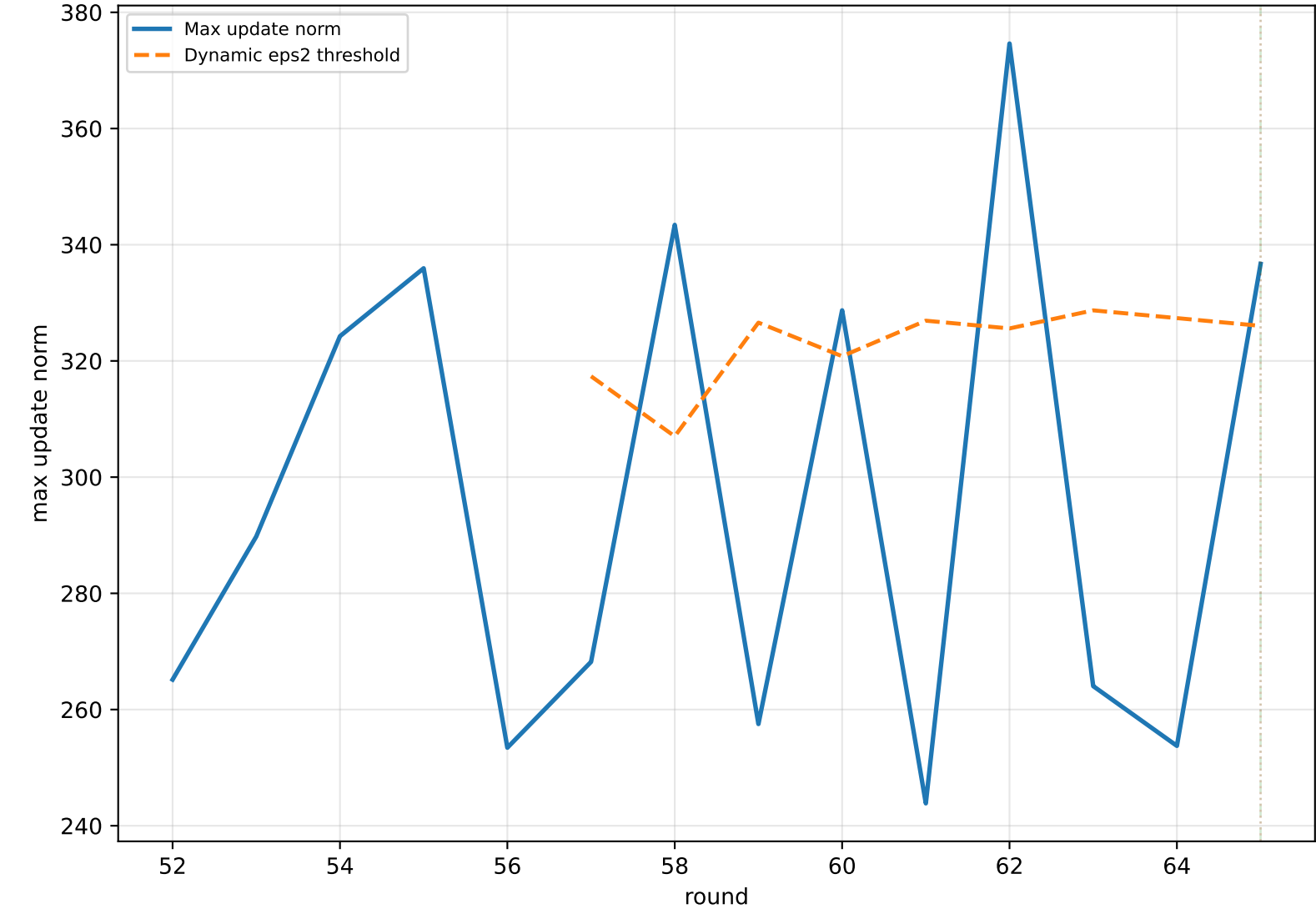
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



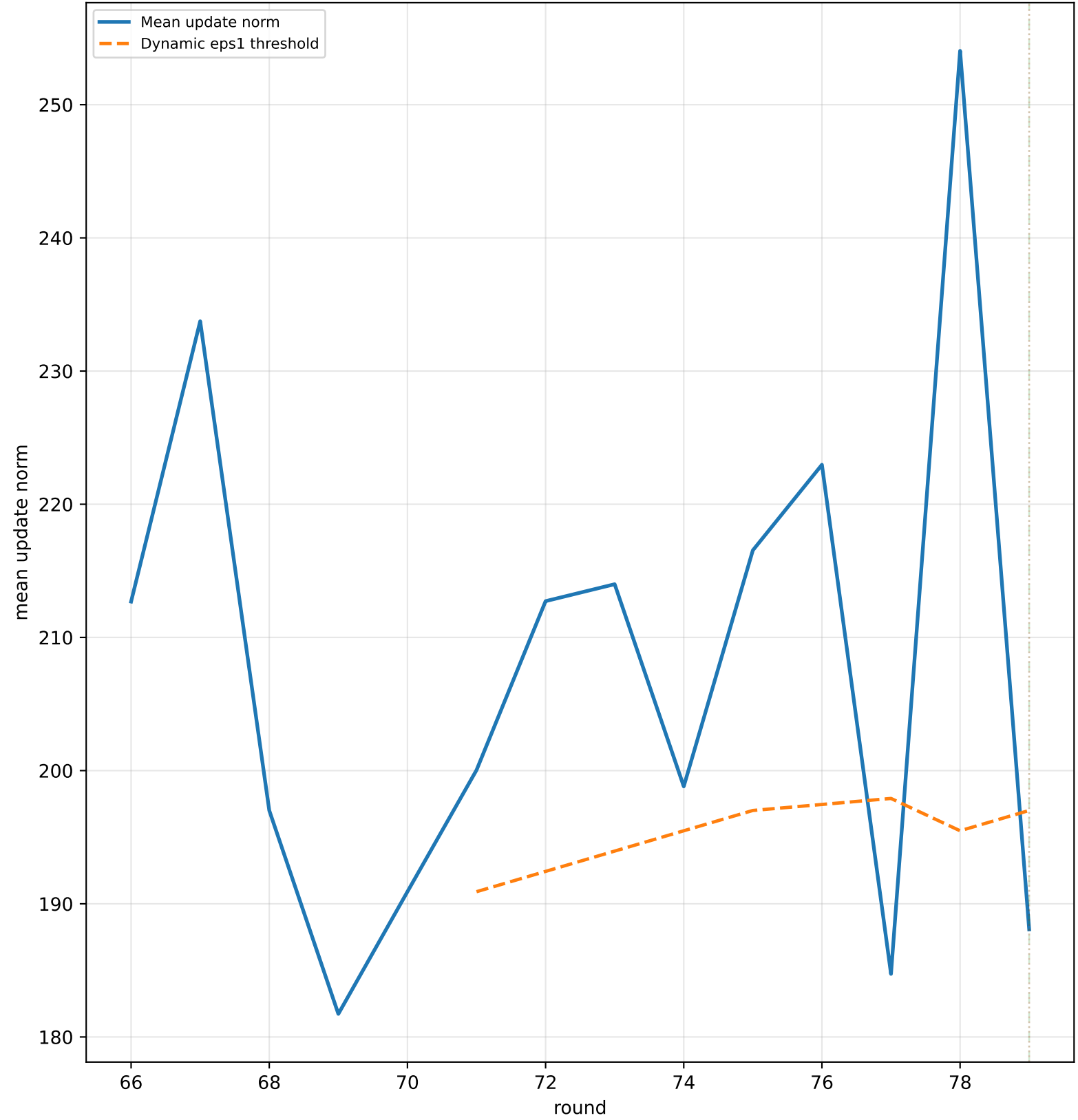
Cluster 2 | Max update norm



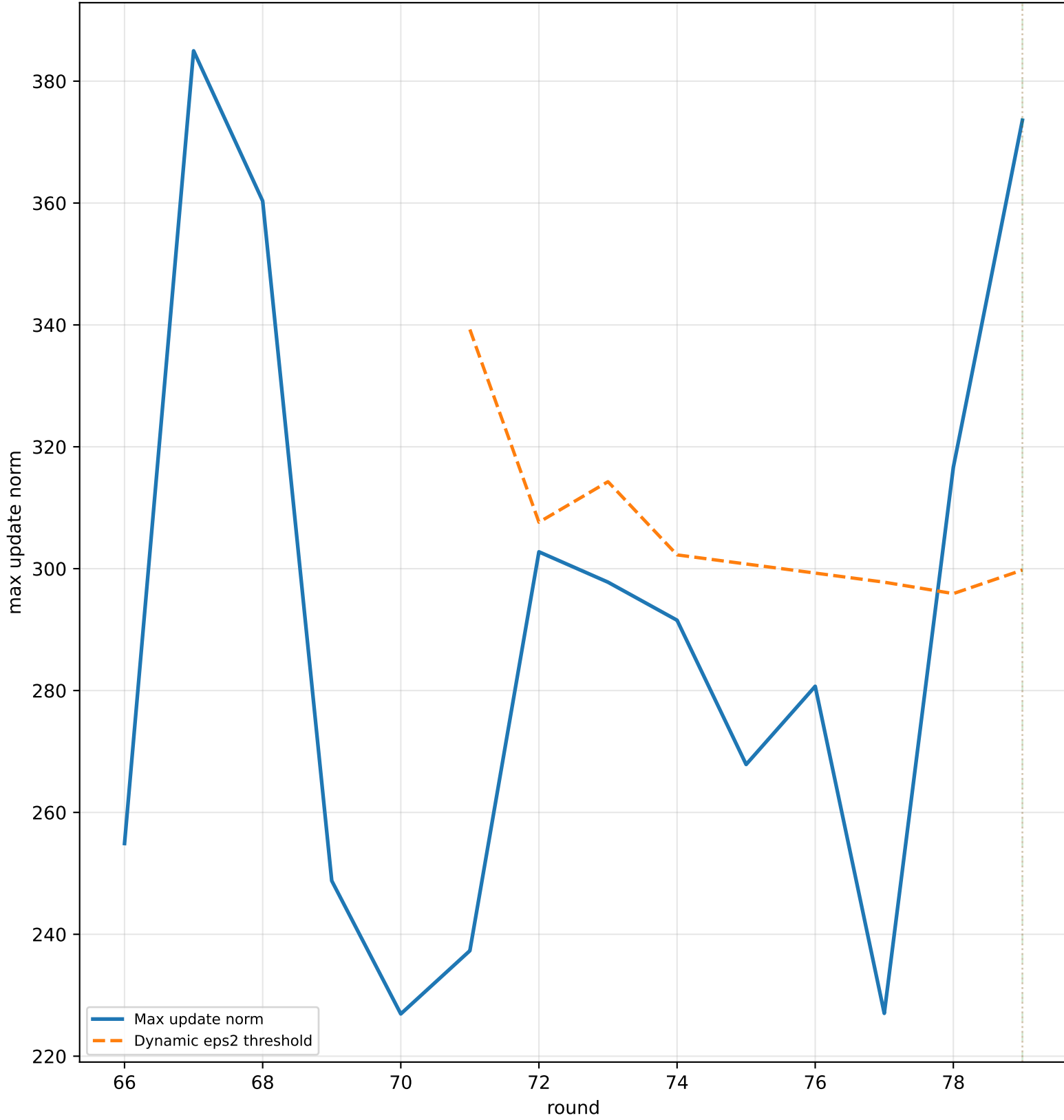
Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (4-4)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Cluster 4 | Mean update norm



Cluster 4 | Max update norm



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

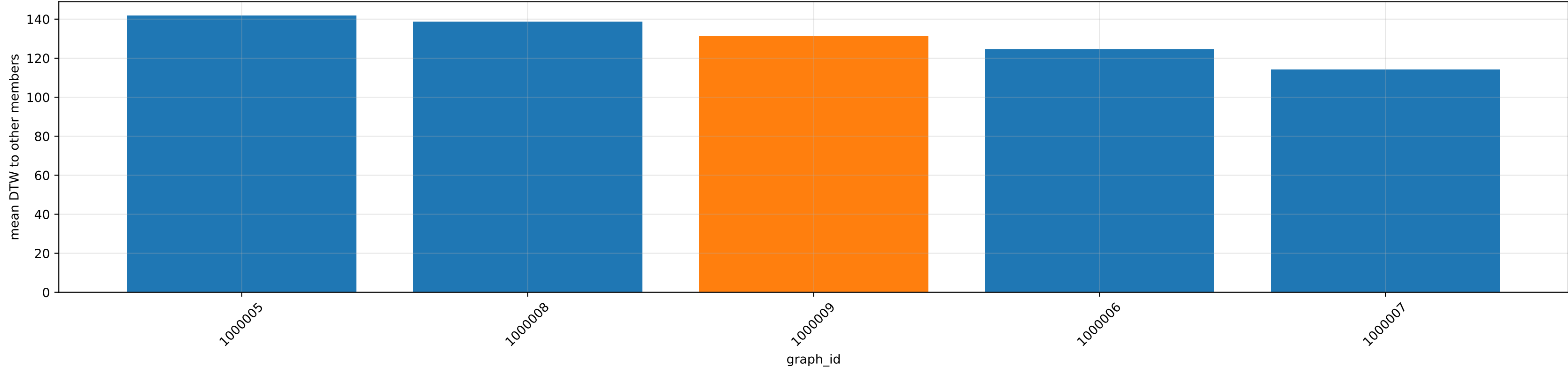
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	53	0	1 vs 4	1000009	1000005	no

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
131.206	nan	129.326

Per-member mean DTW



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

First applied split pairwise DTW metadata | seed=1

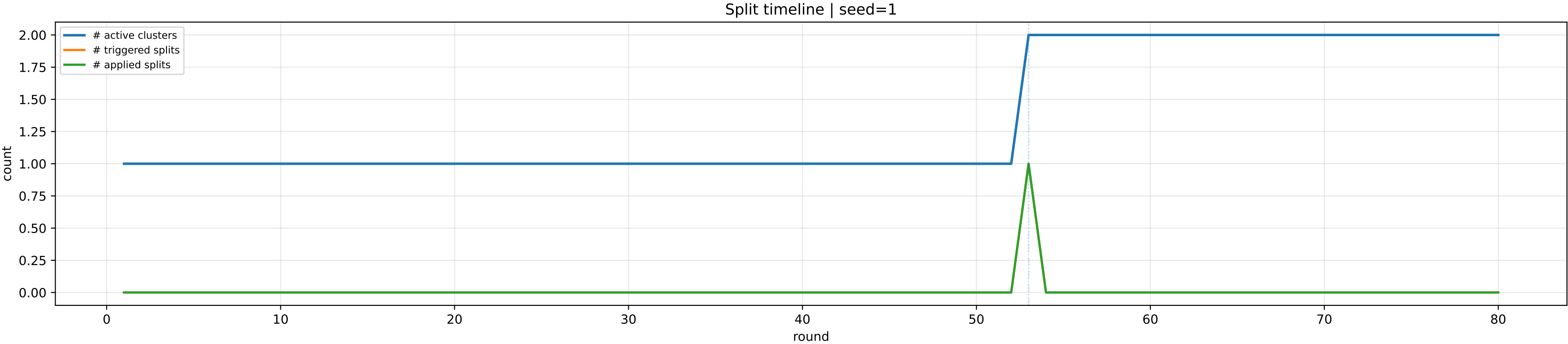
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	53	0	5	1000009.0	0.031	131.206	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 202.1 at (1000005, 1000006)

graph_id	1000005	1000006	1000007	1000008	1000009
1000005	0.0	202.1	70.1	186.3	109.0
1000006	202.1	0.0	126.7	52.9	116.5
1000007	70.1	126.7	0.0	137.8	121.6
1000008	186.3	52.9	137.8	0.0	177.8
1000009	109.0	116.5	121.6	177.8	0.0

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028



Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	53	53	1, 2	1:1, 2:4

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Split lineage story | seed=1
round 53

Cluster 0
5 clients

Cluster 2
4 clients

↓
split off -> Cluster 1
1 client
specialized_cycle6:1

Applied split story | seed=1

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
53	0	5	179.635 < 182.870 , 337.378 > 333.439	C1 (1 client)	specialized_cycle6:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

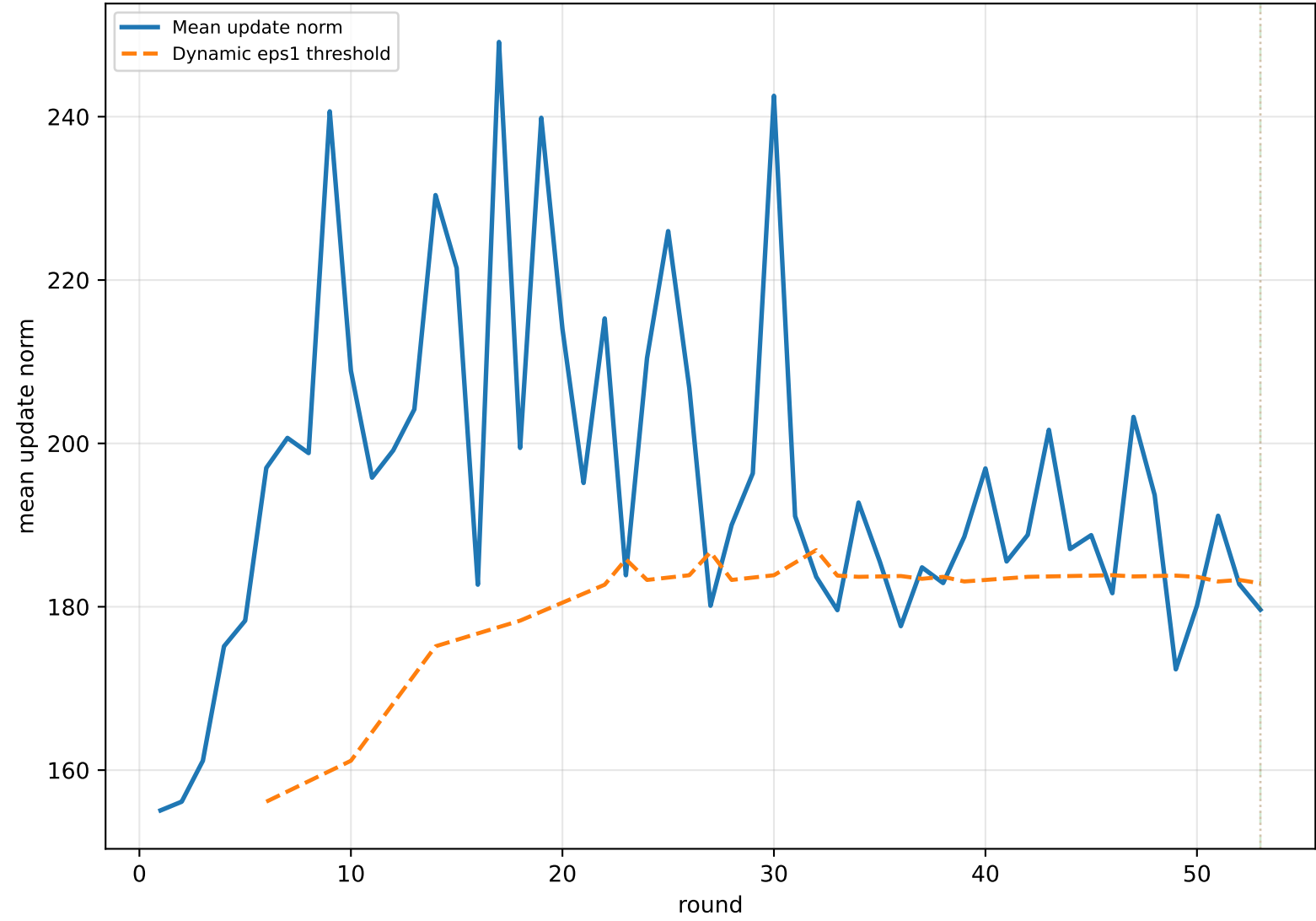
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000009	4	specialized_cycle6:1
2	4	1000005, 1000006, 1000007, 1000008	0, 1, 2, 3	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

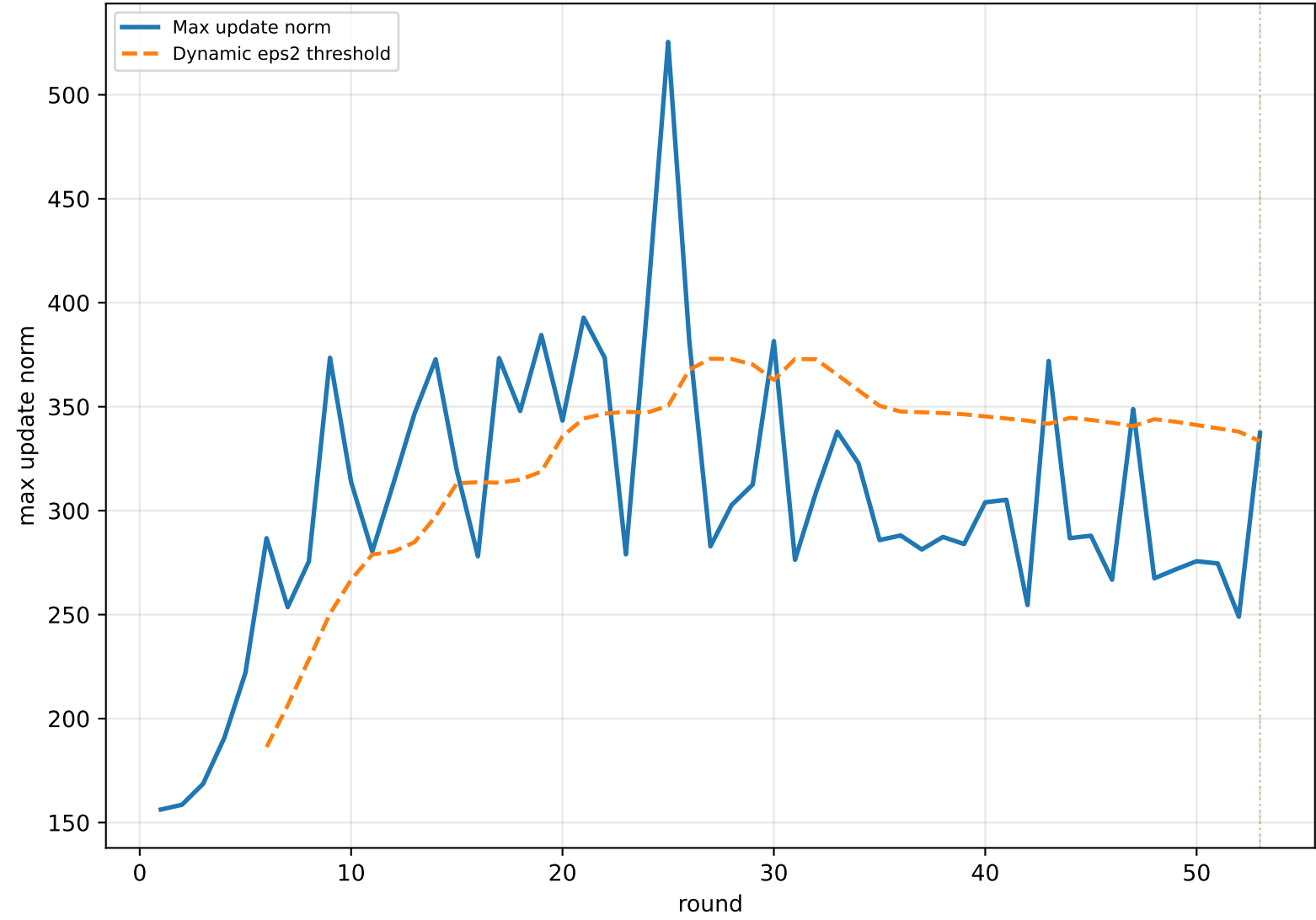
Controlled task-specialization level: Mid | GCFL+ aggregated / pooled over 5 clients | seed=1 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

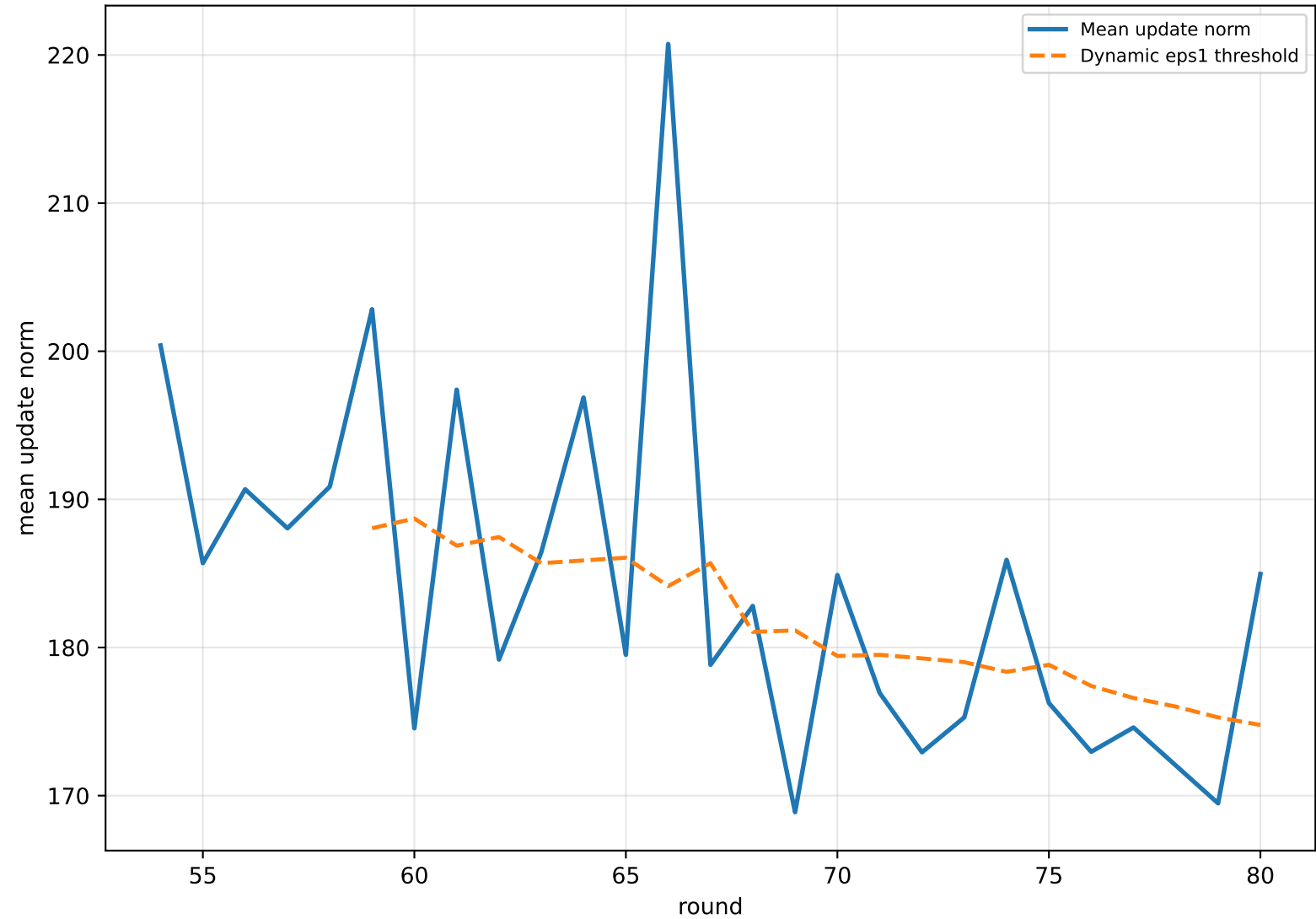
Cluster 0 | Mean update norm



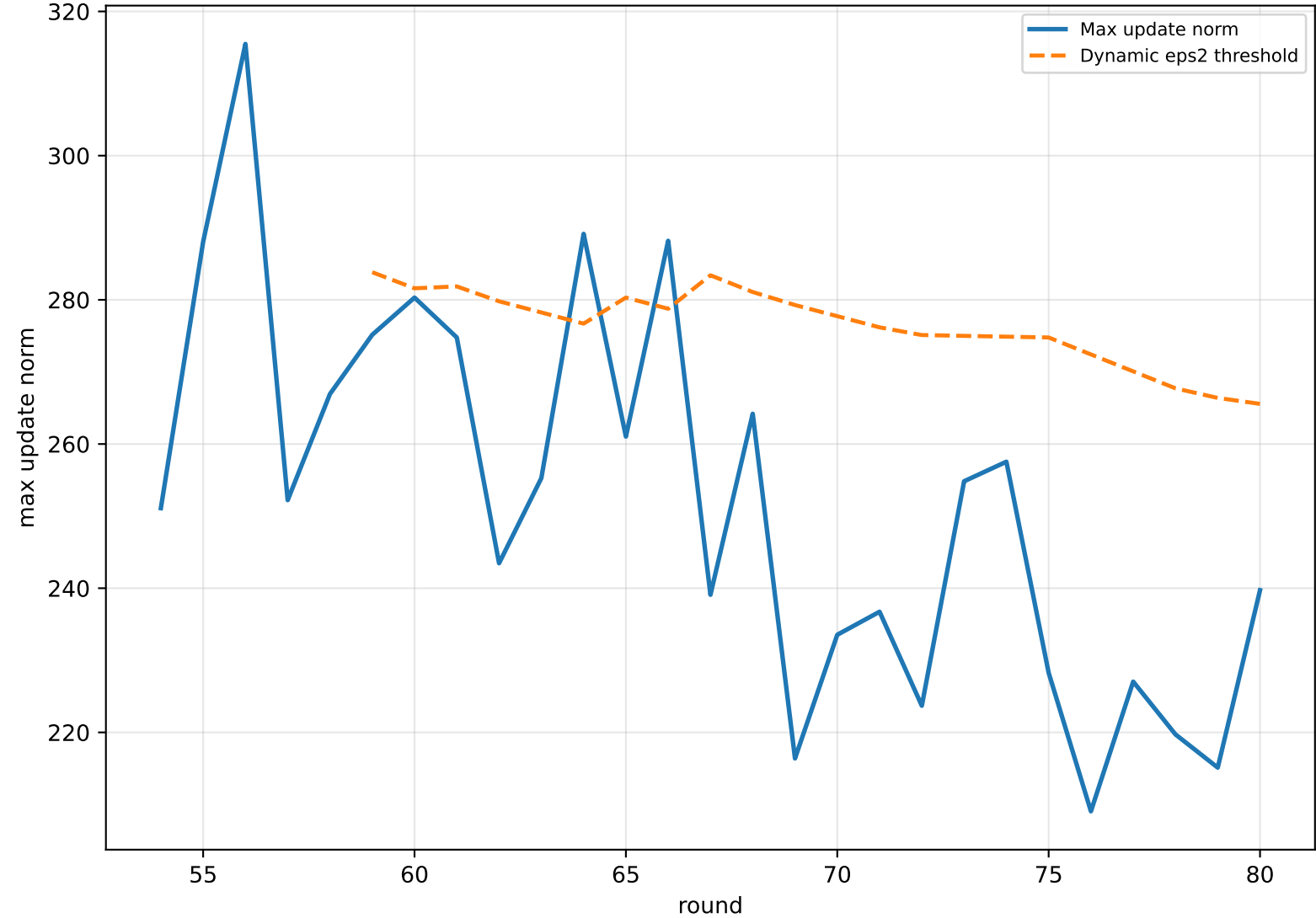
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

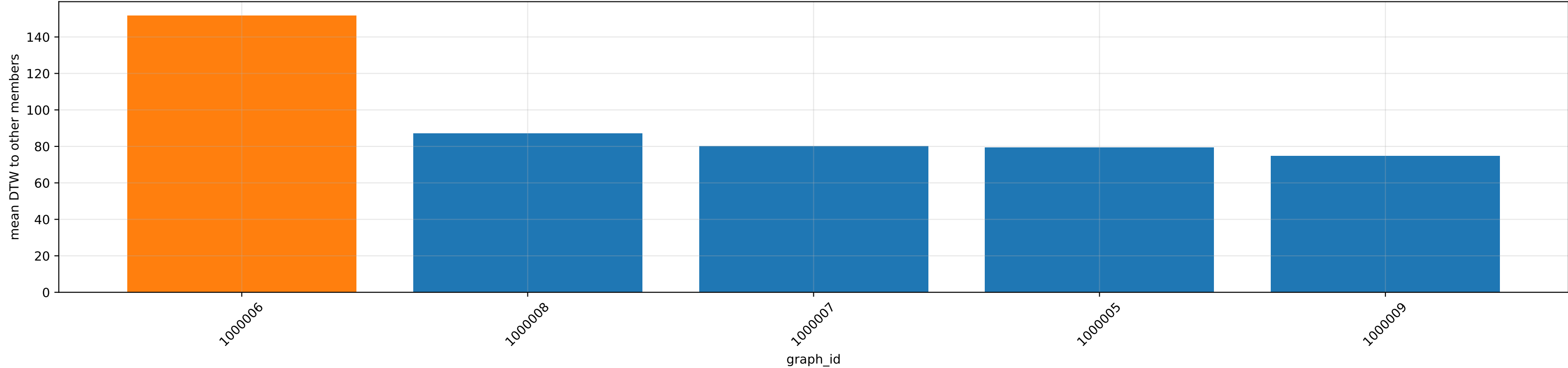
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	63	0	1 vs 4	1000006	1000006	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
151.764	nan	56.554

Per-member mean DTW



Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

First applied split pairwise DTW metadata | seed=2

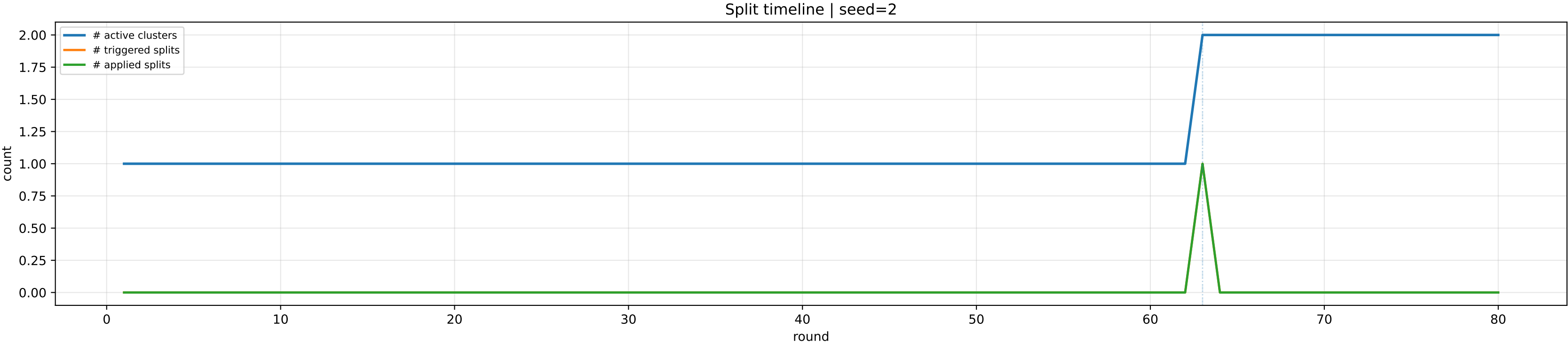
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	63	0	5	1000006.0	0.026	151.764	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 170.6 at (1000006, 1000007)

graph_id	1000005	1000006	1000007	1000008	1000009
1000005	0.0	154.9	34.2	75.9	52.8
1000006	154.9	0.0	170.6	153.9	127.6
1000007	34.2	170.6	0.0	57.8	57.7
1000008	75.9	153.9	57.8	0.0	61.0
1000009	52.8	127.6	57.7	61.0	0.0

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028



Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	63	63	1, 2	1:1, 2:4

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Split lineage story | seed=2
round 63

Cluster 0
5 clients

Cluster 2
4 clients

split off -> Cluster 1
1 client
specialized_cycle3:1

Applied split story | seed=2

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
63	0	5	173.527 < 174.477 , 309.102 > 277.497	C1 (1 client)	specialized_cycle3:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1

Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

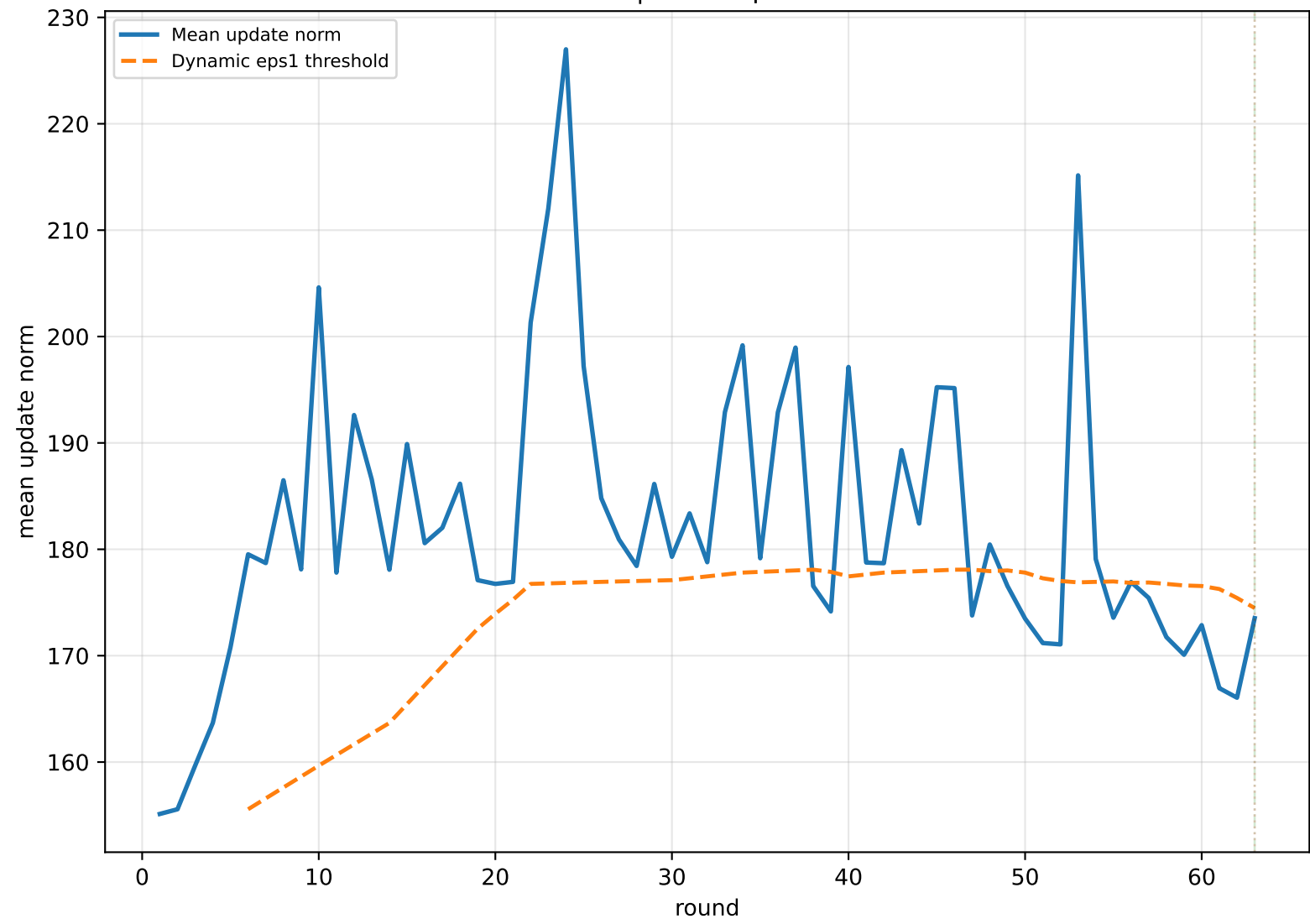
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000006	1	specialized_cycle3:1
2	4	1000005, 1000007, 1000008, 1000009	0, 2, 3, 4	specialized_cycle2:1 , specialized_cycle4:1 , specialized_cycle5:1 , specialized_cycle6:1

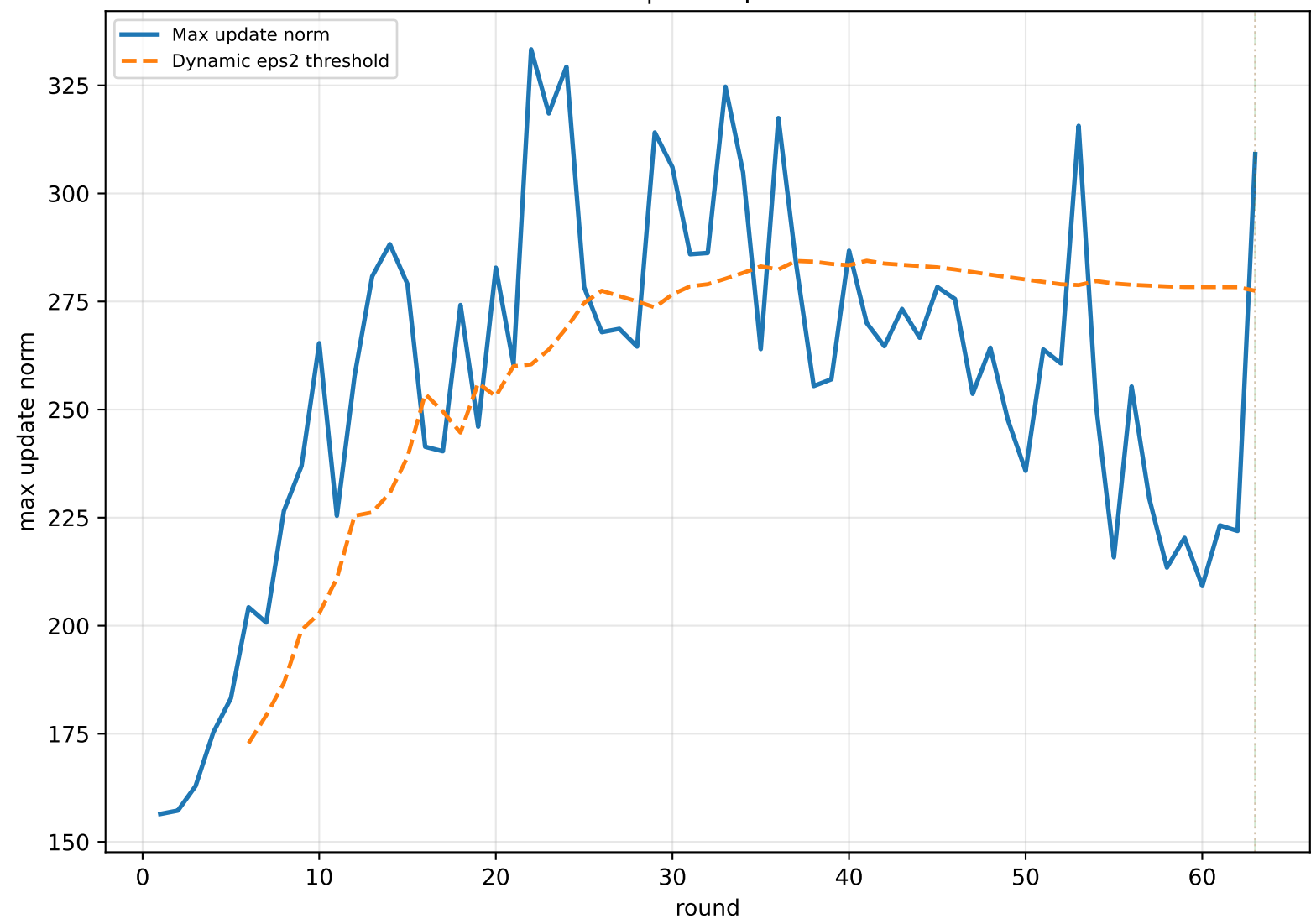
Controlled task-specialization level: Mid | GCFL+ specialized | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 12.8%±0.0% / true 25.7% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 9.7%±0.0% / true 19.4% | cycle6=visible 11.8%±0.0% / true 23.6%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

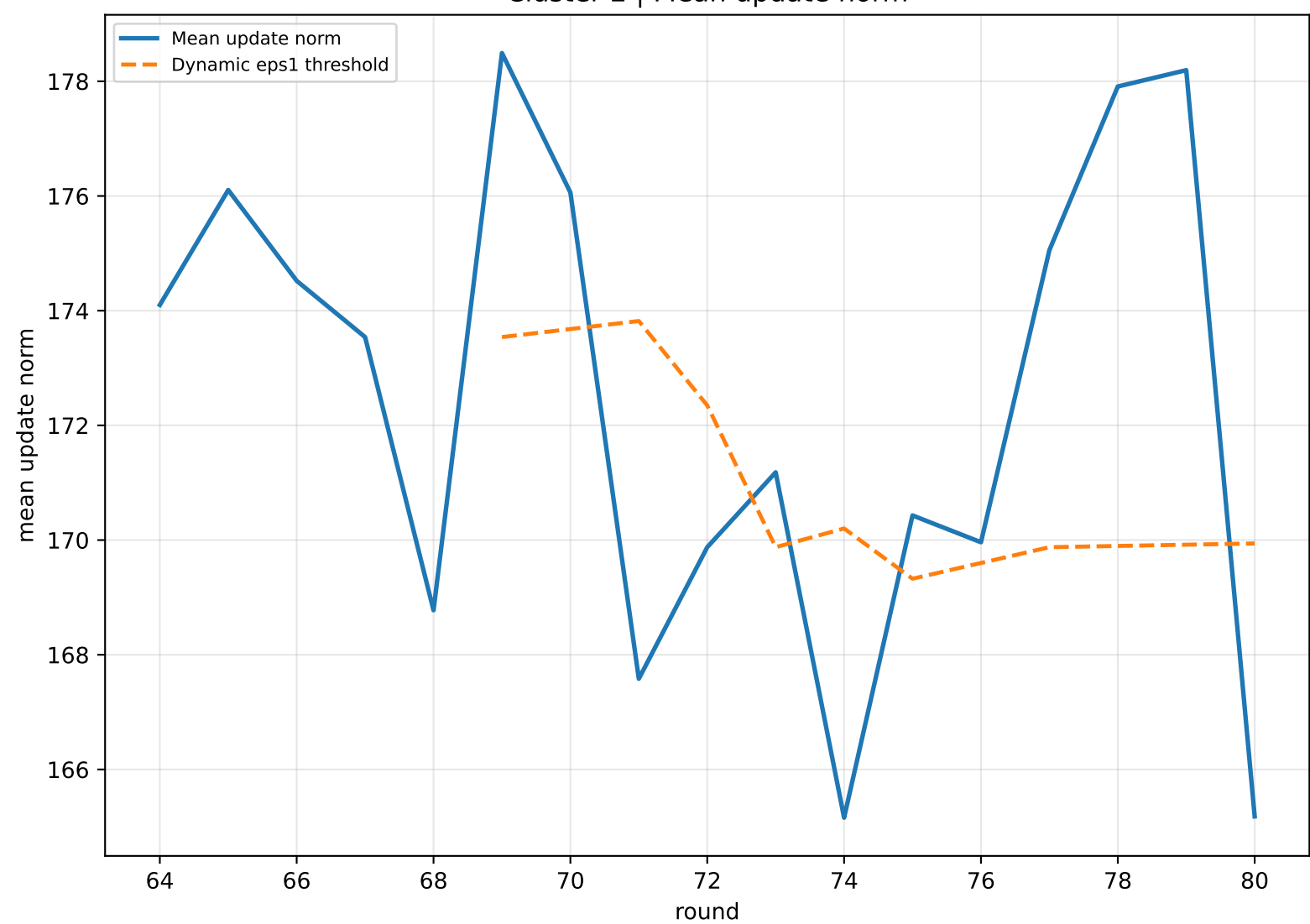
Cluster 0 | Mean update norm



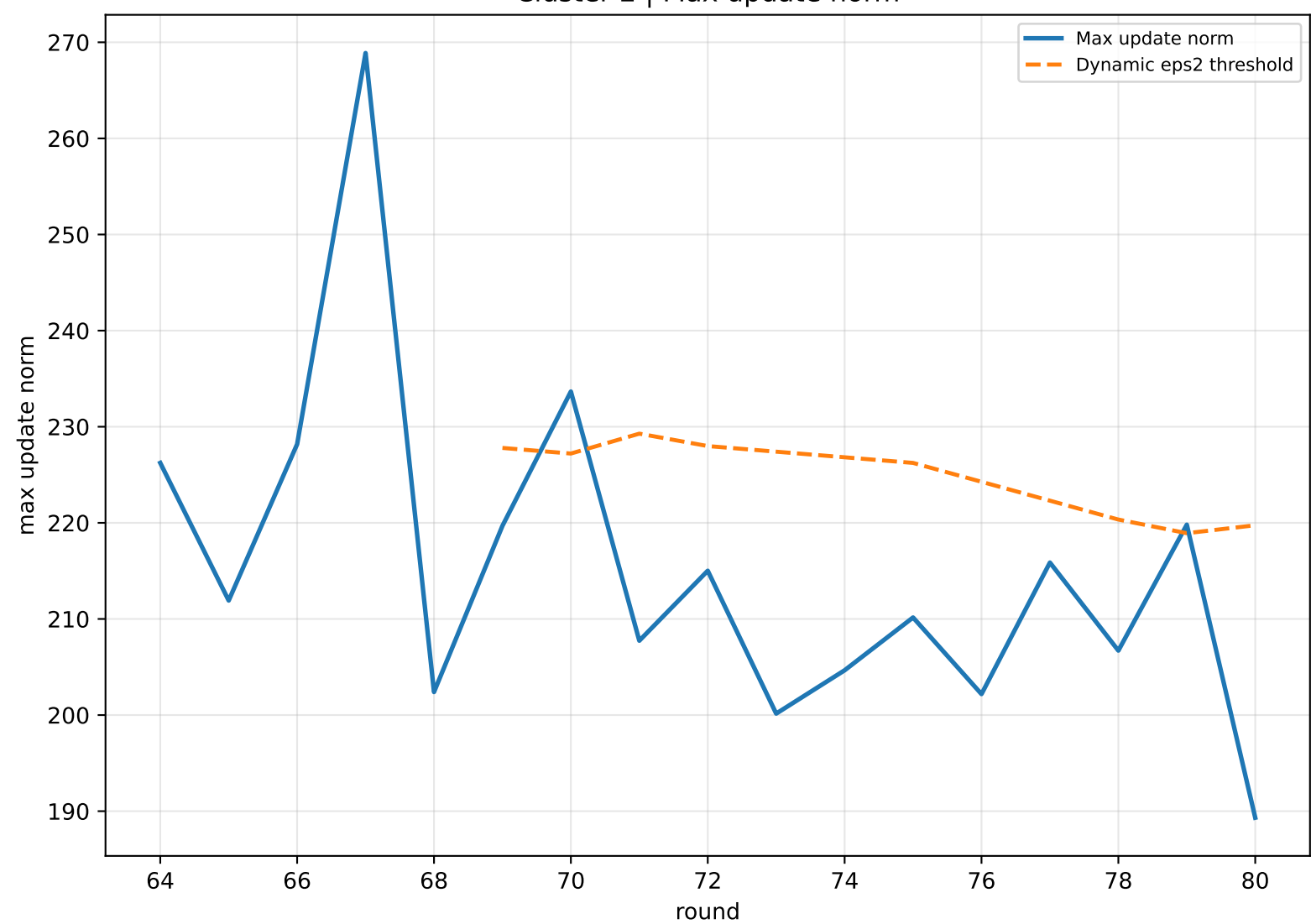
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



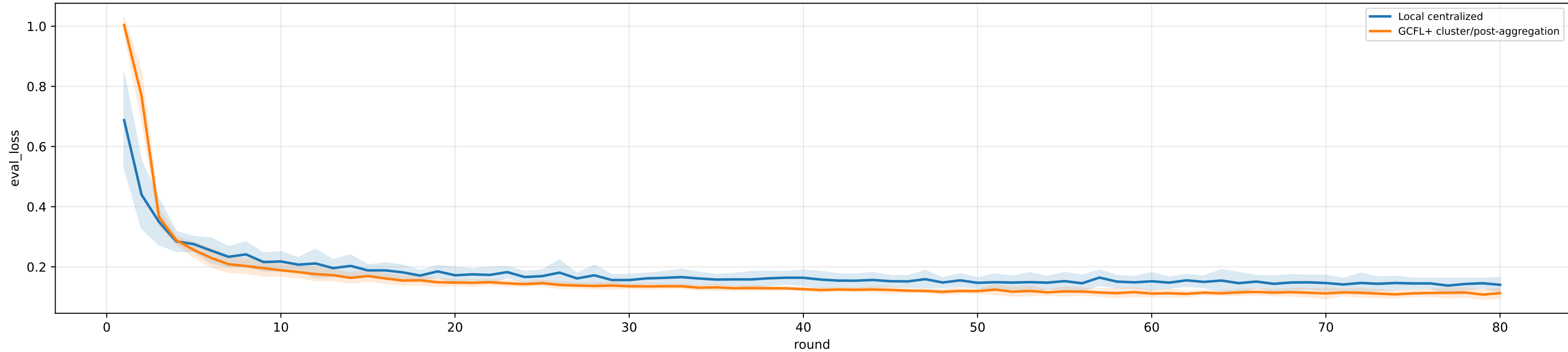
Controlled task-specialization level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

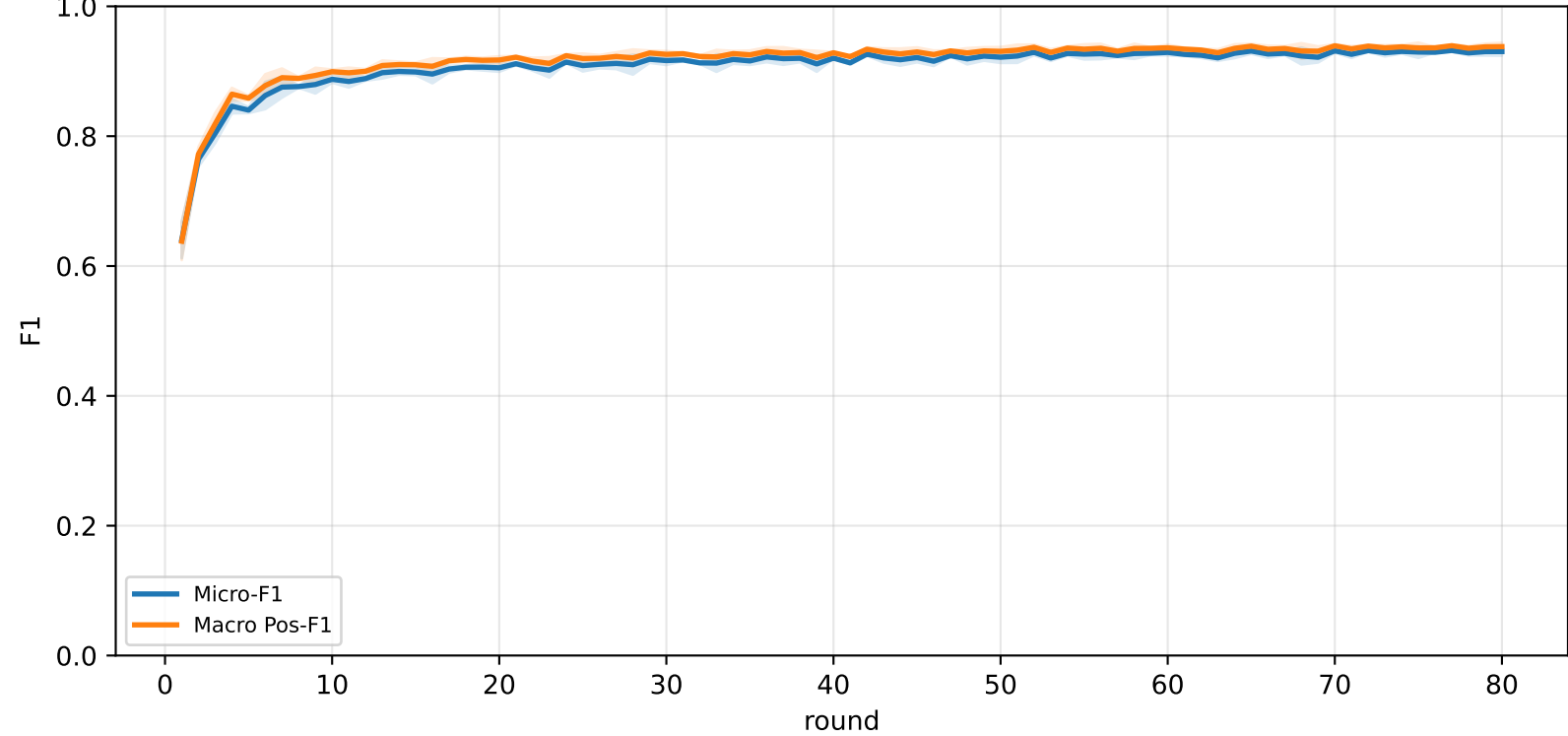
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 0.3%	93.5% ± 0.2%	98.9% ± 0.3%	98.0% ± 0.1%	95.7% ± 0.7%	91.4% ± 0.5%	83.5% ± 0.9%
GCFL+	93.9% ± 0.6%	94.5% ± 0.5%	98.9% ± 0.1%	98.3% ± 0.1%	95.1% ± 0.3%	93.1% ± 1.0%	87.2% ± 1.2%

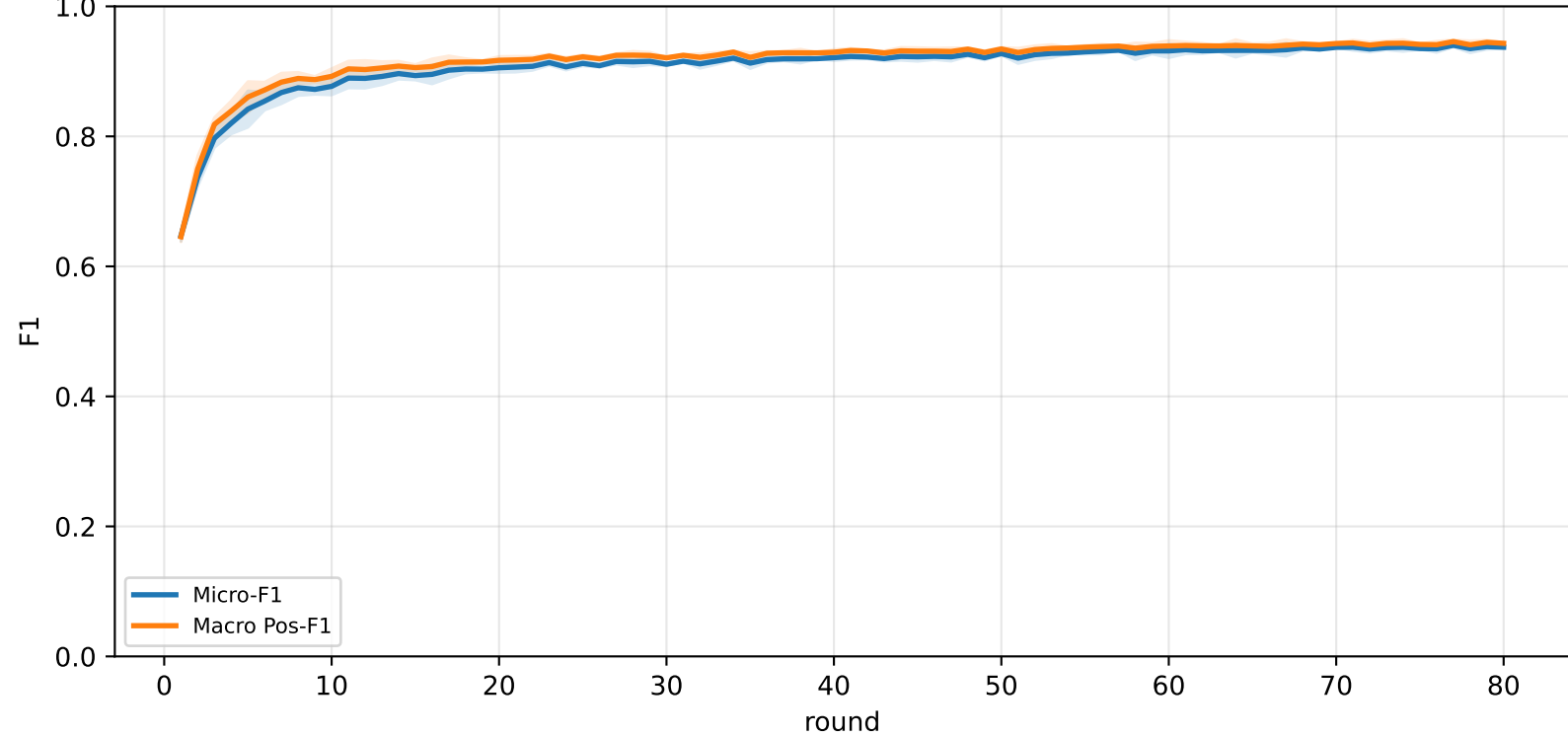
Validation loss comparison



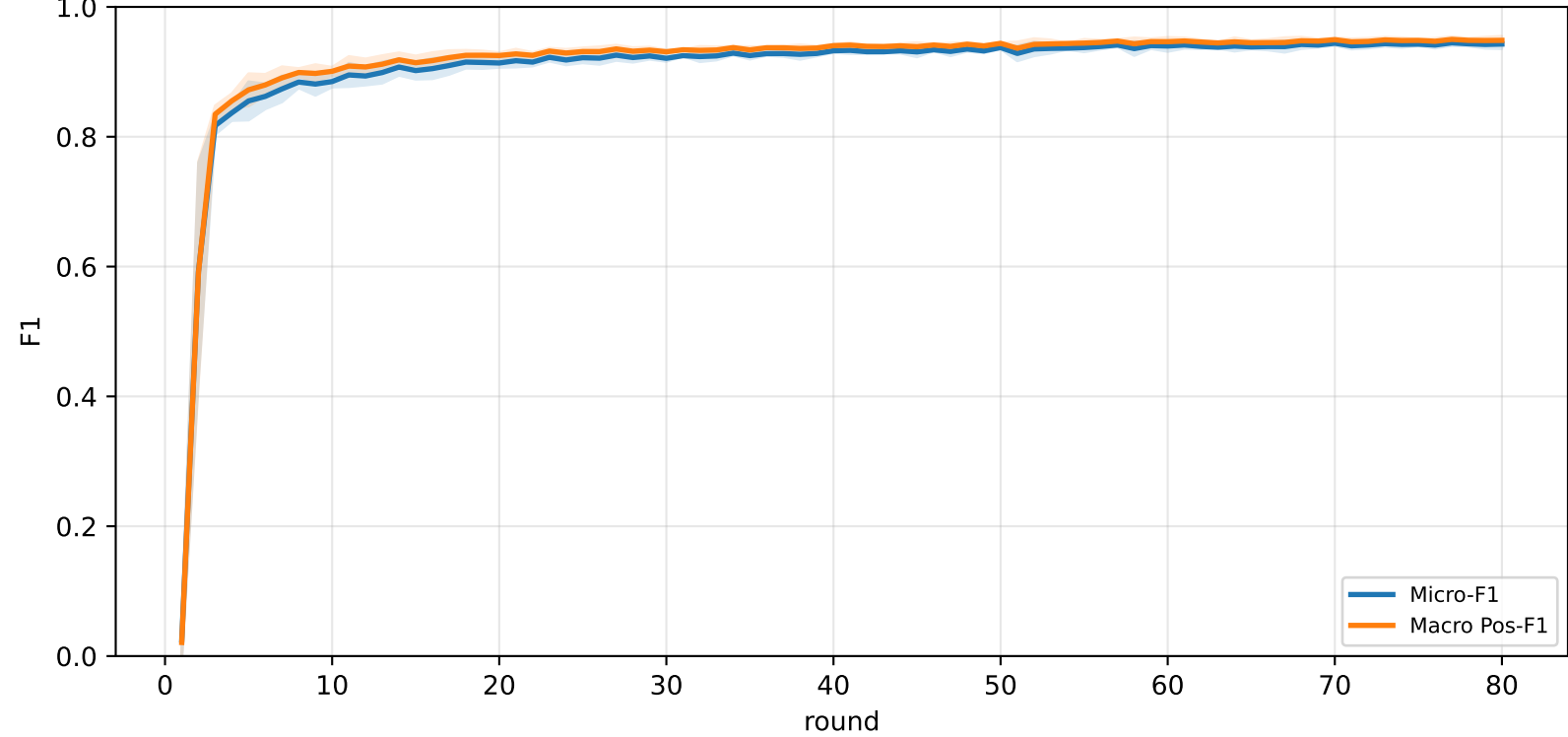
Pooled local centralized micro / macro F1



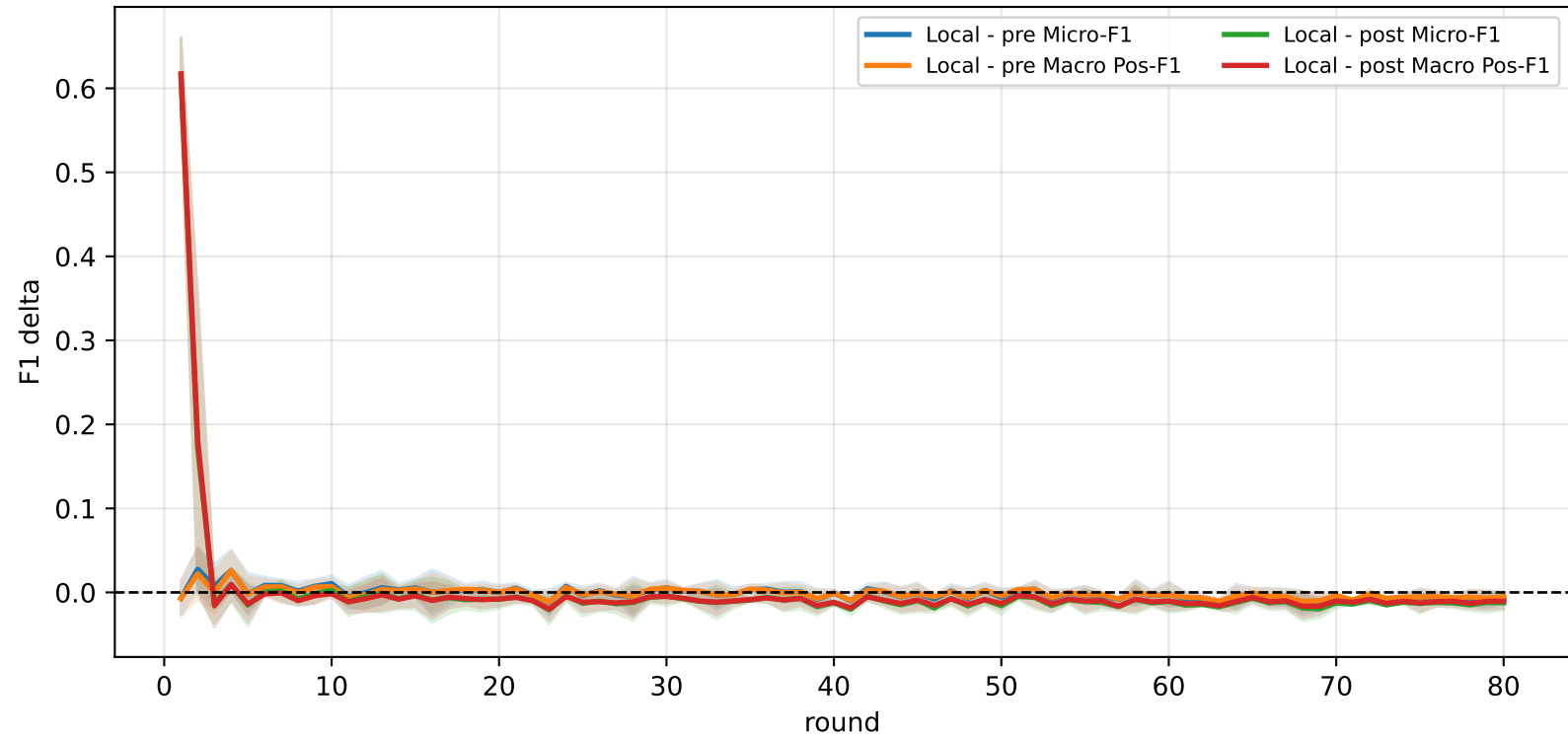
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

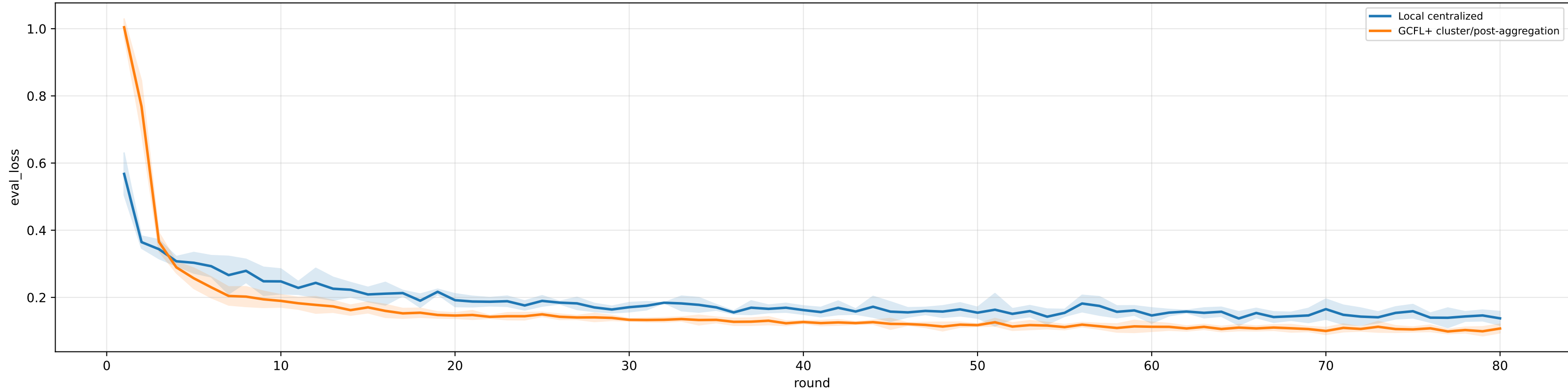


family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

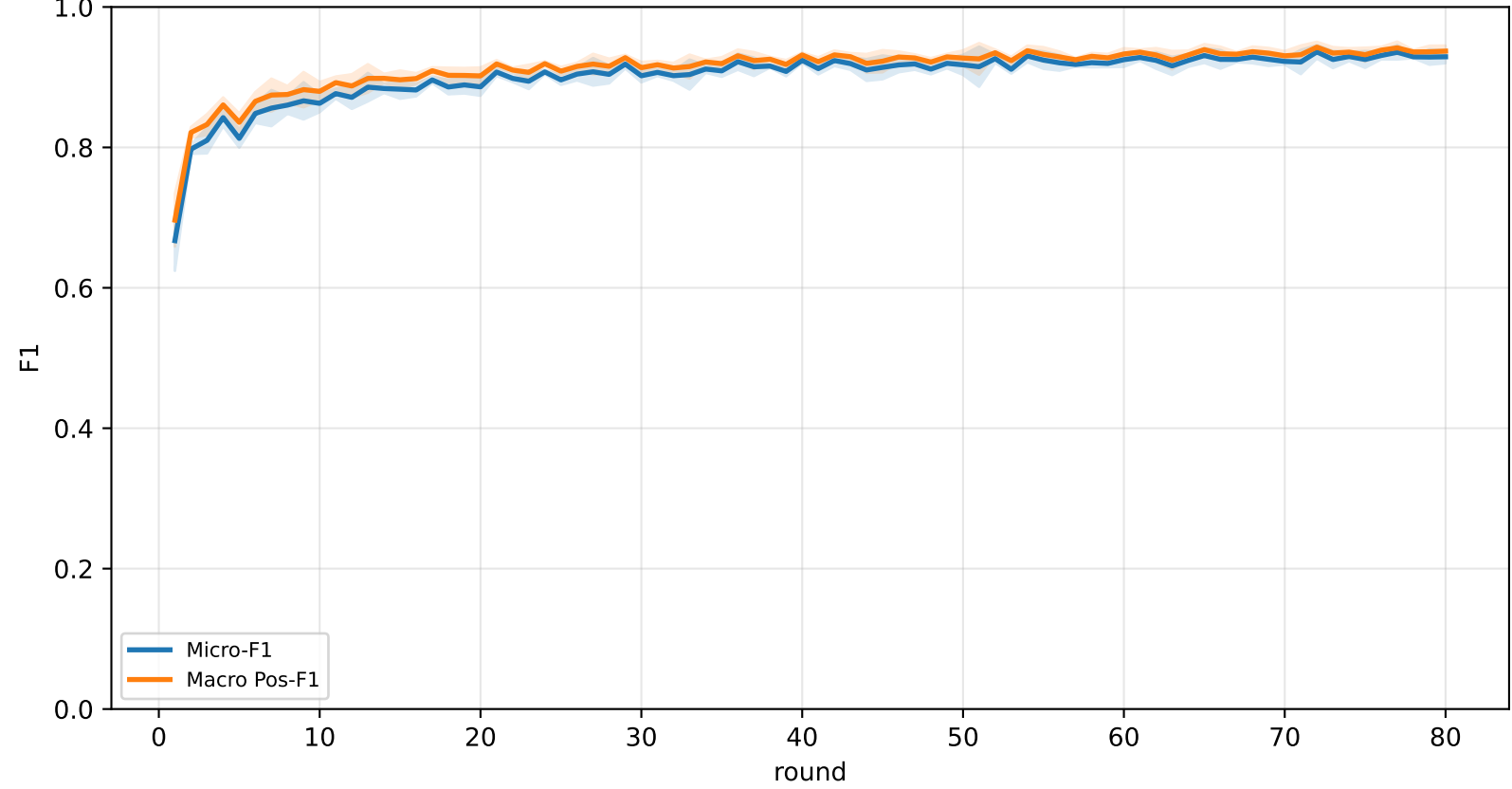
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 1.0%	93.5% ± 0.8%	99.1% ± 0.2%	98.1% ± 0.1%	95.2% ± 0.9%	91.2% ± 1.4%	84.0% ± 3.0%
GCFL+	93.9% ± 0.8%	94.5% ± 0.7%	98.9% ± 0.4%	98.2% ± 0.3%	94.9% ± 0.3%	93.2% ± 1.3%	87.0% ± 1.4%

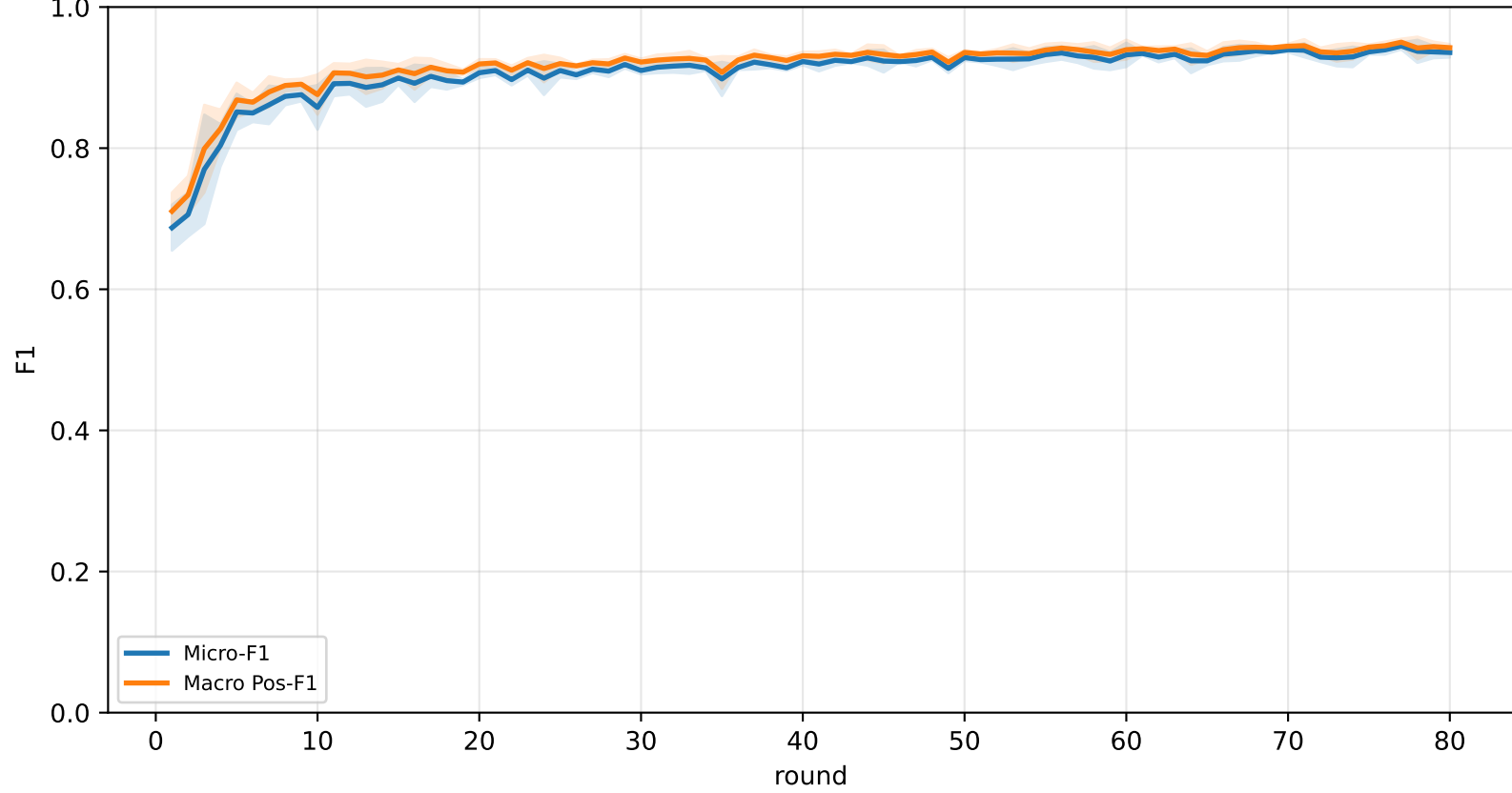
Validation loss comparison



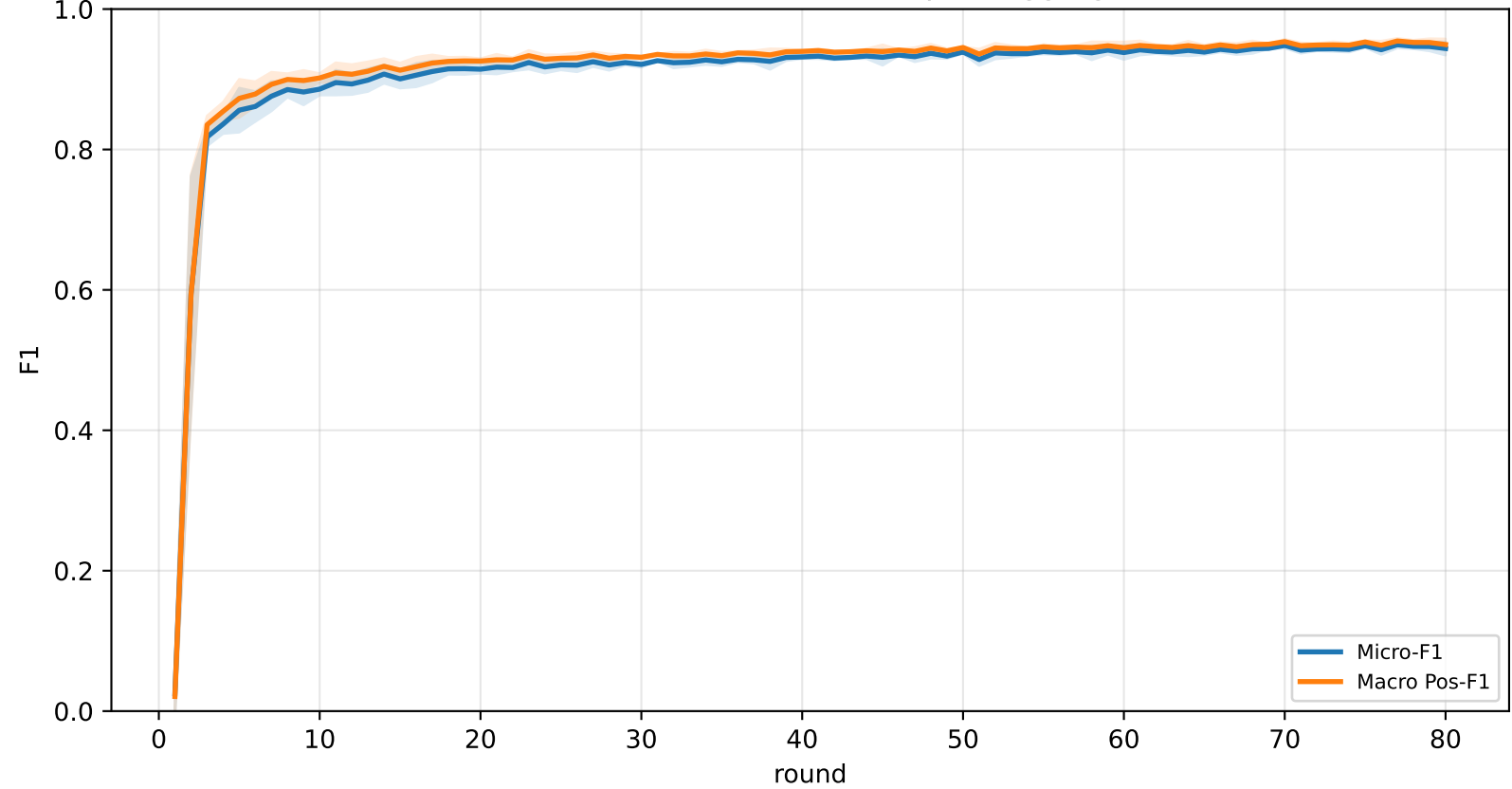
Pooled local centralized micro / macro F1



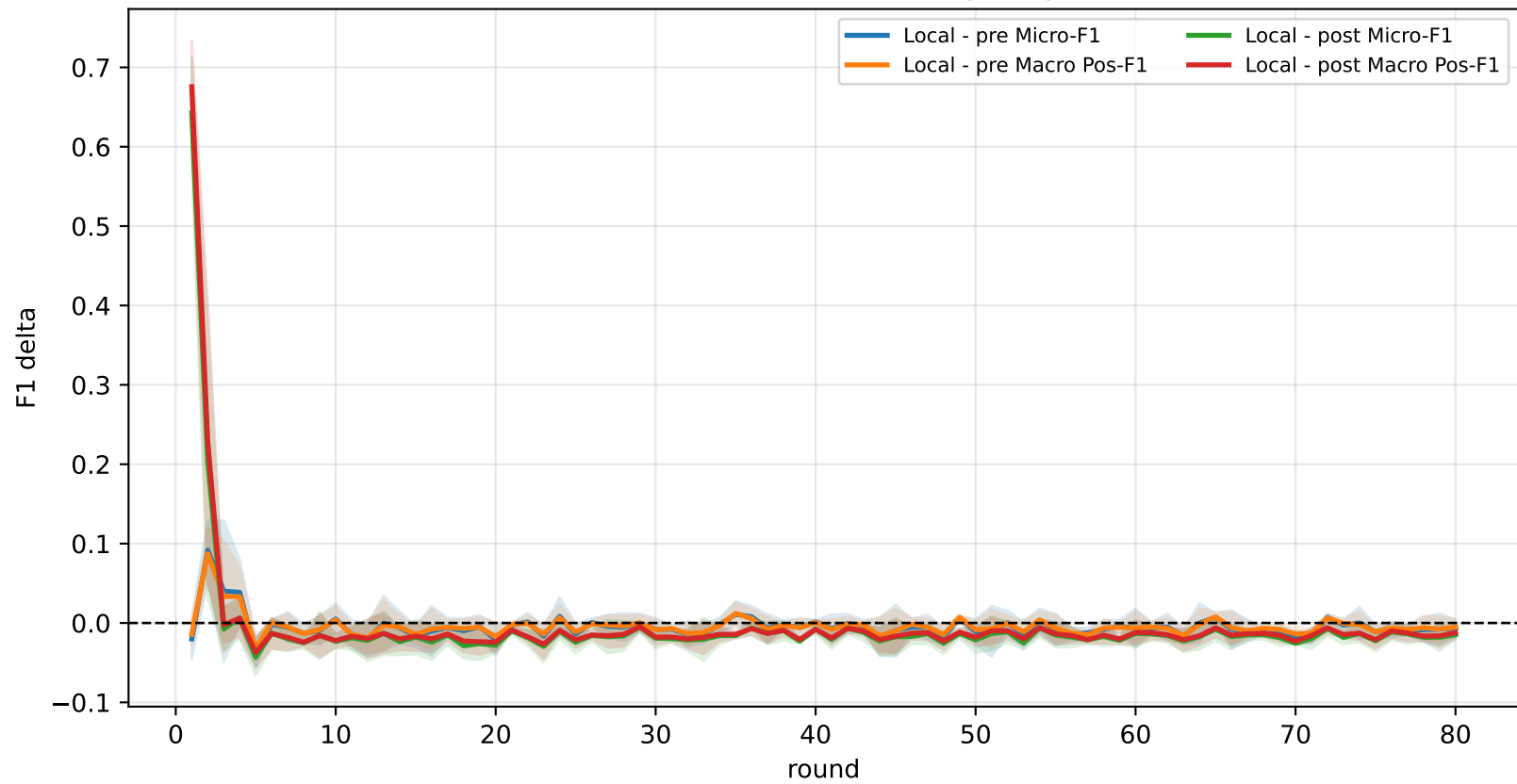
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

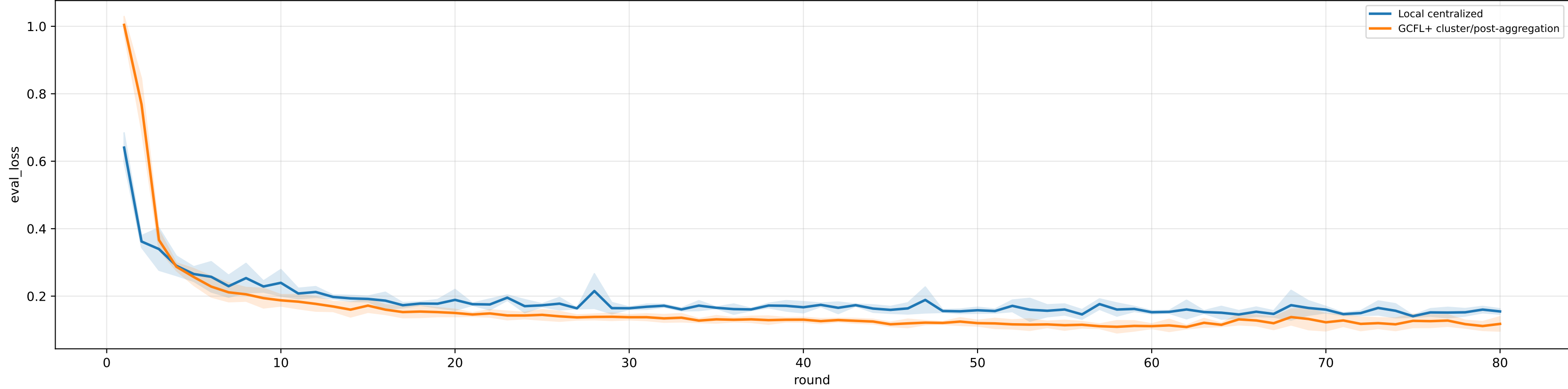


family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

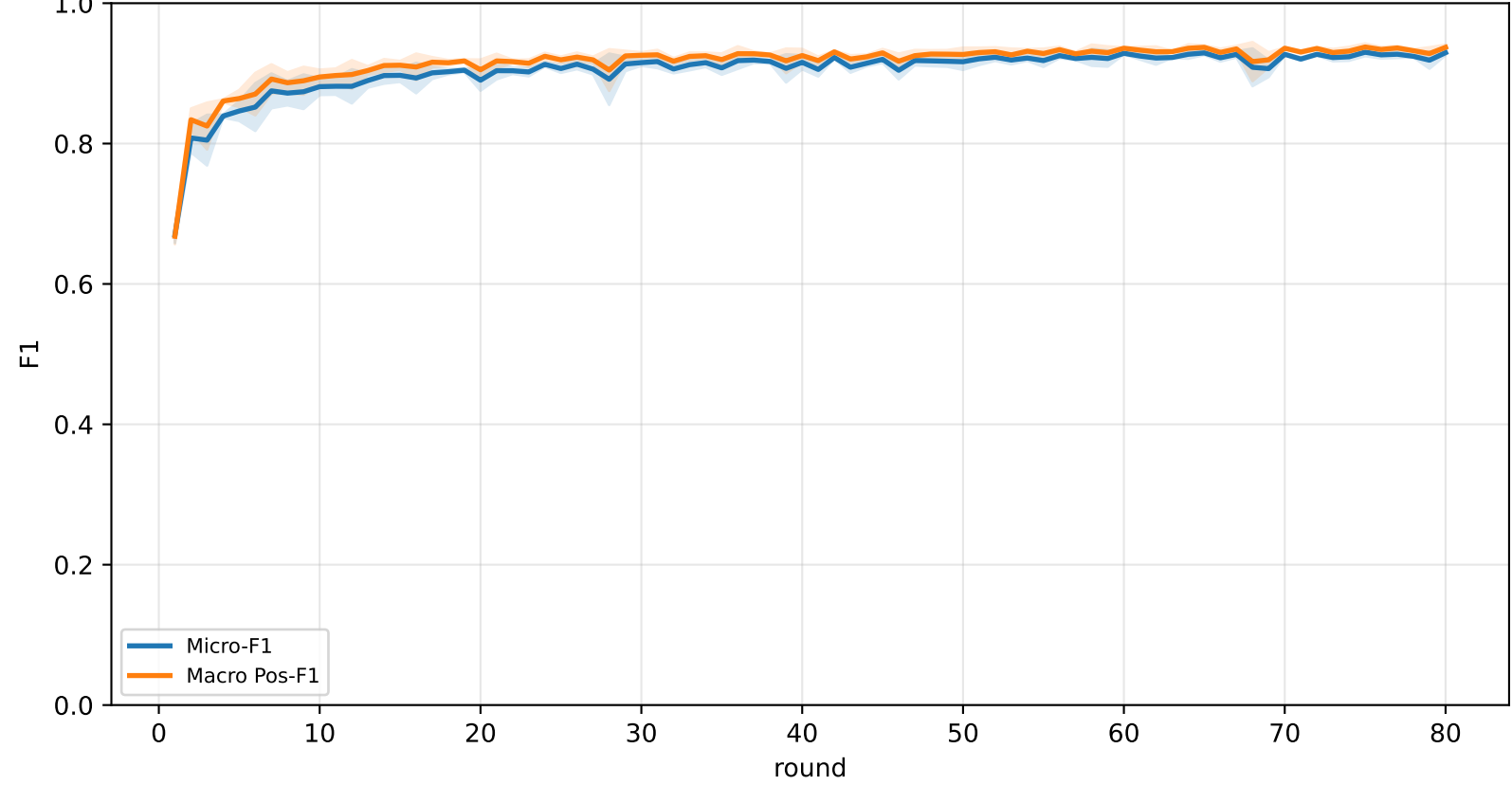
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.7% ± 0.2%	92.7% ± 0.2%	98.9% ± 0.1%	97.9% ± 0.1%	94.7% ± 1.2%	90.1% ± 0.6%	81.9% ± 0.3%
GCFL+	93.5% ± 0.7%	94.1% ± 0.6%	98.6% ± 0.5%	98.5% ± 0.1%	94.4% ± 0.6%	92.8% ± 1.3%	86.3% ± 1.0%

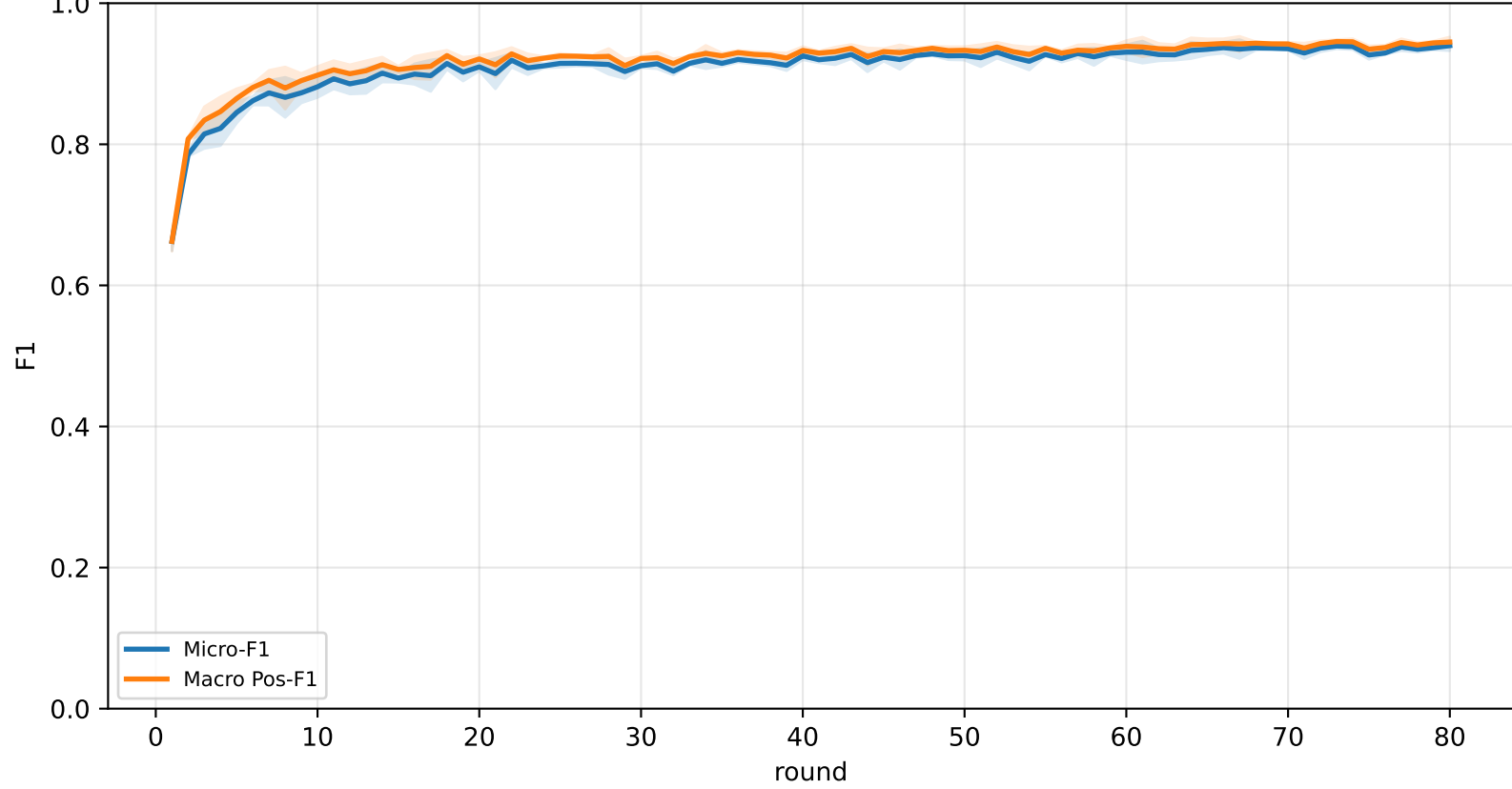
Validation loss comparison



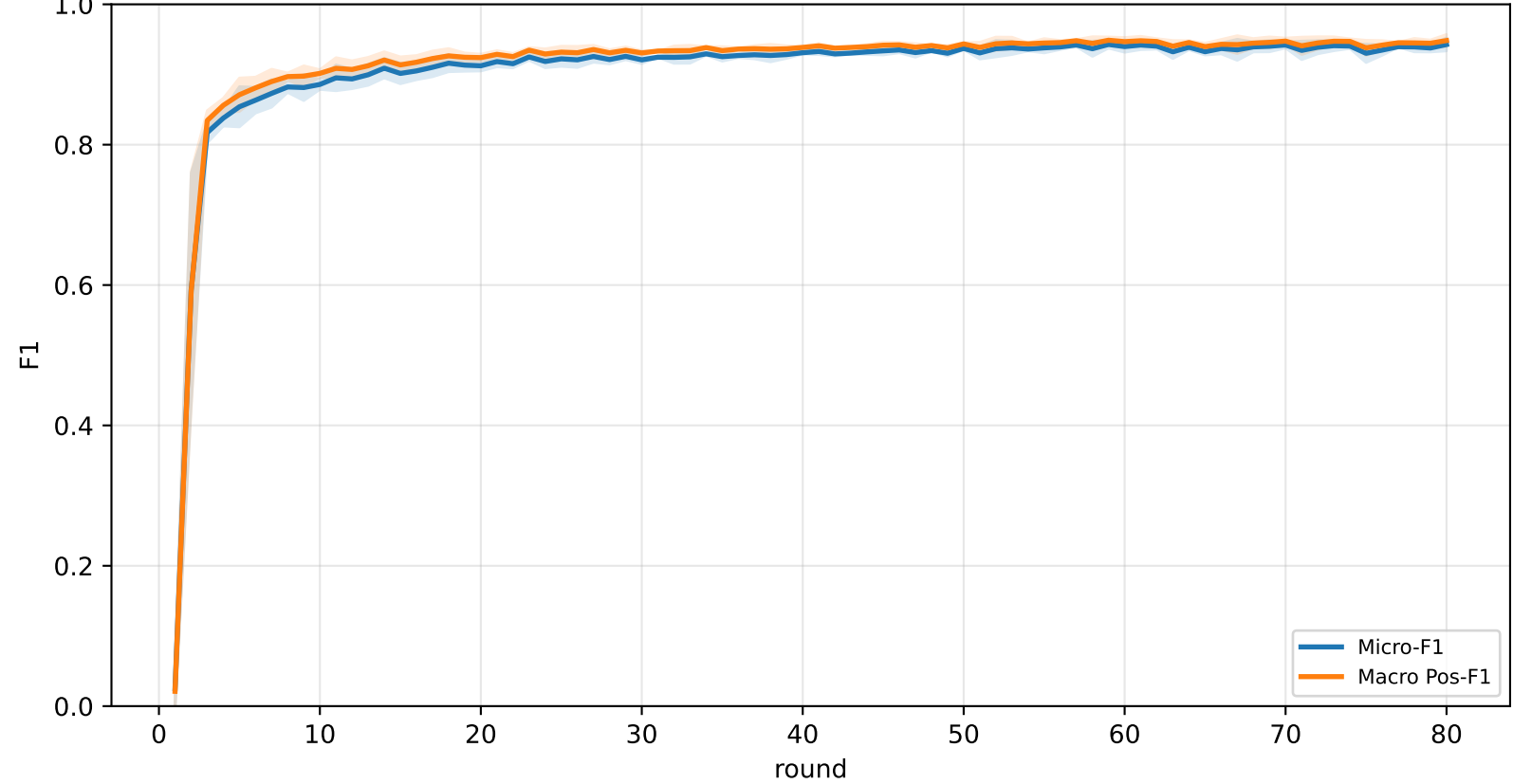
Pooled local centralized micro / macro F1



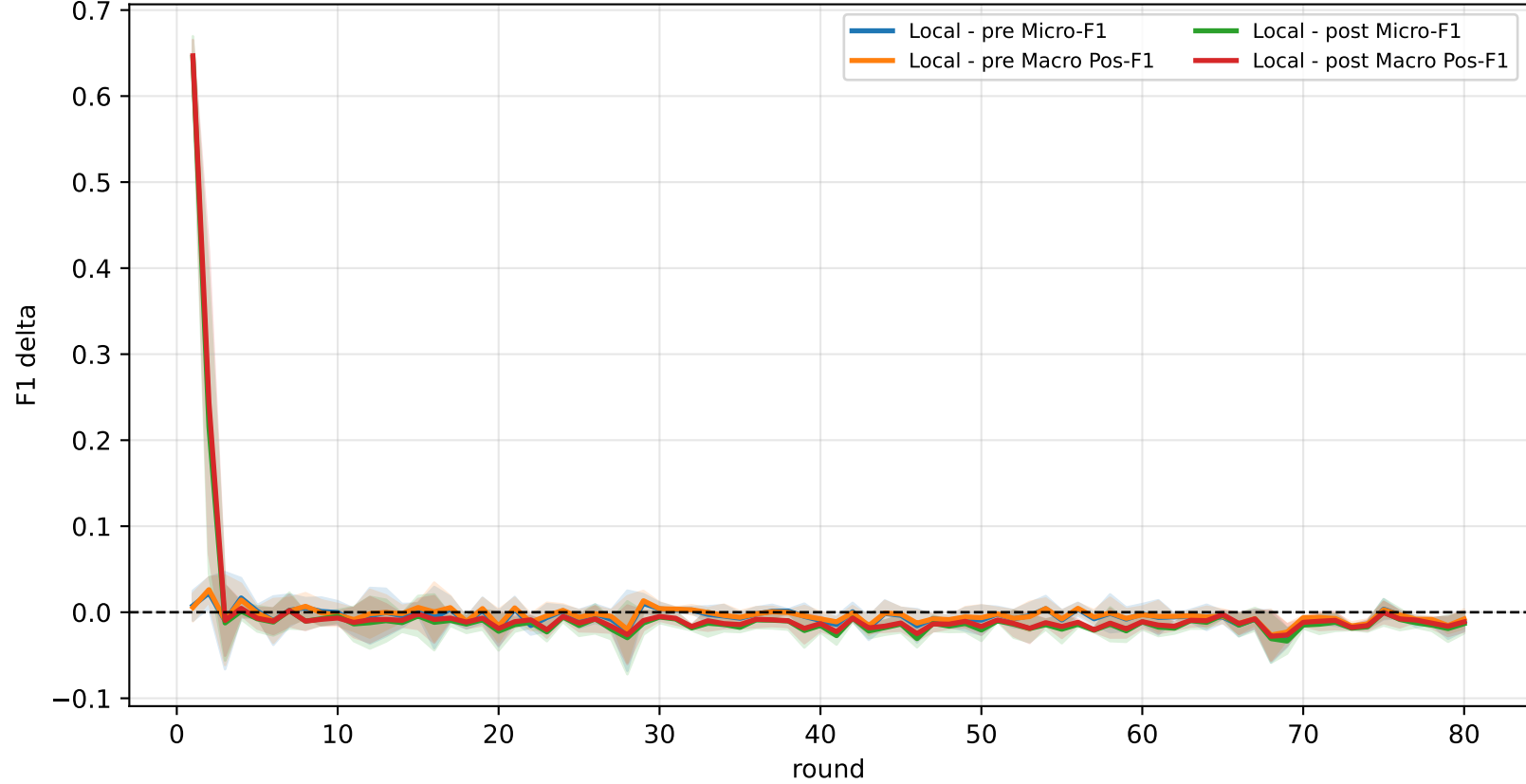
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

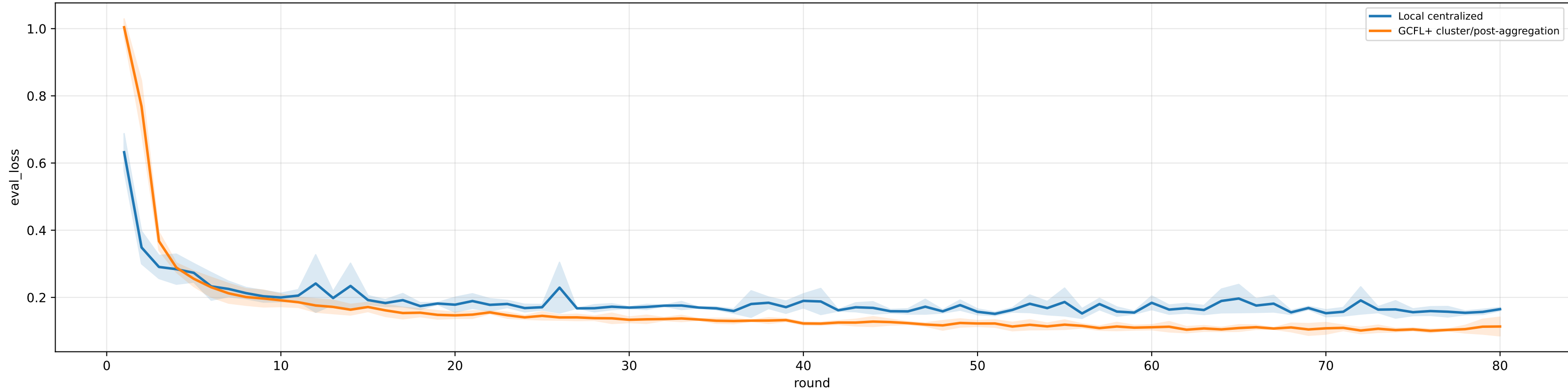


family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

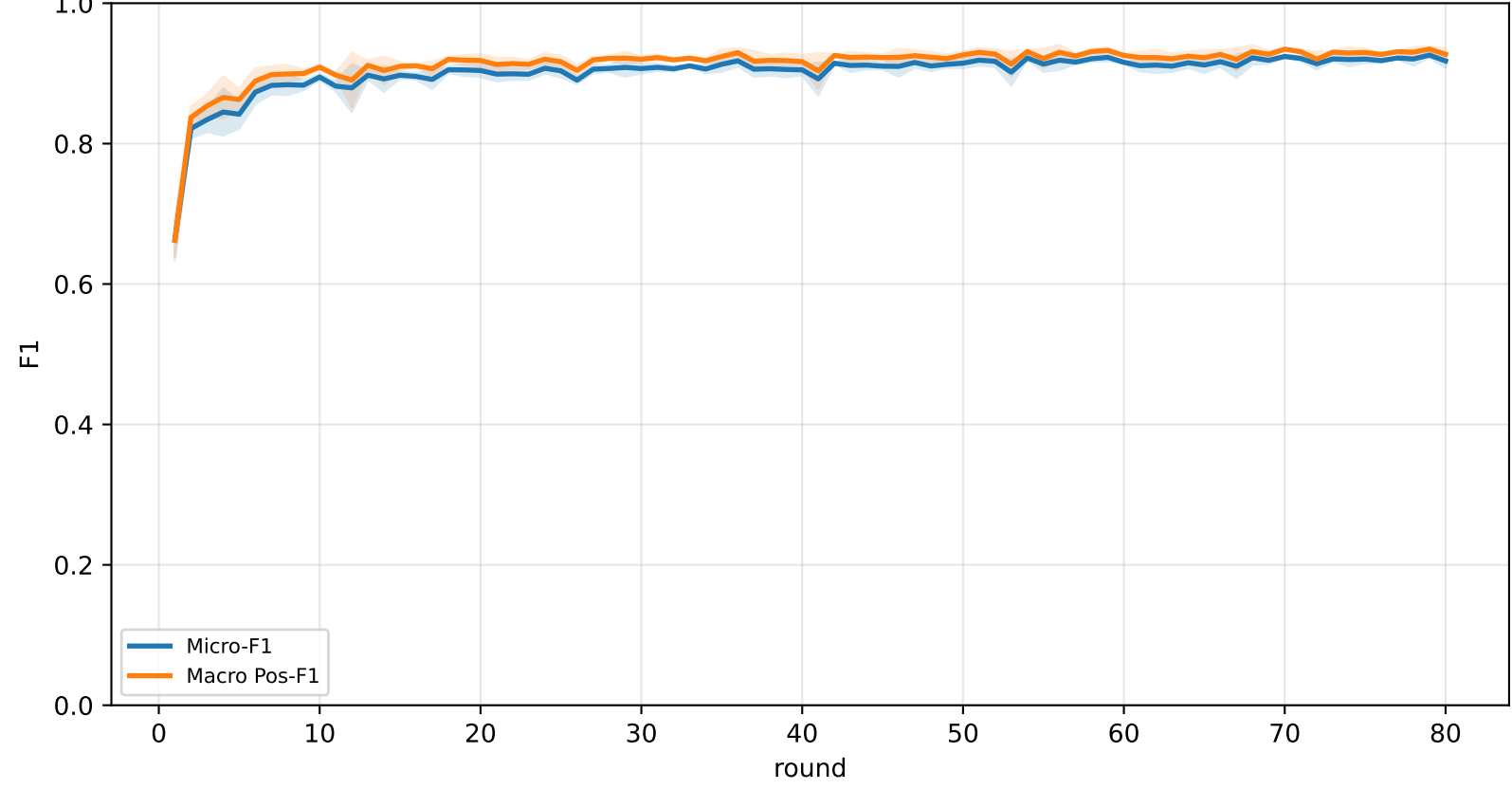
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.5% ± 0.4%	92.6% ± 0.2%	98.9% ± 0.4%	98.2% ± 0.1%	96.7% ± 0.5%	89.2% ± 0.2%	80.2% ± 1.5%
GCFL+	94.1% ± 0.8%	94.7% ± 0.6%	99.2% ± 0.2%	98.4% ± 0.4%	95.3% ± 0.2%	93.5% ± 1.4%	87.2% ± 1.6%

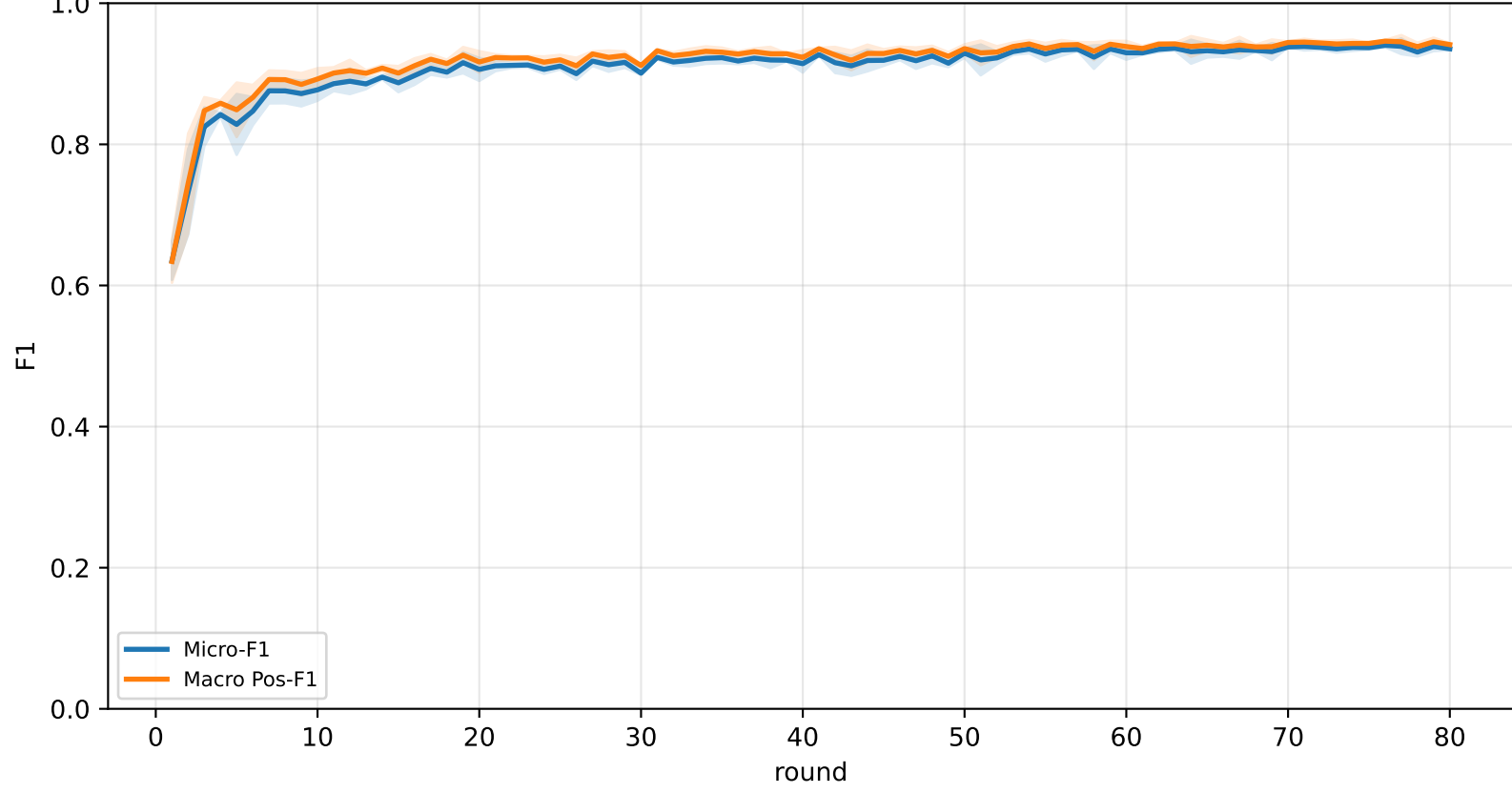
Validation loss comparison



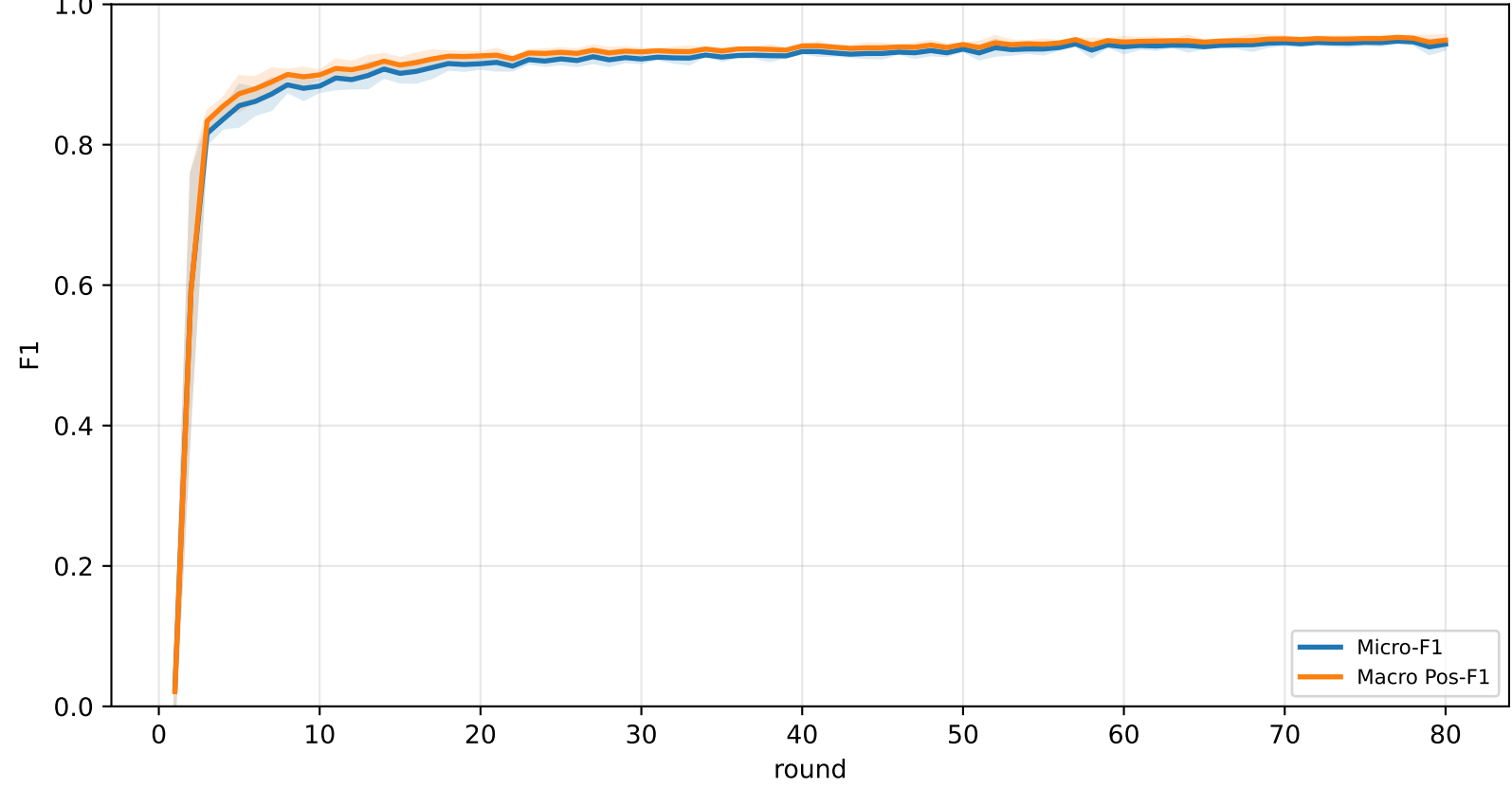
Pooled local centralized micro / macro F1



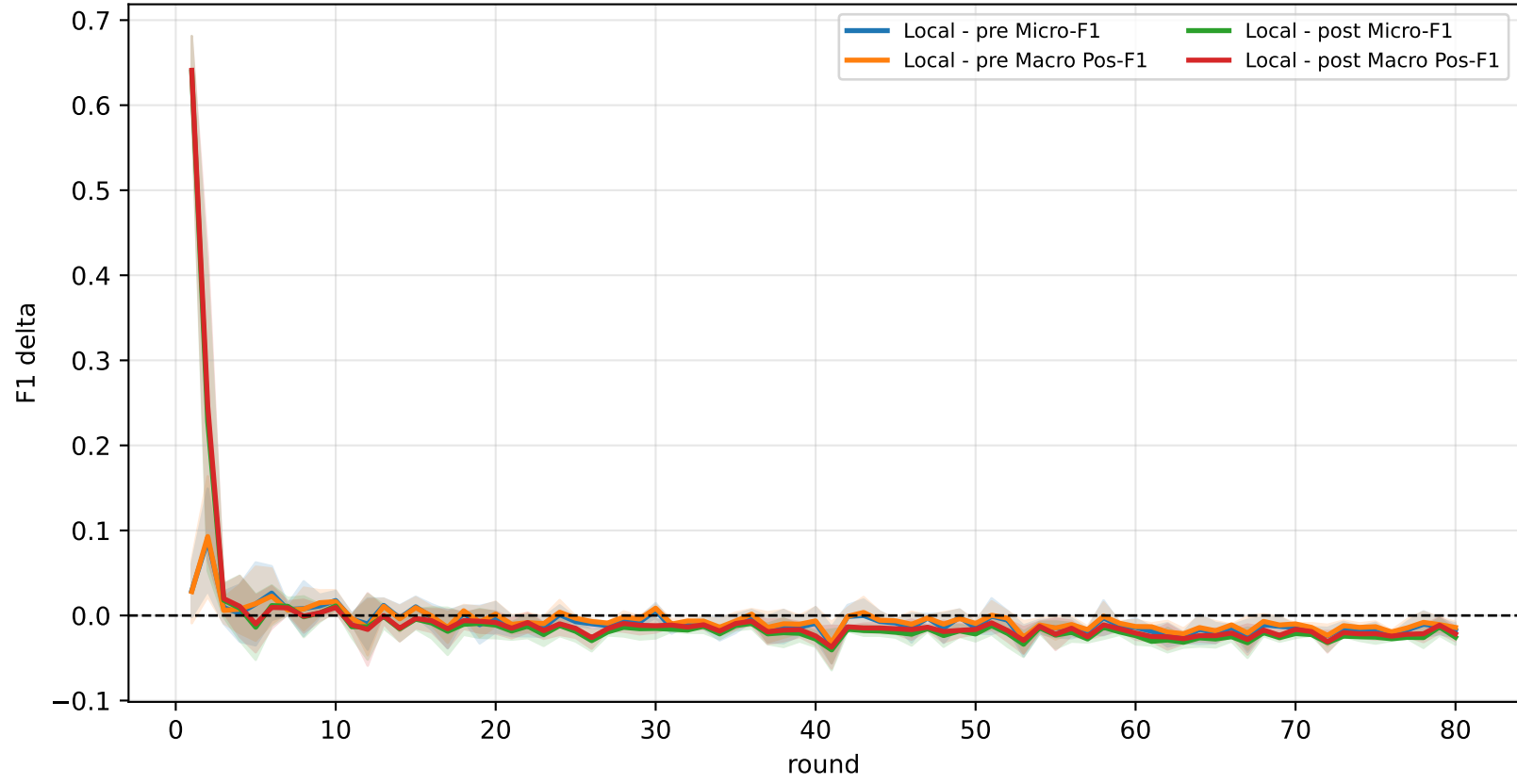
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

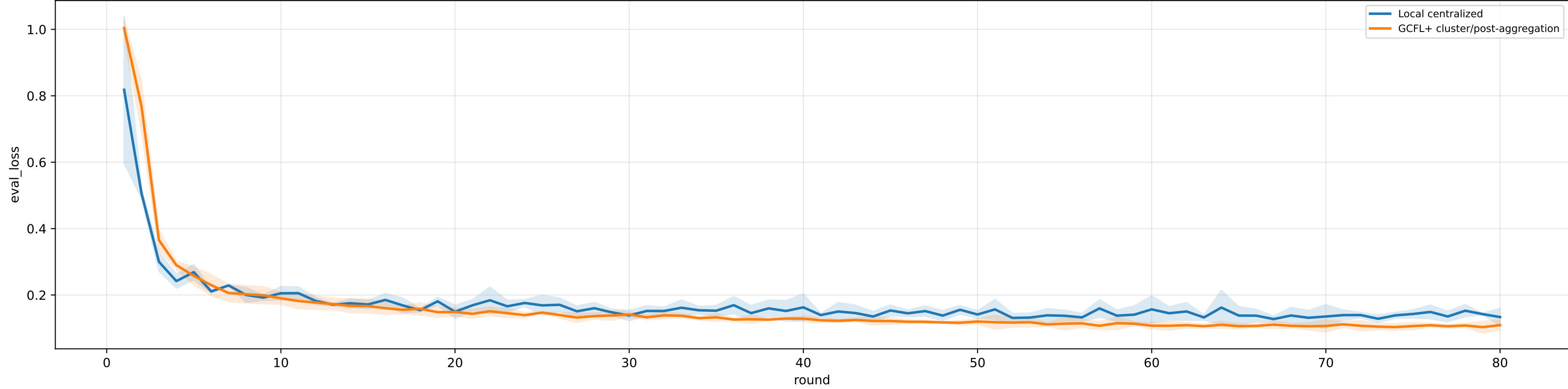


family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

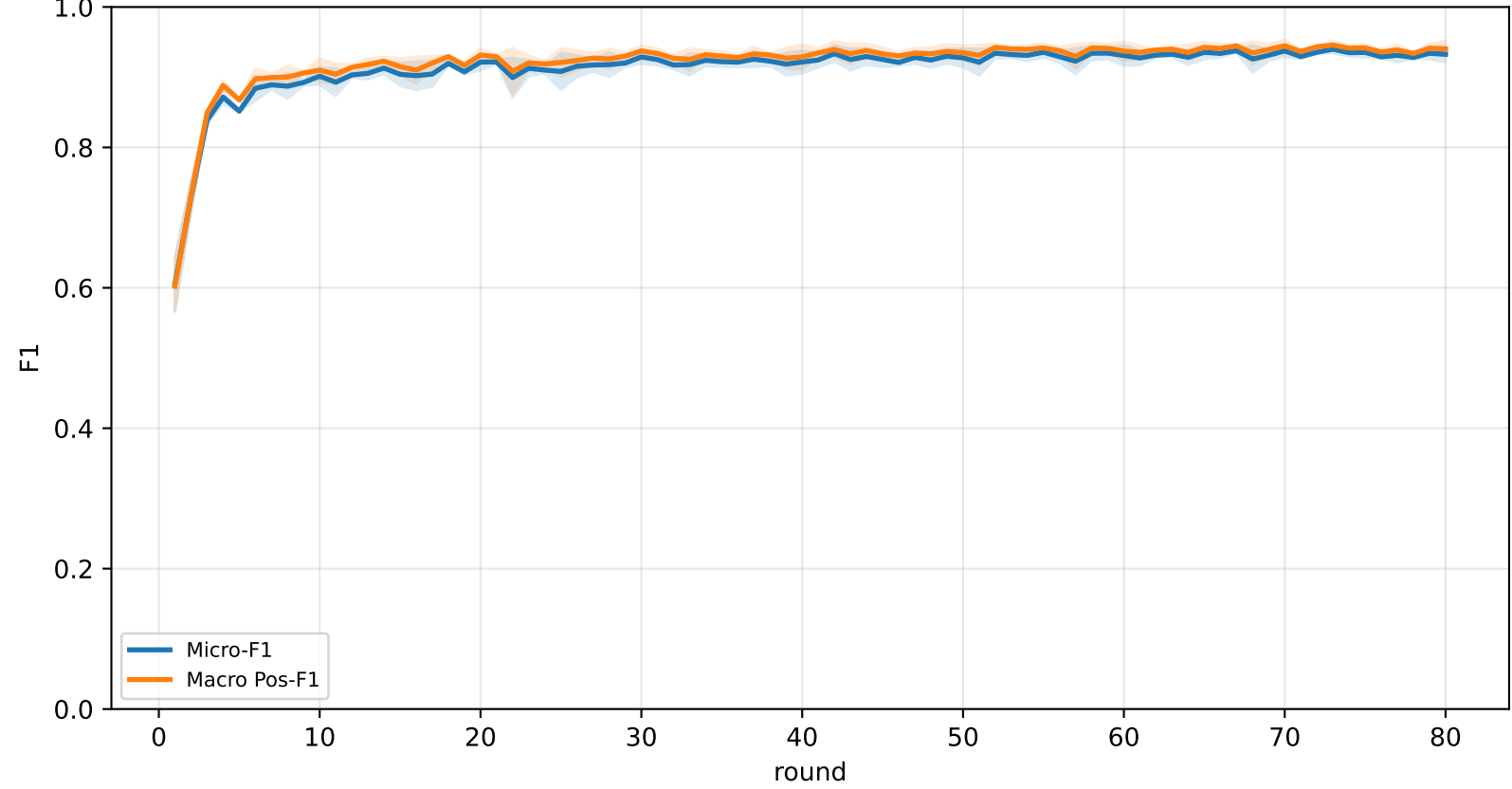
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.2% ± 0.7%	94.0% ± 0.6%	98.6% ± 0.5%	97.9% ± 0.1%	96.2% ± 0.5%	94.4% ± 0.3%	82.9% ± 2.4%
GCFL+	94.1% ± 0.8%	94.7% ± 0.6%	99.0% ± 0.1%	98.5% ± 0.2%	95.3% ± 0.8%	93.3% ± 1.1%	87.6% ± 1.2%

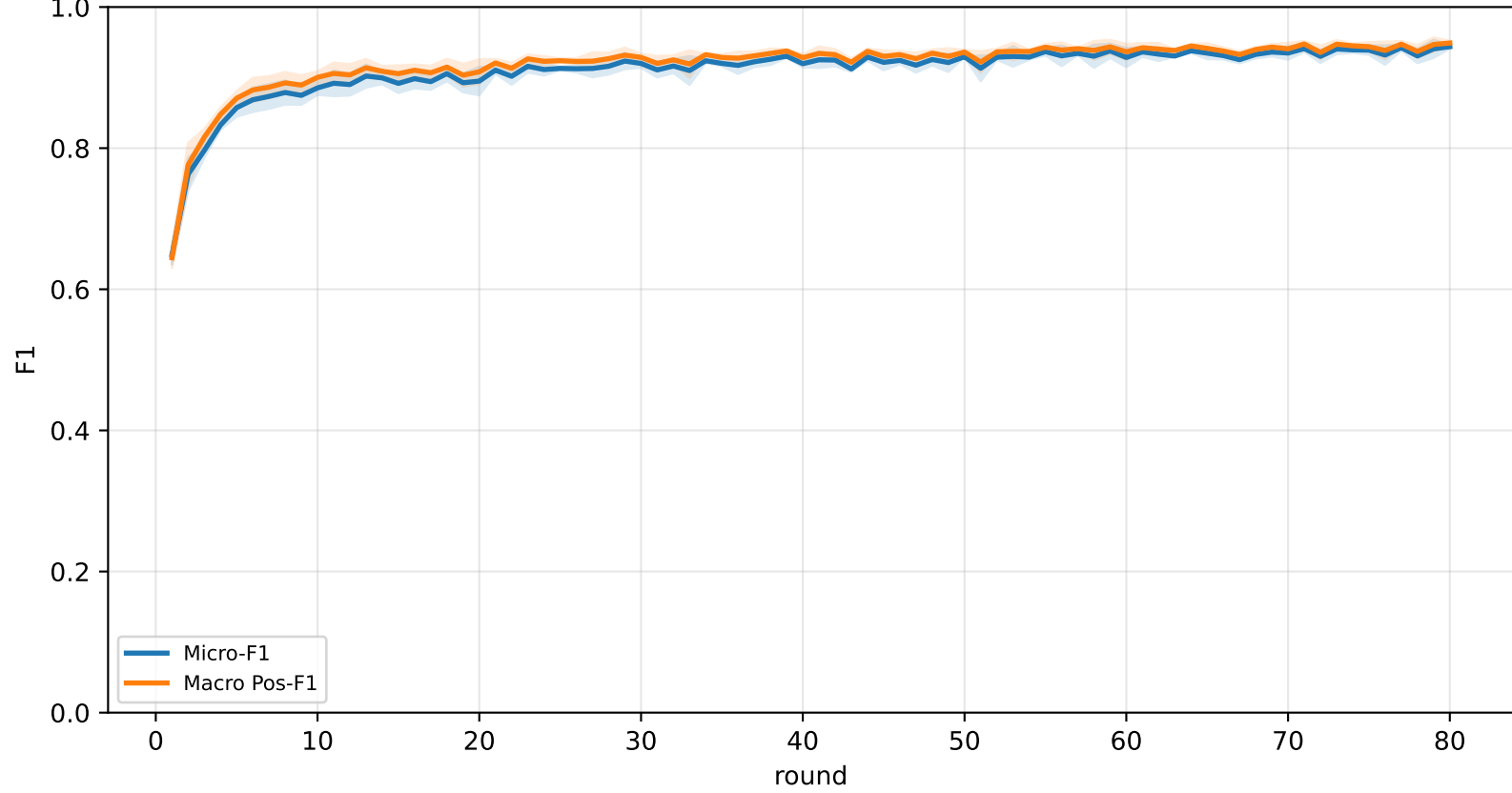
Validation loss comparison



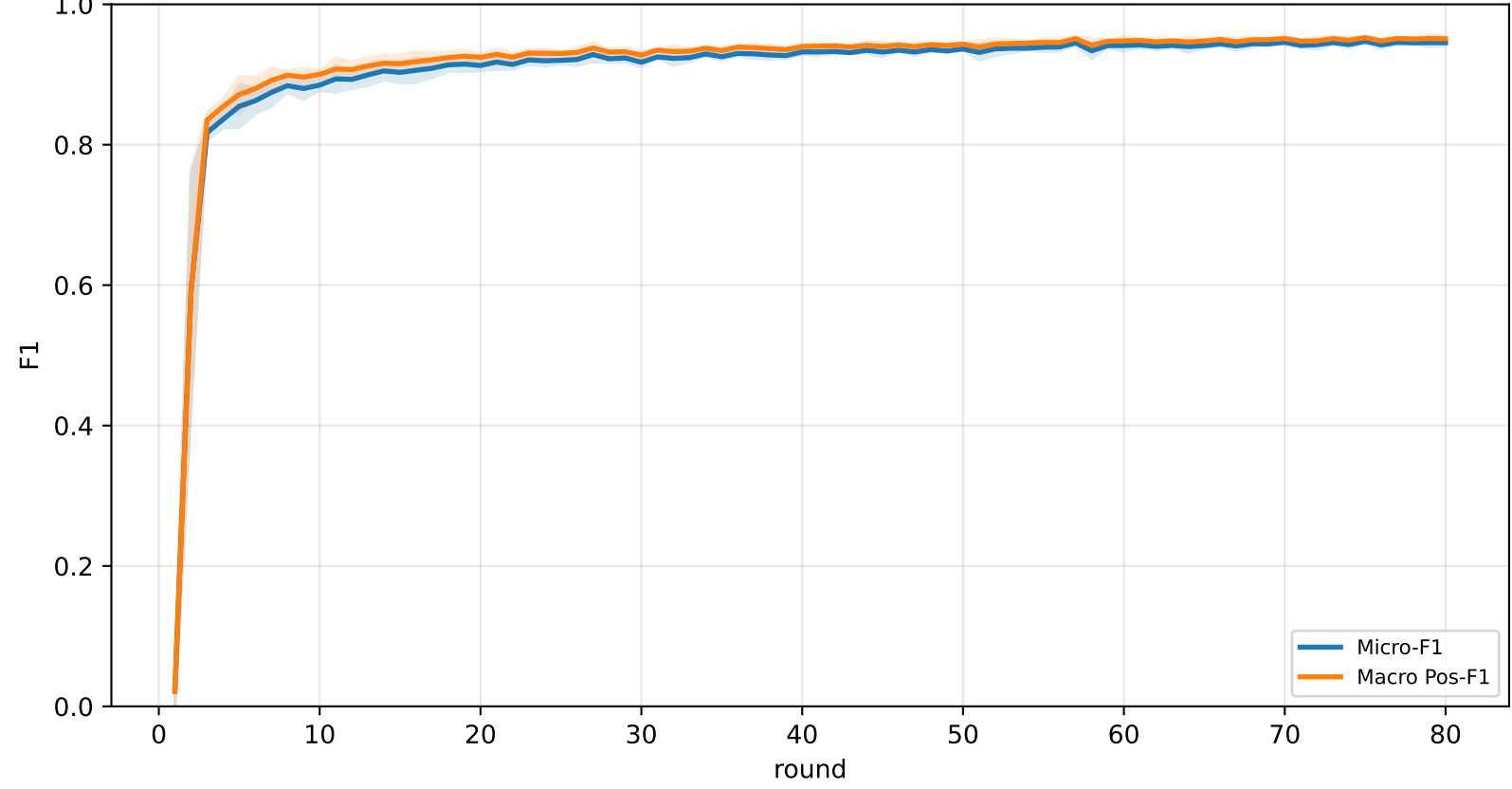
Pooled local centralized micro / macro F1



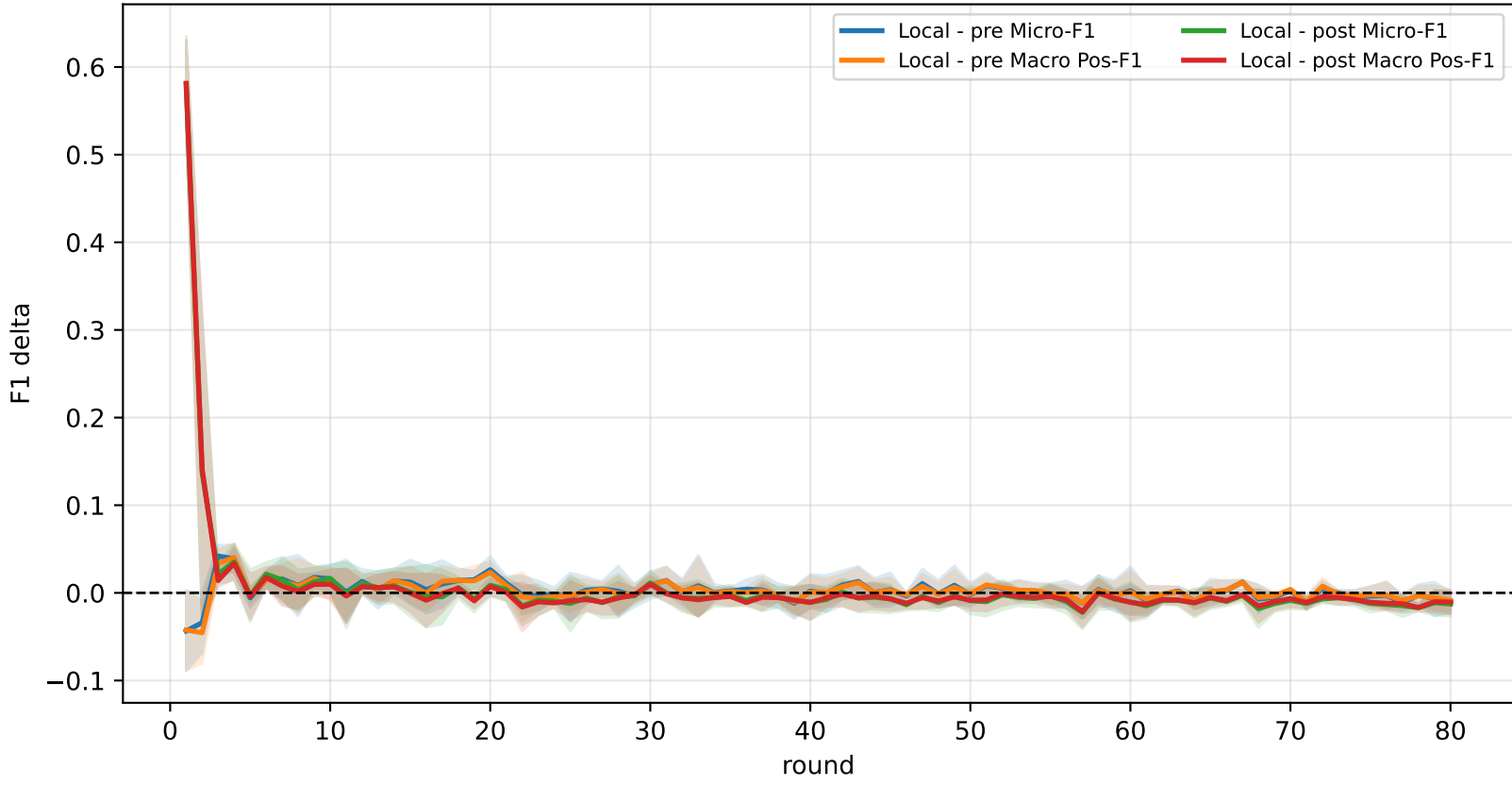
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)

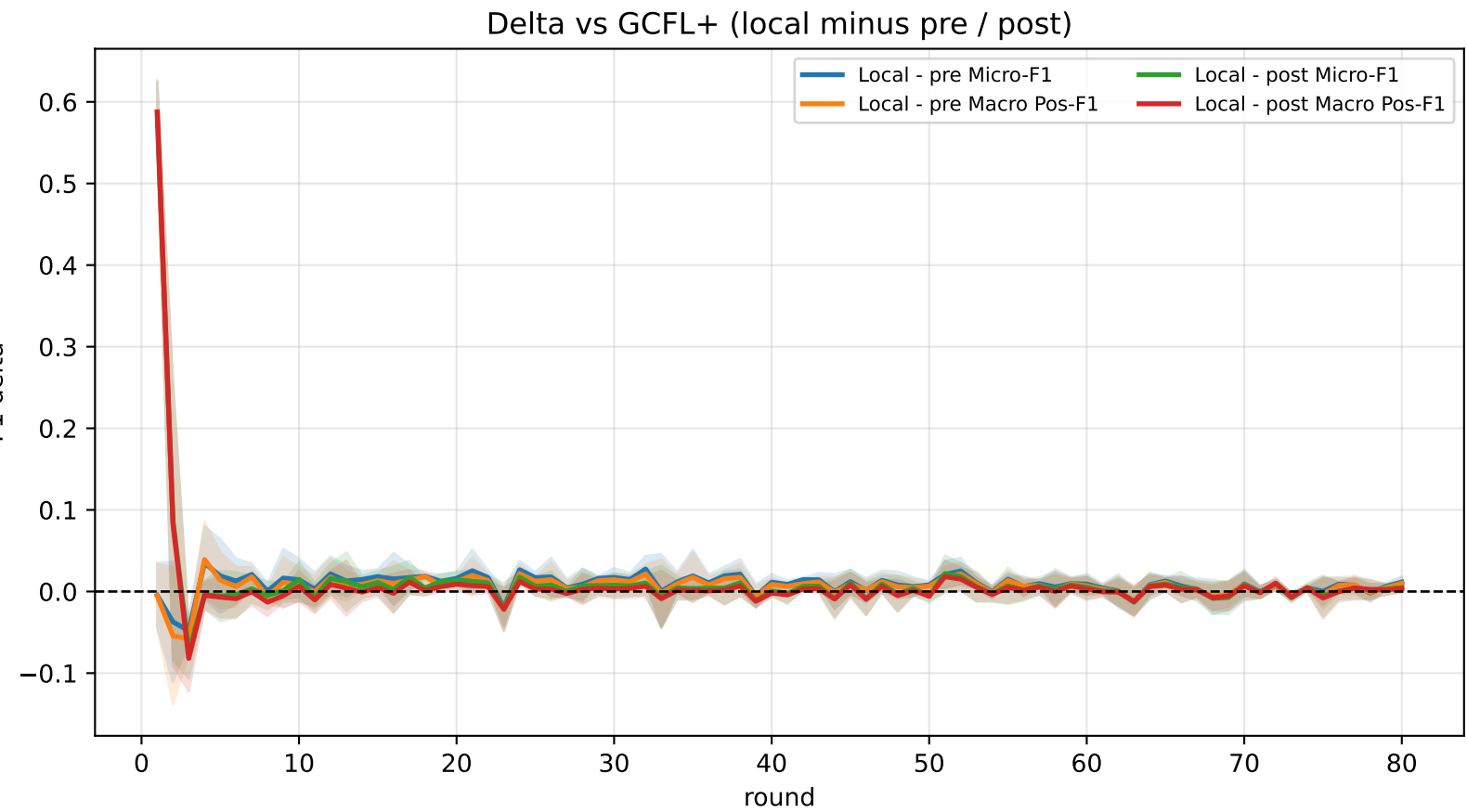
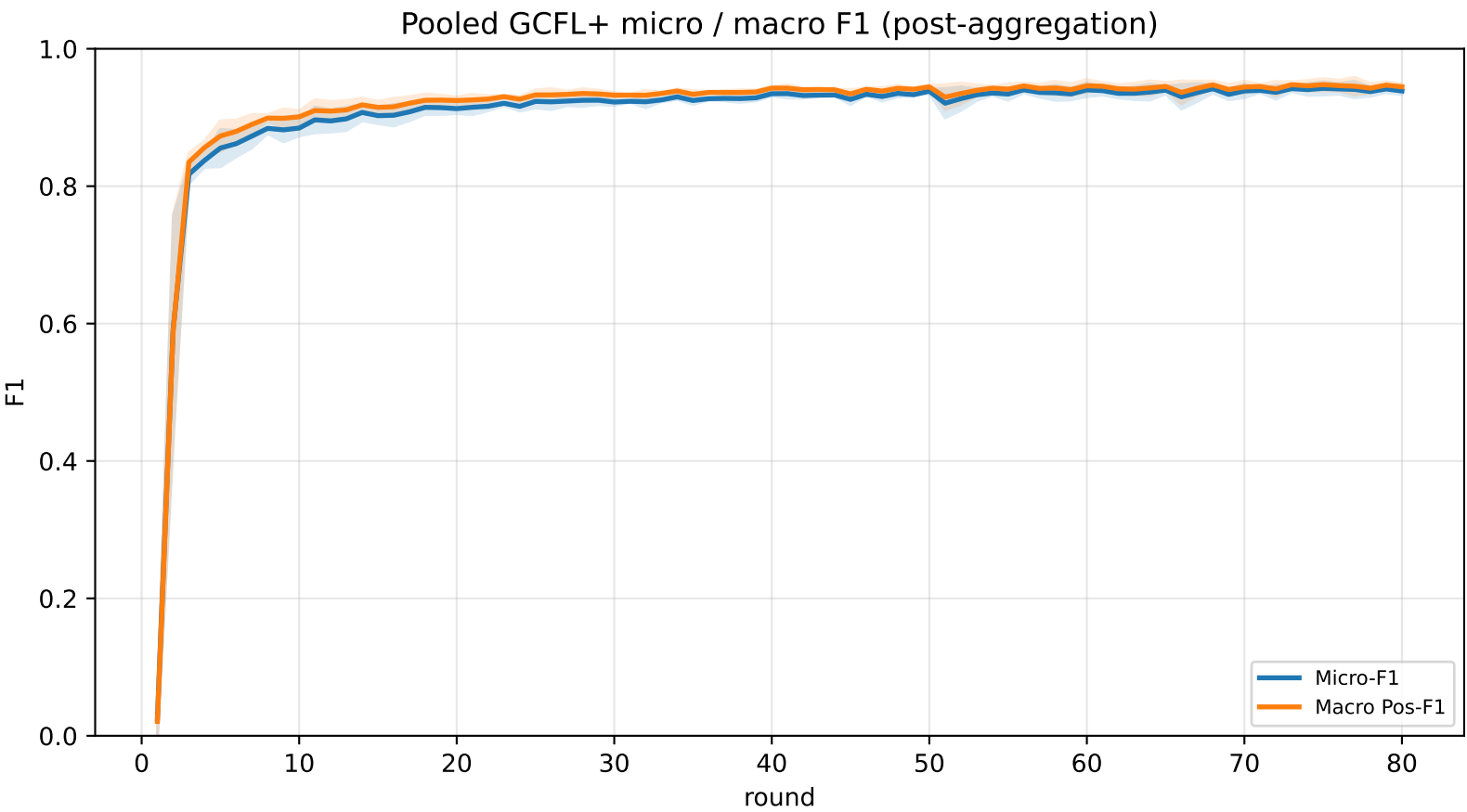
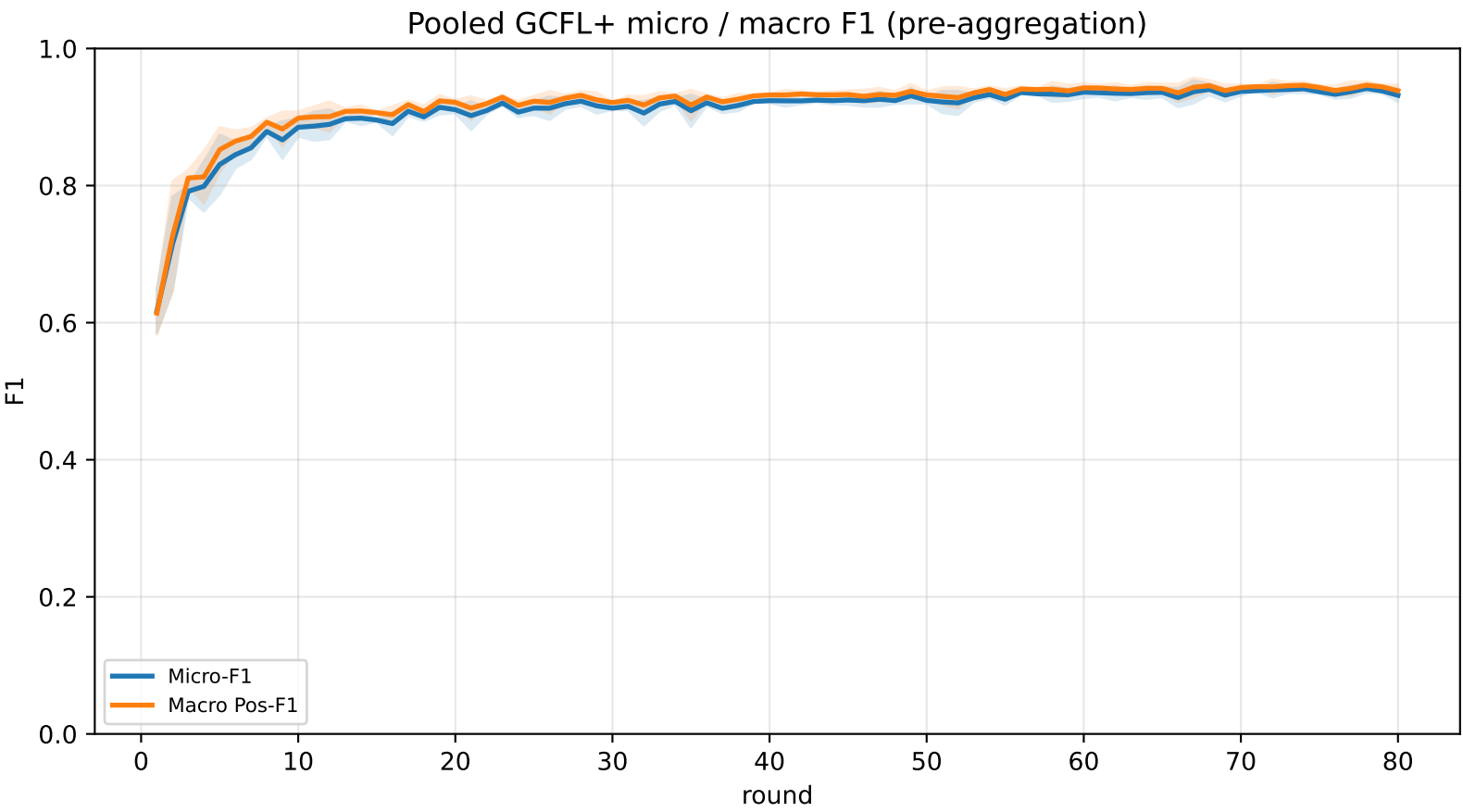
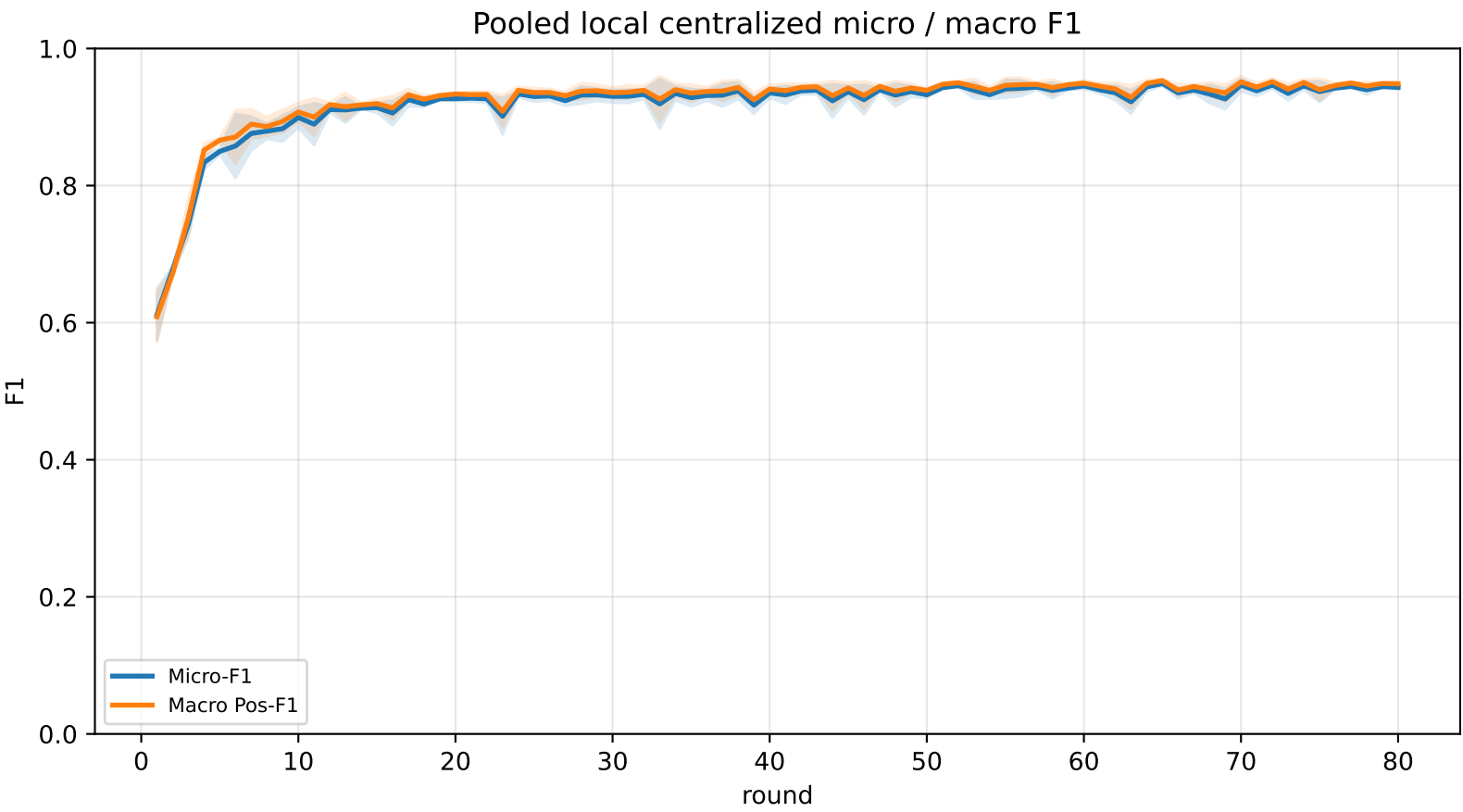
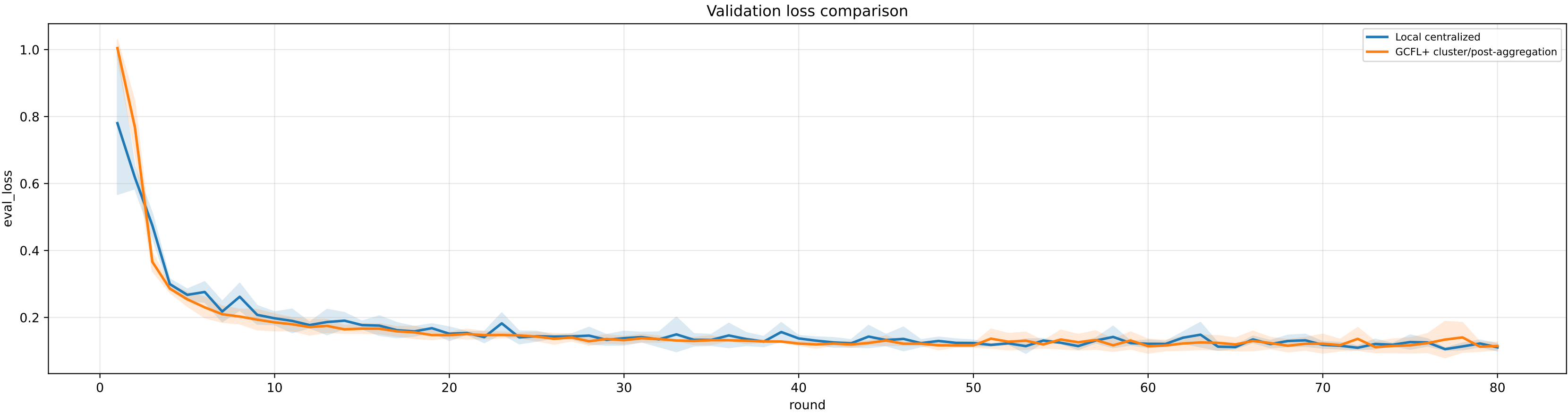


Delta vs GCFL+ (local minus pre / post)



family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.5%	94.7% ± 0.4%	98.9% ± 0.3%	97.9% ± 0.3%	95.5% ± 0.4%	92.4% ± 1.0%	88.7% ± 1.2%
GCFL+	94.1% ± 0.6%	94.7% ± 0.6%	98.9% ± 0.4%	98.1% ± 0.1%	95.7% ± 0.5%	92.8% ± 1.6%	87.9% ± 1.2%



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

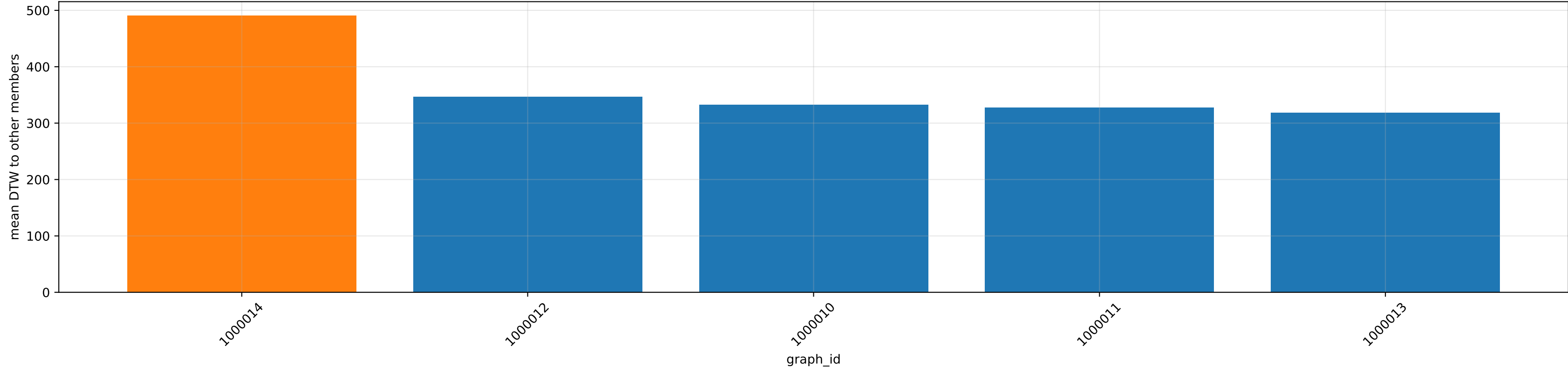
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	48	0	1 vs 4	1000014	1000014	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
490.853	nan	278.005

Per-member mean DTW



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

First applied split pairwise DTW metadata | seed=0

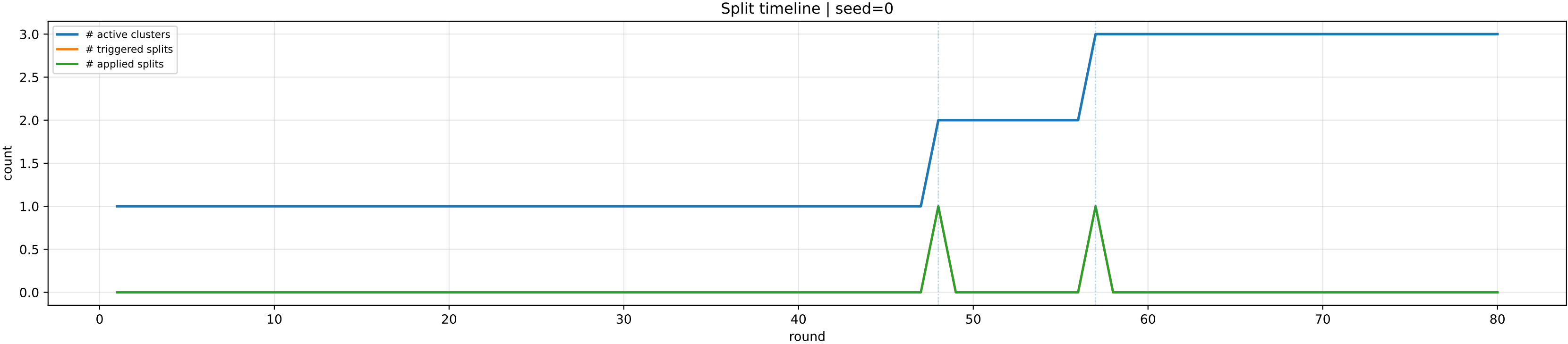
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	48	0	5	1000014.0	0.009	490.853	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 606.5 at (1000011, 1000014)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	73.2	360.7	316.6	578.0
1000011	73.2	0.0	334.7	294.5	606.5
1000012	360.7	334.7	0.0	288.4	403.6
1000013	316.6	294.5	288.4	0.0	375.3
1000014	578.0	606.5	403.6	375.3	0.0

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161



Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	48, 57	48, 57	1, 3, 4	1:1, 3:1, 4:3

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Split lineage story | seed=0

Cluster 0
5 clients

round 48

Cluster 2
4 clients

round 57

Cluster 4
3 clients

split off -> Cluster 1
1 client
specialized_cycle6:1

split off -> Cluster 3
1 client
specialized_cycle4:1

Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
48	0	5	189.117 < 201.557 , 557.820 > 482.478	C1 (1 client)	specialized_cycle6:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1
57	2	4	215.941 < 229.909 , 413.141 > 392.197	C3 (1 client)	specialized_cycle4:1	C4 (3 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle5:1

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

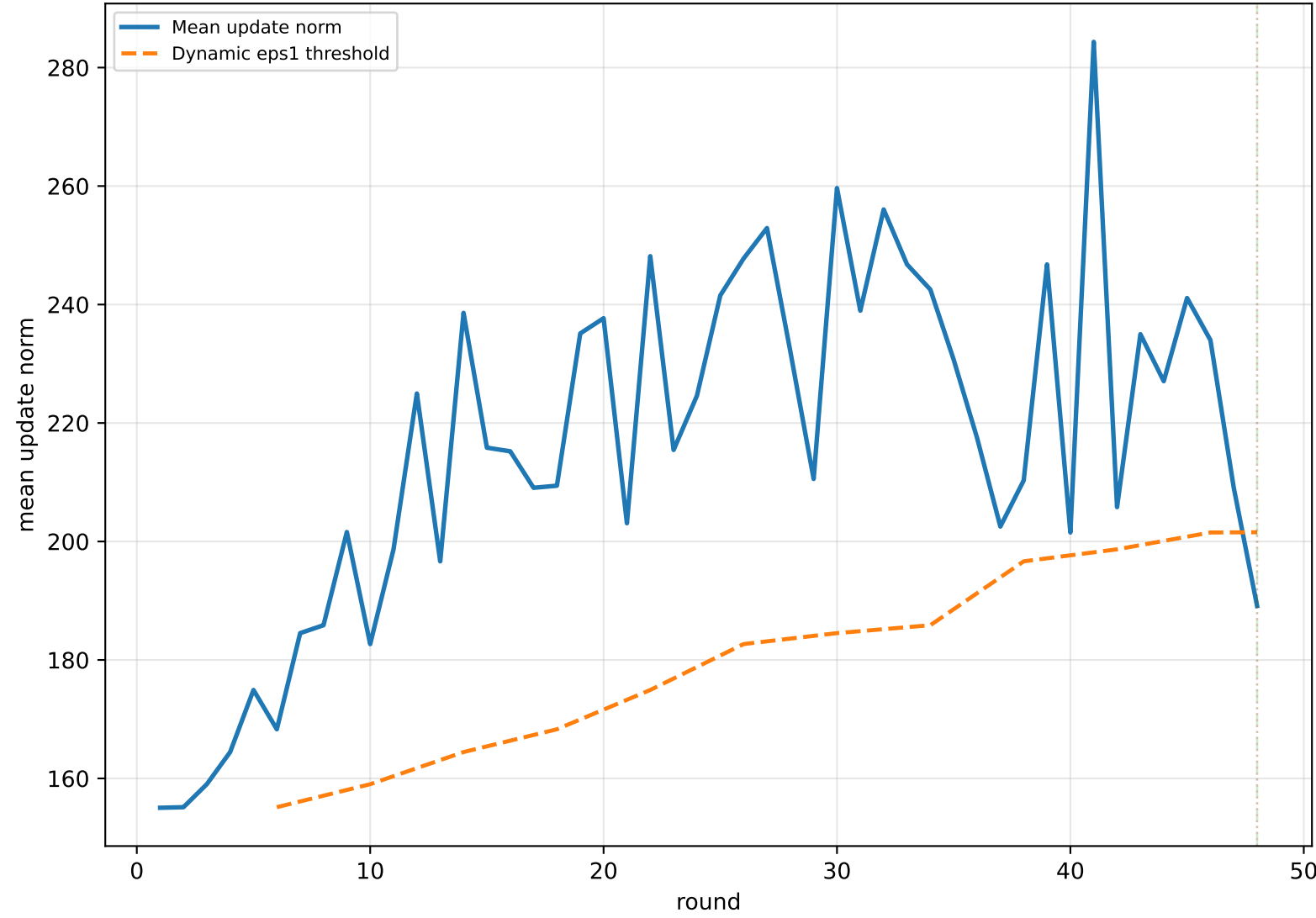
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000014	4	specialized_cycle6:1
3	1	1000012	2	specialized_cycle4:1
4	3	1000010, 1000011, 1000013	0, 1, 3	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle5:1

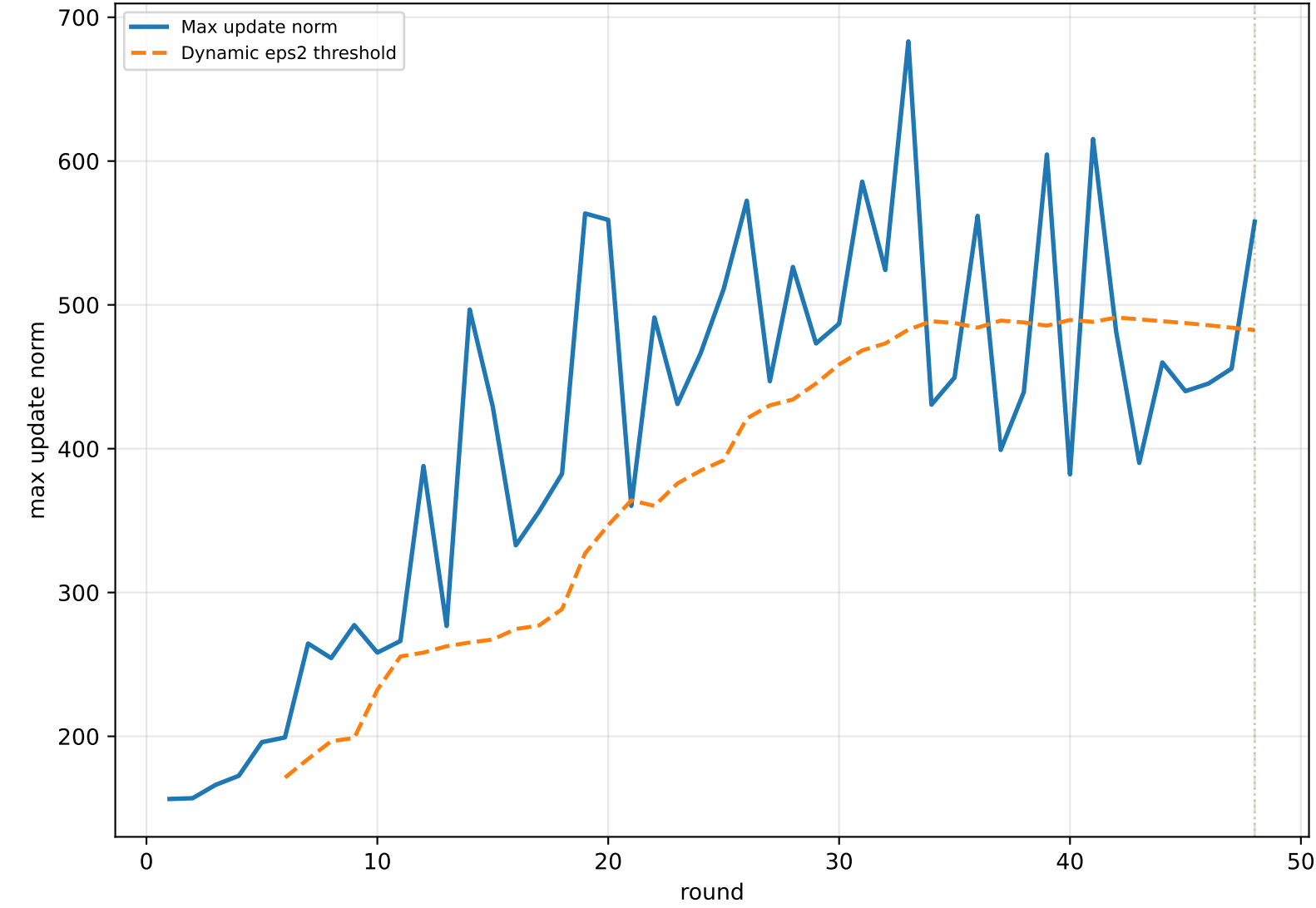
Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

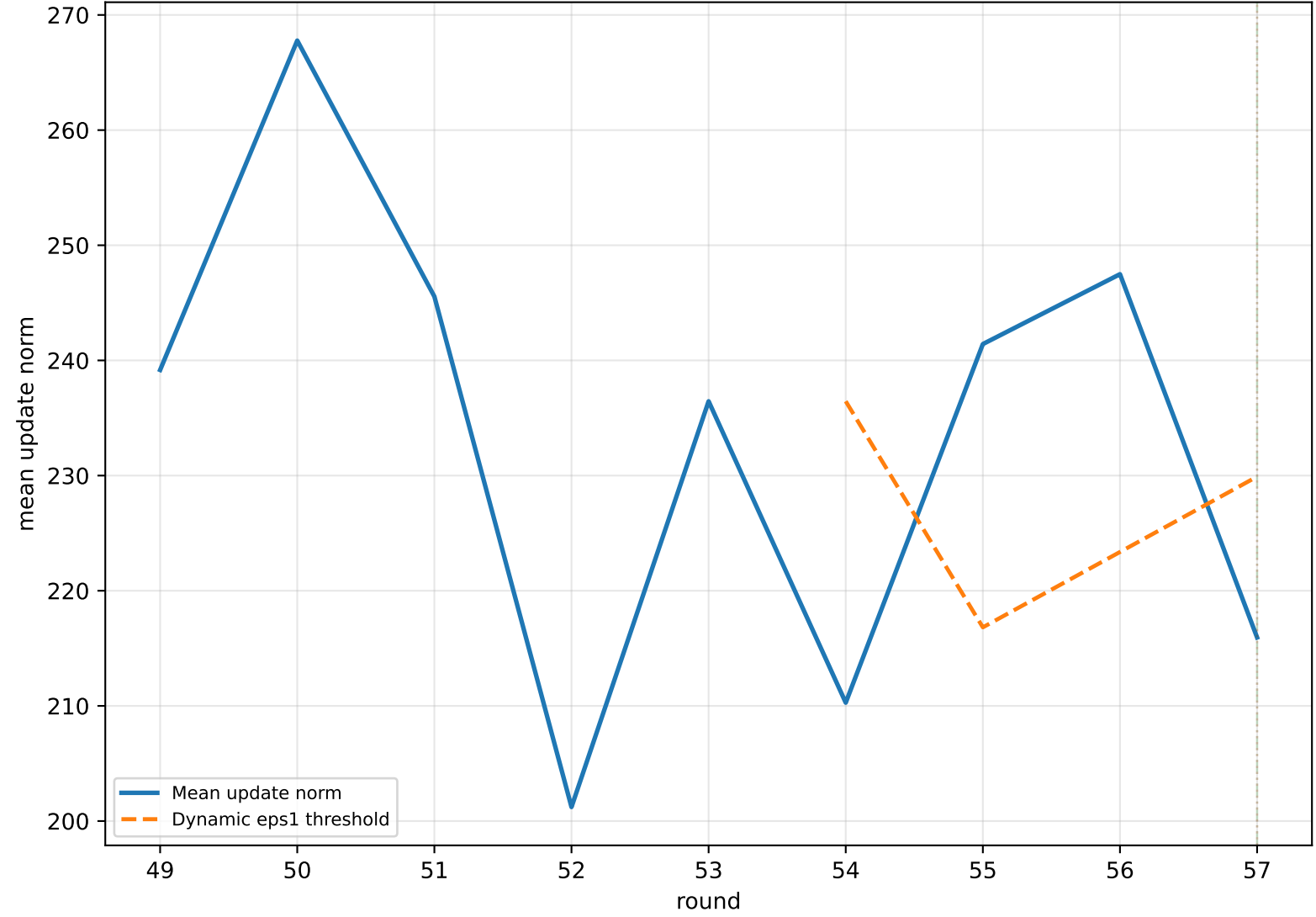
Cluster 0 | Mean update norm



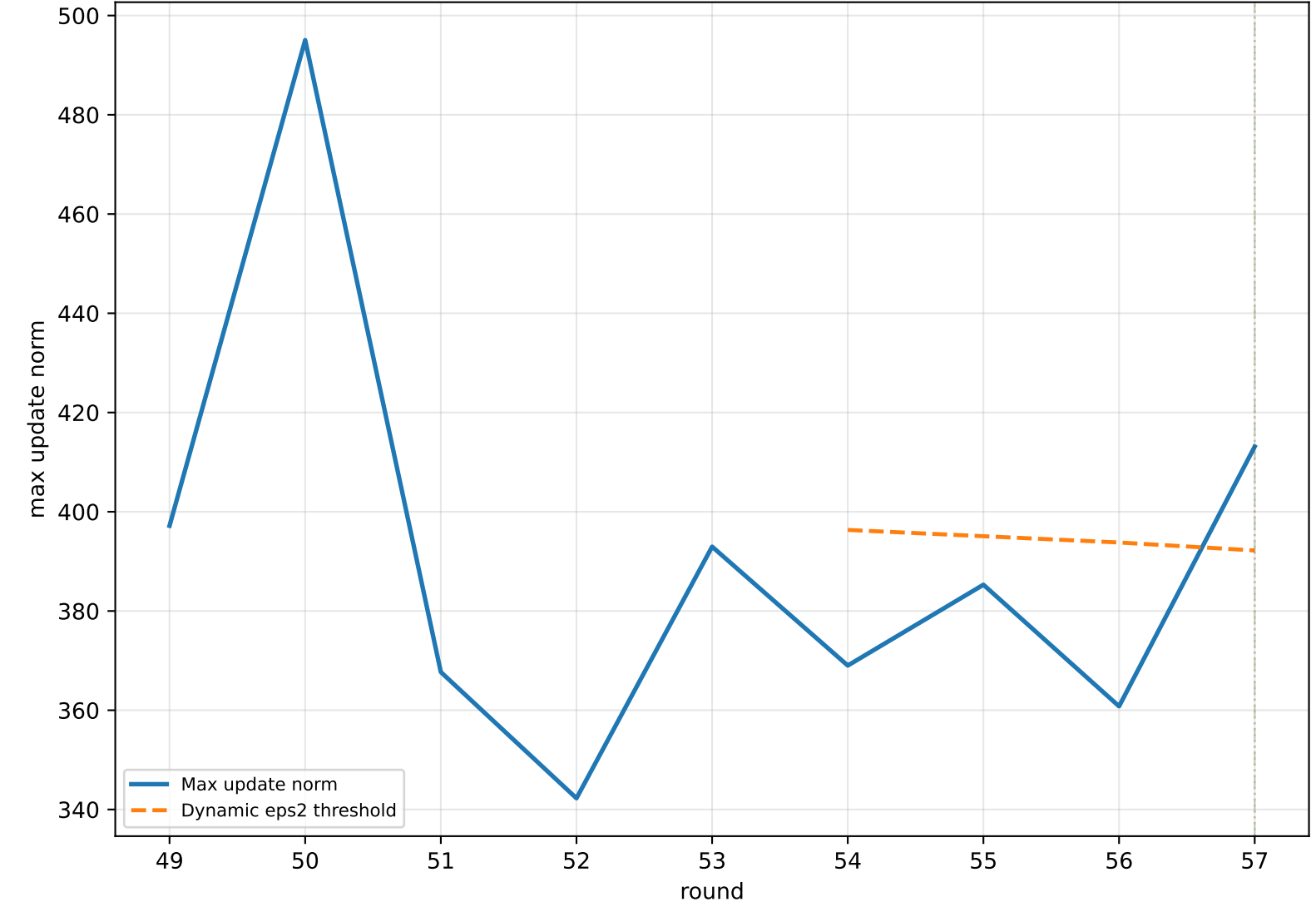
Cluster 0 | Max update norm



Cluster 2 | Mean update norm

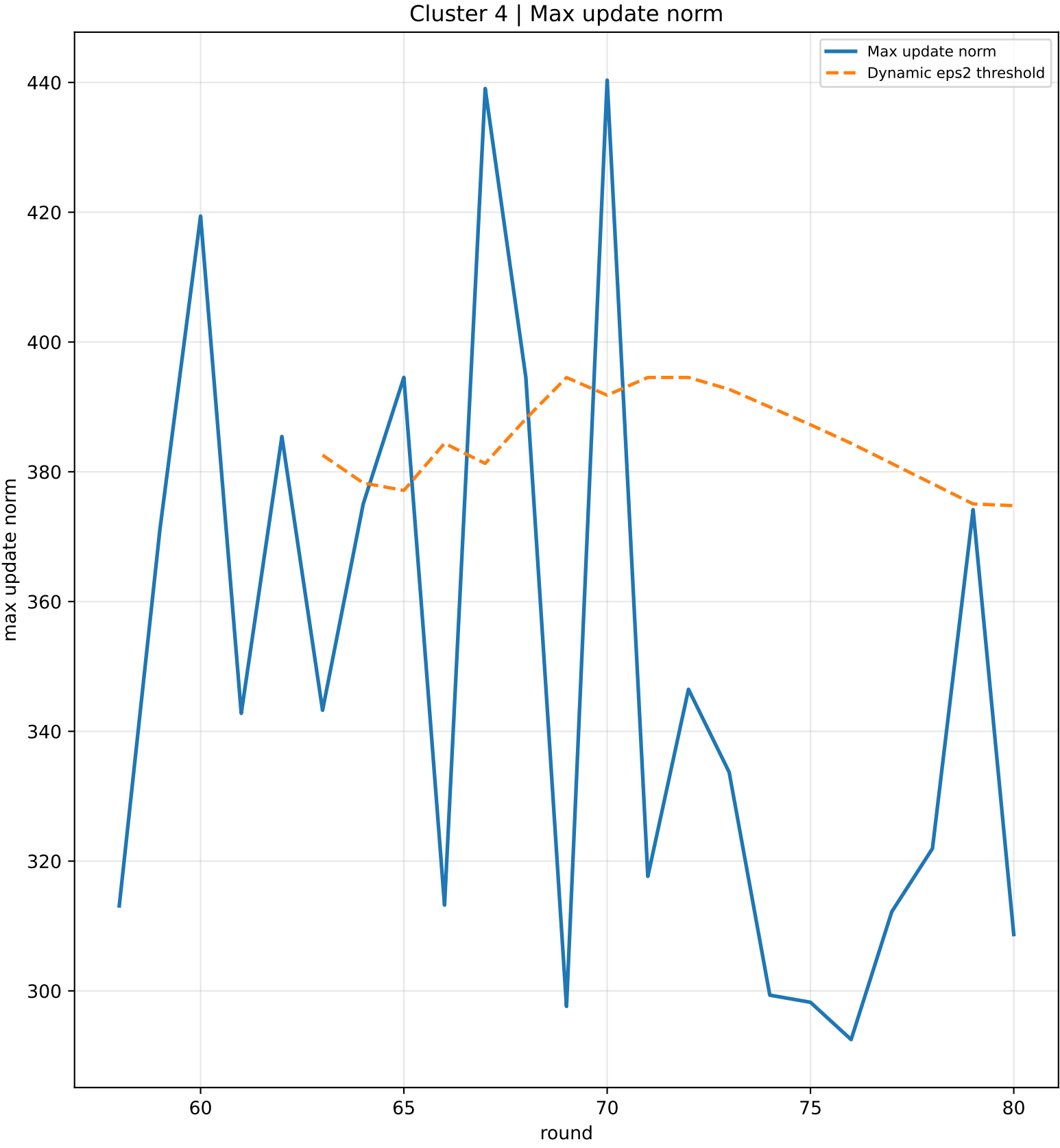
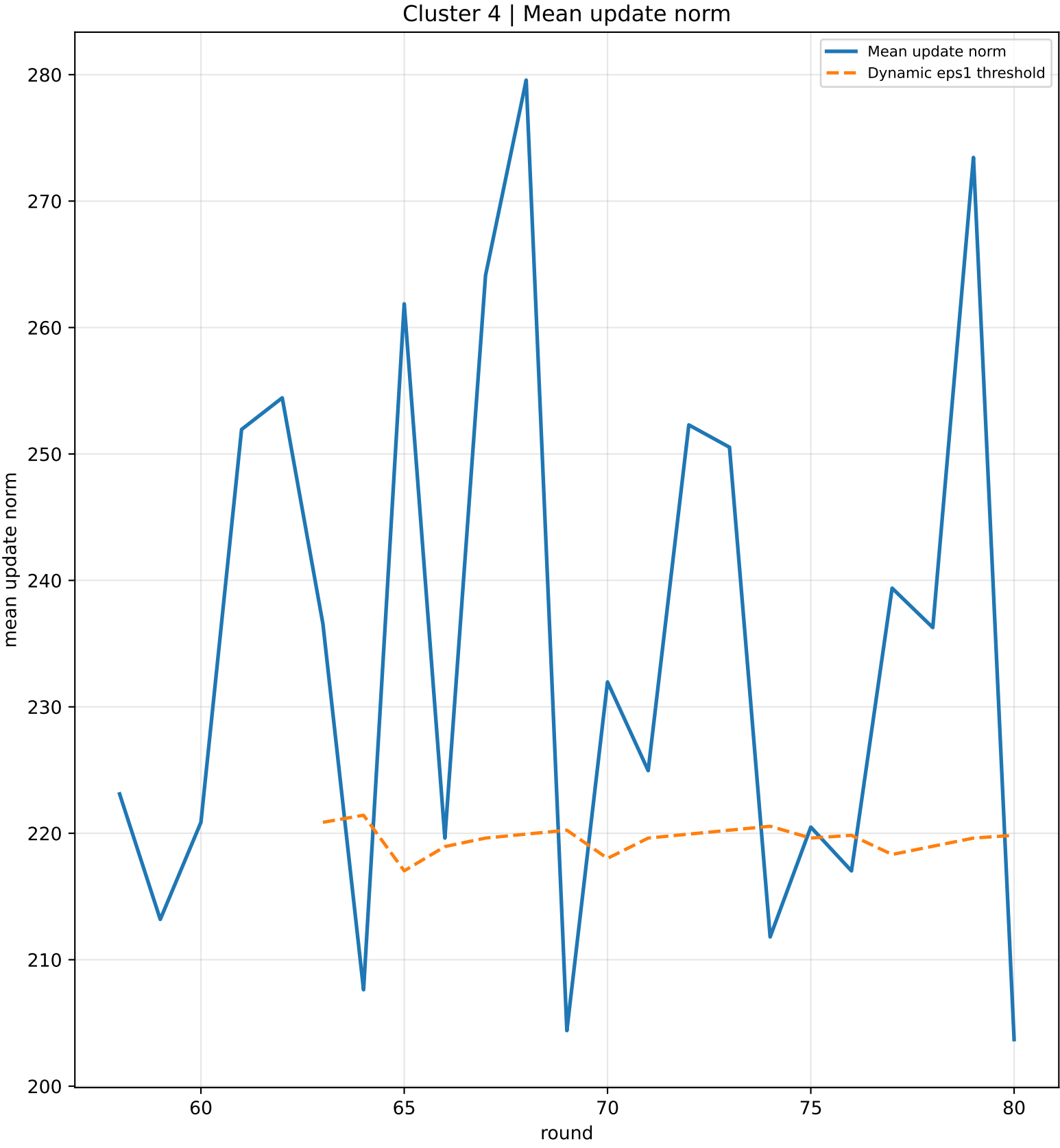


Cluster 2 | Max update norm



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (4-4)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

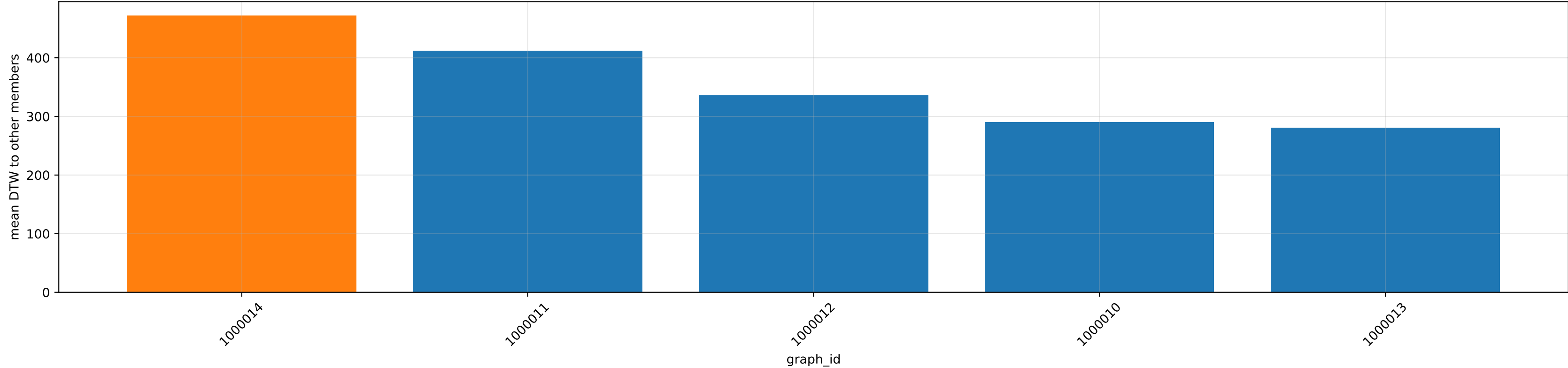
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	71	0	1 vs 4	1000014	1000014	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
472.127	nan	282.260

Per-member mean DTW



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

First applied split pairwise DTW metadata | seed=1

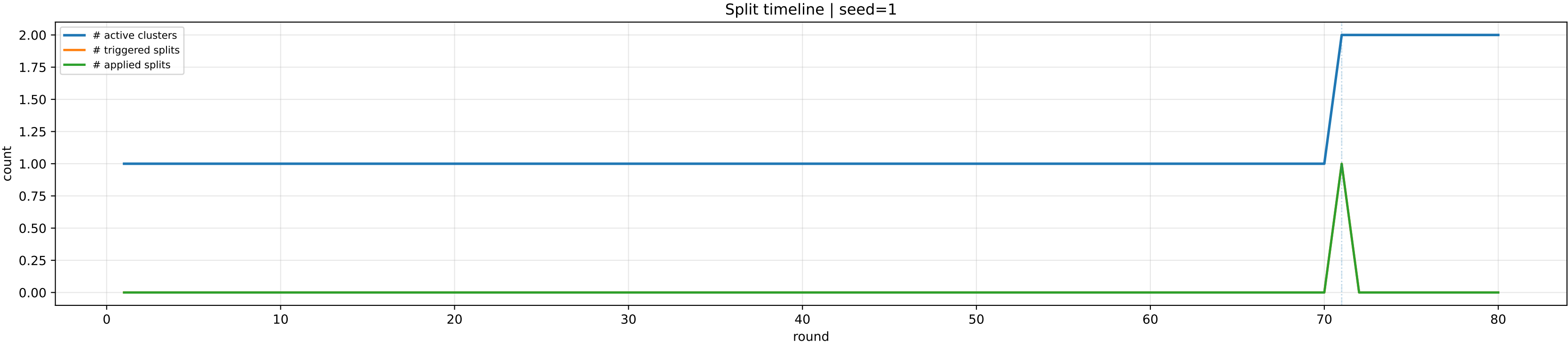
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	71	0	5	1000014.0	0.009	472.127	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 609.0 at (1000011, 1000014)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	338.7	258.9	111.1	453.5
1000011	338.7	0.0	390.3	309.9	609.0
1000012	258.9	390.3	0.0	284.7	408.6
1000013	111.1	309.9	284.7	0.0	417.4
1000014	453.5	609.0	408.6	417.4	0.0

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161



Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	71	71	1, 2	1:1, 2:4

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Split lineage story | seed=1
round 71

Cluster 0
5 clients

Cluster 2
4 clients

↓
split off -> Cluster 1
1 client
specialized_cycle6:1

Applied split story | seed=1

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
71	0	5	202.013 < 220.832 , 534.563 > 525.341	C1 (1 client)	specialized_cycle6:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

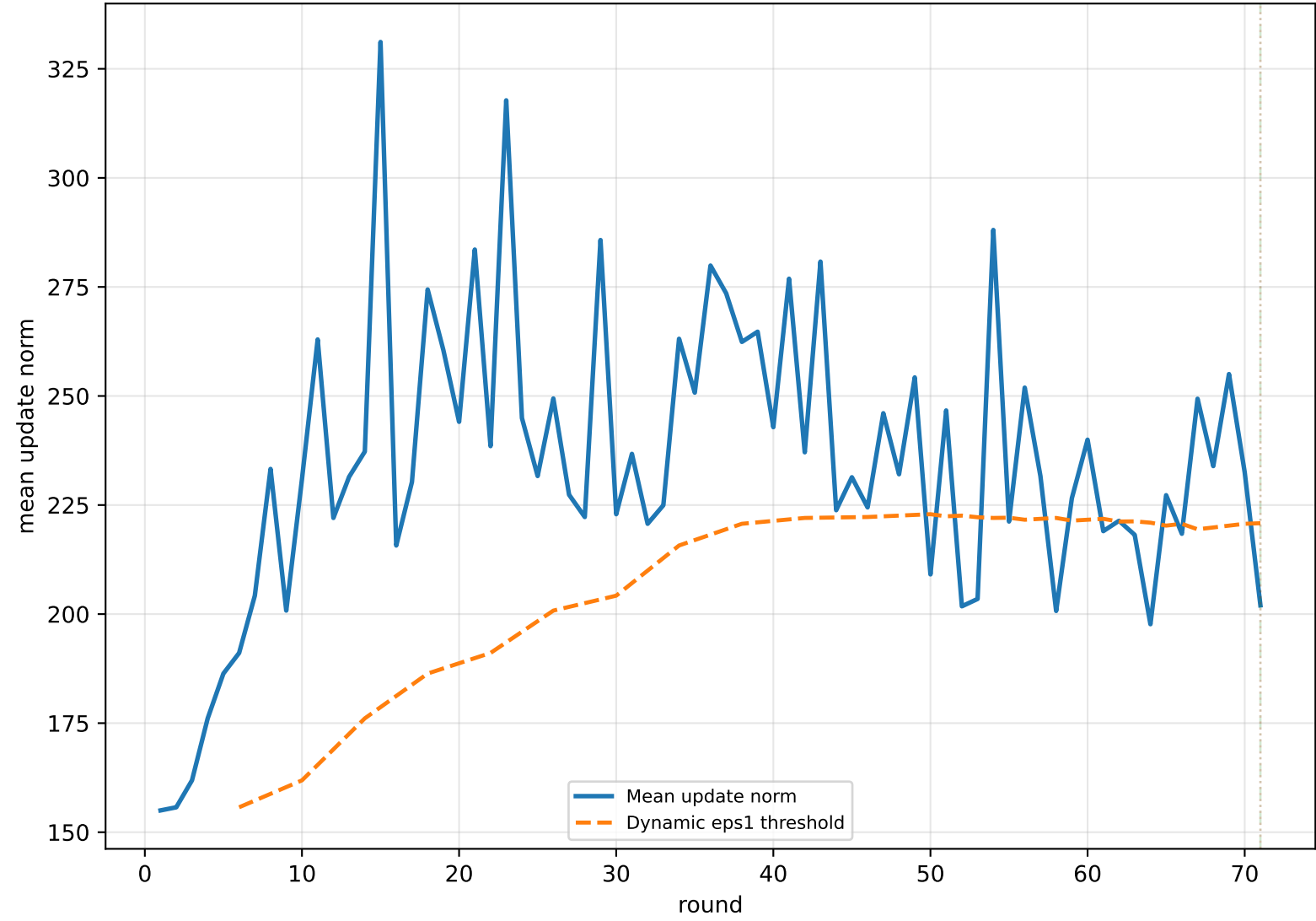
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000014	4	specialized_cycle6:1
2	4	1000010, 1000011, 1000012, 1000013	0, 1, 2, 3	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle5:1

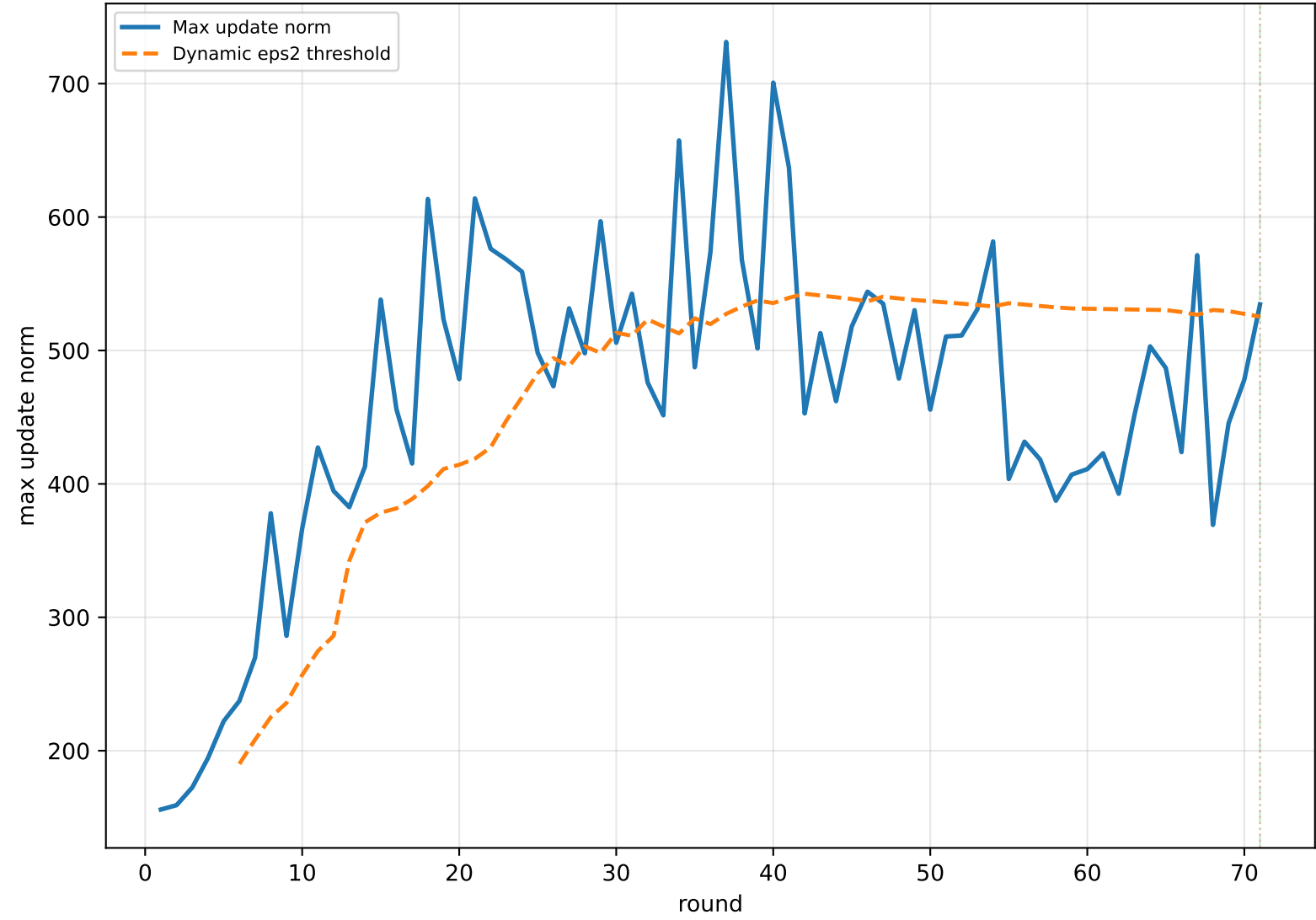
Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

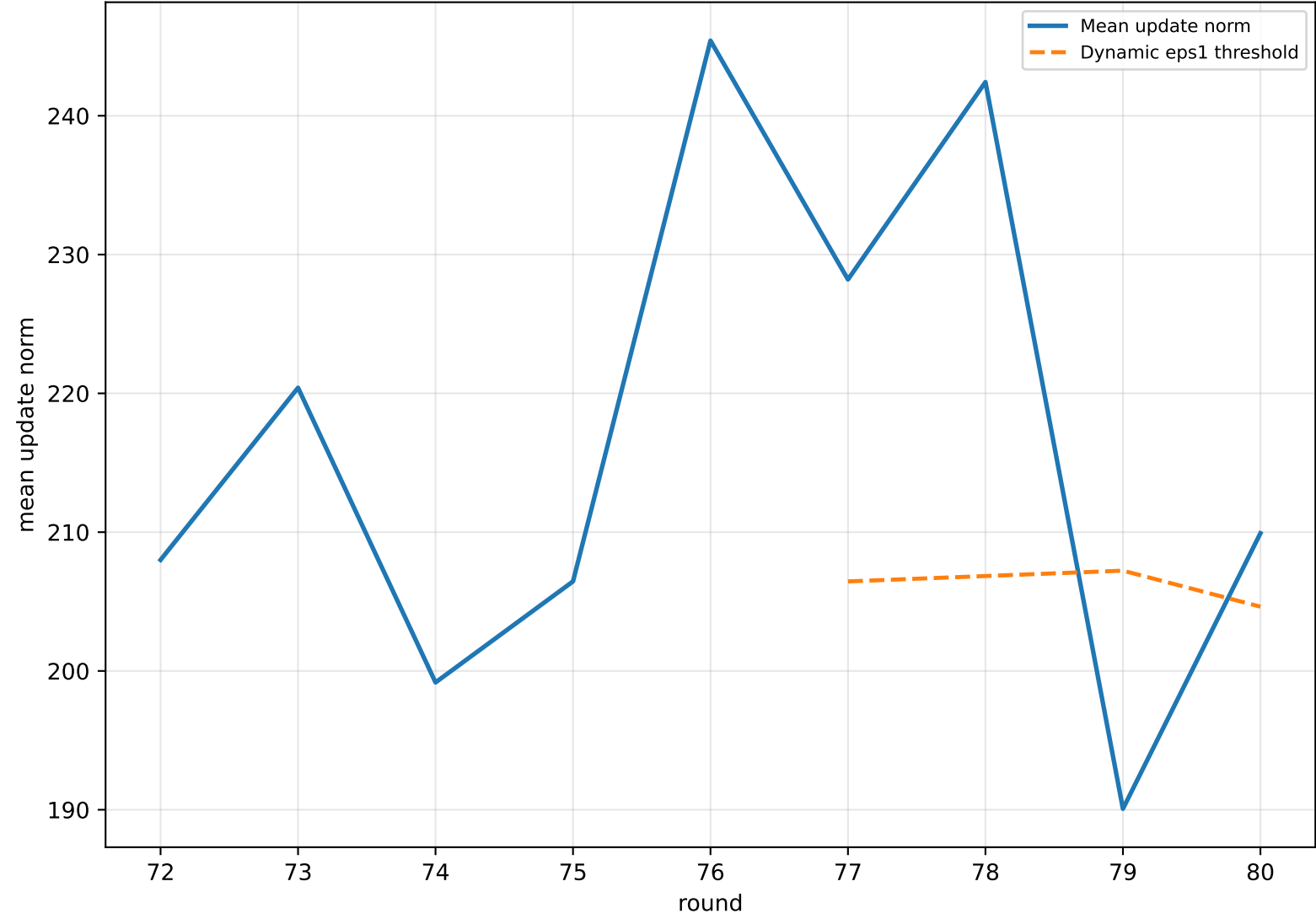
Cluster 0 | Mean update norm



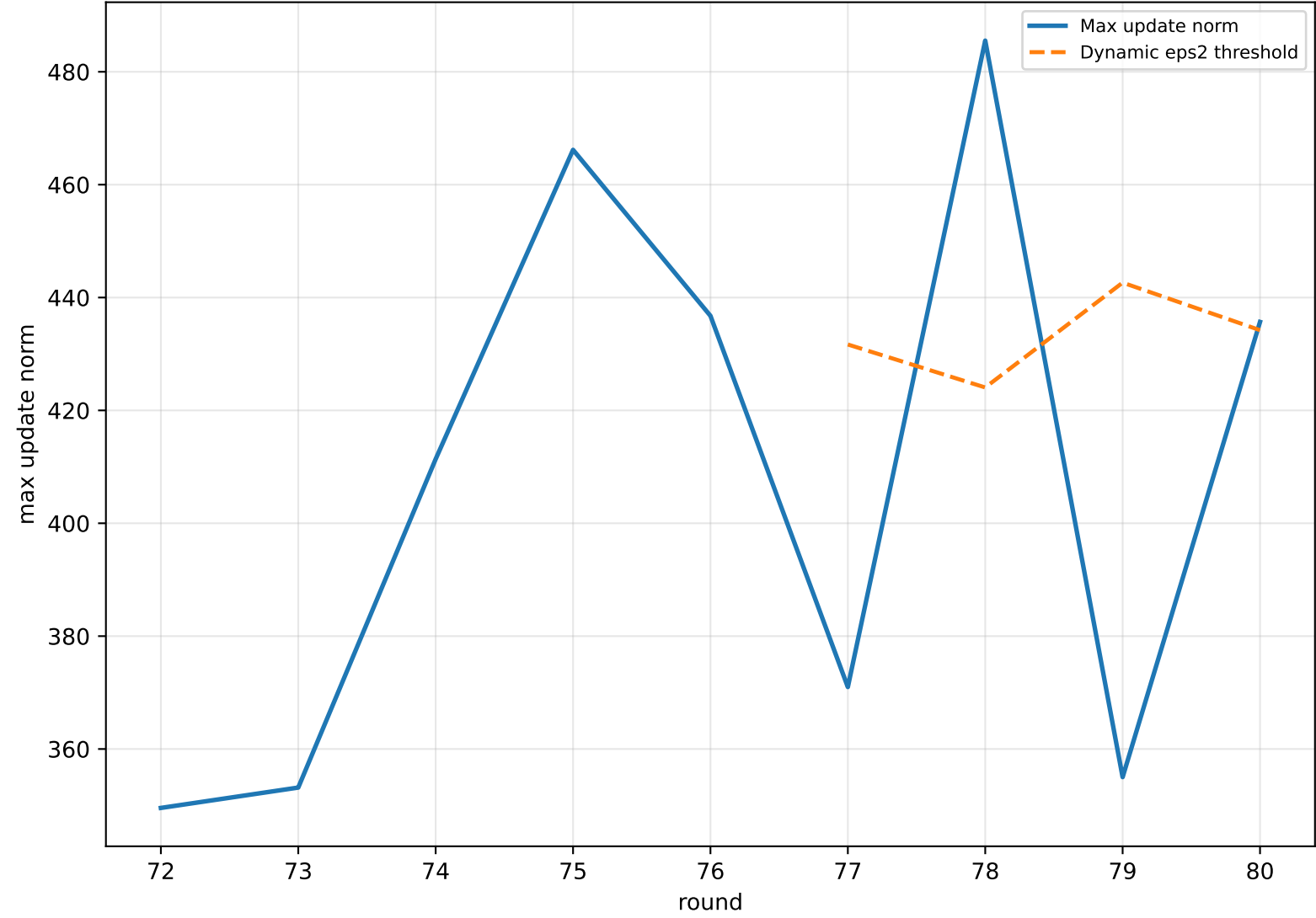
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

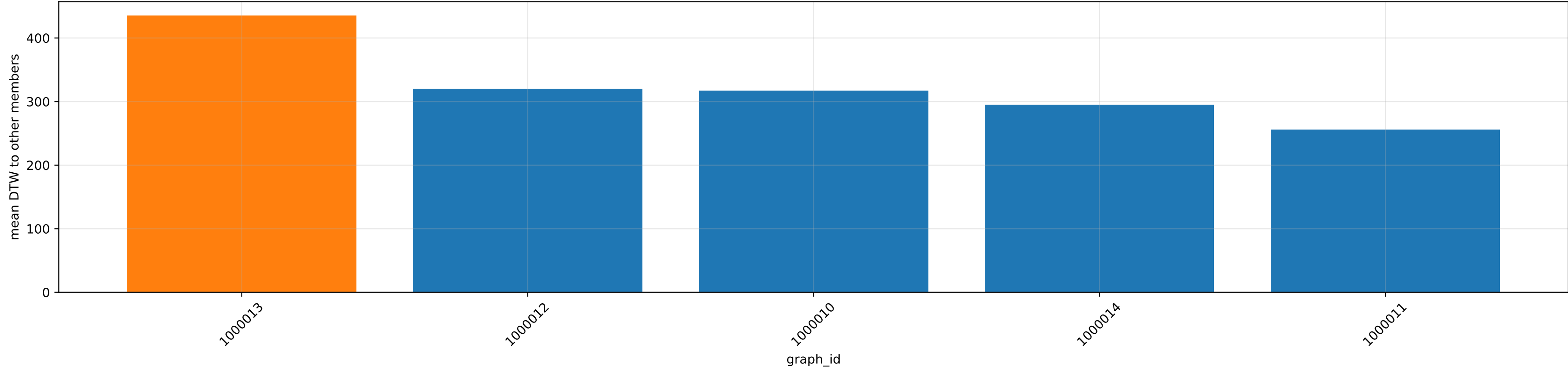
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	55	0	1 vs 4	1000013	1000013	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
435.348	nan	250.975

Per-member mean DTW



Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

First applied split pairwise DTW metadata | seed=2

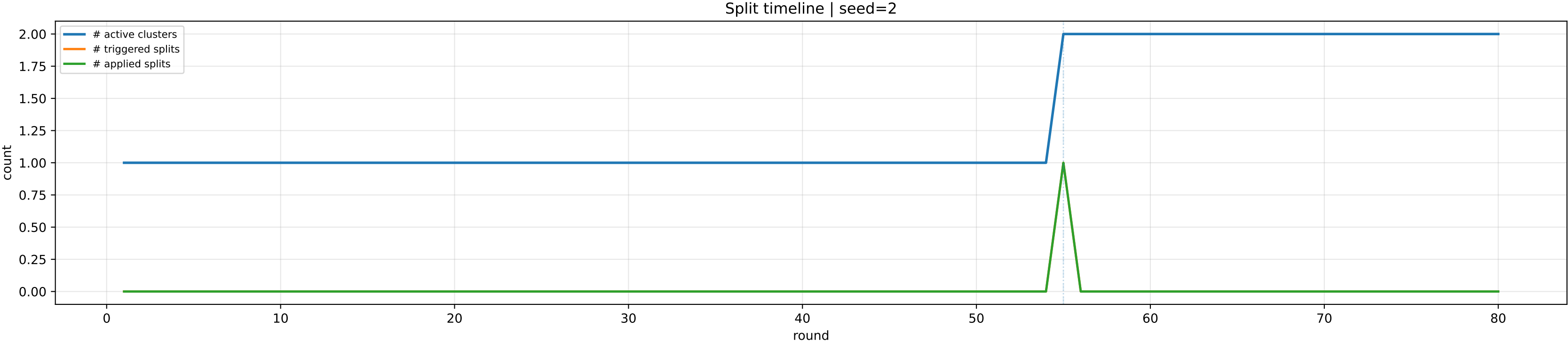
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	55	0	5	1000013.0	0.010	435.348	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 514.3 at (1000012, 1000013)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	253.4	399.3	307.3	309.4
1000011	253.4	0.0	164.2	428.7	176.3
1000012	399.3	164.2	0.0	514.3	203.1
1000013	307.3	428.7	514.3	0.0	491.1
1000014	309.4	176.3	203.1	491.1	0.0

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161



Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	55	55	1, 2	1:1, 2:4

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Split lineage story | seed=2
round 55

Cluster 0
5 clients

Cluster 2
4 clients

↓
split off -> Cluster 1
1 client
specialized_cycle5:1

Applied split story | seed=2

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
55	0	5	200.942 < 201.509 , 476.550 > 468.164	C1 (1 client)	specialized_cycle5:1	C2 (4 clients)	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle6:1

Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

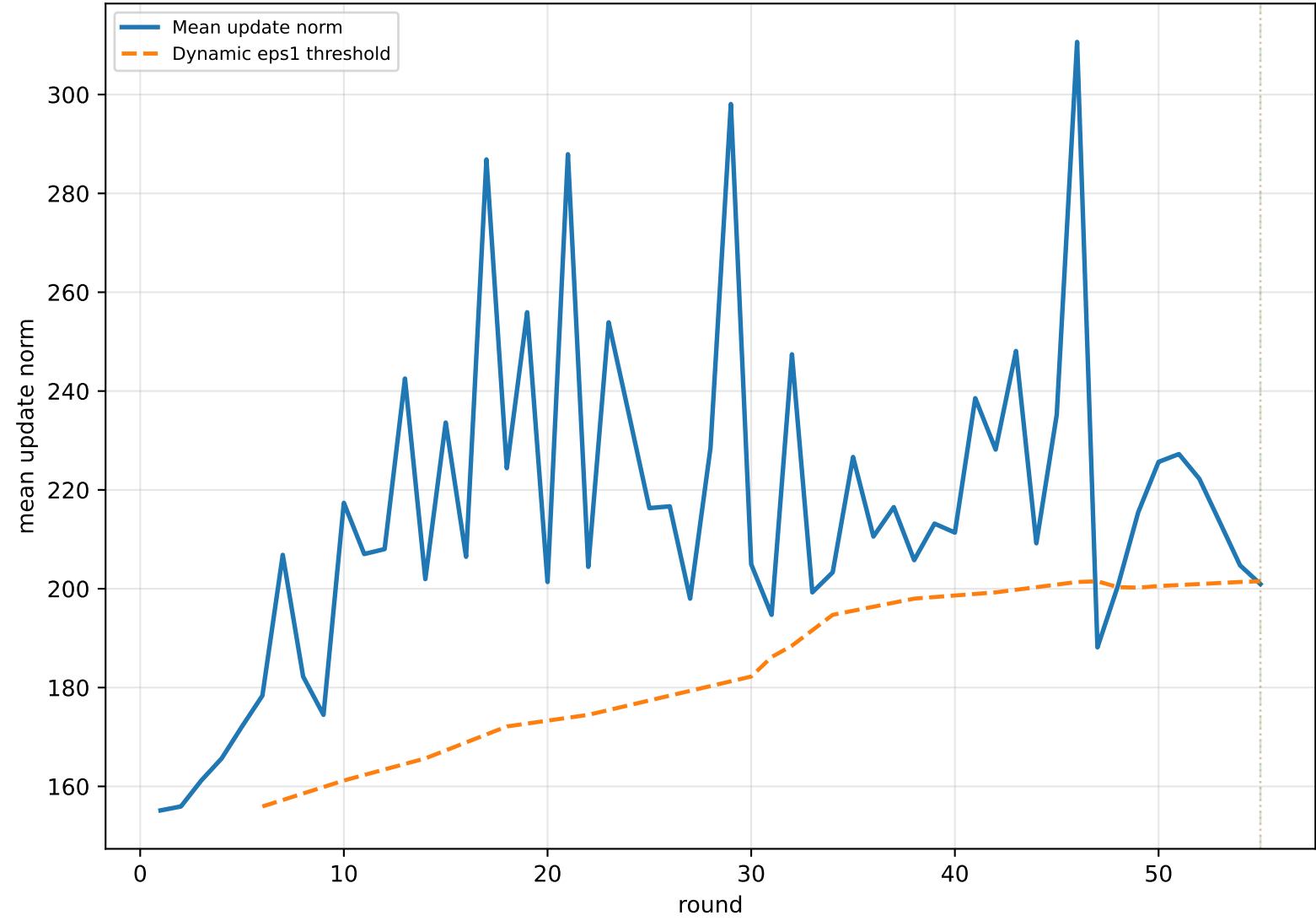
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000013	3	specialized_cycle5:1
2	4	1000010, 1000011, 1000012, 1000014	0, 1, 2, 4	specialized_cycle2:1 , specialized_cycle3:1 , specialized_cycle4:1 , specialized_cycle6:1

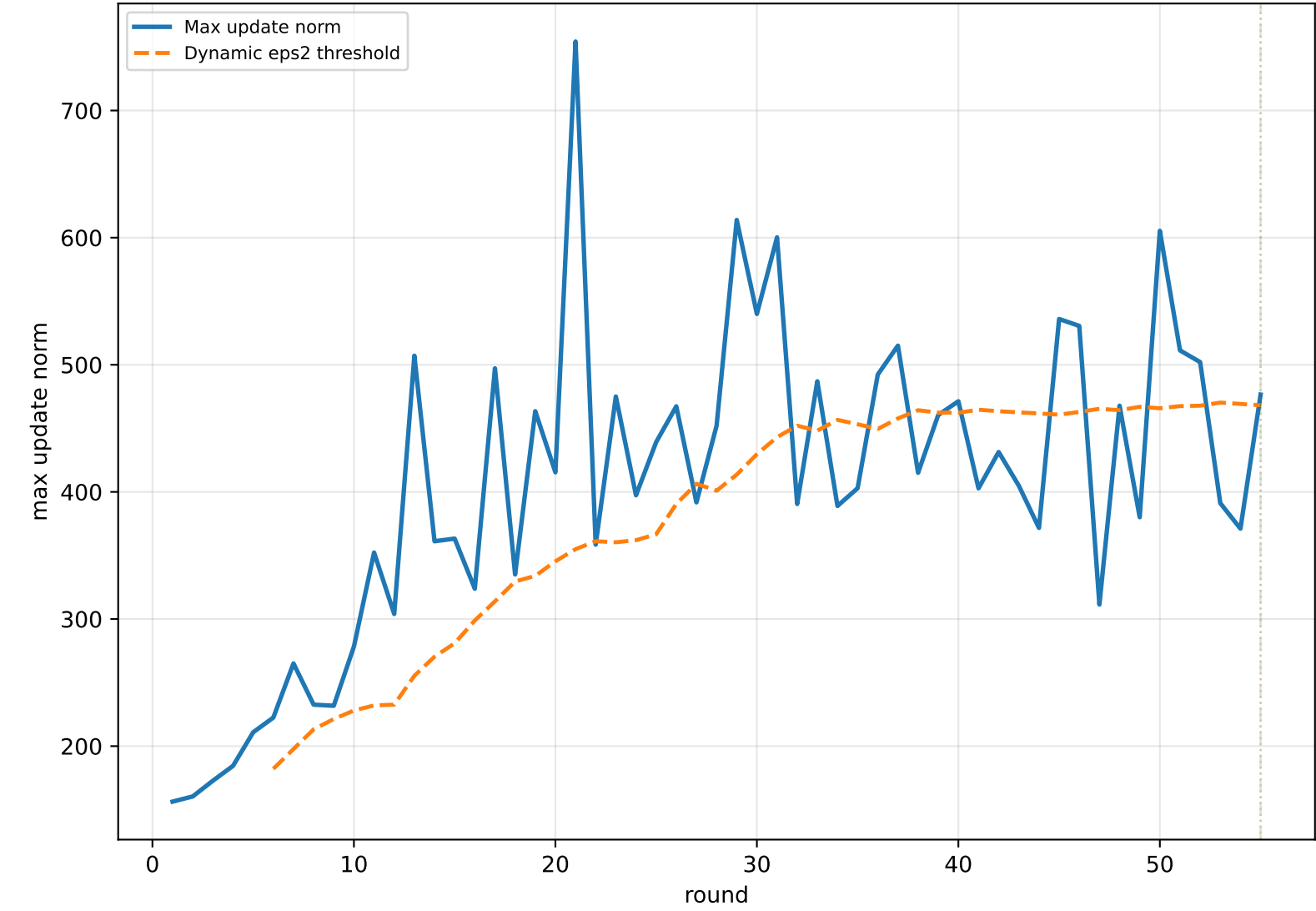
Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

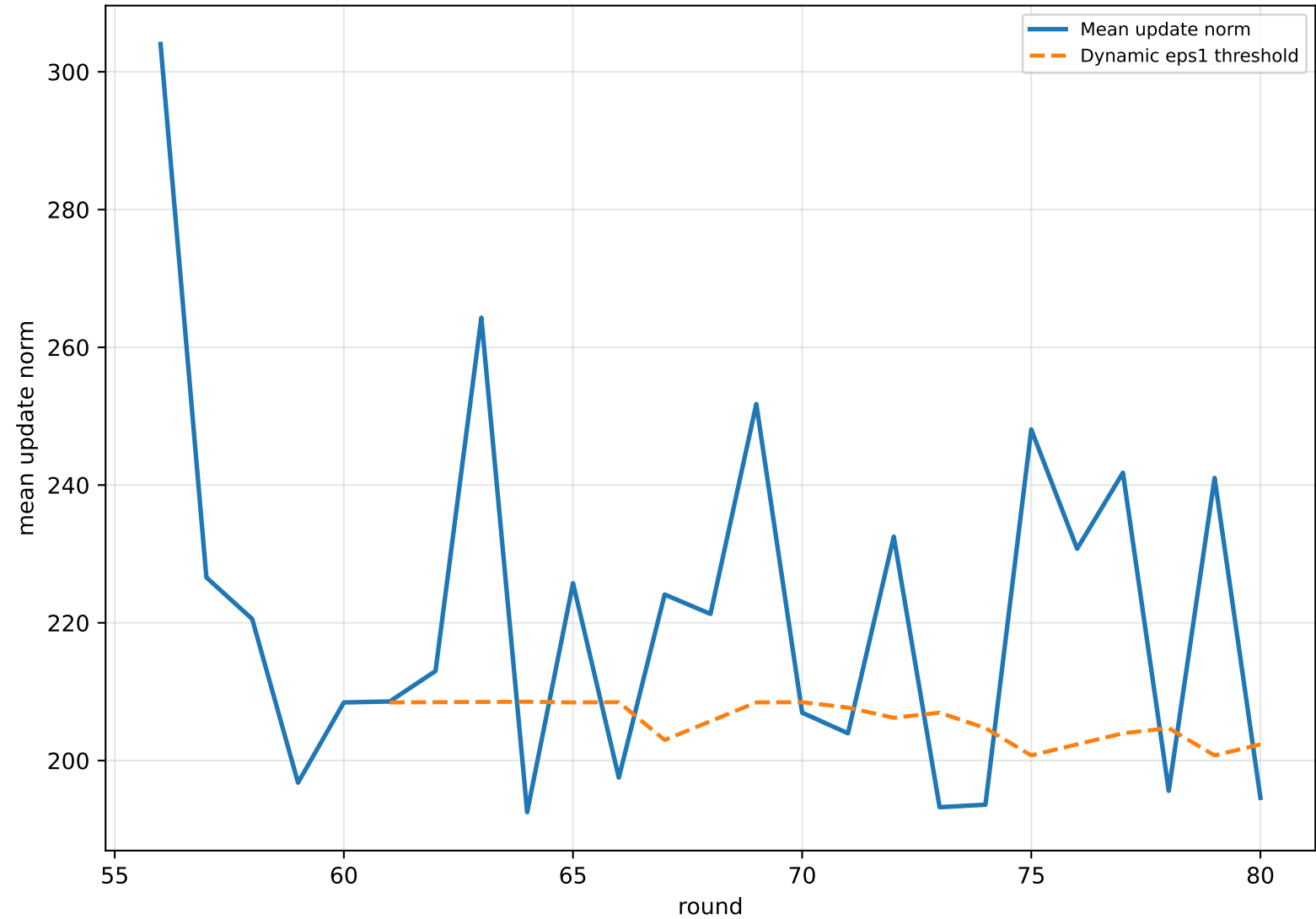
Cluster 0 | Mean update norm



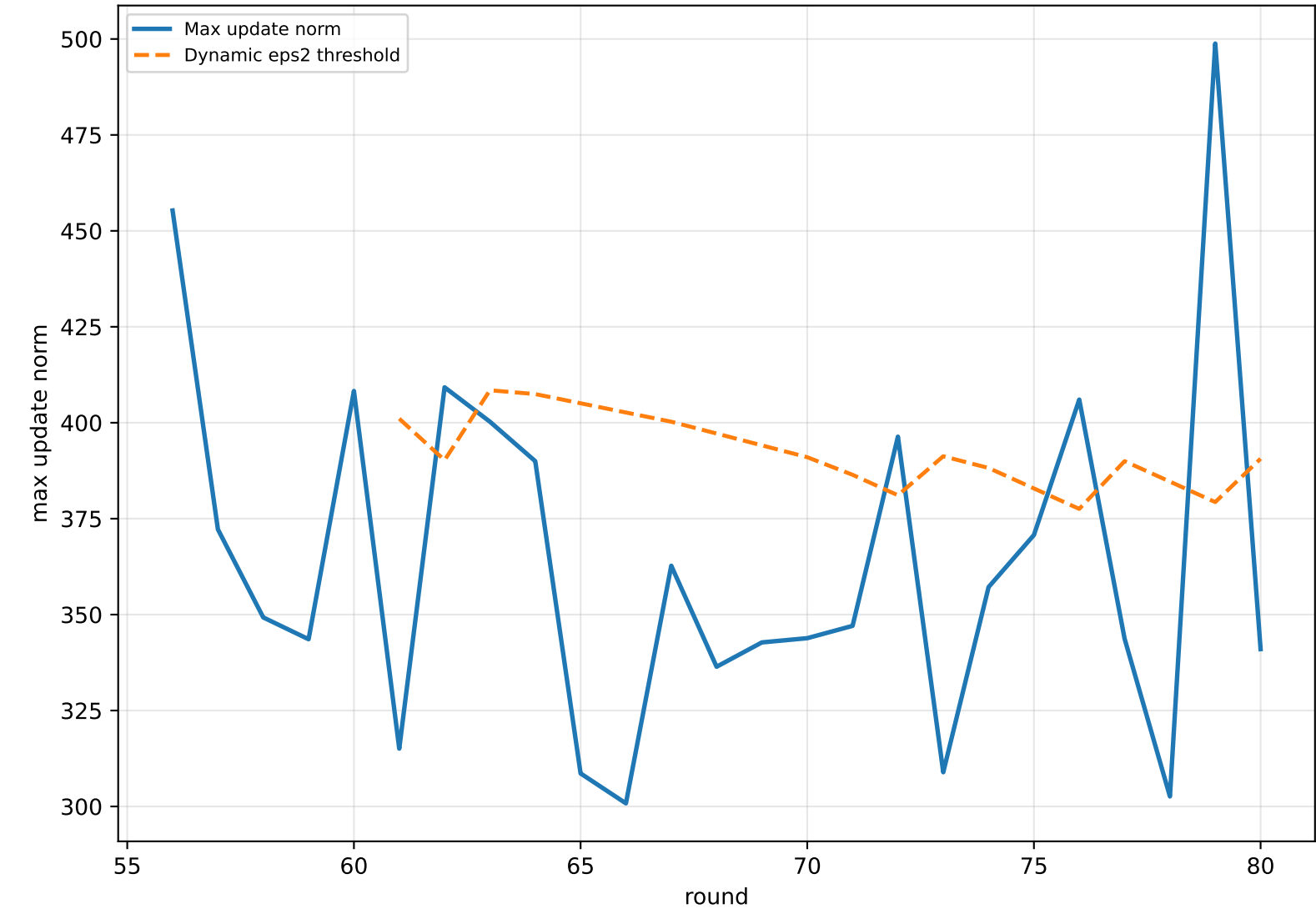
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



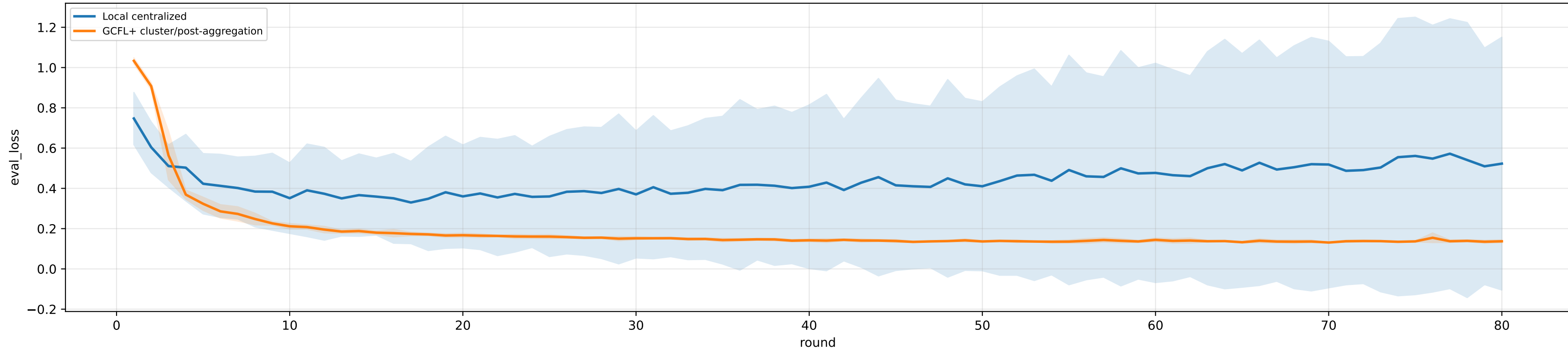
Controlled task-specialization level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

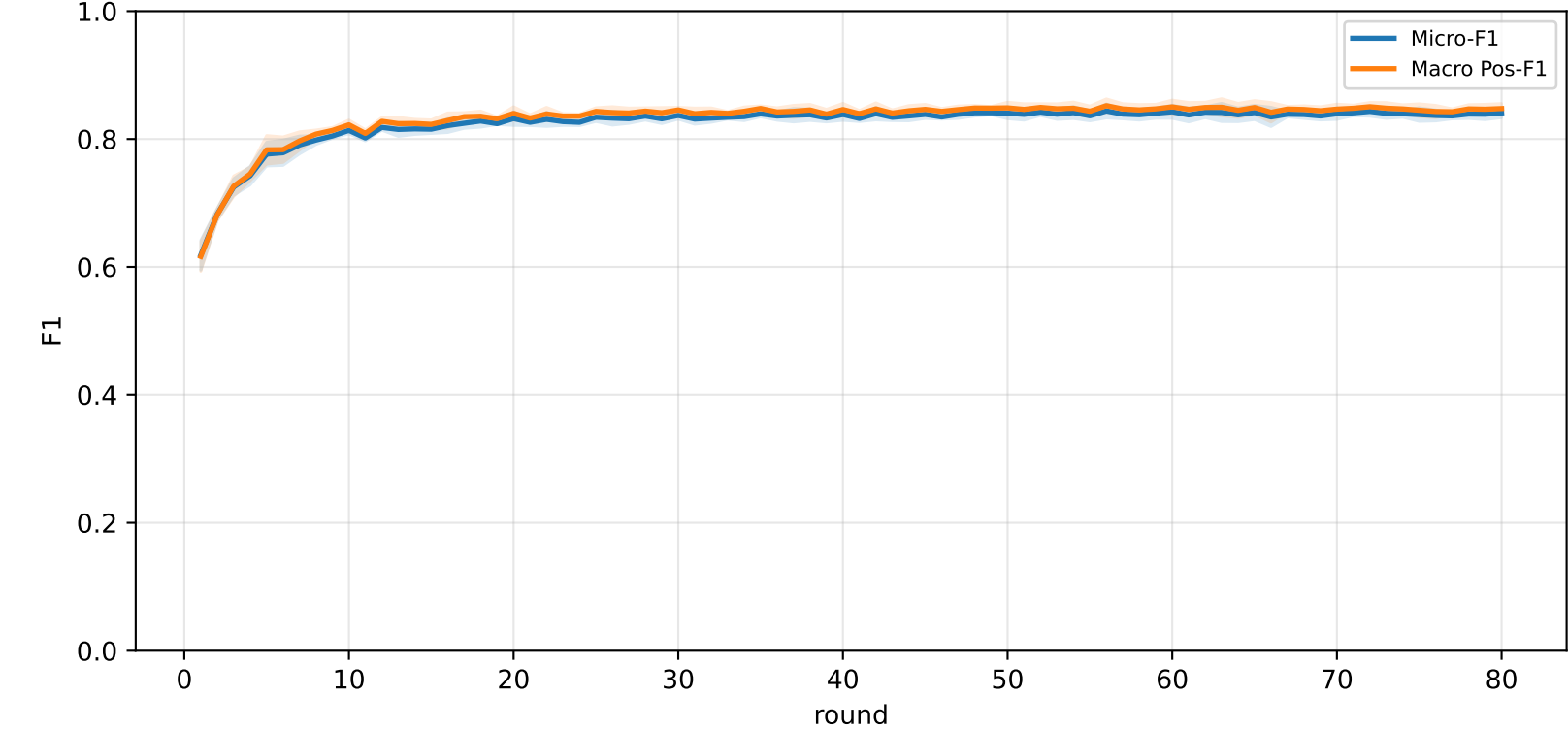
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.1% ± 0.3%	85.3% ± 0.3%	87.0% ± 1.6%	88.2% ± 0.8%	86.8% ± 0.5%	83.1% ± 1.1%	73.7% ± 0.3%
GCFL+	92.3% ± 0.2%	93.2% ± 0.2%	98.7% ± 0.1%	97.9% ± 0.2%	94.3% ± 1.4%	91.8% ± 1.0%	83.2% ± 1.9%

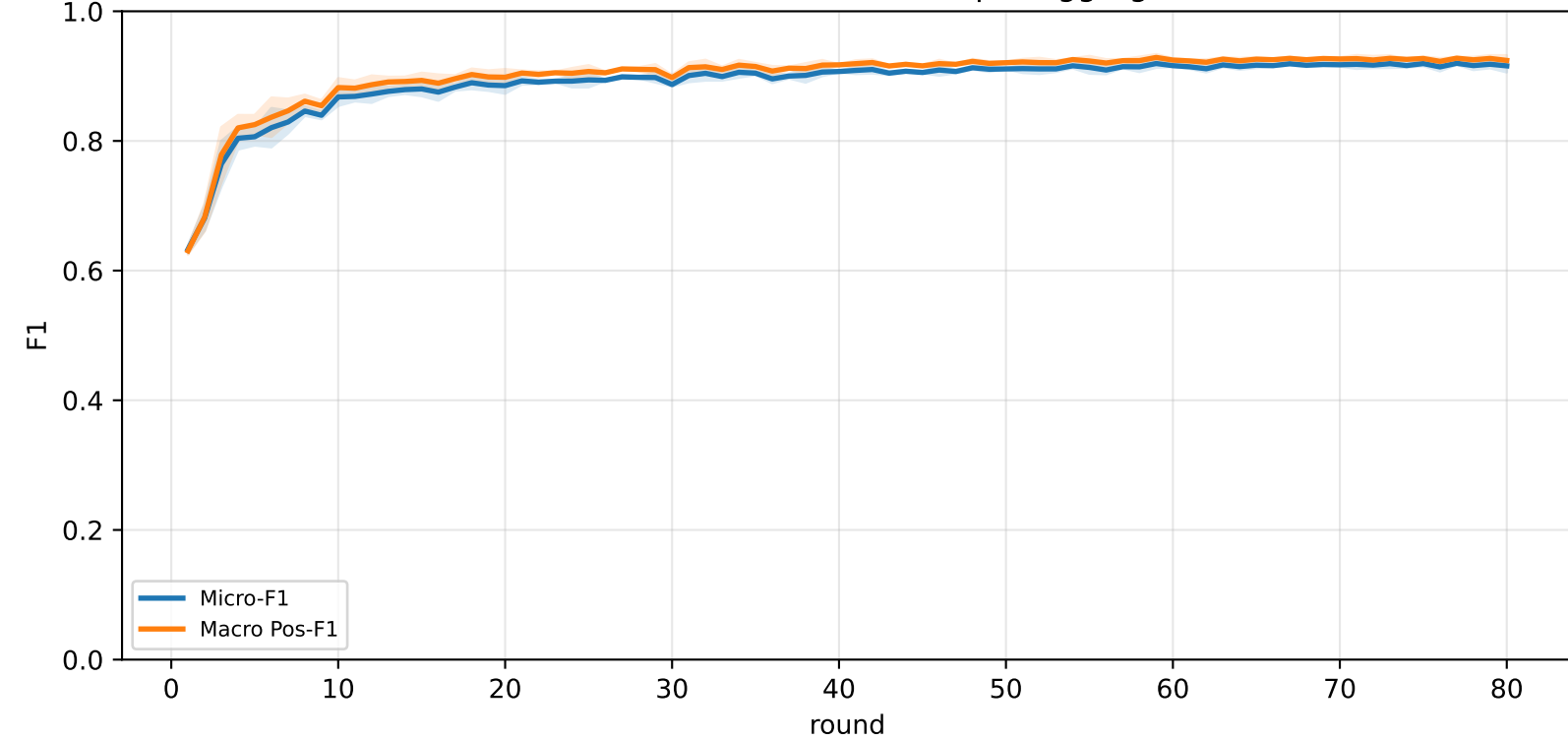
Validation loss comparison



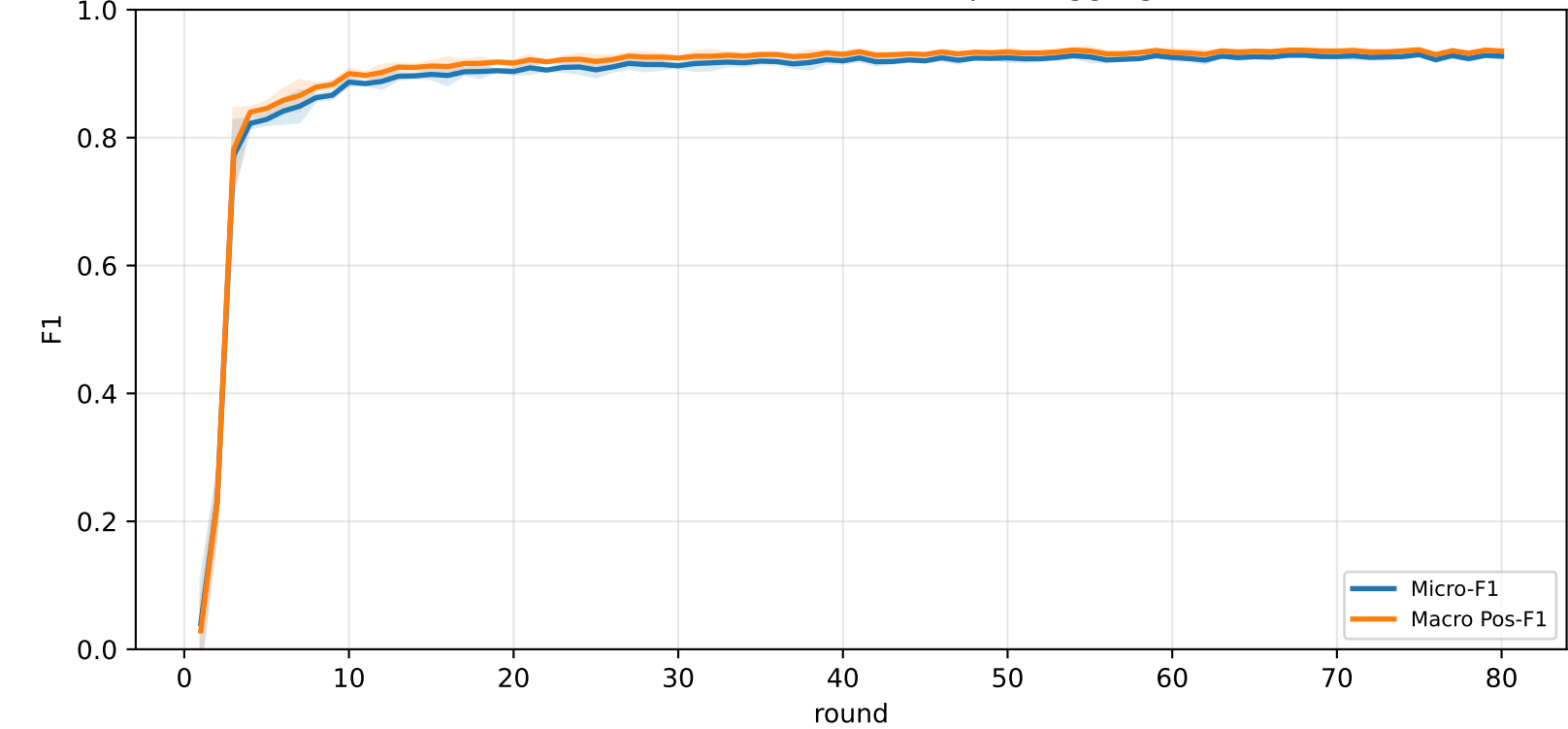
Pooled local centralized micro / macro F1



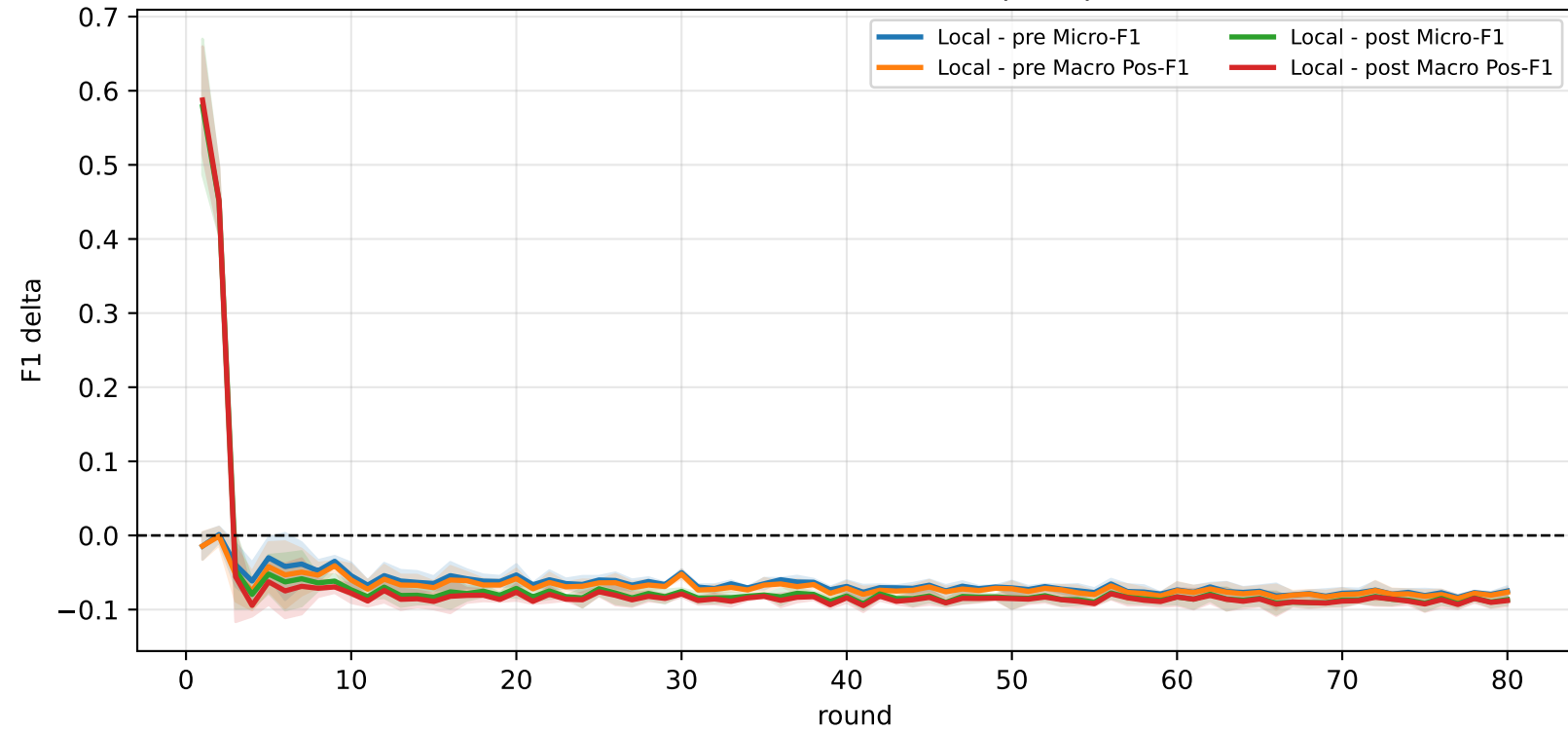
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

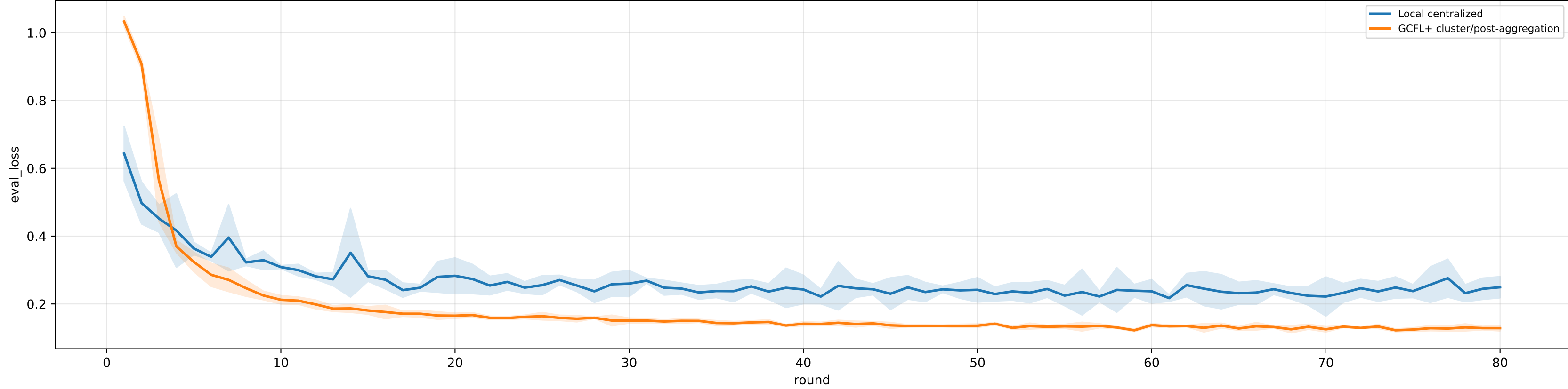


family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

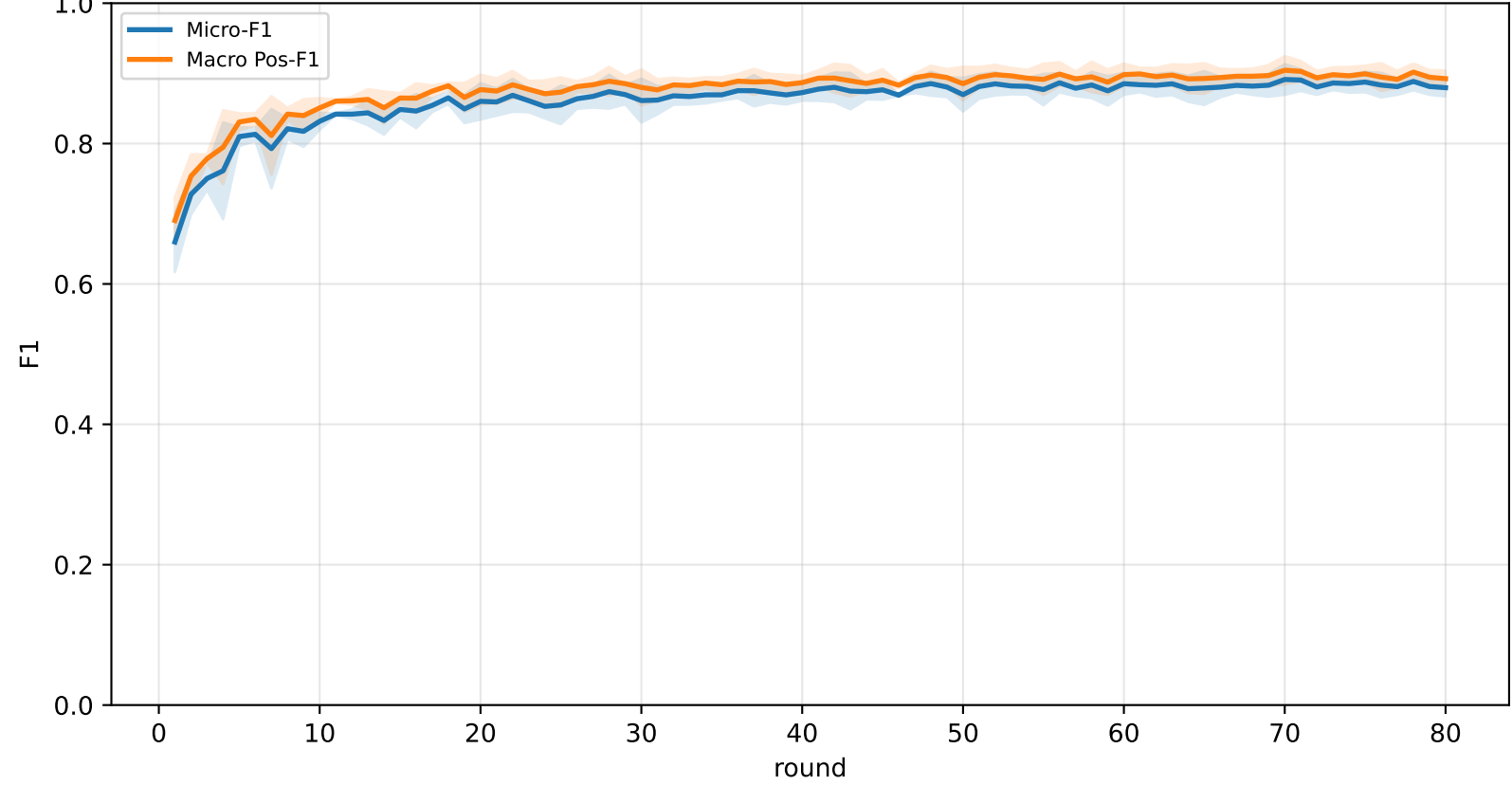
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.3% ± 1.5%	89.8% ± 1.2%	99.0% ± 0.2%	97.2% ± 0.4%	93.2% ± 0.6%	83.8% ± 5.3%	75.6% ± 1.0%
GCFL+	92.5% ± 0.1%	93.3% ± 0.1%	98.8% ± 0.2%	98.2% ± 0.2%	93.9% ± 1.8%	92.0% ± 1.2%	83.8% ± 1.6%

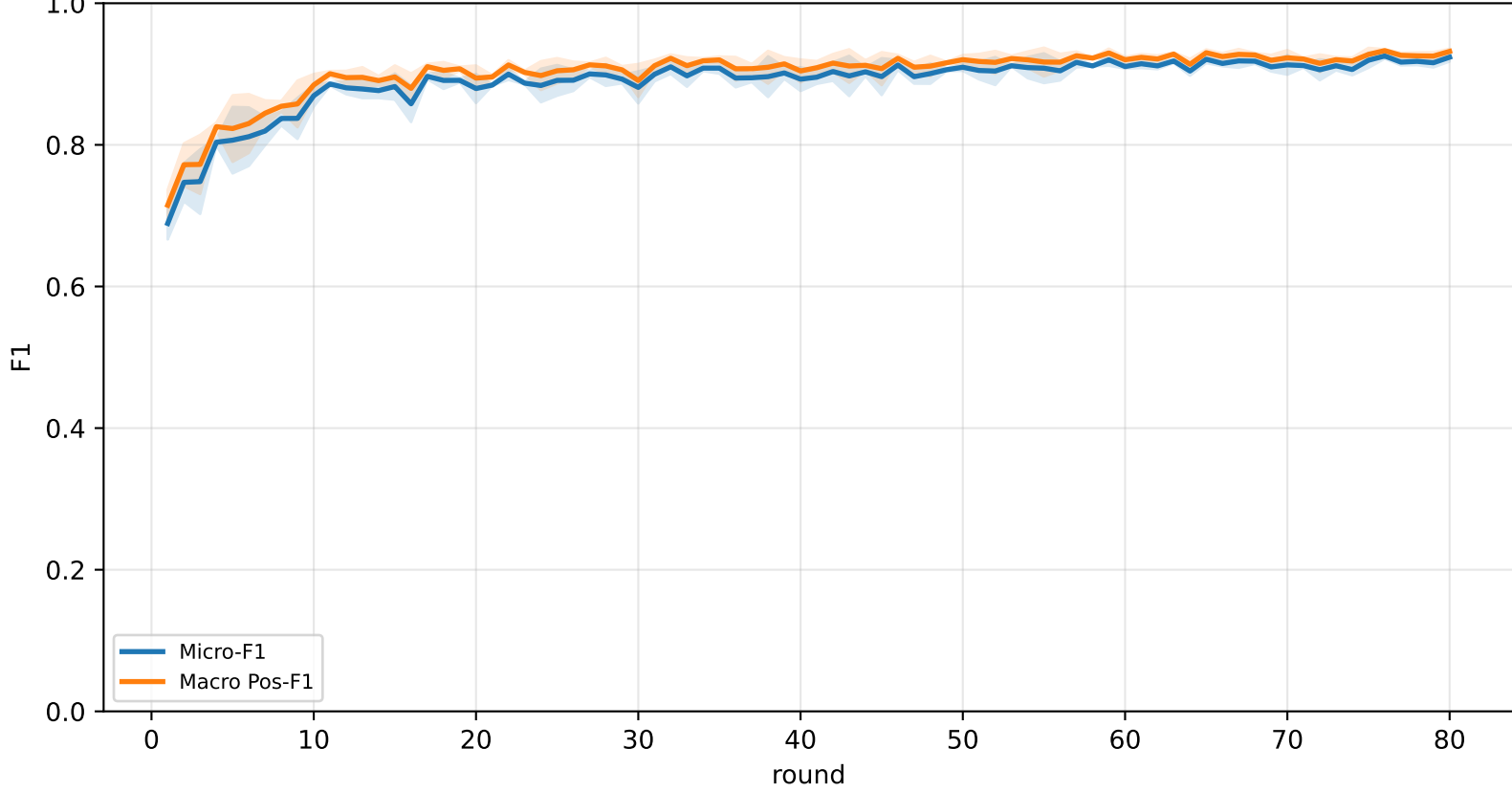
Validation loss comparison



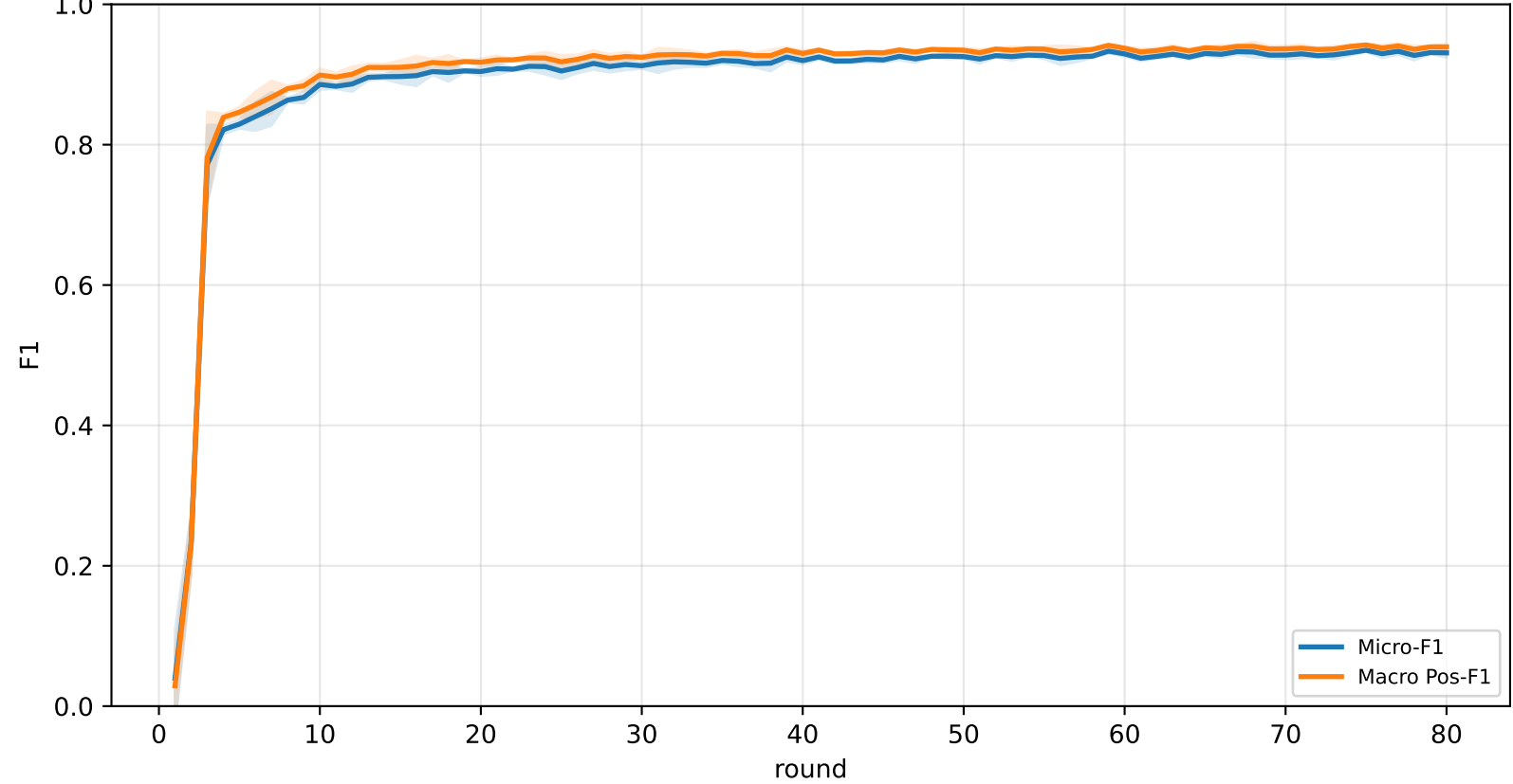
Pooled local centralized micro / macro F1



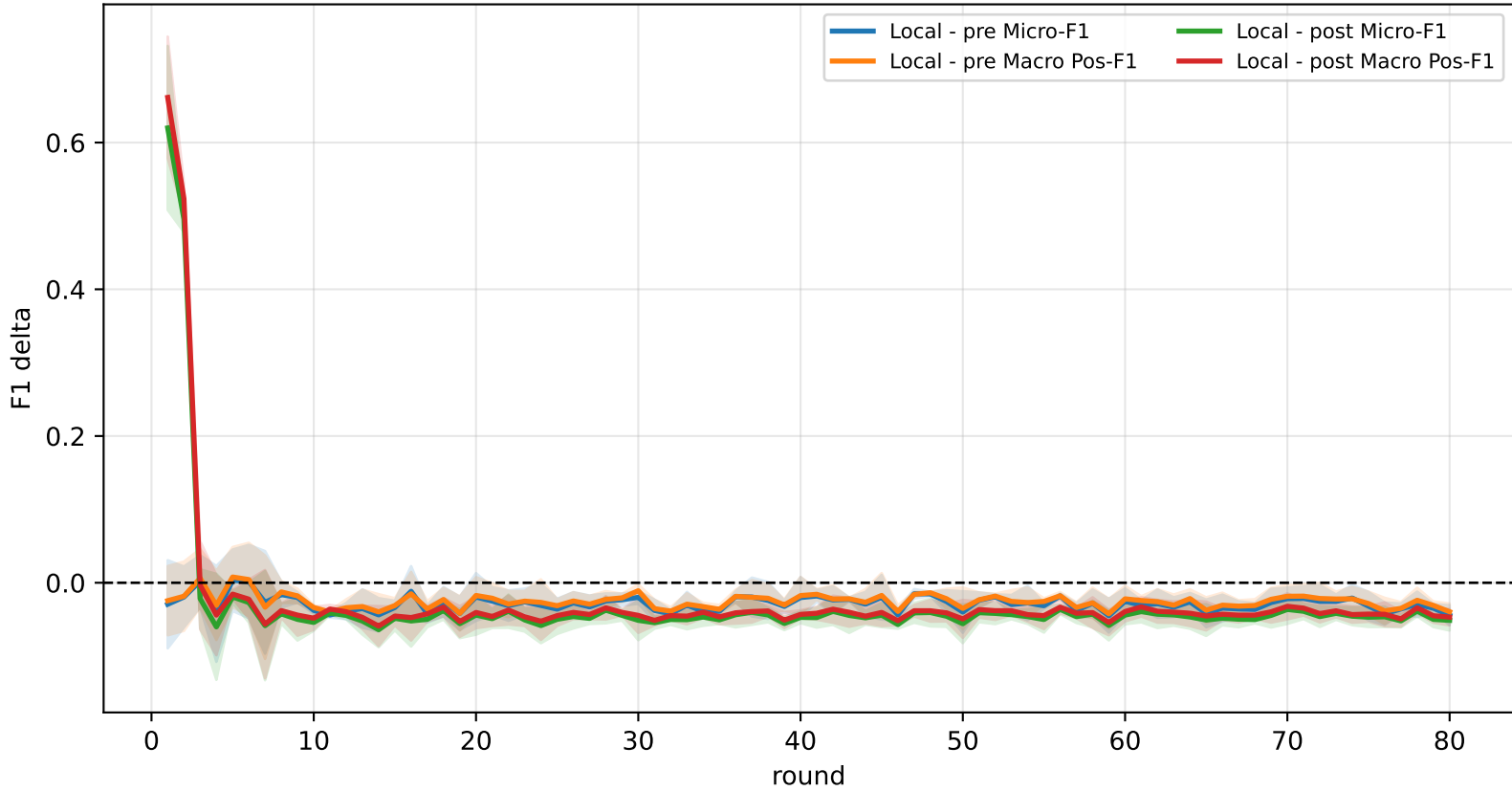
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)



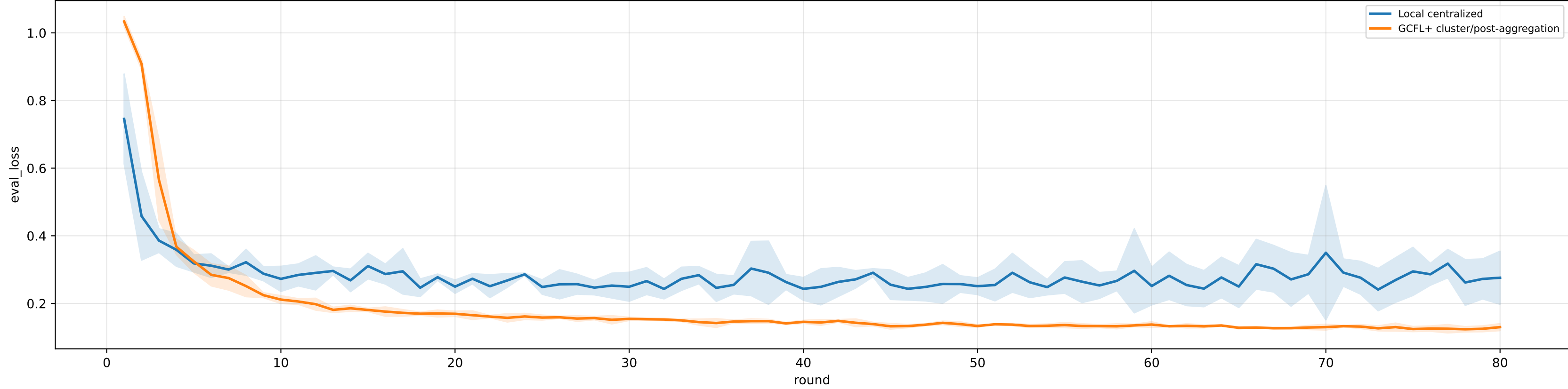
Controlled task-specialization level: High | GCFL+ diagnostics | family=specialized_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

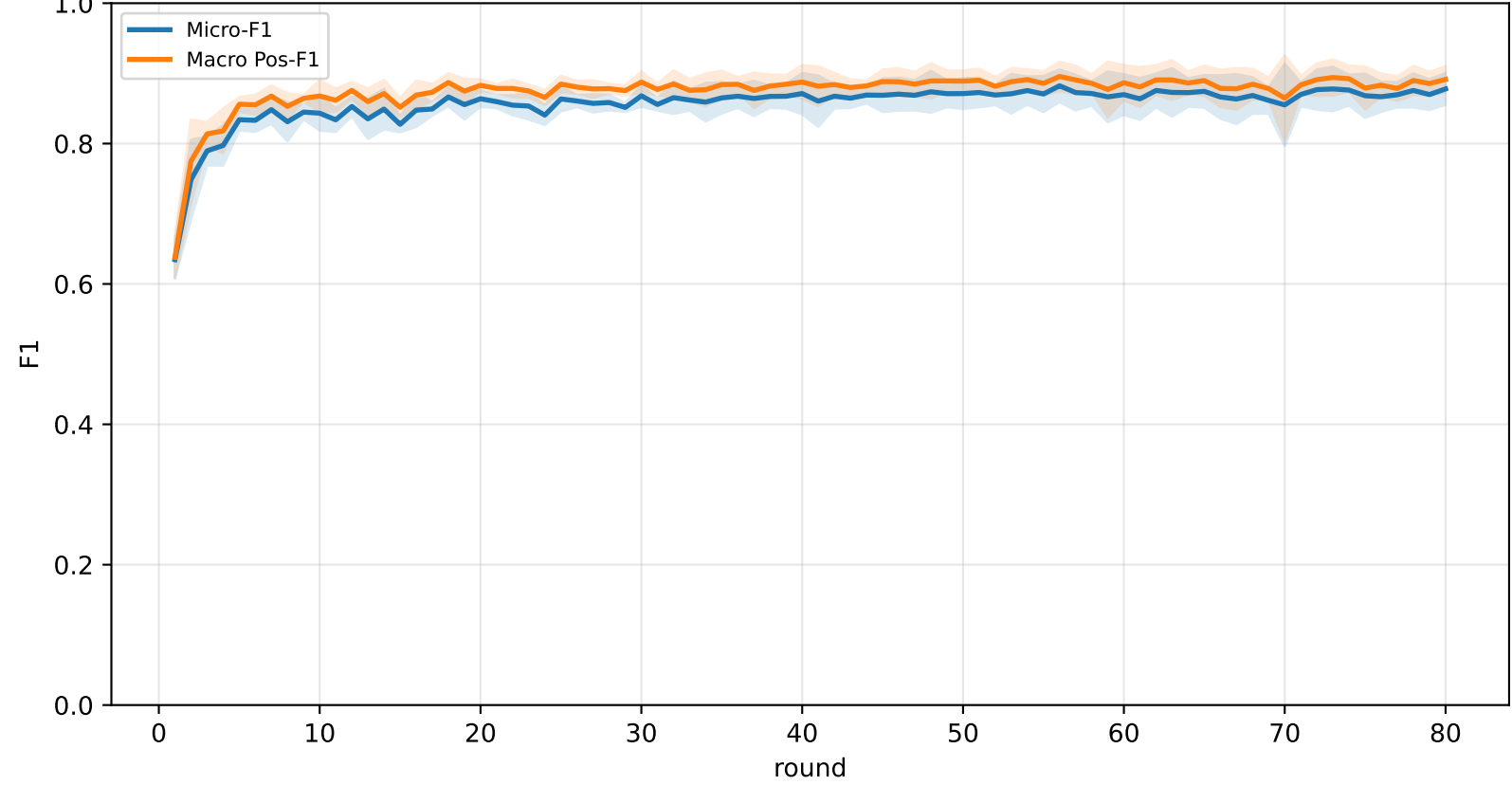
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.2% ± 2.5%	89.2% ± 1.7%	98.4% ± 0.6%	98.4% ± 0.3%	94.1% ± 0.6%	82.5% ± 4.4%	72.3% ± 4.5%
GCFL+	92.4% ± 0.2%	93.2% ± 0.1%	98.7% ± 0.4%	98.1% ± 0.1%	93.8% ± 1.6%	92.0% ± 1.1%	83.5% ± 1.8%

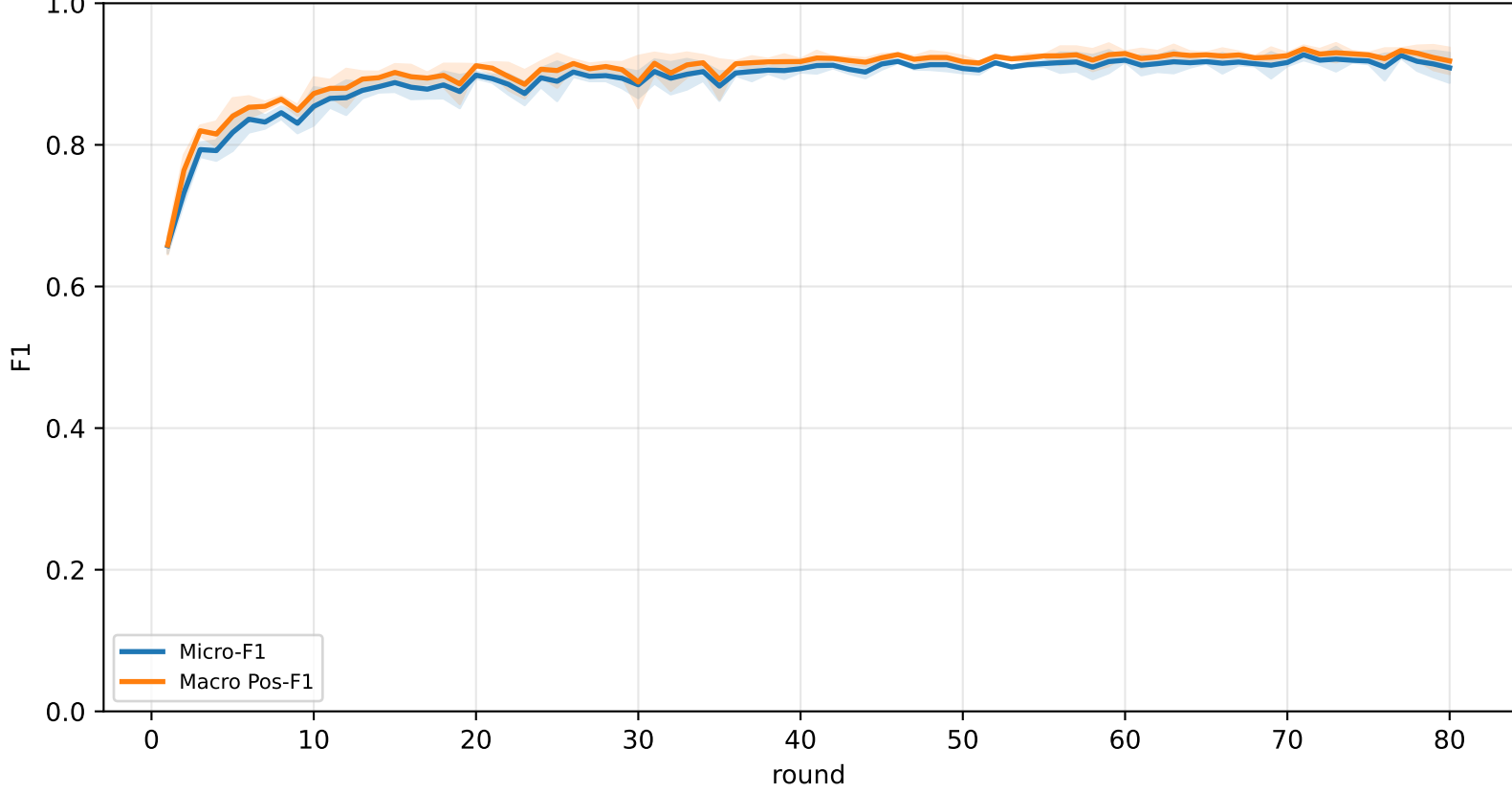
Validation loss comparison



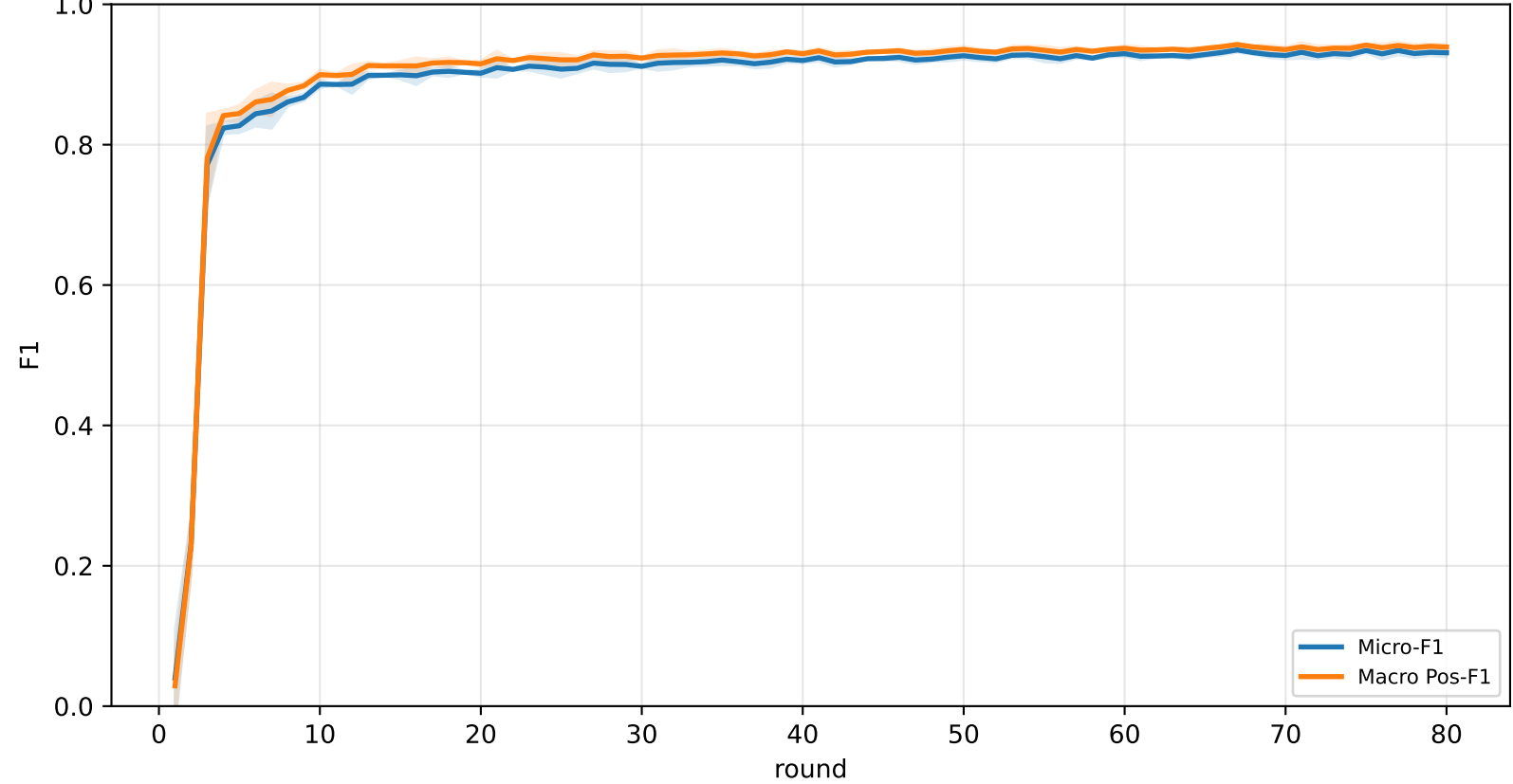
Pooled local centralized micro / macro F1



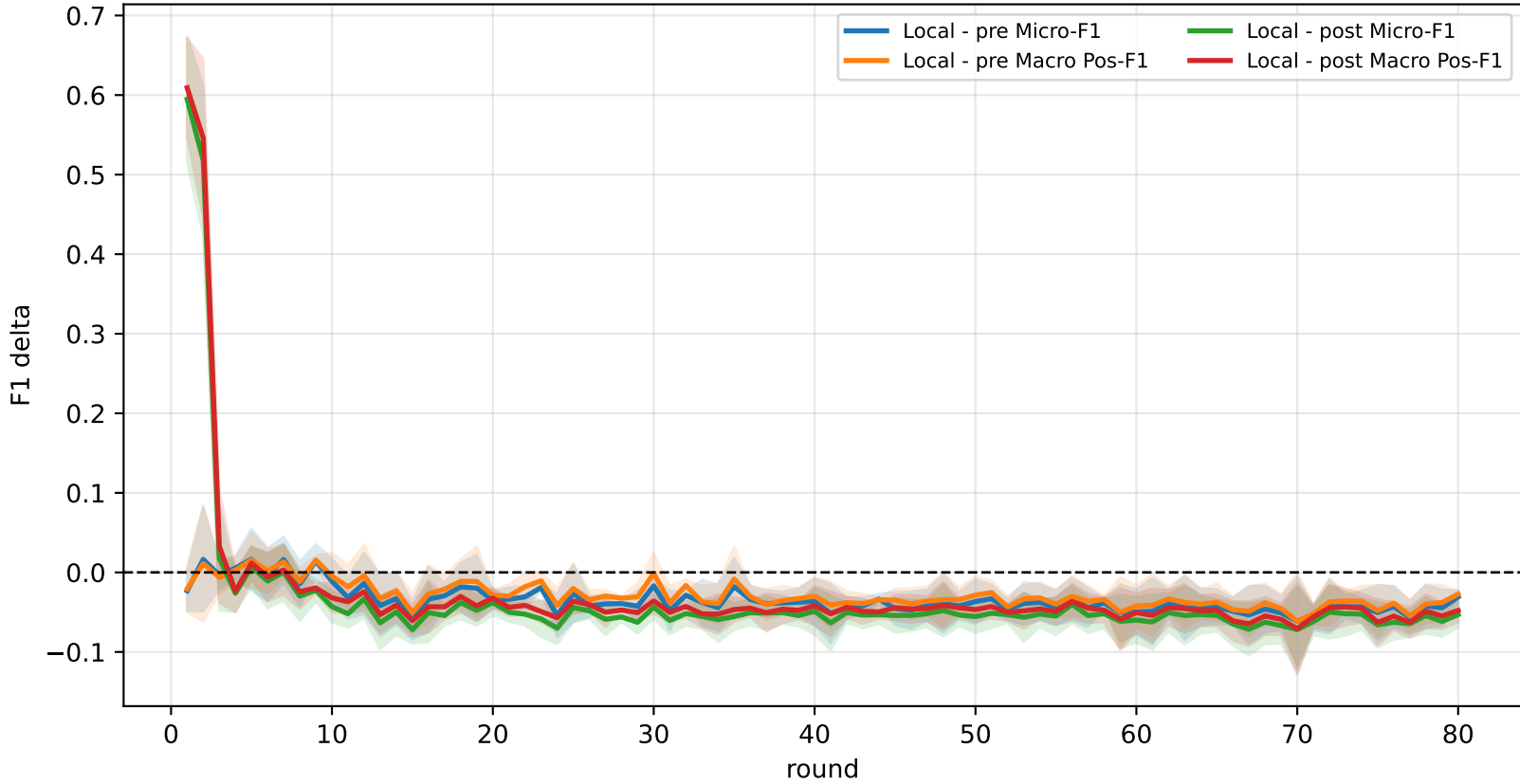
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)

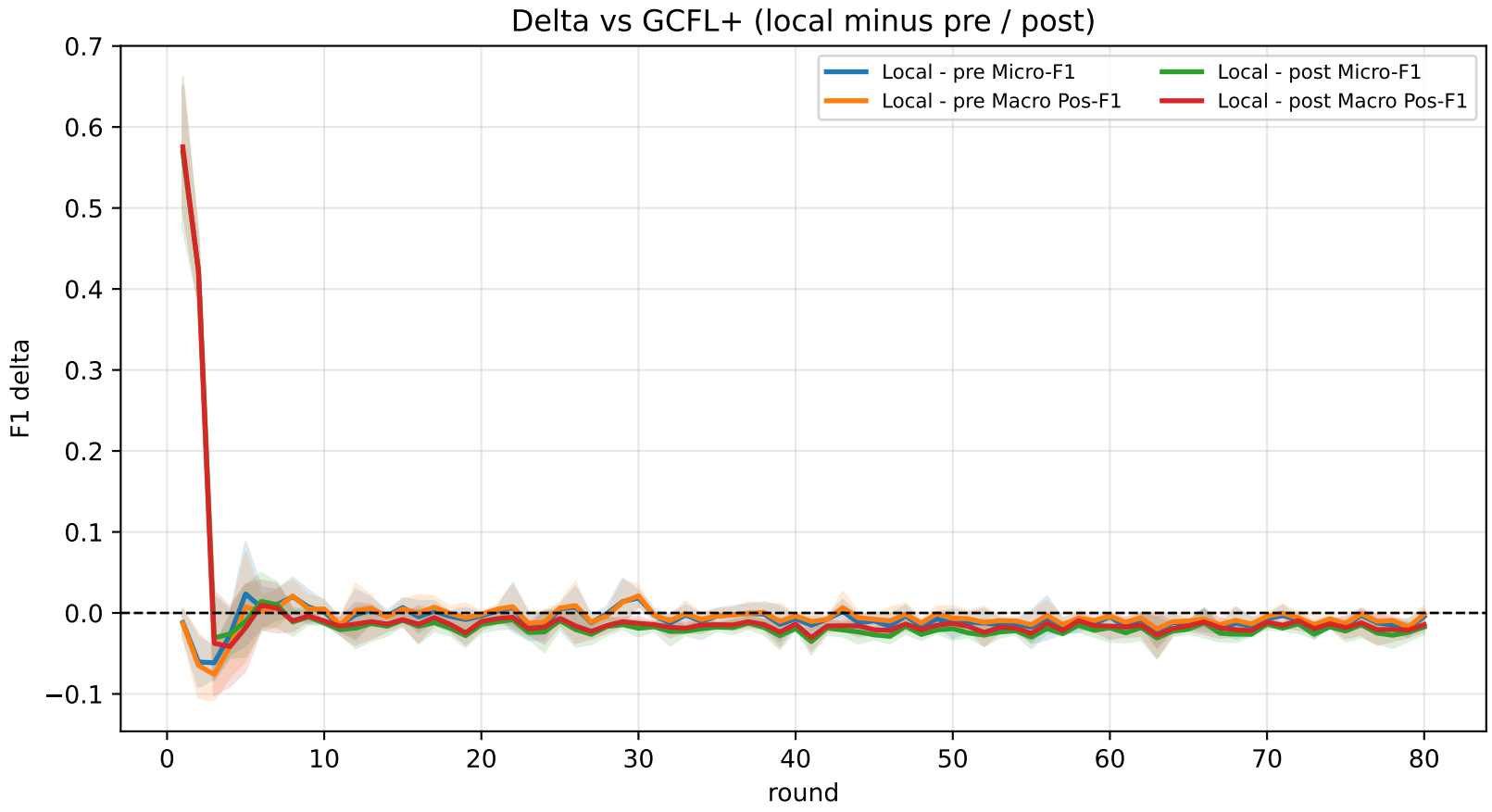
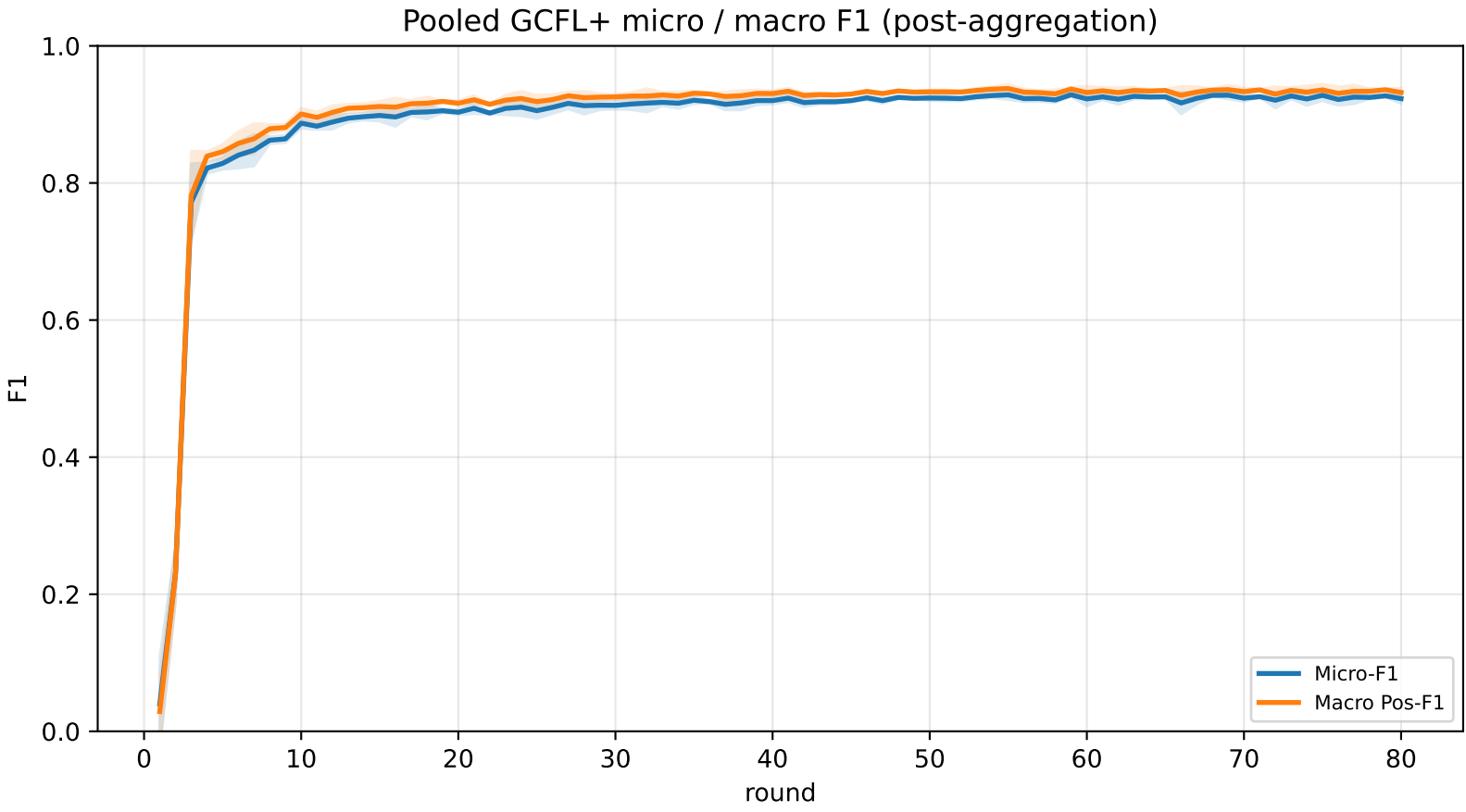
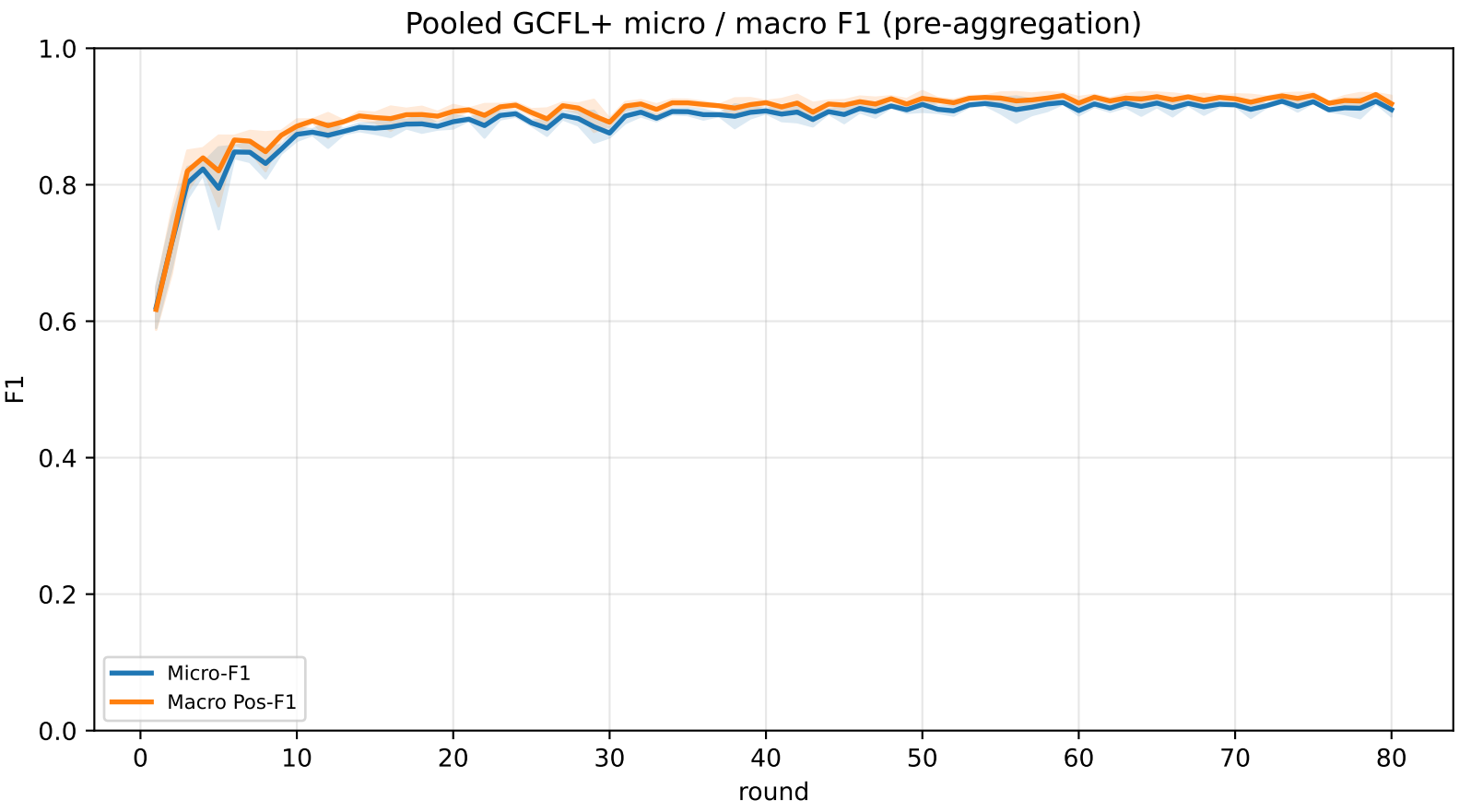
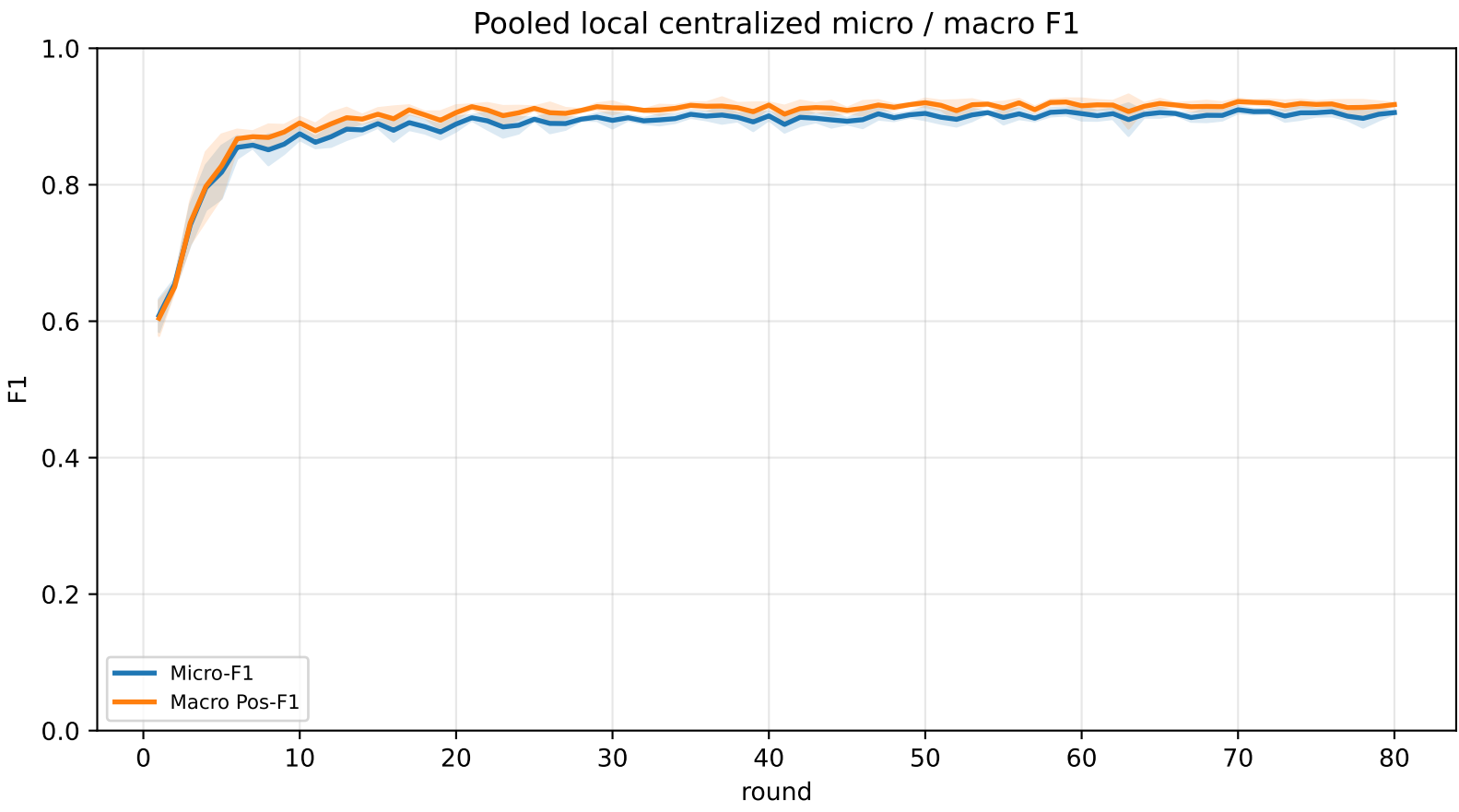
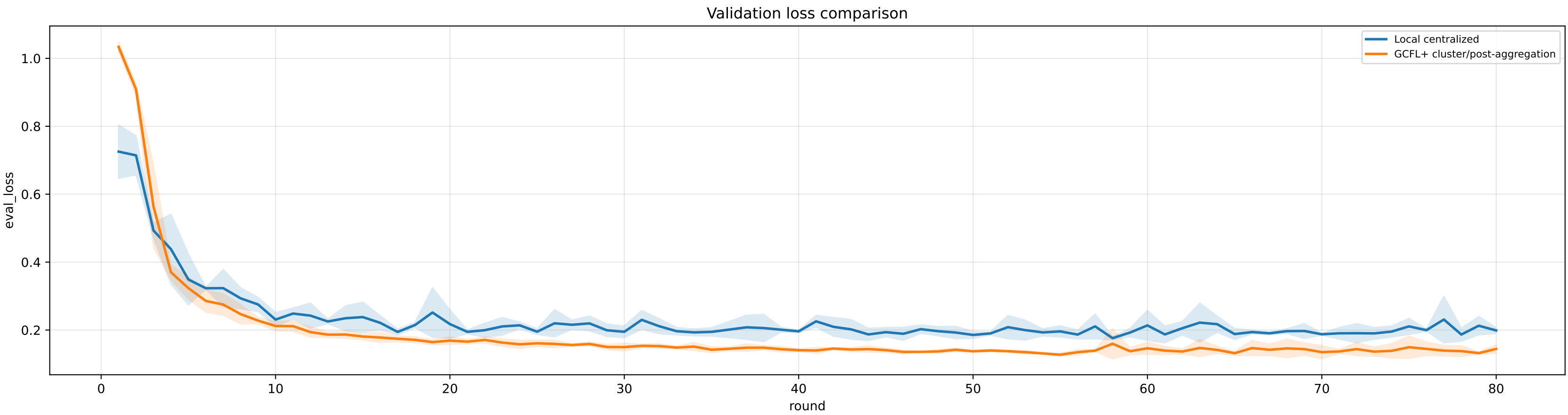


Delta vs GCFL+ (local minus pre / post)



family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.8% ± 0.2%	91.4% ± 0.1%	99.0% ± 0.3%	97.8% ± 0.2%	97.0% ± 0.3%	89.3% ± 0.6%	74.0% ± 0.3%
GCFL+	92.2% ± 0.0%	93.1% ± 0.1%	98.7% ± 0.6%	97.7% ± 0.3%	95.3% ± 0.6%	91.4% ± 1.2%	82.5% ± 1.5%

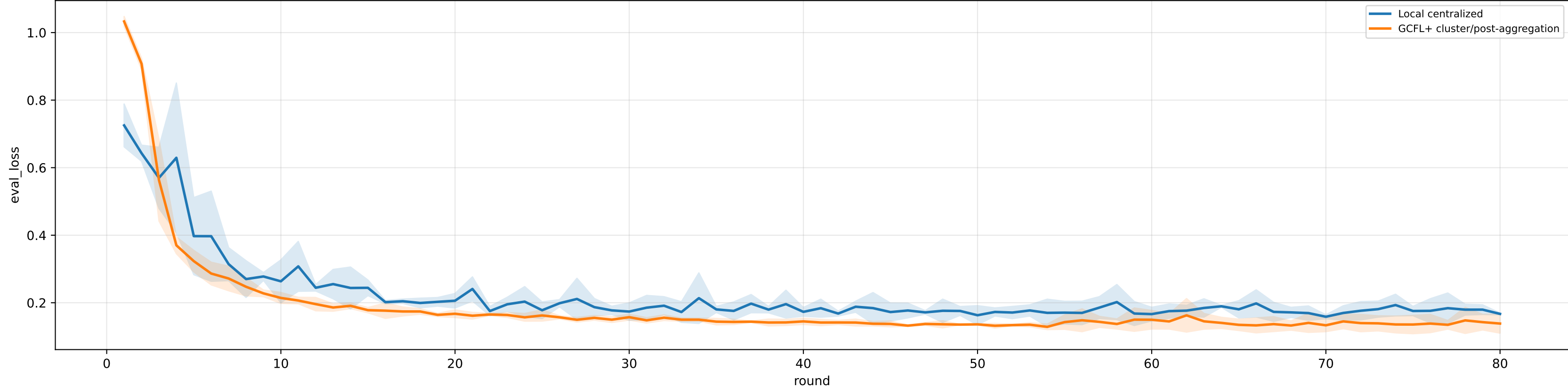


family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

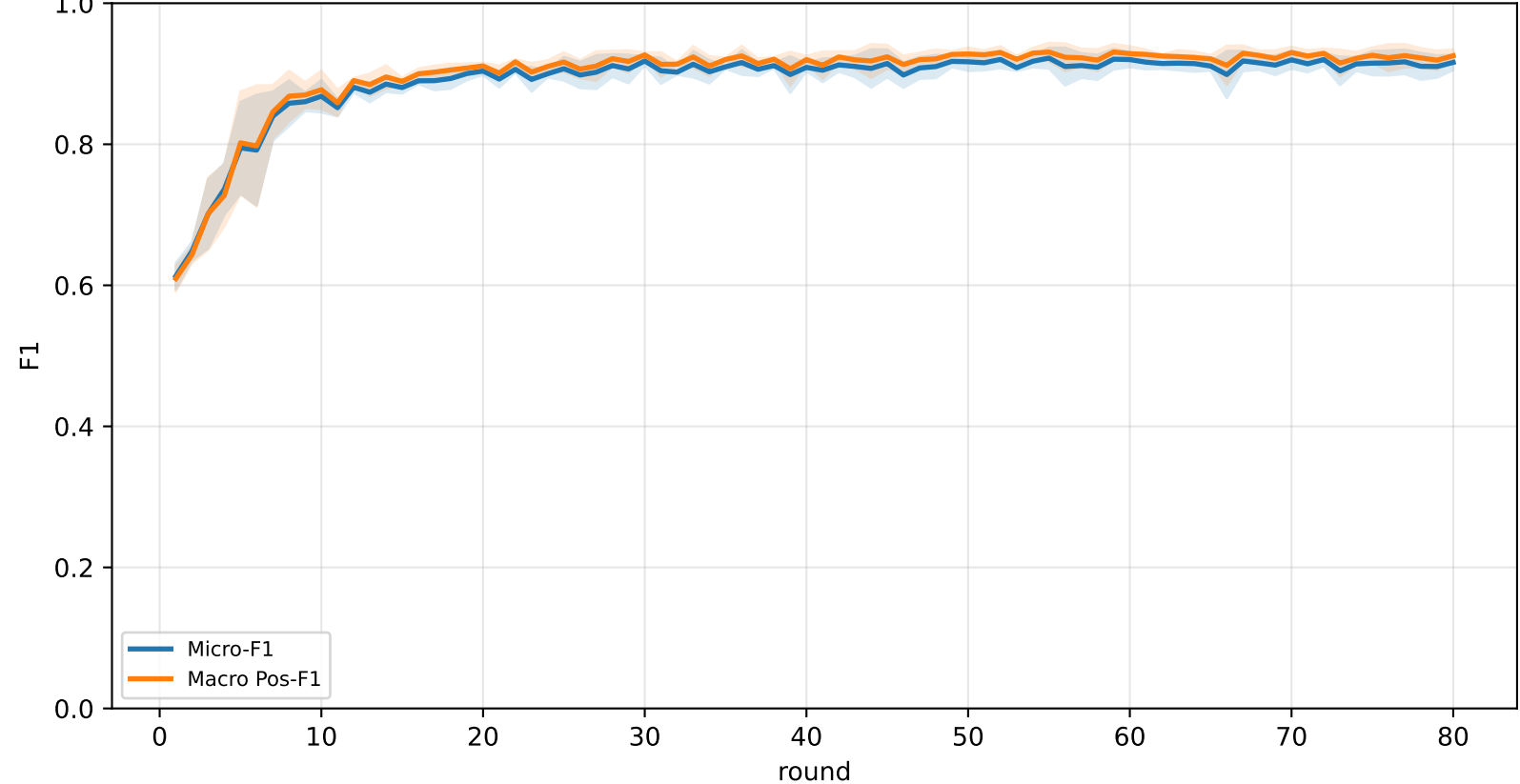
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.3% ± 1.0%	92.4% ± 0.7%	98.4% ± 0.5%	97.7% ± 0.5%	94.7% ± 1.2%	93.5% ± 0.9%	77.8% ± 3.3%
GCFL+	92.6% ± 0.5%	93.4% ± 0.5%	98.9% ± 0.3%	98.0% ± 0.9%	94.0% ± 1.8%	93.2% ± 0.2%	82.8% ± 2.4%

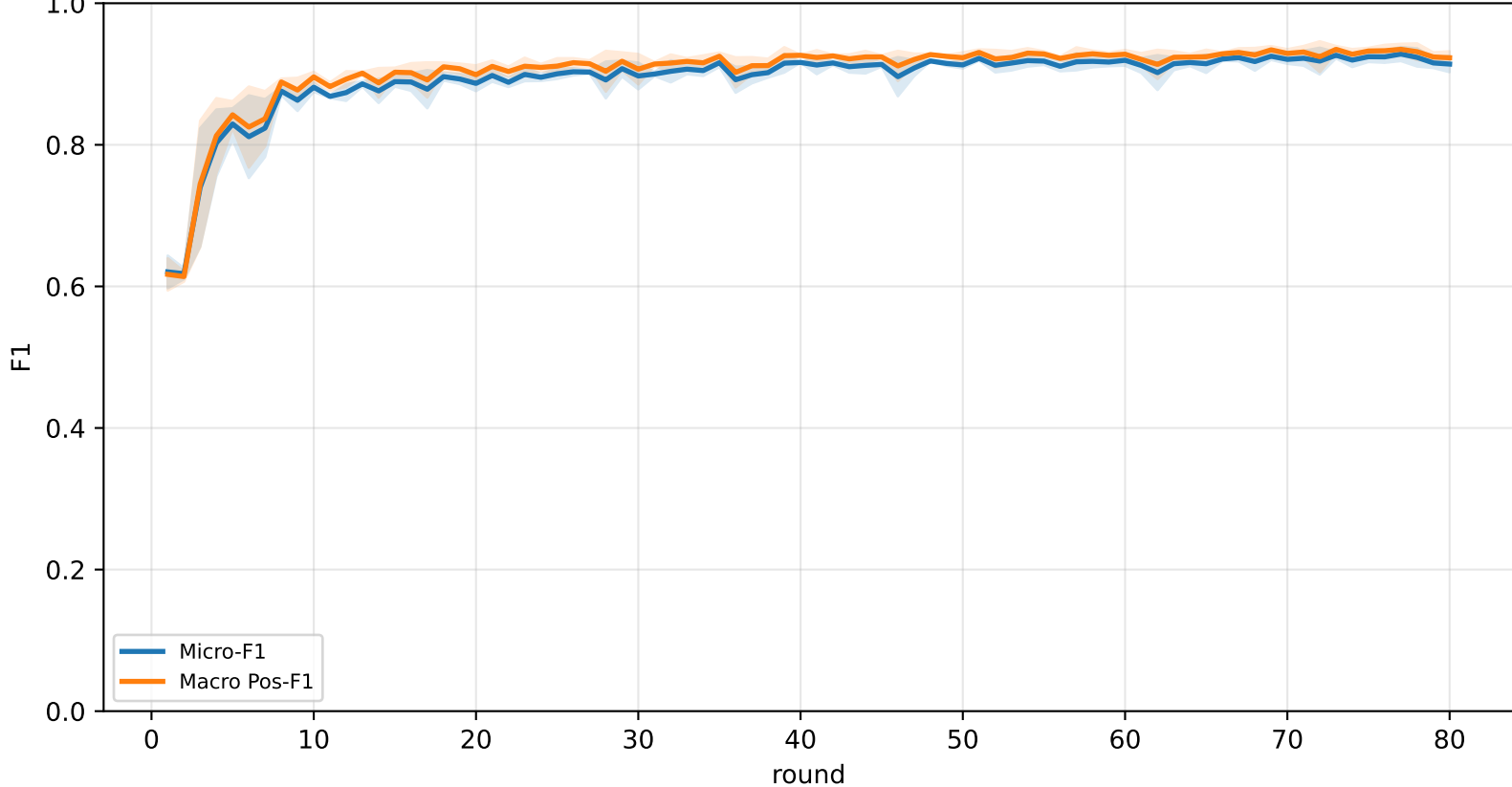
Validation loss comparison



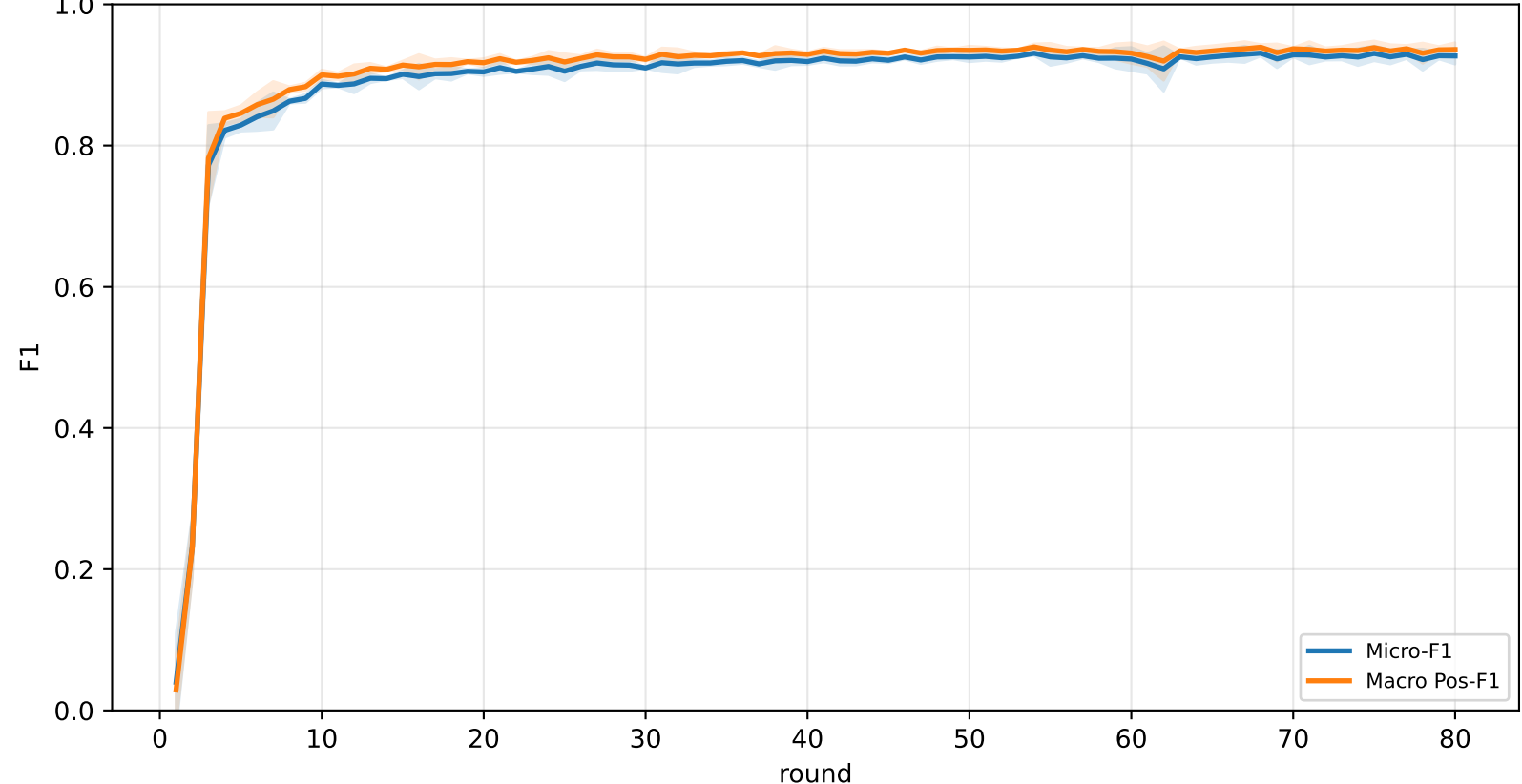
Pooled local centralized micro / macro F1



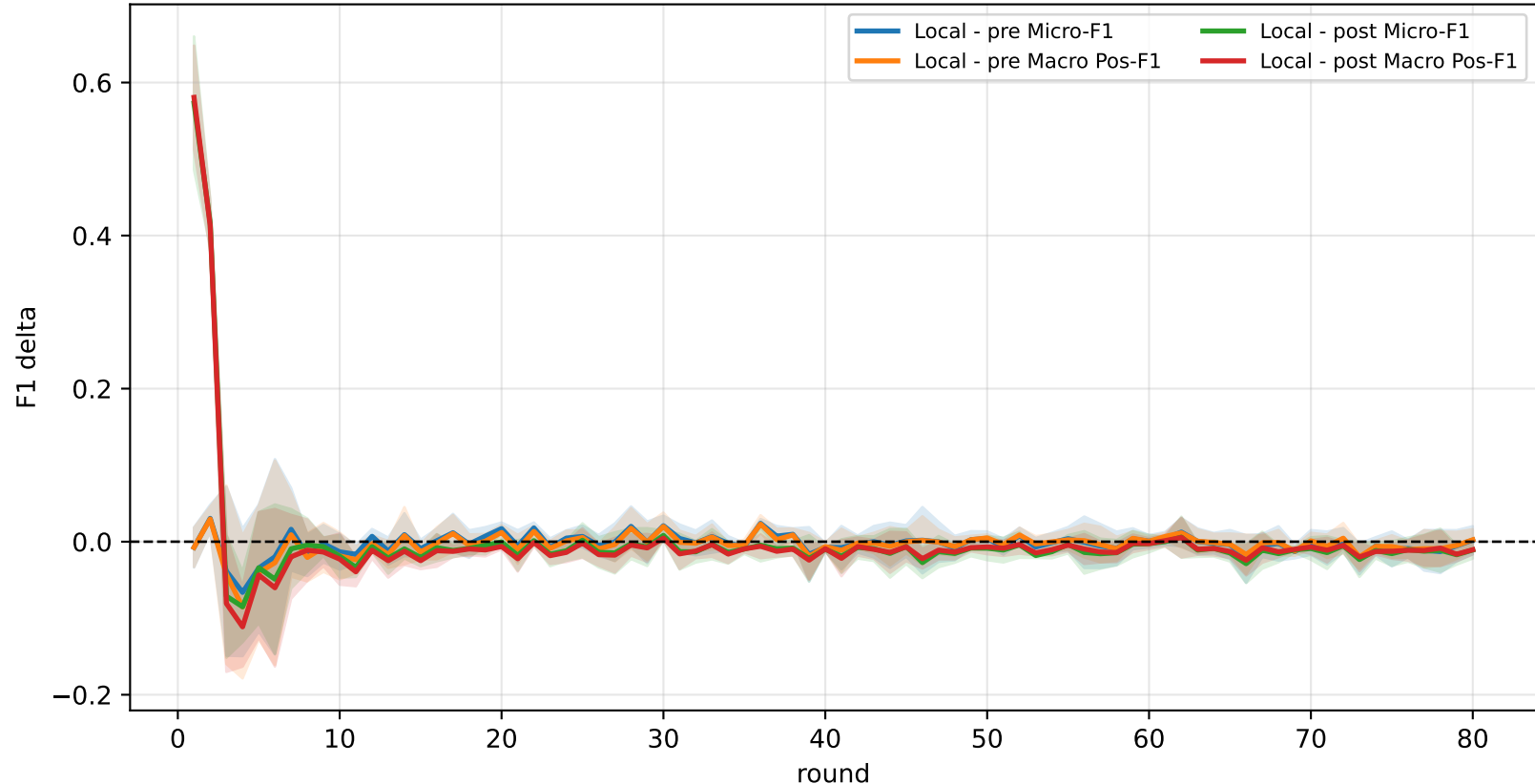
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

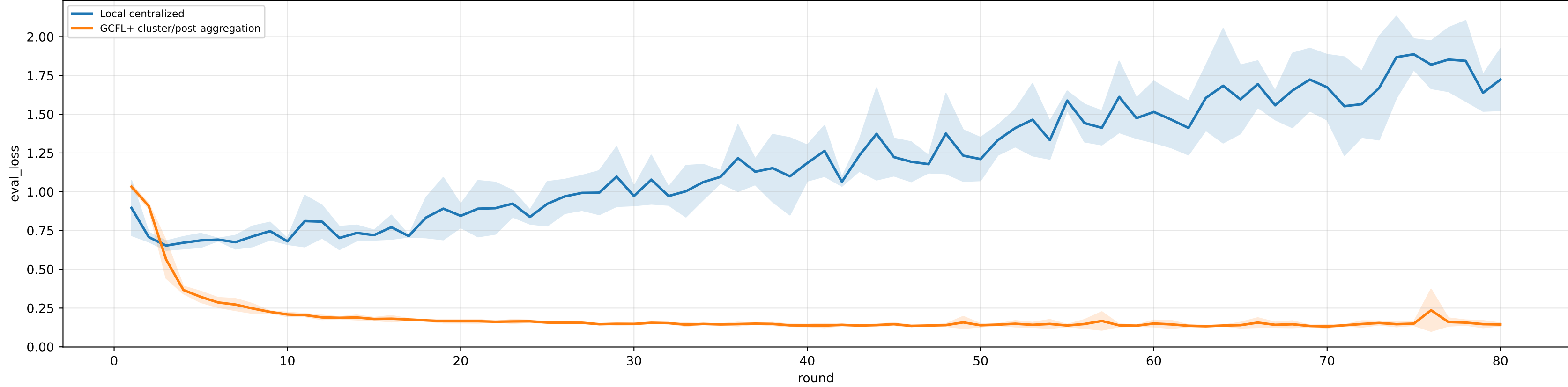


family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

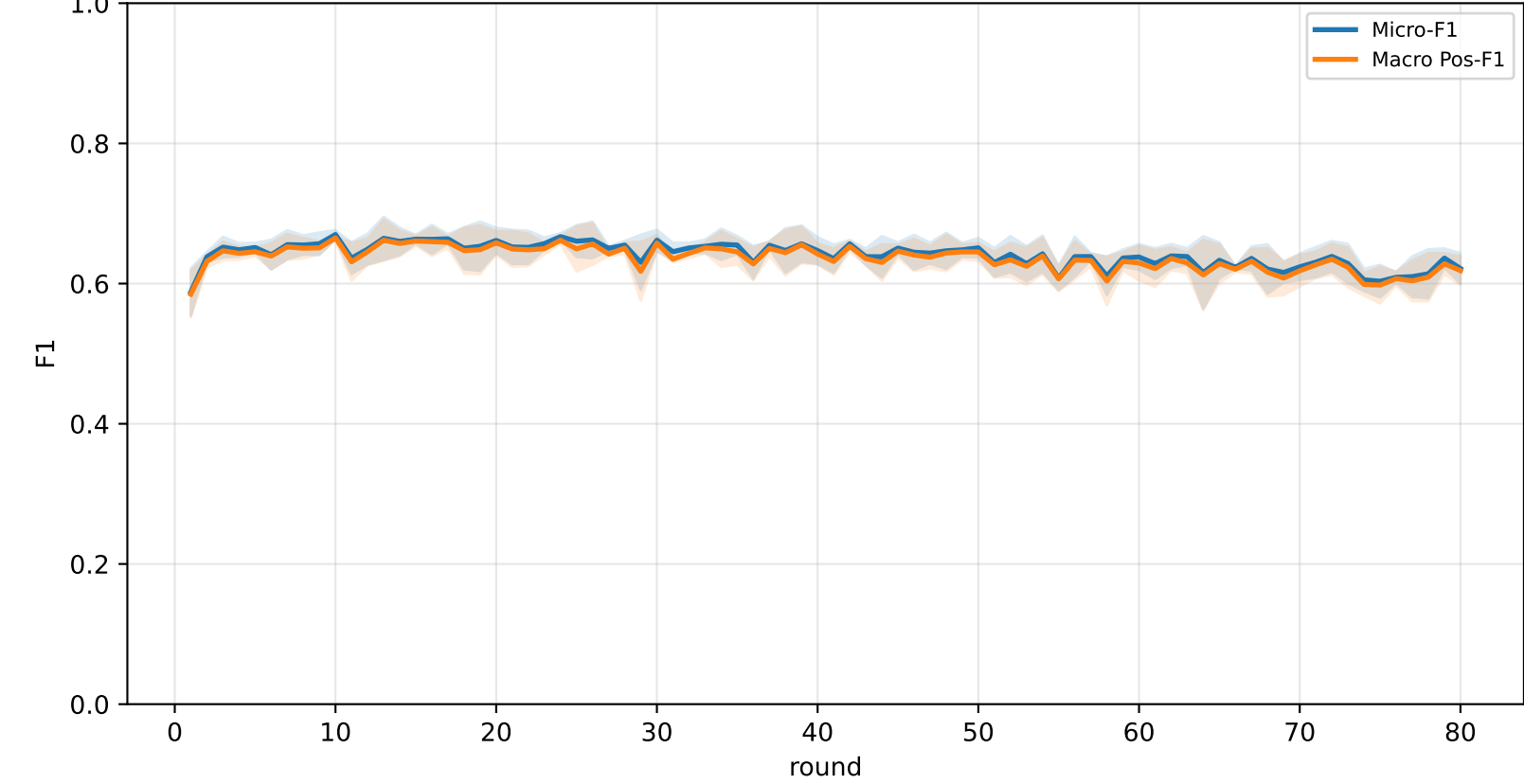
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.1% ± 1.1%	63.5% ± 1.0%	53.6% ± 1.8%	60.5% ± 0.7%	63.8% ± 0.9%	69.7% ± 0.8%	70.1% ± 1.9%
GCFL+	92.0% ± 0.3%	92.9% ± 0.3%	98.3% ± 0.4%	97.4% ± 0.9%	94.4% ± 1.5%	90.7% ± 1.7%	83.5% ± 2.3%

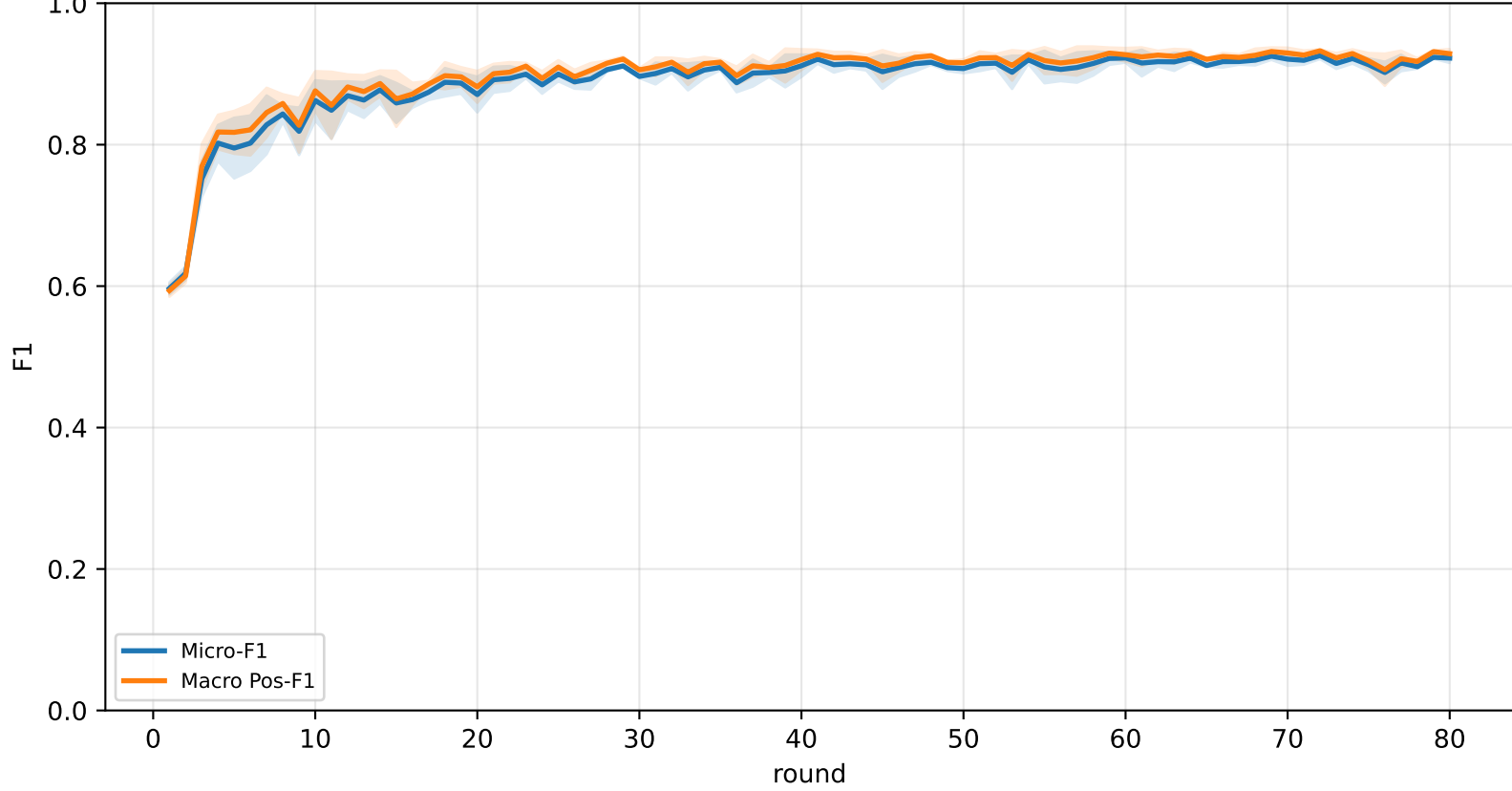
Validation loss comparison



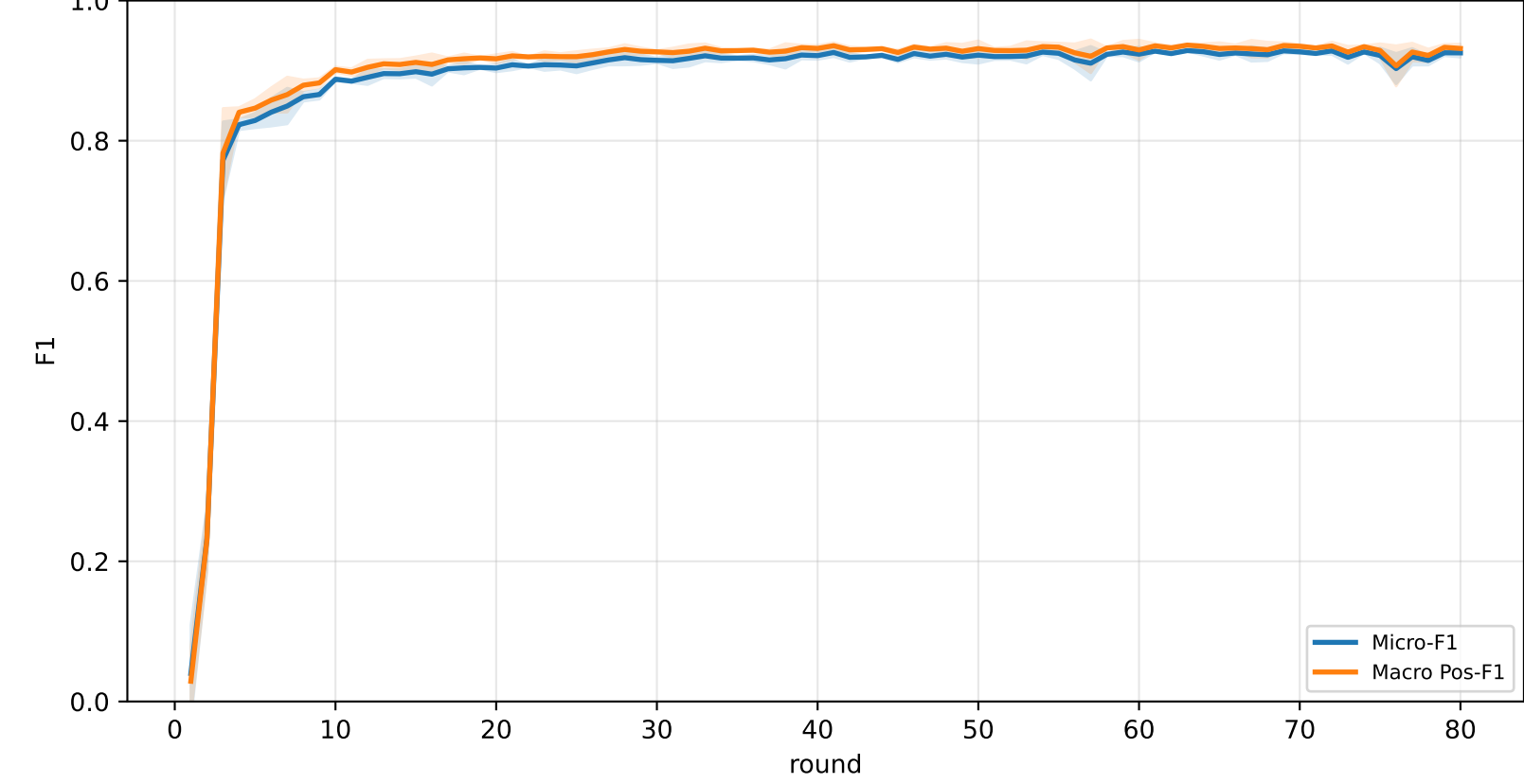
Pooled local centralized micro / macro F1



Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

